

RightShip Inspection Ship Questionnaire (RISQ)

Please send an email to risq@rightship.com if you have questions regarding the RISQ.



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Revision history table

Version Number	Revision Description	Revision Date
0	Original Release	01 May 2021
1	Key Changes Summary • Scope & Timing: Added requirement for the inspector to use an Australian ladder when accessing the cargo hold. • Section 1: Removed extra option from Q1.7. • Section 4: Added new question for life buoys & pyrotechnics. • Updated guides for lifeboat/raft launching and fire control plan. • Section 5: Updated ballast control panel requirement and Q5.9 guide. • Section 10: Updated working access arrangement requirements (Q10.15 & Q10.18). • Section 13: Revised Q13.24 for sounding pipes. • Added engine control console & steering gear system to Q13.33. • Section 14: Added steam lines to deck pipe list. New question for walkways and ladders. • Section 15: Updated inspection guide for Q15.2	18 October 2021
2.0	Key Changes Summary • Glossary: Updated definition of "Competent Person." • Q2.3, Q2.5, Q2.15: Added checks for emergency overrides and operational controls. • Q9.12, Q9.15: Hatch cover practices updated. • Q13.41: Included manual pump inspection. • Q4.21: Incorporated MLC/ILO standards. • Q13.17, Q15.5, Q15.9: Updated to include industry recommendations. • Q4.3, Q4.5: Guidance for confined space entry. • Q4.8: Tag & Lockout procedures added. • Q4.14: Emergency power sources included. • Q4.37, Q4.39: Ladder safety and pilot ladder requirements updated. • Q8.8: Added cargo damage checks. • Q8.9: Risk of partially open hatch covers emphasized. • Q10.6: Clear marking of tools following technical requirements. • Multiple questions (e.g., Q14.12, Q14.15): New instructions for inspectors added. • Q13.14, Q13.17: Updated to reflect manufacturer's maintenance guidelines. • Q4.43: Added guidelines for onboarding new personnel.	01 June 2022
3.0	Key Changes Summary • Added Code of Conduct to reduce bribery; replaced "non-conformity/N-C" with "Finding." • Included class survey status in Q2.1; moved MLC-related Q2.16 to health & welfare section. • Added radar familiarity guideline (Q3.3); removed VDR recordkeeping requirement (Q3.15); clarified BAWAS enablement (Q3.12). • Updated enclosed space risks (Q4.5), hot work permit info (Q4.7), incident report review (Q4.10), gas detector tests (Q4.16), and radio efficacy checks (Q4.30). • Added coal temp monitoring (Q8.18), fumigation certificate clarity (Q8.25), crane cabinet inspection (Q9.20), towing requirement (Q10.23). • Updated standards (12.7), securing steel plates (13.28), biofuel management (14.1), portable light voltage checks (14.10), and clarified Q14.12. • Integrated ITF requirements (15.1), potable water checks (15.5), asbestos reporting (15.13), added section 16 applicability instructions; updated reference list.	28 February 2023
3.1	Key Changes Summary • Glossary of Terms: Added new terminologies and clarified signage definitions. • Q1.12: Added inspection guide for selecting primary and secondary Class Societies for dual-class ships. • Q1.17: Inspectors must include EEDI and EEXI details from IEEC in comments. • New Section 7C: Ships using alternative fuel – Methanol (11 questions). • New Section 7D: Ships using alternative fuel – Ammonia (4 questions). • New Section 17: Ship-to-Ship Operations (10 questions). • Q2.9: Updated drug and alcohol policy requirement aligned with OCIMF standards. • Q3.3: Updated manoeuvring poster to include engine power limitations. • Q3.6: Clarified transfer of comms between ECDIS and radar. • Q4.5: Added enclosed space register and safety measures for shore-side personnel. • Q4.11: Added requirement for fall protection device inspection. • Q8.30: Added requirement for carrying battery-powered vehicles onboard. • Q8.9 & Q8.20: Updated conveyor belt monitoring and skirt board standards. • Q9.2: Revised hatch cover training and clarified operational procedures. • Q14.19: Updated securing arrangements for deck cargo.	01 May 2024
3.2	Please refer to Annex A for a detailed summary of the changes introduced in RISQ 3.2.	01 Nov 2025

Glossary Of Terms

Alternative Fuel:

"Alternative fuels" refer to fuel or power sources that have the potential to replace fossil fuels fully or partially in the energy supply chain for transportation. Examples of such alternative fuels include Liquefied Natural Gas (LNG), Liquefied Petroleum Gas (LPG), Methanol, Biofuel, Ammonia, and Hydrogen.

At Sea Operation:

Indicates offshore waters or partially sheltered waters where transfers may be undertaken between vessels underway or at anchor.

Best Practices:

Are approaches, procedures or tools that ship managers use to operate their vessels more safely, greener, smarter and above minimum compliance.

Broken Stowage:

Broken stowage is lost cargo space in the holds of a vessel due to the contour of the hull and / or the shape of the cargo. Dunnage, ladders, and stanchions are example of broken stowage. Broken stowage is shown as a percentage figure, which is an estimation of the space that will be lost.

Capesize:

Bulk carriers with deadweight of above 90,000 tonnes.

Cargo Unit:

Cargo unit means a vehicle, container, flat, pallet, portable tank, packaged unit, or any other entity, etc., and loading equipment, or any part thereof, which belongs to the ship but is not fixed to the ship (Assembly resolution A.489 (XI)).

Company:

The company means the owner of the ship, or any other organisation such as a ship manager or bareboat charterer who has assumed the responsibility for operation of the ship from the owner of the ship, including the duties and responsibilities imposed by the International Safety management (ISM) Code. This company would normally be the name recorded on the ship's Document of Compliance. May also referred to as vessel's manager (SOLAS) Chapter IX/1, 2020

Competent Authority:

A minister, government department or other authority empowered to issue regulations, orders or other instructions having the force of law.

Competent Person:

A person who has, through a combination of training, education and experience, acquired knowledge and skills enabling that person to correctly perform a specific task.

Dunnage:

Materials of various types, often timber or matting, placed among the cargo for separation, to increase the friction between the base of the cargo unit and the deck. This spreads the load of cargo unit across the deck, and hence provides ventilation; protection from damage and, with certain cargoes, provides space in which the fork lift's tynes truck may be inserted.

Ex- Rated Equipment:

Equipment that has been classified as safe for use in hazardous areas.

Flag Administration:

The maritime administration of a vessel's country of registry.

Flag State:

The Government of the nation whose flag a vessel is entitled to fly.

Fumigator-In-Charge:

A person designated by a fumigation company, government agency or appropriate authority.

Gravity-Based Self-Unloading Vessel:

A bulk carrier equipped with a self-Unloading system that includes hoppers hold, gravity gate, belt conveyor, elevating system and discharge boom. This type of vessel delivers free-flowing dry bulk commodities. This system has the ability to discharge on shore or to an offshore facility.

Handymax:

Bulk carriers with a deadweight of up to 60,000 tonnes.

Heavy-Lift:

There is no standard definition of a heavy lift in weight terms, although the cargo insurance policy may set a weight figure as part of the critical item criteria. However, for the purpose of this inspection assessment and questionnaire the term 'heavy lift' means a cargo that weighs more than 50 tonnes.

High Modulus Synthetic Fibre:

Manmade, continuous filament synthetic fibre with modulus in the range of 50-150 GPa.

Hybrid Self-Unloading Vessel:

A bulk carrier equipped with both deck cranes and belt conveyor on-board. The hybrid self-unloading system includes conventional cargo hold, deck crane, hopper, belt conveyor and discharge boom. This system has the ability to discharge on-shore or to an off-shore facility.

Industry Recommendations:

RightShip supports and endorses particular methods of working or procedure.

Line Design Break Force (LDBF):

Is the minimum force that a new, dry, spliced mooring line will break at when tested according to appendix B of Mooring Equipment Guidelines (MEG4). This is for all mooring line and tail materials, except those manufactured from nylon which are tested wet and spliced. This value is declared by the manufacturer on each line's mooring line certificate and is stated on a manufacturer's line data sheet.

Loose Gear:

An item of equipment that can be used to attach a load to a lifting appliance but does not form an integral part of the appliance or load. This includes a block, shackle, hook, swivel, connecting plate, ring, chain block or hoist, chain or overhauling weight.

Mental Health:

According to the World Health Organization, mental health is "a state of well-being in which every individual realises his or her own potential, can cope with the normal stresses of life, can work productively and fruitfully, and is able to make a contribution to her or his community."

Mental Disorders:

Mental disorders comprise a broad range of problems, with different symptoms. However, they are generally characterized by some combination of abnormal thoughts, emotions, behaviour and relationships with others. Examples are schizophrenia, depression, intellectual disabilities and disorders due to drug abuse. Most of these disorders can be successfully treated.

Out of Gauge:

The term out-of-gauge refers to any cargo that has dimensions that exceed the normal dimensions of a standard shipping container.

Panamax:

Bulk carriers of between 60,000 to 90,000 deadweight tonnes.

Planned Maintenance System (PMS):

The parts(s) of the company's Safety Management System (SMS) that address inspection, maintenance and repair of the ship.

Ro-Ro Ship:

A ship which has one or more decks (either closed or open), not normally subdivided in any way and generally running the entire length of the ship in which goods (packaged or in bulk; in / on road vehicles – including road tank vehicles – trailers, containers, pallets, demountable or portable tanks; or in / on similar cargo transport units or other receptacles) can be loaded or unloaded normally in a horizontal direction.

Safety Management System (SMS):

The Company's documented quality management system provided on board the vessel and in the office which addresses the requirements of the IMO ISM Code.

Ship to Ship Operation (STS):

Includes the Lightering/topping-off to and from Barges as well as the transshipment of cargoes at sea, involving specialized vessels with conveyors/ cranes or the use of offshore floating cranes.

STS Organiser:

An STS Organiser is a shore-based operator, any of the cargo owners or charterers, responsible for arranging an STS transfer operation. The Organiser may delegate his duties to an STS Service Provide.

Ship Design Minimum Breaking Load (Ship design MBL):

Is the stated value around which a ship's mooring system is designed and established at the ship design stage. The minimum breaking load of new, dry mooring lines for which a ship's mooring system is designed, to meet IACS standard environmental criteria restraint requirements. The ship design MBL is the core parameter against which all the other components of a ship's mooring system are sized and designed with defined tolerances.

Statutory Requirements:

Statutory requirements are those that are required by law. These requirements are non-negotiable and must be complied with. The following are agreed definitions for terms used within this questionnaire.

Signage*:

Danger Signs: These indicate a hazardous situation that, if not avoided, will result in fatal consequences or potentially severe injury. The signal word 'DANGER' should be reserved for the most extreme situations.

Warning Signs: This signifies a hazardous situation that, if not mitigated, could lead to severe injury or even potential fatality.

Caution Signs: These signify a potentially hazardous situation that, if not properly managed, could lead to minor or moderate injuries.

Notice Signs: These are utilized to convey information that is deemed significant, although it does not pertain to hazardous conditions.

Sub-Freezing Temperature:

Colder than the temperature at which water freezes (i.e. colder than 32°F or 0°C)

Working Load Limit:

The maximum load that a mooring line should be subjected to in operational service, calculated from the standard environmental criteria. The WLL is expressed as a percentage of ship design MBL and should be used as a limiting value in both ship design and operational mooring analyses. During operation, the WLL should not be exceeded.

In the same way that SWL is a limit for fixed equipment, the WLL value is used as a limit with the standard environmental criteria and mooring layout when establishing mooring system designs. Steel wire ropes have a WLL of 55% of the ship design MBL and all other cordage (synthetic) have a WLL of 50% of the ship design MBL.

Lead Inspector:

In a dual inspection format, the lead inspector is one of the nominated inspectors responsible for coordinating the inspection process. This includes determining how the RISQ sections are distributed between inspectors and selecting which parts of the inspection each inspector will conduct. The lead inspector oversees the execution of the inspection and ensures consistency, accuracy, and adherence to RightShip's inspection protocols.

Co-Inspector:

The co-inspector is the second inspector participating in the dual inspection. This individual conducts the assigned portions of the inspection as designated by the lead inspector, based on the nominated sections of the RISQ. The co-inspector supports the inspection process and contributes to the overall assessment in accordance with the lead inspector's direction.

**At present, there is no universally accepted standard for marine safety signage. A variety of signs have been introduced globally, and numerous vessels indicate potential risks through painted warning notices or stencils. RightShip strongly urges companies to carefully consider the definitions of these signs to guarantee their appropriate usage.*

Abbreviations

AFV:	Alternative Fuel Vehicles
AIS:	Automatic Identification System
AMSA:	Australian Maritime Safety Authority
ABMP	Ammonia Bunkering Management Plan
AEGL	Acute Exposure Guideline Level
ARMS	Ammonia Risk Mitigation (Management) System
BAC:	Blood Alcohol Concentration
BEV:	Battery Electric Vehicles
BFO:	Bunkering Facility Organisation
BLU Code:	The Code of Practice for the Safe Loading and Unloading of Bulk Carriers
BNWAS:	Bridge Navigational Watch Alarm System
BWM:	Ballast Water Management
BYOD:	Bring Your Own Device
BAC:	Break Away Coupling
BCOs	Ballast Control Operators
BLEVE	Boiling Liquid Expanding Vapour Explosion
BMP	Bunker Management Plan
BOG	Boil Of Gas
BS	Bunker Station
CAP:	Condition Assessment Program
CATZOC:	Category Zone of Confidence
CBA:	Collective Bargaining Agreements
CBM:	Condition Based Maintenance
CBO:	Condition Based Overhaul
CBT:	Computer Based Training
CCTV:	Closed-Circuit Television
CII:	Carbon Intensity Indicator
CMS:	Continuous Machinery Survey
CoC:	Confirmation of Compliance
CoP:	Certificate of Proficiency
CPA:	Closest Point of Approach
CPP:	Controllable Pitch Propeller
CRA:	Certificate of Receipt of Application
CSM:	Cargo Securing Manual
CSO:	Company's Security Officer
CSS Code:	Cargo Stowage and Securing Code
CTF:	Coating Technical File
DG:	Dangerous Good
DGNSS:	Differential Global Navigation Satellite System
DRI:	Direct Reduced Iron
DSC:	Digital Selective Calling

DUKC:	Dynamic Under Keel Clearance
ECA:	Emission Control Area
ECDIS:	Electronic Chart Display and Information System
EEBD:	Emergency Escape Breathing Devices
EGCS:	Exhaust Gas Cleaning System
ENC:	Electronic Navigational Charts
EPIRB:	Emergency Position Indicating Radio Beacon
ESD:	Emergency shutdown
ERS:	Emergency Release System
ERC:	Emergency Release Coupling
ETB:	Emergency Towing Booklet
EV:	Electric Vehicle
ELINCS	European List of Notified Chemical Substances
FML:	Flow Moisture Limit
FOSFA:	Federation of Oils, Seeds and Fat Associations
FPR	Fuel Preparation Room
GAFTA:	Grain and Feed Trade Association
GVU	Gas Valve Unit
GMDSS:	Global Maritime Distress and Safety System
GNSS:	Global Navigation Satellite System
GPS:	Global Positioning System
GRB:	Garbage Record Book
HAZOP:	Hazard and Operability Analysis
HDOP:	Horizontal Dilution of Precision
HIMP:	Hull Inspection and Maintenance Program
HLS:	Helicopter Landing Site
HME:	Harmful to the Marine Environment
HMSF:	High Modulus Synthetic Fibre
IACS:	International Association of Classification Societies
IAMSAR:	International Aeronautical and Maritime Search and Rescue
IGC	International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk
IAPH:	International Association of Ports and Harbors
ICS:	International Chamber of Shipping
IEC:	International Electro- technical Commission.
IEE:	International Energy Efficiency
IEEC	International Energy Efficiency Certificate
IGF Code:	The International Code of Safety for Ships using Gases or other Low-flashpoint Fuels
IHO:	International Hydrographic Organization
ILO:	International Labour Organization
IMDG Code:	International Maritime Dangerous Goods Code

IMFO:	International Maritime Fumigation Organisation
IMO:	International Maritime Organisation
IMO DCS:	IMO Data Collection System
IMSBC:	International Maritime Solid Bulk Cargoes
IOPPC:	International Oil Pollution Prevention Certificate
IS Code:	International Code on Intact Stability
ISM Code:	International Safety Management Code
ISPS:	International Ship and Port Facility Security
JPO:	Joint Plan of Operation
LDBF:	Line Design Break Force
LMP:	Line Management Plan
LNG:	Liquefied Natural Gas
LOTOTO:	Lock-Out, Tag-Out, Try-Out
LSA:	International Life-Saving Appliance
MARPOL:	The International Convention for the Prevention of Pollution from Ships, 1973 as modified by the Protocol of 1978
MBL:	Minimum Breaking Load
MEG4:	Mooring Equipment Guidelines Edition 4
MFAG:	Medical First Aid Guide for Use in Accidents Involving Dangerous Goods
MHB:	Material Hazardous only in Bulk
MLC:	Maritime Labour Convention
MMSI:	Maritime Mobile Service Identity
MPX:	Master Pilot Exchange
MSL:	Maximum Securing Load
MOC	Management Of Change
MSDS:	Material Safety Data Sheet
NATO:	North Atlantic Treaty Organization
NOx:	Nitrogen Oxides
OCIMF:	Oil Companies International Maritime Forum
OCM:	Oil Content Meter/Monitor
OHS:	Occupational Health and Safety
OMM:	Operating and Maintenance Manual
OOG:	Out of Gauge
OOW:	Officer of the Watch
OWS:	Oily Water Separator
P&I Club:	Protection and Indemnity Club
PFSOs:	Port Facility Security Officers
PIC:	Person in Charge
PMS:	Planned Maintenance System
PPE:	Personal Protective Equipment
PRVs:	Pressure Relief Valves

PPU:	Power Pack Unit
PTB:	Personal Transfer Basket
PWOM:	Polar Water Operation Manual
QRA	Quantitative Risk Assessment
RCDS:	Raster Chart Display System
RPE:	Respiratory Protective Equipment
SART:	Search and Rescue Transponder
SCAMIN:	Scale Minimum
SCBA:	Self-Contained Breathing Apparatus
SCR:	Selective Catalytic Reduction
SDMBL:	Ship Design MBL
SDS:	Safety Data Sheet
SEA:	Seafarers' Employment Agreement
SEEMP:	Ship Energy Efficiency Management Plan
SMS:	Safety Management System
SOLAS:	International Convention for the Safety of Life at Sea.
SOPEP:	Shipboard Oil Pollution Emergency Plan
SOx:	Sulphur Oxides
SRIM:	Security Related Information to Mariners
SoC:	Statement of Compliance
SSO:	Ship Security Officer
SSP:	Ship Security Plan
STCW:	Standards of Training, Certification and Watch keeping
SWBM:	Still Water Bending Moment
SWL:	Safe Working Load
SWSF:	Still Water Shear Force
T&P NMs:	Temporary and Preliminary Notices to Mariners
TCPA:	Time to Closest Point of Approach
TDBF:	Tail Design Break Force
TMC:	Transmitting Magnetic Compass
TML:	Transportable Moisture Limit
UKC:	Under Keel Clearance
UKHO:	United Kingdom Hydrographic Office
UMS:	Unattended Machinery Space
V/V:	Volume of fumigant per total volume of gas
VDR:	Voyage Data Recorder
VGM:	Verified Gross Mass
WF:	Solids that evolve flammable gas when wet
WIDS:	Water Ingress Detector Systems
WLL:	Working Load Limit
XTC:	Cross-Track Corridors

Objective of the RightShip dry inspection

The objective of the RightShip inspection is to assess the quality of ships, verify the familiarity and compliance of ship's crew with the safety, statutory requirements, industry recommendations, best practices and required items within the RightShip Inspection Ship Questionnaire. The outcome of the RightShip Dry Inspection will reflect the actual condition and standard of operation of the vessel at the time of inspection.

The RightShip Dry Inspection allows the identification and assessment of risk that the use of vessel may transfer to our clients and external stakeholders.

RightShip inspection ship questionnaire

The RightShip Inspection Ship Questionnaire covers a series of questions related to safety, environmental protection, maintenance, industry recommendations and good practices. For inspection purposes the vessels covered in RightShip inspection are grouped into four categories:

1. Bulk carriers that are carrying solid bulk cargoes other than grain
2. Bulk carriers that are carrying grain cargoes
3. General cargo ships that are carrying general and/or container cargoes
4. Container ships that are carrying container cargoes

There are two options for the completion of an inspection using the Rightship Ship Inspection Questionnaire. The first is in the traditional approach where all questions are answered during a physical inspection. The second is a hybrid approach where those questions denoted with the letter (M) can be completed on review of documentation provided by the vessel managers in advance of a physical inspection, with the remaining questions denoted with a letter (V) completed during a subsequent shortened physical inspection of the vessel. For either approach all questions must be answered.

How to answer the RightShip ship inspection questionnaire

The questions in each section may be accompanied by a "Guide to Inspection". The Guide to Inspection assists the vessel's manager in preparing the vessel for inspection and the inspector in answering the questions and completing the inspection report.

The questions in each section must be answered by the inspector. The inspector must answer the question on the basis of the "Guide to Inspection" and any reference sources.

The inspector has an option to select one of four (4) responses for each question:

- ☐ YES: The "Yes" box should be checked when inspector, on the basis of the "Guide to Inspection" and other industry references, concludes that the answer to the question is "Yes".
- ☐ NO: The "No" box should be checked when inspector, on the basis of the "Guide to Inspection" and other industry references concludes that the answer to question is "No".
- ☐ N/A (Not Applicable): The "N/A" box should be checked when the subject matter in question does not apply to the vessel. If the inspector selects N/A on the basis of his / her judgment, a supplementary comment should be added, and they should state the reasons the "N/A" box was selected.
- ☐ N/V (Not Viewed): The "Not Viewed" box should be checked if the subject matter in question was not checked by the inspector. A supplementary comment should be added, and they should state the reasons the N/V box was selected.

Finding is a failure to meet a requirement which is a need, expectation, or obligation. It can be a deficiency in characteristic, documentation, or procedure, (including work practice) through finding physical defects, test failures, incorrect or inadequate documentation and maintenance,

a deviation from testing and inspection, or non-compliance with the industry good practices and recommendations. A Finding means an observed situation where objective evidence indicates the non-fulfilment of a specified requirement.

- > The inspector must record a Finding in the Finding box (the "Action" box in the inspection template) when the answer "No" is checked. The Finding must specify and explain the reason the negative response is made. The inspector must not include the solution for fixing the Finding.
- > The inspector shall make supplementary comments in the "Comments" section, when required by the "Guideline to Inspection" or when an additional clarification is required to understand the matter related to a specific question.
- > The inspector must not check the "Yes" box when the inspector's comments contain negative elements. When comments contain negative elements, the "No" box must be checked. The inspector must respond to all questions and each question must have one of its check boxes marked. The inspection report will be rejected and returned if all questions have not been answered. When the "Yes" box is checked, the inspector may amplify the answer to the question by adding positive comments in the comments box. Objective evidence must be used by the inspector when answering the question. The word of ship staff alone shall not be considered as sufficient evidence when answering a question. Crew's familiarity with a task and ability to demonstrate a task is considered an objective evidence.
- > In the "Supplementary Comments" section at the end of each section, the inspector may add comments related to the section or a subject related to the section that has not been covered by the question. All dates should be entered in the format DD/MMM/YYYY.

Inspection procedure

The inspector must conduct the RightShip Dry Inspection as per the following mandatory requirements.

Boarding the vessel and opening meeting

The Inspector must show a valid identification card upon boarding the vessel.

The inspector must always wear appropriate PPE and must set a good example in all respects by maintaining the highest standard of ethical behaviour throughout the inspection.

The inspector must have an opening meeting with the Master or Master's representative. The inspector is required to introduce themselves and outline the objectives, requirements, and plan for the inspection. The inspector and the Master or Master's representative should agree on the sequence for the inspection.

The sequence of inspection must not affect the safe operation of the ship or effect the rest hours of the ship's personnel.

The minimum PPE for Rightship inspectors includes, safety shoes, overalls, safety helmet, hearing protection, gloves, safety glasses and the Rightship Hi Vis vest.

During inspection

The inspector must always be accompanied by a qualified and responsible Officer.

The actual tank and hold access for physical assessment of the condition of ballast tanks, void spaces and cargo holds can be made only with the authority of the Master and provided that port and terminal regulations allow. In all cases, the enclosed space entry procedures set out in Resolution A.1050 (27) (Revised Recommendations for Entering Enclosed Spaces Aboard Ships) must be strictly observed.

The inspector must be an observer throughout the inspection and must not be involved or interfere with the operation and/or operate any items on board the ship. However, the inspector must notify the ship's staff when observing any unsafe conditions or operations being carried out.

The inspection shall not take place at night unless it has been authorised by RightShip and agreed by the vessel's manager. If the inspector notes any Finding, then they must be pointed out and discussed at the time and the location, with the person accompanying the inspector. In this way, the nature of the Finding can be more easily understood by the ship's staff and this will reduce the duration of the closing meeting.

Closing meeting

The inspector must not provide any advice or suggestion on how to rectify any Findings. The inspector must not give any verbal indication about the overall inspection result.

A copy of the list of Findings must be provided to the Master. The inspector must record any Findings, on which action was taken to rectify while he or she was on board.

Type, Scope and guide to timing of inspection

RightShip inspections are conducted in three formats:

1. Standard Inspection

A comprehensive onboard inspection carried out by a single inspector. The process is expected to take between 10 and 12 hours to complete.

2. Hybrid Inspection

The vessel manager must submit all required documentation at least 72 hours before the inspector boards the vessel. Provided the documentation is complete, submitted on time and the vessel is physically prepared for inspection in accordance with RISQ, the inspection can be concluded in 8-10 hours.

3. Dual Inspection

Two inspectors are assigned to the vessel. The lead inspector is responsible for dividing the inspection tasks between them. This format is designed to be completed within 6 to 7 hours.

Note: The durations stated above refer to the net inspection time and do not include interruptions. Examples of such interruptions may include, but are not limited to, meal breaks, Port State Control (PSC) inspections, and Flag State inspections.

Completion of the list of Findings and inspection report

On completion of the inspection, the list of Findings shall be submitted to RightShip immediately after the inspection. The inspector must then complete the inspection report and send the completed ship inspection report to the RightShip within 72 hours of departing the vessel. If for any reason the 72 hours deadline cannot be achieved, the inspector must advise RightShip of the reason why and when the report can be expected to be sent.

The inspector must avoid subjective comments and Findings which are based on assumptions, beliefs and opinions or influenced by emotions or personal feelings. The inspector's Findings and comments must be based on facts, observations, and valid references within the industry.

If the inspector made any subjective comments or expressed an opinion, they must give the factual basis and specific reasons why such a comment / opinion was recorded.

Vessel's Manager

The vessel's manager is responsible for ensuring the records relating to the officers on board the vessel (Deck and Engineers Matrix) is up-to-date and send to RightShip prior to inspection.

The time taken for inspection can be greatly reduced by the state of preparedness of the ship. The latest edition of the RightShip Inspection Ship Questionnaire should be on board and, as applicable, the RightShip Questionnaire should have been completed. To expedite the inspection, the vessel's Manager may consider having a representative on board during the inspection. However, any representative shall not interfere with the inspection unless there are safety concerns.

Electronic certificates

Where the vessel is issued with electronic certificates, Rightship recommends the vessel's manager provide Inspectors a temporary access to their online electronic certificate system at least two(2) days before the inspection. This will facilitate the inspectors to minimise the time of inspection on board the vessel.

Remote review of digital documentation

Rightship may ask for remote review of digital documentation to minimise the time of inspection on board the vessel. If agreeable by the vessel's manager, the additional digital documents must be securely sent to the appointed inspector 72 hours prior to inspection.

Root Cause Analysis

The vessel's manager may send written comments relating to the report, to RightShip.

The vessel's manager is required to provide a meaningful root-cause analysis including, corrective actions and sustainable long term preventative actions for each Finding within 15 days of physical inspection of the vessel.

Please note that the inspection process cannot be completed until a satisfactory response has been received. If a satisfactory response is not received within 15 days, the inspection will be invalid.

Assessment of the vessel's management begins when the physical inspection of the ship is completed. The quality of the vessel's management system will be evaluated by the quality of the replies that the vessel's manager makes to the Findings recorded during the RightShip inspection. The inspection outcome will be determined by the RightShip Dry inspections team.

Maintaining a safe and high-quality fleet of vessels that not only complies with the statutory requirements but also complies with industry good practices, is operated by well-trained ship's crew and has an effective management system, creates a distinctive competitive advantage for the ship owner.

Engaging with the Inspector – Code of Conduct

RightShip Inspectors adhere to the RightShip code of conduct and will uphold professionalism and integrity while carrying out the inspection. Any attempts to coerce or offer bribes at any point during the inspection will not be tolerated. Please note that RightShip Inspectors will report any cases of attempted coercion or attempts to manipulate Inspection outcomes. In these cases, the Inspection will likely be deemed unacceptable.

RightShip Inspection Enhanced Training Requirements

RightShip has implemented a series of enhanced training requirements under its Inspection Program, aimed at complementing the mandatory training crew members receive through their STCW certification.

Training may be delivered through various formats, including classroom-based instruction, online learning modules, and computer-based training. Where an IMO Model Course is available, any related training must be structured, developed, and delivered in accordance with the relevant IMO guidelines.

In the absence of an IMO Model Course, training must be developed in close collaboration with Subject Matter Experts (SME) to ensure it is formally structured and directly aligned with the specific instructional area. Any course completion certificate issued must explicitly reference compliance with the applicable syllabus.

All formal training must include:

- > A clearly defined syllabus
- > Explicit learning objectives
- > An assessment component
- > Issuance of a certificate upon successful completion

Additionally, instructors must have relevant experience in training delivery and hold a recognised 'Train the Trainer' certification.

A Subject Matter Expert (SME) is a professional with deep and specialised knowledge in a specific scientific discipline, function, process, technology, machine, material, or piece of equipment.

At RightShip, SMEs are recognised as playing a critical role in improving the quality, relevance, and effectiveness of training programs.

Section 1: General Information

- 1.1 Vessel's name as it appears on the Certificate of xxx
- 1.2 Vessel's IMO Number: (M)
- 1.3 Flag: (M)
- 1.4 Date the vessel was delivered: (M)

Guide to Inspection

Date of delivery can be found either in form A of the International Oil Pollution Prevention (IOPP) Certificate or Safety Construction Certificate.

- 1.5 This question has been removed from the current version of the document.

- 1.6 Maximum assigned deadweight (metric tonnes):

- 1.7 Vessel type: (M)

- ☐ Bulk carrier - carrying solid bulk cargos other than grain
- ☐ Bulk carrier - carrying grain cargos
- ☐ General cargo ship - carrying general and/or container cargos
- ☐ Container ship - carrying container cargos

- 1.8 Hull type: (M)

- ☐ Double Bottom-Single Skin Side
- ☐ Double Hull

- 1.9 Vessel's operation at the time of inspection:

- ☐ Loading
- ☐ Discharging
- ☐ Bunkering
- ☐ At anchor
- ☐ Idle
- ☐ In dry dock / shipyard
- ☐ At sea/river transit
- ☐ Repairs afloat

- 1.10 Name of cargo being handled:

N/A ☐

Guide to Inspection

This question should be answered N/A when vessel is in ballast condition.
Refer to shipper declaration and/or dangerous goods declaration to determine the correct name of the cargo.

- 1.11 Details of Port State Control inspection history for the last 12 months: (M)

N/A ☐

Guide to Inspection

Inspector shall record the summary of significant deficiencies and, if the vessel was detained, detainable deficiencies. The records of Port State inspection should be retained on board for at least two years.
If the vessel's name and/or the vessel's manager changed after the inspection, record the name of vessel and/or vessel's manager at the time of inspection.

1.12 Name of Classification Society: (M)

- ☐ IACS- Classification Society
- ☐ Non-IACS- Classification Society

Guide to Inspection

If the vessel has dual class, select the name of the classification society issuing the statutory certificates from the drop-down list and the name of the second society in comments.

If the vessel has changed class within the past 6 months, record the previous classification society and the date of change as a comment.

1.13 Expiry date of Class Certificate: (M)**1.14 Date the last Special Survey was completed: (M)** N/A ☐**1.15 Date of last routine dry dock: (M)** N/A ☐**1.16 Date of unscheduled repair / and or last dry dock: (M)** N/A ☐**Guide to Inspection**

Record the reason for unscheduled repair and/or dry dock.

1.17.1 Attained EEDI: (M) N/A ☐**1.17.2 Carbon Intensity Indicator (CII): (M)** N/A ☐**Guide to Inspection**

The Carbon Intensity Indicator (CII) is a measure of how efficiently a ship transports goods or passengers and is given in grams of CO2 emitted per cargo-carrying capacity and nautical mile. The ship is then given an annual rating ranging from A to E, whereby the rating thresholds will become increasingly stringent towards 2030.

The CII applies to all cargo, RoPax and cruise ships above 5,000 GT. The yearly CII is calculated based on reported IMO DCS data and the ship is given a rating from A to E.

The Attained EEDI value is listed in Section 3.1 of the supplement to the International Energy Efficiency (IEE) Certificate.

The Attained CII and its resulting rating (A-E) are recorded on the Data Collection System (DCS) Statement of Compliance (SoC).

1.18 This question has been removed from the current version of the document.**1.19 Name of the DoC Manager: (M)****Guide to Inspection**

The name of the DoC Manager is recorded in the vessel's Document of Compliance.

1.20 Date the current DoC Manager took over the vessel:**Guide to Inspection**

The date of the vessel's manager taking over the vessel is recorded in the Continuous Synopsis Record.

1.21 Dates of last two visits of the vessel's manager: (M)

1st Visit: ☐N/A ☐2nd Visit: ☐N/A ☐

Guide to Inspection

Record the position of ship's manager attending i.e. Marine Superintendent, Engineer Superintendent or Naval Architect. The inspector should, under Q 4.3, record a finding if the vessel had not been inspected through a combination of marine and technical superintendent visits at regular intervals, as required by the inspection guide.

1.22 Standard/Hybrid Inspection ☐ Name of the Rightship inspector:Dual Inspection ☐ Name of lead inspector:

Name of Co-Inspector:

1.23 Port of inspection: (V)

1.24.1 Standard ☐ Date and time inspector arrived at the vessel: (V)Hybrid: ☐ Time spent reviewing documentation prior to boarding: (V)

Date and time inspector arrived at the vessel: (V)

Dual: ☐ Date and time inspectors arrived at the vessel:

1.24.2 Date and time of starting and completion of opening meeting: (V)

1.24.3 Date and time of start and completion of closing meeting: (V)

Guide to Inspection

If this was a hybrid inspection, the inspector shall record in the comments the reason for reviewing the documents on board, if applicable.

1.25 Date and time Rightship inspector left the vessel: (V)

Guide to Inspection

Record the arrival and departure time/dates for each session of the inspection (when the inspection was carried out in more than one session) or when the inspection was carried out by more than one inspector.

1.26 Total time taken for inspection: (V)

1.27 Date the inspection was completed: (V)

Guide to Inspection

Record the actual time of inspection and exclude suspension of inspection for any reason i.e., meals, PSC inspection etc.

1.28 Name of the ship's P&I club:(M)

- ☐ International Group of P&I
- ☐ Non-International Group of P&I

Guide to Inspection

The current list of P&I Clubs that are member of the International Group is provided below.

- > American Steamship Owners Mutual Protection and Indemnity Association, Inc.
- > American Club (Europe)
- > The Britannia Steam Ship Insurance Association Limited
- > The Britannia Steam Ship Insurance Association Europe
- > Gard P&I. (Bermuda) Limited
- > Assuranceforeningen Gard (Gjensidig)
- > The Japan Ship Owners' Mutual Protection & Indemnity Association
- > The London Steam-Ship Owners' Mutual Insurance Association Limited
- > The London P&I Insurance Company (Europe) Limited
- > NorthStandard Limited
- > NorthStandard EU DAC
- > The Standard Club UK Ltd
- > The Standard Club Asia Ltd
- > The Standard Club Ireland DAC
- > The Shipowners' Mutual Protection & Indemnity Association (Luxembourg)
- > Assuranceforeningen Skuld (Gjensidig)
- > Skuld Mutual Protection and Indemnity Association (Bermuda) Ltd.
- > The Steamship Mutual Underwriting Association (Bermuda) Limited
- > Steamship Mutual Underwriting Association Ltd.
- > Steamship Mutual Underwriting Association (Europe) Limited
- > Sveriges Ångfartygs Assurans Förening / The Swedish Club
- > The United Kingdom Mutual Steam Ship Assurance Association Limited
- > UK P&I (Bermuda)
- > UK P&I (UKNV)
- > The West of England Ship Owners Mutual Insurance Association (Luxembourg)

1.29.1 Is the vessel manager a subscriber to the Dry Bulk Centre of Excellence (DBCE)?

Yes ☐ No Knowledge of This ☐ No ☐

1.29.2 Has a DryBMS self-assessment been completed?

Yes ☐ No Knowledge of This ☐ No ☐

Guide to Inspection

The inspector is not required to request or review any documentation to validate the Master's response.

Section 2: Certification and personnel management

2.1 Is the latest Class Survey Status available, are all Statutory Certificates listed within it valid, is the vessel free of any overdue Conditions of Class, and are all classification and statutory surveys up to date?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should attach a PDF copy of the Class Survey Status (the version reviewed and evaluated during the inspection) to the inspection report.

A finding should be recorded in the following cases:

1. A short-term certificate has been issued due to any defect or damage to the ship's structure, machinery, or equipment.
2. The vessel has any overdue Conditions of Class.

In the comments section, include details if the vessel has any Condition of Class, Significant Recommendation, or Memorandum.

If the ship has been provided with a letter of dispensation by the Flag State, record the following in the comments:

1. Expiry date of the dispensation letter
2. Nature of the deficiency that led to the issuance of the letter
3. Corrective actions mandated by the Flag State to ensure compliance.

The class survey status shall be available on board and should be dated not more than seven (7) days prior to the date of the inspection. Record a Finding if an up-to-date class survey status was not available on board.

The Inspector should accept electronic certificates containing the features below:

1. Validity and consistency with the format and content required by the relevant international convention or instrument, as applicable
2. Protected from edits, modifications, or revisions other than those authorised by the issuer or the Administration
3. A unique tracking number, and
4. A printable and visible symbol that confirms the source of issuance

(GUIDELINES FOR THE USE OF ELECTRONIC CERTIFICATES, 2016)

The inspector may request the Master to demonstrate the validity of the electronic certificate following the instructions available on board the ship.

If the Master fails to demonstrate, to the satisfaction of the inspector, that an electronic certificate meets the requirements, the inspector shall record a Finding.

The IMSBC Code fitness certificate in accordance with IMSBC Code (2020 Edition) may be issued upon request from owners/shipbuilders on voluntary basis from 1 January 2020.

For cargoes listed in Table G1 (Cargo newly added and requirements on construction/equipment (IMSBC Code-4th amendment) as 'Group A and B' or 'Group B', IMSBC Code (2020 Edition) a fitness certificate will be issued in cases where ships comply with requirements in Table G1.

2.2 Has the vessel been provided with certificates of financial security for seafarers? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

From 18 January 2017, all ships which are subject to MLC have been required to carry and display on board two certificates confirming that financial security is in place for:

- (a) shipowners' liabilities for repatriation of crew, essential needs such as food, accommodation, medical care and up to four months' outstanding contractual wages and entitlements in the event of abandonment (Regulation 2.5, Standard A2.5.2 Paragraph 9)
 - (b) contractual payments for death or long-term disability due to an occupational injury, illness or hazard set out in the employment agreement or collective agreement (Regulation 4.2, Standard A4.2.1 paragraph 1(b))
- (FAQs: Maritime Labour Convention 2006 As Amended Financial Security Requirements - The Shipowners' Club, 2020)

2.3 Can all crew communicate effectively in the working language of the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record the common language and the level of English proficiency of the crew on board the vessel.

On all ships, to ensure effective crew performance in safety matters, a working language shall be established and recorded in the ship's logbook. The company, as defined in regulation IX/1, or the Master, as appropriate, shall determine the appropriate working language. Each seafarer shall be required to understand and, where appropriate, give orders and instructions and to report back in that language. If the working language is not an official language of the State whose Flag the ship is entitled to fly, all plans and lists required to be posted shall include a translation into the working language.

On ships to which SOLAS chapter I applies, English must be used on the bridge as the working language for bridge-to-bridge and bridge-to-shore safety communications as well as for communications on board between the pilot and bridge watchkeeping personnel, unless those directly involved in the communication speak a common language other than English.

(SOLAS 74, 2020)

2.4 Is the vessel's manning in compliance with the Safe Manning Certificate? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments the actual and required manning of the vessel.

Minimum safe manning is the level of manning that will ensure that a ship is sufficiently, effectively, and efficiently manned to provide safety and security of the ship, safe navigation and operations at sea, safe operations in port, prevention of human injury or loss of life, the avoidance of damage to the marine environment and property, and to ensure the welfare and health of seafarers through the avoidance of fatigue.

Except in ships of limited size or propulsion power (which are not quantified), the determination of the minimum safe manning level should also consider the provision of qualified officers to ensure that it is not necessary for the Master or Chief Engineer to keep regular watches by adopting a three-watch system.

(PRINCIPLES OF SAFE MANNING, IMO Resolution 1047(27), 2000)

Inspectors should review the crew list and, considering the level of operation at sea and port, assess if there are enough personnel on board to fulfil the following principles of safe manning:

- > Maintain safe navigation by adequate manning of bridge throughout the passage.
- > Mooring, tending mooring at port and unmooring the ship safely.
- > Effective performance of cargo operation to ensure safe carriage of cargo during transit.
- > Performance of on-board functions such as drills, ship security issues, equipment maintenance.
- > Manning levels should be such as to ensure that the time and place available for taking rest periods are appropriate for achieving a good quality of rest.

2.5 Do all personnel maintain rest period/work hours and are the rest hours in compliance with STCW or MLC requirements? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if:

- > There are two or more consecutive violations by any seafarer on-board in any 30 day period.
- > The vessel's manager has not been informed at least monthly of compliance levels on board.
- > The work hour records are not to ILO format - Inspector should refer to the IMO/ILO guideline "Guidelines for the Development of Tables of Seafarers' Shipboard Working Arrangements and Formats of Records of Seafarers' Hours of Work or Hours of Rest".

Each Administration shall, for the purpose of preventing fatigue:

- > Establish and enforce rest periods for watchkeeping personnel and those whose duties involve designated safety, security and prevention of pollution duties in accordance with the provisions of section A-VIII/1 of the STCW Code.
- > All persons who are assigned duty as officer in charge of a watch or as a rating forming part of a watch and those whose duties involve designated safety, prevention of pollutions and security duties shall be provided with a rest period of not less than:

1. minimum of 10 hours of rest in any 24-hour period; and
 2. 77 hours in any 7-day period.
 3. The hours of rest may be divided into no more than two periods, one of which shall be at least 6 hours in length, and the intervals between consecutive periods of rest shall not exceed 14 hours.
- > Parties may allow exceptions from the required periods of rest in paragraphs 2 and 3 above provided that the rest period is not less than 70 hours in any 7-day period.

Exception from the weekly rest period provided for in paragraph 2 shall not be allowed for more than two consecutive weeks. The intervals between the two periods of exceptions onboard shall not be less than twice the duration of the exception.

The hours of rest provided for in paragraph 1 may be divided into no more than three periods, one of which shall be at least six hours in length, and neither of the other two shall be less than one hour in length. The intervals between consecutive periods of rest shall not exceed 14 hours. Exceptions shall not extend beyond two 24-hour periods in any 7-day period. Exceptions shall, as far as possible, take into account the guidance regarding prevention of fatigue in section B-VIII/1. (Regulation VIII/1 Fitness for duty STCW 2017)

2.6 Has the Master been provided with relevant ship handling training? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

This question applies to large ships or vessels with considerable length or deadweight. If the Master has accumulated 36 months or more of sea service in rank and has served on the same type of vessel for at least the past 12 months, the response to this question should be "Not Applicable".

Masters with less than 36 months of sea service in rank, or those newly promoted, should complete a ship handling training course aligned with the objectives of STCW Code B-V/a.

- > It is important that Masters and Chief Officer have had relevant experience and training before assuming duties of Master or Chief Officer of large ships or ships having unusual manoeuvring and handling characteristics significantly different from those in which they have recently served. Such characteristics will generally be found in ships which are of considerable deadweight or length or of special design or of high speed.
- > Before initially assuming command of one of the ships referred to above, the prospective Master should have sufficient and appropriate general experience as Master or Chief Officer, and either:
 - Have sufficient and appropriate experience manoeuvring the same ship under supervision or in manoeuvring a ship having similar characteristics or,
 - Have attended an approved ship handling simulator course on an installation capable of simulating the manoeuvring characteristics of such a ship.

Training may be conducted:

- > Onboard company vessels under formal supervision, documented through a structured competency development process, or
- > At an approved shore-based training facility.

When training is conducted during a parallel voyage prior to assuming command of a vessel with significantly different handling characteristics, the vessel manager should:

- > Define the required training activities,
- > Ensure these activities are completed by the incoming Master, and
- > Have them assessed and documented by the current Master or a suitably qualified Superintendent before the handover of command.

The vessel manager is responsible for evaluating the handling characteristics of all vessel types under their management and identifying when additional ship handling training is necessary—particularly when a Master is reassigned to a vessel with significantly different characteristics from those previously served on.

Evaluation of vessel handling characteristics should consider:

- > Number and type of propellers/drives,
- > Number and type of rudders,
- > Number and type of thrusters,
- > And not be based solely on vessel size.

Where shore-based ship handling training is required, the validity period of the course should be defined, specifying the maximum time before assignment to a vessel type after which the training should be refreshed or repeated.

RightShip defines a vessel as 'large' or having 'considerable deadweight or length' if it exceeds either:

- > 270 meters in length overall (LOA), or
- > 140,000 deadweight tons (DWT).

2.7 Have officers and ratings responsible for cargo handling on ships carrying dangerous and hazardous substances in solid form in bulk, undergone formal training? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the ship is not carrying solid bulk cargo, the question should be answered N/A.

The course should be conducted according to IMO Model Course 1.45, titled 'Safe Handling and Transport of Solid Bulk Cargoes'. The objectives of the course include ensuring that trainees understand the potential dangers that may occur during the loading, carrying, and unloading of solid bulk cargoes. It aims to enhance their awareness of safety and environmental protection, improve their ability to use the IMSBC Code, and help them learn standardized operations to reduce potential safety hazards. This is to ensure the safety of personnel, ships, and cargoes, and to prevent marine environmental pollution.

A trainee successfully completing this course will be able to:

- > Identify risks associated with the carriage of solid bulk cargoes by sea;
- > Assess the acceptability of consignments for safe shipment, especially Group A solid bulk cargoes;
- > Observe safe working practices in the loading and carriage of solid bulk cargoes;
- > Observe ship and port security regulations related to the loading, carriage, and unloading of solid bulk cargoes; and
- > Supervise the operation process of certain cargoes following regulations in the IMSBC Code.

Upon successful completion of the course, an appropriate certificate or similar document should be issued to the trainees to prove that they have received training in using the IMSBC Code in maritime practice in accordance with IMO Model Course 1.45.

2.8 Have officers and ratings responsible for cargo handling on ships carrying dangerous and hazardous substances in packaged form, undergone formal training (V)?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the vessel is not handling packaged dangerous, hazardous, and harmful cargoes (dangerous goods) or was not carrying such cargo at the time of inspection or during the previous voyage, respond to this question with N/A.

The course should enable trainees to contribute effectively to the preparation and execution of the safe carriage of dangerous goods and marine pollutants by sea. Trainees should understand the legal implications and correctly apply or verify compliance with detailed instructions, including safe packaging, handling, stowage, and segregation of dangerous, hazardous, and harmful cargoes. Additionally, they should be aware of the necessary precautions in relation to other cargoes as outlined in the IMDG Code and its supplement, as well as the International Convention for Safe Containers (CSC), 1972, insofar as they affect the transport of dangerous goods. The course should be designed and delivered to meet the requirements of the IMO model course 1.10, "Dangerous, Hazardous, and Harmful Cargoes."

2.9

Has an SMS policy and procedure been established to enforce the STCW Convention and Code requirements for the purpose of preventing drug and alcohol abuse? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments, the date of the last drug and alcohol test that was carried out on board either by an independent agency or under controlled conditions by ship's personnel with specimens being forwarded to an independent agency. Record in comments the date of the last unannounced alcohol test conducted on-board.

The drug and alcohol policy should define who it applies to. In most circumstances this should be everyone in the organisation, including contractor personnel. The company should ensure that its contractors' policies on drugs and alcohol align with company policy. There may also be reasons to make reference to safety-critical positions/roles or safety-critical work/operations. It should also specify what substances or categories of substance are prohibited under the policy. This should include any psychoactive substance (i.e. affecting cognitive function), regardless of circumstances of use. Individual countries and jurisdictions may have differing regulations on the import, stocking, possession, or use of drugs or alcohol, and there may be legal requirements for testing. The policy will need to take this into account and make appropriate stipulations and requirements, especially with respect to prohibited behaviours, sampling and testing. These requirements should be made clear to personnel. The policy should also refer to those situations where the use of substances is permitted, specifying what the company will allow and the company's expectations and requirements with respect to usage and fitness to work. Local regulations should always be considered and should take precedence when defining specific policy parameters. Permitted use may include prescription drugs brought onboard a ship, with evidence of a valid prescription from a qualified registered medical practitioner or issued from the ship's medical store or the terminal's clinic.

This does not preclude a company from having a zero-alcohol and a zero-drug policy on company premises, ships, etc.

A drug and alcohol policy should set out clear and consistent expectations regarding conditions of employment, employee behaviour and the consequences of not meeting these expectations. It is recommended that companies set out in their policy a clear and consistent legally compliant stand-down procedure for those cases where the drug and/or alcohol test is confirmed positive. Employees are expected to follow stand-down orders. Depending on company, role and situation, this could mean standing down from all duties, or standing down from safety critical duties only.

The policy should highlight and describe those situations or behaviours where disciplinary or other action will be taken, up to and including termination of employment.

Typical examples may include:

- > Using, distributing/selling, or possessing alcohol or drugs at work other than where use is permitted.
- > Non-compliance with Standards of Training, Certification and Watchkeeping (STCW) guidance.
- > Being under the influence at work, or otherwise unfit for work, due to any prohibited substance.
- > Not complying with stand-down orders given after a positive test result.
- > Refusal to test, interfering with or obstructing a sampling and testing process or otherwise failing to cooperate or comply.

Consequences should be consistent with local company policy and the law. Since many companies will have operations spanning different countries, it may be necessary to adapt the drug and alcohol policy according to legal advice from those jurisdictions. Companies should have a fair review process, ensuring personnel can challenge a positive result. It is recommended to choose drug and alcohol testing programmes with the most appropriate technology for the specific type of organisation. Considerations may include available tests, cost, gender sensitivity, any medical advice, availability of competent persons and laboratories to carry out relevant tests, and the practicalities of the sampling processes.

Specific legal requirements of the jurisdiction may determine what testing can be done and how it must be done. Provided the legal requirements are met, the procedures should include:

- > Employees must consent to being tested for drugs and alcohol for practical and legal reasons. Should they refuse when an employer has grounds for testing, they should be made aware that they may face disciplinary action.
- > Testing must strictly follow validated and authorised processes, to ensure samples cannot be contaminated or tampered with, and that testing procedures and analyses are accurate. It is recommended that companies follow a legally defensible process. This must include adequate training of all parties from sample collection to issuing the validated results, a robust chain of custody process, validated testing methods or laboratories, and an MRO process for non-negative (laboratory positive) samples.
- > Unannounced testing can be done at the workplace and can be done by a trained company collecting officer or an external party – for both drugs and alcohol.
- > Unannounced alcohol tests conducted on-board ships or in terminals can be initiated by the ship's Master, the Terminal Manager or the company. Initiation by the company is important to ensure there is the ability to test the Master or the Terminal Manager.

Alcohol Limits under the STCW Code: Understanding Blood vs. Breath Measurements

The STCW Code stipulates:

- 1- A limit of not greater than 0.05% BAC (blood), or
2- 0.25 mg/L alcohol in breath.

Blood and breath alcohol concentrations are measured in different biological mediums (blood vs. exhaled air). While there is a common conversion ratio (approximately 2100:1), it is not exact and can vary by individual physiology, temperature, and testing method. Relying on conversions can lead to compliance gaps or legal disputes, especially during audits or investigations.

It is recommended that the company's alcohol and drug policy clearly define separate limits for blood and breath alcohol concentration in line with STCW A-VIII/1 requirements, rather than relying on conversions between measurement units. Additionally, guidance should be provided for interpreting urine analysis results in consultation with the testing contractor, recognising the current absence of statutory limits for this medium.

The following types of drug and alcohol screening at a minimum are recommended:

Types of testing	Frequency
Pre-employment	As required
For cause/reasonable suspicion/post-accident or incident	As required
*Unannounced random/periodical	Once per year OR sufficient to serve as a deterrent to misuse
Return to work/post-treatment	As required

***Legal requirements:**

The inspector is not required to verify whether the vessel's manager has sought legal advice in the preparation of the company's drug and alcohol policy and procedures during the physical inspection.

Ship manager with offices in multiple jurisdictions are advised to seek legal advice, as there may be country-specific restrictions on the importation, possession, storage, or use of drugs and alcohol. Additionally, legal obligations concerning drug and alcohol testing may vary by location. Non-compliance with such regulations can result in legal consequences for both the employer and the employee.

Given that drug and alcohol policies and testing programmes may have implications under human rights, employment, criminal, occupational health and safety, and data protection laws, it is strongly recommended that such policies be developed with input from legal professionals, drug and alcohol programme specialists, human resources personnel, and occupational health or medical experts.

- 2.10 This question has been removed from the current version of the document.
- 2.11 This question has been removed from the current version of the document.
- 2.12 This question has been removed from the current version of the document.
- 2.13 Is the officer matrix accurately completed and does it reflect the information on officers and engineers on board the vessel at the time of inspection? (V)
- ☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should record a finding when the officer matrix has not been updated to reflect the officers who were on board at the time of the inspection. An allowance should be made for any officer who had changed within the previous seven days. If there is a documented procedure in place and senior officers have been transferred between sister ships, the inspector must not record a Finding for a lack of overlap or parallel sailing period.

The vessel's manager shall provide sufficient overlap for Master / Chief Officer and Chief Engineer / Second Engineer to ensure that they are familiar with the vessel's operation before taking charge, and both senior officers and senior engineers are not changed at the same time.

The vessel's manager is responsible to maintain up-to-date records relating to the officers and engineers on board the vessel at the time of inspection. The inspector should have a copy of the updated officer matrix and check the tour on board, qualifications and experience of officers and engineers against the crew list and seaman books. The actual details of Master, Chief Engineer, Chief Officer and Second Engineer / First Engineer must be checked against the data contained in the matrix and a Finding shall be recorded for inaccurate updates.

Random checks must be made of the actual records applicable to junior officers and junior engineers.

A seafarer may hold a Certificate of Receipt of Application (CRA) and a valid national STCW Certificate, for a period not exceeding three (3) months while an application for the STCW Endorsement Certificate is being processed. The inspector shall check the validity of CRAs.

2.14

If ECDIS is fitted on board, has the Master and Navigation Officers been familiarised with the ECDIS equipment installed on board, and were objective evidences of this familiarisation available? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Inspector to record how the familiarisation training was carried out. ECDIS familiarisation should be provided to all on-signing deck officers before they take an independent navigation watch, and each time they join any vessel. (Recommendations on Usage of ECDIS and Preventing Incident, 2020)

The STCW Code contains requirements for approved training on ECDIS. In cases where the approved training has not been completed, a limitation shall be included on the certificate and endorsements issued to the seafarer.

Where such a limitation is not specified, the certificate and endorsements are evidence of having successfully completed the required approved training and that the standard of competence has been achieved.

No requirement exists for the approved training on ECDIS equipment to be type specific. The knowledge, understanding and proficiency required to be demonstrated is generalized to ensure seafarers have the necessary skills for basic operation of all types of equipment.

In accordance with regulation, I/14, companies are responsible for ensuring that seafarers employed on their ships are familiarized with the installed equipment, including ECDIS.

It is agreed that seafarers required to have training in the use of ECDIS:

1. Should not be required to provide documentation of training in ECDIS that is specific to the installed equipment; and
2. Are required to be familiarised with the ECDIS equipment installed on board.

(STCW.7/Circ.24/Rev.1, 2017)

Deck officers who hold a Certificate of Competency with validity over 01 January 2017, in accordance to regulations II/1 and II/2 of the annex to the STCW-Convention and without an ECDIS limitation, fulfil the requirement of generic ECDIS-training.

The vessel's manager can consider a wide variety of options for achieving familiarisation both on-board and ashore. These include but are not limited to:

- > Shore based manufacturer training followed by installation-specific familiarisation on-board.
- > Independent training on specific systems followed by installation specific familiarisation.
- > Computer Based Training (CBT), followed by installation-specific familiarisation on-board.
- > Internet / Intranet Based Training (eLearning) followed by installation specific Familiarisation on-board.
- > On-board training by appropriately trained crew or training personnel.
- > Manufacturer provided training mode on the ECDIS, followed by installation-specific familiarisation on-board.
- > Company bridge procedures and manuals.

Regardless of the method(s) used, it is essential that all watch keeping officers are competent in the use of the on-board ECDIS prior to taking charge of a navigational watch and remain so thereafter.

(Industry Recommendations for ECDIS Familiarisation, 2012)

The SMS should include a procedure to ensure the Master and all watchkeeping officers familiarise themselves with the use of the ECDIS equipment installed on board prior to taking an independent navigational watch. The procedures for familiarisation should include:

- > The time scale for the familiarisation.
- > The method(s) of familiarisation with the ECDIS equipment.
- > The location of the familiarisation, whether on board or ashore.
- > The identity of the appropriately trained crew or training personnel authorised to deliver the familiarisation.
- > The method of evaluation of the familiarisation upon completion, and before taking an independent navigational watch.
- > The records to be maintained.

2.15 This question has been removed from the current version of the document.

Section 3: Navigation

3.1 Is practical guidance on navigational safety incorporated in the vessel manager's navigation instruction / procedures and are officer's familiar with the company's navigation procedures? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Masters, Officers, and Engineers should be familiar with the utilisation of the override function of engine power limiters. They should also understand that the activation of this override may be necessary while navigating in pilotage waters.

The practical guidance on navigational safety shall include the following:

- > Allocation of bridge watch keeping duties and responsibilities.
- > Procedures for passage planning and navigation, including departures from the passage plan.
- > Chart and nautical publication update and correction procedures.
- > ECDIS procedure (including chart and software updates).
- > Procedures to ensure that all essential navigation equipment and main and auxiliary machinery are available and fully operational.
- > Ship position reporting procedures.
- > Accident and near miss reporting procedures.
- > Recording of relevant events and Voyage Data Recorder (VDR) policy.
- > Use of Bridge Navigational Watch Alarm System (BNWAS) modes (automatic, manual, on and off) and procedures for ensuring correct operation.
- > Bridge access and distraction prevention procedures.
- > Procedures for familiarisation and effective handover when crew changes occur.
- > Training and drill requirements.
- > A system for identifying particular training needs.
- > A procedure for when to call the Master to the bridge.

(Bridge Procedures Guide 2022)

The ECDIS procedure should include the following:

- > Voyage planning and execution
- > Watch-keeping with ECDIS
- > Ensuring against over-reliance on ECDIS
- > Chart Maintenance
- > Departure and Arrival checks
- > ECDIS failure and backup system
- > Safety settings
- > The use and reliability of CATZOC
- > ECDIS display layers for various navigation conditions.
- > Managing manual layers to ensure current important information is available and out-of-date material is archived or removed.
- > Display T&Ps NMs and use of AIO function.
- > Where there is no appropriate safety contour available on the ENC
- > Depth contour shading: two shade versus four-shade
- > Define the XTC for various sea area, such as pilotage water, confined waters, coastal waters, and open water, for each leg of voyage.
- > Post voyage review, so that any hazards or useful information discovered can be incorporated into future passage plans.
- > The route validation
- > A protocol for naming and identifying saved routes to avoid selecting and incorrect route.
- > The frequency of, and preferred method for, position verification while using ECDIS.

Where ECDIS is being used as the primary means of navigation it must be clearly stated as such by the company and a policy in the SMS. (Admiralty guide to ECDIS implementation, policy, and procedures, 2016)

A checklist should be established with clear instructions on how to deal with sensory input failure of ECDIS and how it may affect safe navigation. This checklist should be kept in bridge.

(ECDIS LTD, 2019)

Anchoring procedures must be incorporated in the navigation procedure and shall provide guidance on the following:

- > How to select a good anchorage location, planning the anchoring position and approach in different weathers and visibility condition; bridge team management; traffic density, negotiating overcrowded anchorages with additional risks of collision; safety of swing room, under keel clearance
- > Keeping a safe anchor watch, including position-keeping, proper use of radar and GPS guard rings/alarms. OOW use of main engine.
- > The minimum requirement for the Master's Bridge Orders.
- > When to have the engineers on stand-by, the engine room manned, and the main engines on standby or ready for immediate use.

- > Amount of cable, scope, holding ground, anchor holding power, proximity of shoreline, dangers of dragging anchor, and risk of collision and grounding.
- > When the vessel is in ballast condition, the use of additional ballast.
- > The use of two anchors
- > The limitations on the anchoring equipment under heavy stress
- > The use of anchors in an emergency
- > Deep water anchoring
- > Recognising when a dangerous situation is developing when at anchor and when to move
- > Taking early and effective action
- > Factors affecting a vessel when at anchor in heavy weather, including yawing and snatching
- > Putting to sea in the advent of adverse and severe weather

(Standard Safety Bulletin on Safe Anchoring, 2008)

Special consideration should be taken to create a backup of ECDIS data on a regular basis so any part of the passage could be reviewed. The company SMS should include frequency and arrangement of ECDIS data backup.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

The SMS should incorporate a procedure to ensure the integration of the Pilot, or similar roles such as the Mooring Master, Lightering Master, Marine Advisor, Deep Sea Pilot, etc., into the bridge team. This procedure should involve the documentation and discussion of specific information before any transfer of conn or responsibility takes place. The information should include, but not be limited to:

- > The vessel's particulars and maneuvering characteristics.
- > The planned passage.
- > The review of the ECDIS and the status of safety depth alarms and layers in use.
- > The use of any navigational aids provided by the pilot.
- > Mooring and/or anchoring requirements.
- > Towage and/or tug assistance.
- > Under keel clearance.
- > Relevant defects and/or constraints.

The Master and the Pilot exchange should be documented, discussed, and agreed upon before any transfer of conn or responsibility occurs. This Master/Pilot exchange should be repeated whenever there is a change of pilot.

Navigation officers should be aware of the actions required in the event of single or multiple ECDIS failures. This awareness can be achieved by incorporating familiarisation with such contingency plans into the ECDIS familiarisation checklist or by conducting routine drills. Inspectors should record a finding if the bridge team are unfamiliar with the company's contingency plans for ECDIS failures.

3.2 Are the requirements of the Master's standing orders explained to the deck officers, and are bridge order books (Night Order) being completed by the Master and countersigned by the officers? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

For the purpose of this question, the term bridge team refers specifically to the Master and the Officers on navigation watch. Record a Finding if the detail of visibility criteria, calling the Master, minimum CPA and ECDIS display layers for various navigation conditions was not incorporated in the Master's standing order. The Master shall clearly highlight the potential safety risks involved in VHF radio communication between vessels and reliance on AIS communication information, for the purpose of collision avoidance. The VHF or AIS text facility should not be used for collision avoidance purpose. Master shall be called if the vessel is needed to exit the XTC.

The Master shall ensure that all situations requiring the Master's call are documented in the Master standing order in line with the Bridge Procedures Guide checklist C2.17 "calling the Master". The Master should explain particular requirements to the Bridge team in the Master's Standing Orders. These orders should be drafted to support the SMS.

Company and Masters' Standing Orders should be read by all Bridge Team members upon joining the ship, signed, and dated. A copy of the orders should be available on the bridge for reference.

In addition to Master's Standing Orders, specific instructions will be needed. At least at daily intervals, the Master should write in the bridge order book what is expected of the OOW for that period. These orders should be signed by each OOW when taking over a watch, to confirm that they have read, understood, and will comply with the orders.

The OOW should brief other members of the Bridge Team, as appropriate, on any activities or requirements for the forthcoming watch. The Master may also issue night orders for periods when the Master is resting, and specific information about the current leg of the passage should be included in them.

(Bridge Procedures Guide, 2022)

There is an expectation that bridge order book entries are made by the Master at least daily when the vessel is at sea.

AIS information overlaid on ECDIS should be used as an identification tool and not as a collision avoidance tool.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

3.3

Is the manoeuvring information for the vessel displayed on the bridge and are bridge logbooks, bell book, radar performance book, and Change of Watch at Sea check list being correctly maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The OOW shall be familiar with the difference between X-Band (3 cm) and S-Band (10 cm) radars, as well as their characteristics and limitations, such as the impact of different weather conditions on their performance and shadow and blind sectors. For all ships of 100 metres in length and over and all chemical tankers and gas carriers regardless of size, a pilot card, wheelhouse poster and manoeuvring booklet should be provided.

(Provision and display of manoeuvring information on board ships, 2011)

The OOW should be familiar with the handling characteristics and stopping distances of the ship. In addition, the OOW should know how these characteristics are affected by the current and anticipated machinery status. Information regarding the manoeuvring characteristics should be recorded on the Pilot Card and on the Wheelhouse Poster and the manoeuvring booklet. Please refer to Bridge Procedures Guide Checklists C1.2 and C1.3.

(Bridge Procedures Guide, 2022)

All ships engaged on international voyages shall keep on board a record of navigational activities and incidents including drills and pre-departure tests. When such information is not maintained in the ship's logbook, it shall be maintained in another form approved by the Administration.

(SOLAS 1974, regulations V/26 and V/28.1)

The quality of the radar picture needs to be checked regularly. This may be done automatically using a performance monitor.

(Bridge Procedures Guide, 2022)

The following should be recorded in the bridge logbook:

- > Navigational information including positions at regular intervals and method of position fixing, courses steered, allowances made for compass error, leeway and set.
- > Record of course, distance and speed made good, and course and distance to go should be completed daily.
- > Full set of routine weather observations, with a report of sea and swell conditions, should be entered at the end of each watch.
- > Details of severe weather met during the voyage, and the action taken should be recorded.
- > Full details of any matters which might affect the cargo and its condition.

RightShip recommends that the performance of the radar(s) when operational should be checked and recorded by the OOW at the end of each watch unless this contradicts the makers recommendations. A numeric, percentage, graphical, or other measurement value should be recorded.

Before taking over a navigation watch, the incoming officer should positively confirm the ECDIS configuration against the passage plan requirement. The outgoing officer should highlight any changes to the ECDIS configuration outside the passage plan parameters.

If an ECDIS alarm must be disabled for any reason, this should be recorded on a formal tracking form to be handed over to subsequent watches and approved by the Master.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

The communicated ECDIS configuration by the officer of watch should be documented.

Change of Watch at Sea checklist should be used as per section C2 (Checklist C2.16) of the Bridge Procedures Guide and at any other time required by the SMS.

Rightship recommends that the ECDIS display setting should be incorporated into the Change of Watch at Sea checklist.

The following documents described in the appendices to Recommendation on the Provision and Display of Manoeuvring Information on Board Ships (annex, resolution A.601(15)) should be updated to include the manoeuvring characteristics of the ship when the ship has all shaft and engine power available, and when shaft or engine power has been limited:

- .1 the Pilot card;
- .2 the wheelhouse poster; and
- .3 the manoeuvring booklet.

(Amendments to the 2021 Guidelines on the Shaft / Engine Power Limitation System to Comply with the EEXI Requirements and Use of a Power Reserve (Resolution MEPC.335(76)), as amended by Resolution MEPC.375(80)) (2024))

3.4 Have operational checks on navigational equipment been done and are checklists being effectively completed when preparing for sea and prior to port entry? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Operational checks on navigational equipment should be undertaken when preparing for sea and prior to port entry as per section C (Checklists C2.1, C2.6 and C2.7) of the Bridge Procedures Guide and at any other time required by the SMS.

Before entering restricted or coastal waters, it is important also to check that full control of engine and steering function is available.
(Bridge Procedures Guide, 2022)

3.5 Are there records indicating that routine tests and checks of bridge equipment are being undertaken regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Daily tests and checks of bridge equipment should be undertaken, including the following:

- > Manual steering should be tested at least once per watch (as per Checklist C2.1 of BPG).
- > Gyro and magnetic compass errors should be checked and recorded at least once a watch when this is possible.
- > The synchronisation of all compass repeaters, including repeaters at the emergency steering position, should be regularly checked.
- > To ensure adequate performance, information from electronic equipment should always be compared and verified against information from different independent sources; and
- > All available positioning systems and sources (GNSS, DGNSS, satellite communications terminals with integrated GNSS, and terrestrial radio navigation aids) should be cross checked.

Checks should confirm that the equipment is functioning properly and that it is successfully communicating with any other bridge system to which it is connected:

- > Built-in test facilities should be used frequently, including alarm self-test functions.
- > Configuration settings should be checked and confirmed to be in accordance with the SMS and the passage plan; and
- > Operational settings and alarms should be correctly set and checked on the equipment and/or the BNWAS.

(Bridge Procedures Guide, 2022)

3.6 Has the Master/Pilot information exchange been taking place effectively and is the standard pilot card being completed as required? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Officer should record the time when the Master/Pilot information exchange is completed. If different, the time of the transfer of conn between the Master and Pilot should also be recorded. Additionally, the time of the transfer of conn between Pilots, and if applicable, between the Pilot and Master on an outbound passage, should be recorded.

The Pilot card should clearly indicate whether a power limiter is engaged, the time necessary to override the power limitation systems, and the ship's maximum power capacity, both with and without the application of the limiter.

While navigating in pilotage waters, the main engine shall be readily available to respond immediately to the full range of manoeuvring commands as outlined in the Pilot card.

The pilot and the Master should exchange information regarding the pilot's intentions, the ship's characteristics, and operational factor as soon as practicable after the pilot has boarded the ship.

For an effective Master/Pilot information exchange, use should be made of the MPX checklist (Checklist C1.1 of Bridge Procedures Guide). It is essential that the MPX result in clear and effective communication and should cover:

- > Presentation of a completed standard Pilot Card (Checklist C1.2 of Bridge Procedures Guide);
- > The pilotage plan and the circumstances when deviation from the plan may be required.
- > Any amendments to the plan should be agreed, and any changes in individual Bridge Team responsibilities made before pilotage commences.
- > Updates on local conditions such as weather, depth of water, tides and tidal streams.
- > An update on traffic conditions.
- > Ship's dimensions and manoeuvring information should be provided in the form of the Wheelhouse Poster (Checklist C1.3 of Bridge Procedures Guide). A manoeuvring booklet containing more detailed information should also be available on the bridge.
- > Any unusual ship handling characteristics and machinery, navigational equipment and crew limitations that could affect the safe conduct of pilotage and berthing.
- > Information on berthing arrangements including the use, characteristics and number of tugs, mooring boats, mooring arrangements and other external facilities.
- > Contingency plans should also be considered. These should identify possible abort points in the event of a malfunction or a shipboard emergency; and
- > Formal confirmation of the working language.

(Bridge Procedures Guide, 2022)

3.7 Has the vessel's manager produced a guideline for under keel clearance and air draft clearance? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Inspector should verify the accuracy of the UKC calculation.

The UKC policy should incorporate the minimum allowed under-keel clearance for both coastal, river navigation and while alongside, including guidance on the action to be taken in shallow water to ensure the minimum clearance is maintained. The required minimum air draft for passing under bridges or overhead cables must be defined by the vessel's manager. The vessel's manager's guidelines shall cover the calculation of Dynamic Under Keel Clearance (DUKC) and air draft. The UKC Calculation on board shall take CATZOC information in the account.

The CATZOC value highlights the accuracy of data presented on charts.
With six categories, it informs the user about how far they can rely on the chart when planning a passage or conducting navigation.

Companies should set out their minimum UKC and procedures for operating within different values of CATZOC in the SMS.
(Bridge Procedures Guide, 2022)

For each Zone of Confidence (ZOC) value, reference shall be made to either Hydrographic publication UKHO NP 5012 or figure 3.8 of the Bridge Procedures Guide 2022.

3.8 Are the fire and safety rounds being conducted at the end of each watch? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

No other activity or duties should be allowed to interfere with keeping a proper look-out. The officer of watch should not be the sole look-out during hours of darkness.

(Bridge Procedures Guide, 2022)

In areas not covered by a fire detection system, regular fire patrols should be conducted. Such patrols should avoid using the bridge lookout during the hours of darkness.

3.9 Does the manning level in the bridge at all stages of the voyage and anchor meet or exceed that required by the Bridge Manning Matrix and are lookout arrangements adequate? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The recommended form and example of the Bridge Manning Matrix is contained in the Bridge Procedures Guide. The Bridge Manning Matrix shall be posted in the Bridge.

Under the STCW Code, the Officer of the Watch (OOV) may, in certain circumstances, be the sole look-out in daylight conditions. Clear guidance on the conduct of sole look-out should be included in the SMS.

(Bridge Procedures Guide, 2022)

3.10 Is navigation equipment in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the magnetron of radar has not been changed as per manufacture recommendation but mitigation measures are in place, such as regular Performance Monitor tests per watch and maintaining a spare magnetron as essential spares, the inspector should not record the finding but should enter a comment detailing when the last magnetron change occurred and what mitigation measures were in place.

The ship-borne navigational systems and equipment shall comply with SOLAS Chapter V Regulation 19.
The navigation equipment when fitted in the bridge, regardless of whether a vessel is required by legislation to carry such equipment, should be operational.

Random checks should be made to ensure that equipment is operational.

3.11 Are navigation lights, emergency navigation lights, shapes and signalling equipment in working order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The OOW is responsible for ensuring that the navigation lights, emergency navigation lights and signalling equipment are in working order and are ready for immediate use at all times. The condition of lights, flags and shapes should be checked at regular intervals. Sound signalling equipment should be checked daily and maintained in an operational condition. (Bridge Procedures Guide, 2022)

A procedure for testing of the navigation light failure alarm should be posted on the bridge.

The signalling lamp should have 3 spare bulbs and a portable battery pack.

(PERFORMANCE STANDARDS FOR DAYLIGHT SIGNALLING LAMPS, 2000)

3.12 Are navigation officers familiar with the company procedure for the use of the Bridge Navigational Watch Alarm System (BNWAS), were records available to demonstrate that it had been operated when the ship was underway and at anchor, and tested in accordance with the procedure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The BNWAS must be operational whenever the ship is underway and should be used at anchor. Use of BNWAS modes (automatic, manual, on and off) and procedures for ensuring correct operation should be incorporated in the company navigation procedure. The operation of the BNWAS should be part of the departure checklist and a key, if supplied, should be kept with the Master when switched on.

(Bridge Procedures Guide, 2022)

The means of selecting the Operational Mode and the duration of the Dormant Period should be security protected so that access to these controls should be restricted to the Master only. The BNWAS should be powered from the ship's main power supply. The malfunction indication, and all elements of the Emergency Call facility, if incorporated, should be powered from a battery-maintained supply.

If a malfunction of, or power supply failure to, the BNWAS is detected, this should be indicated. Means shall be provided to allow the repeat of this indication on a central alarm panel if fitted.

(Resolution MSC.128 (75) Performance Standard For a BNWAS, 2002)

The vessel manager should establish procedures detailing when the BNWAS should be operational, the personnel responsible for its activation, and the activation process itself. Additionally, the procedures should include measures to safeguard the system against unauthorised deactivation. They should also specify the actions to be taken if a BNWAS stage 2 or 3 alarm is triggered, as well as the periodic tests and checks necessary to ensure the system's proper functioning.

3.13 Where fitted are the standard magnetic compass, gyro compass and Global Navigation Satellite System compass, operational, adjusted and properly maintained? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the gyro compass was not serviced as per manufacturer recommendation.

The magnetic compass is generally fitted above the navigating bridge on the centreline and fitted with a periscope so that the compass is readable from the helmsman's position. Where the magnetic compass is needed to provide heading outputs to other bridge systems, a transmitting magnetic compass (TMC) is fitted. TMC outputs should be corrected for compass error and the TMC should be tested once a week.

A compass deviation card should be maintained on the bridge. The deviation will need to be determined and the compass adjusted at intervals during the ship's life, particularly after any major steel conversion work to the ship. Caution should be observed when using the magnetic compass on ships that carry or have recently carried magnetic cargoes such as iron ore and steel.

Compass safe distances are specified on all electrical bridge equipment and provide the minimum distances from the magnetic compass that equipment can be installed.

A TMC may have variation automatically applied. However, this correction will not include deviation. When correcting TMC outputs for compass error, care should be taken to ensure that the correct values for variation and deviation are applied.

The gyro compass should be run continuously. Should a gyro compass stop for any reason, it should be restarted and subsequently regularly checked and only relied on again when it has "settled" and the error is known. Where the gyro has no direct speed log or position input, manual corrections should be made as required. The gyro will support a number of repeaters, including a required repeater at the emergency steering position. Gyro repeaters on the bridge should be checked against the main gyro at least once per watch and after significant manoeuvring. Other repeaters should be checked frequently.

A Global Navigation Satellite System (GNSS) compass provides an alternative to a gyro compass as a non-magnetic transmitting heading device able to provide heading data to AIS, radar and automatic plotting aids. A GNSS compass or equivalent is required on ships navigating in polar waters at latitudes above 80 degrees. (Bridge Procedures Guide, 2022)

If the observations for a magnetic compass on a vessel show a deviation of the compass on any heading of more than 5 degrees, the compass must be adjusted by a qualified compass adjuster or the Master of the vessel to correct the deviation. If the compass is adjusted by the Master, RightShip recommends that the compass adjustment be checked by a qualified compass adjuster at the next available opportunity.

All magnetic compasses shall be swung and adjusted at least:

- > Every two years.
- > After dry docking; or
- > After significant structural work.

(BS ISO 25862:2019, 2019)

Where flag States have their own requirements then these should be followed.

3.14 Where manual steering is engaged, is the change over from auto steering, and vice versa, recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Times and locations of engaging hand steering should be recorded in the deck logbook or bell book.

Manual steering should be used whenever appropriate including in:

- > Areas of high traffic density.
- > Conditions of restricted visibility; and
- > Any other potentially hazardous situations and particularly when an automatic steering system may provide insufficient control.
- > Manual steering should be tested once per watch as per the checklist C2.1 of the Bridge Procedures Guide.

(Bridge Procedures Guide, 2022)

Examples of other potentially hazardous situations are river transits and when navigating through restricted waters.

3.15 Are deck officers familiar with the procedure to preserve the VDR data in the event of an incident and is there a company policy within the SMS relating to the playback of VDR data? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Watch-keeping officers should understand and be familiar with the procedures for preserving records as required by the SMS.

Company policy relating to the playback of VDR data should be contained within the SMS. Playback of VDR data may provide a tool for analysing the performance of the Bridge Team. A mistake as long as it is not intentional or caused by carelessness, should normally be treated as a learning opportunity. A 'just' culture should give personnel the confidence to admit any mistakes or 'near misses', and this leads to a safer working environment.

Testing is required annually and should always be carried out following repair or maintenance work to the VDR or to any source providing data to the VDR.

(Bridge Procedures Guide, 2022)

3.16

Is a chart and publication management system being implemented to ensure that all charts, nautical publications, and other publications on board are current, maintained and up to date? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends that a shore-based company be engaged to provide navigation support services including ENC's, paper charts and marine publications, so as to ensure that those on board are up to date with the latest edition available.

Use of a chart and publication management system will help to ensure that charts and publications are effectively maintained. A management system should record the charts, publications and licences/ permits carried, and when the charts and other publications were last corrected. Licenses and permits are available from the hydrographic office that produced the ENC or RNC. Licensing arrangements usually include:

- > Pre-pay licensing based on intended use. Normally licenses and permits are specific to a ship and typically allow a chart to be viewed for a period of 3,6 or 12 months on that ship; or
- > Dynamic or pay as you sail (PAYS) licensing based on actual passage. Ships have access to all charts for planning purposes but only pay for charts that they use during navigation.

Licenses and permits should be managed using the ship's chart management system.

(Bridge Procedures Guide, 2022)

The Weekly Notice to Mariners Section VIII and the README file contains important safety information relating to ENCs and ECDIS. The file is included on all ENC media but some ECDIS may not be able to display it; it can however be read on any standalone PC. The vessel's officers should all be aware of the recent content of the file and be able to demonstrate the practical application of the information.

(Admiralty Guide to ECDIS Implementation, Policy and Procedures, 2016)

The chart and publication management system shall cover the ENC management and correction process, including safety measures, to avoid viruses. The NP133C Admiralty ENC Maintenance Record book, or equivalent software, should be available on board.

An effective ENC management system should be in place on board to record ECDIS identification numbers and when licences/ permits were received on board and should include a record of when the ENCs were last updated. This is generally part of ECDI software logging.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

3.17

Were appropriate charts and publication used for the previous voyage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Vessel should obtain licences for and use the largest scale of ENCs available for all stages of each passage.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

Only up-to-date official charts and publications should be used for the appraisal, planning, execution, and monitoring of a passage plan.

For coastal and pilotage planning and for plotting each course alteration point, large scale charts should be used. Any additional charts and publications needed for the intended passage should be identified and obtained before departure.

For ocean passage planning and open water legs, the largest scale charts that are appropriate should be used

(Sections 2.3.1 Bridge Procedures Guide, 2016)

Photocopied/scanned copies of official paper charts (whether subsequently corrected to latest notices to mariner or not) are NOT regarded as satisfying the SOLAS chart carriage requirement.

The following publications shall be available on board and referred to when the primary means of navigation is ECDIS:

- > NP 231 Admiralty Guide to the Practical Use of ENC's.
- > NP 5012 Admiralty Guide to ENC Symbols Used in ECDIS

The Seafarers Handbook for Australian Waters (AHP20) is an official nautical publication providing mariners with important maritime information from various Australian government agencies, under the cover of one combined reference. It is expected that all commercial vessels operating in Australian waters carry and refer to the publication.

3.18 Is the ECDIS of an approved type, compliant with SOLAS requirements, and can the Master and watch keeping officers demonstrate familiarity with its use? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Where an ECDIS is being used to meet the chart carriage requirements of SOLAS, it must:

- > Be type approved.
- > Use up-to-date electronic nautical charts (ENC);
- > Be maintained so as to be compatible with the latest applicable International Hydrographic Organisation (IHO) standards; and
- > Have adequate, independent back-up arrangements in place.

According to SOLAS regulation V/18, ECDIS units on board ships must be type approved. Type approval is the certification process that ECDIS equipment must undergo before it can be considered as complying with IMO performance standards. The process is carried out by flag Administration-accredited type-approval organisations or marine classification societies in accordance with the relevant test standards developed by, inter alia, the International Electro- technical Commission (IEC) (e.g., IEC 61174).

(MSC.1/Circ.1503/Rev.1, ECDIS – GUIDANCE FOR GOOD PRACTICE, 2017)

Information related to current standards and latest software related to ENC and ECDIS are available on the IHO web site. The IHO has issued a new version of the ECDIS presentation library edition 4.0. There will be no need to run the IHO ENC/ ECDIS data presentation and performance checks on the ECDIS.

The inspector should check the version of the IHO Standards installed on the ECDIS to confirm it is current.

Several ECDIS manufactures recommend having particular ECDIS spare parts on board.

If the ECDIS manufacturer handbook includes a list of spare parts that should be stored on board, it is critical that these parts be available on board for replacement anytime they are required.

The Master and watch keepers should be able to demonstrate their competency with the operation of ECDIS. This can be established by requesting use of basic functionality of the ECDIS in the presence of the inspector. These functions may include:

- > Safety setting
- > Setting voyage plan
- > Checking voyage plan
- > Interrogating chart updates
- > ENC symbol identification
- > Manual position fixing (NP5012/NP232)
- > AIS and or Radar overlay if fitted
- > Understanding the limitations of operating in RCDS mode
- > Knowledge of SCAMIN and how it is displayed
- > Knowledge of CATZOCs
- > Familiarity of deck officers with contingency action in case of ECDIS failure.
- > Setting of safety frame/safety cone
- > Creating parallel index lines
- > Route checking and management of alarms
- > Handling unresolved ENC update errors

3.19 This question has been removed from the current version of the document.

3.20 Are T&P NMs and navigation warnings being used correctly in voyage planning and monitoring? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Some ECDIS have a feature to automatically import navigational warning from Sat-C or NAVTEX terminals. Navigation Officers should verify that navigation warning information is currently displayed.

Specific details of a critical navigational warning should be plotted and made alarmable by using the look-ahead feature to highlight the navigational hazard for the Officer of the Watch.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

Ensure the vessel has access to all necessary T&P NM information and that this is documented.

Where relevant to the voyage plan, T&P corrections should be inserted on the ECDIS display using manual corrections. The ADMIRALTY Information overlay (AIO) provides easy reference to T&P information; this can be displayed on a range of ECDIS or on back of bridge systems such as ADMIRALTY e-Navigator.

(Admiralty Guide to ECDIS Implementation, Policy and Procedures, 2016)

Inspectors should check if the system is installed and verify if relevant notices are effectively managed.

The overlay is displayed as a single layer on top of the basic ENC. This ensures that users have the most up to date T&P information available regardless of where they are in the world.

T&P NMs are delivered on a weekly basis on the update DVD or with the online/email updates, depending on requirements.

The information contained in the Overlay is important navigational information that should be used when planning a voyage and may be referred to when navigating. The Admiralty Information Overlay contains all Admiralty T&P NMs in force worldwide and additional ENC P (EP) NMs, which relate specifically to ENCs. (Admiralty Guide to ECDIS Implementation, Policy and Procedures, 2016)

Navigation officer should not entirely rely on AIO as they may not be updated, and applicable T&P notices should be verified against weekly notices to mariners.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

3.21 Has the vessel been safely navigated in compliance with international and inland regulations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

ENC's/Charts of the last voyage should be checked by the inspector to assess whether the vessel has been safely navigated. The inspector shall consider following when assessing the last voyage charts:

- > Largest scale charts to be available with route plotted.
- > Record of weather forecast.
- > Appropriate measures to be taken to comply with environmental requirements and regulations.
- > Safety and alarm setting of ECDIS.
- > Maintenance of safe distance off the coast, from prohibited area and dangerous wrecks.
- > Adequate bridge manning to ensure a proper look-out.
- > Ship's position confirmation at appropriate intervals.
- > Weather monitoring by making regular barometer observations.
- > NAVAREA navigational warning broadcasts where applicable checked.
- > Participation in area reporting systems; and
- > Gyro and magnetic compass errors and radar performance checked properly.
- > Correct minimum layers of ECDIS according to the company SMS.

If an appropriate safety contour is not available on the ENC, a manual alarmable contour should be drawn as a manual layer on the ENC that should always be selected and displayed during the passage.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

3.22 Are records available to show that the echo-sounder recorder is being switched on prior to each approach to shallow water, port entry and departure and has the echo sounder remained in operation while the vessel has been transiting in shallow waters? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The echo sounder should always be used when making a landfall and kept switched on in coastal and pilotage waters. If the echo sounder is fitted with a shallow water alarm, the alarm should be set to an appropriate safe depth to warn of approaching shallow water. It is important to check the units of soundings on the echo sounder are the same as those on the chart in use. The depth alarm on the echo sounder should not be set to a value lower than the ship's sailing draft.

(Bridge Procedures Guide, 2022)

The date and time of switching on should be marked on the recorder chart.

The echo sounders may have an internal memory and record data from the past 24 hours, in which case the recorder is not required.

3.23 Was the berth-to-berth passage plan of the previous voyage comprehensive and approved by the Master? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

To comply with the international, local regulations and industry recommendations, the vessel manager should establish a procedure for onboard officers to appraise, plan, execute and monitor every passage berth to berth including anchorage.

The format of the passage plan should include, but not be limited to, the minimum information that must be included:

- > Parallel indexing: Parallel indexing should not be performed from floating objects unless they have first been checked for position.
- > Chart changes.
- > Methods and frequency of position fixing and/or position verification.
- > Prominent navigation and radar marks.
- > No-go areas.
- > Landfall targets and lights.
- > Clearing lines and bearings.
- > Transits, heading marks and leading lines.
- > Significant tides or current.
- > Safe speed and necessary speed alterations.
- > Changes in machinery space status, i.e. manned/unmanned.
- > Changes in machinery status, i.e. standby for manoeuvring.
- > Changes in bridge watch composition.
- > Changes to fuel and/or scrubber use.
- > Changes in security arrangements.
- > Minimum under keel clearance.
- > Positions where the echo sounder should be activated.
- > Crossing and high-density traffic areas.
- > Safe distance off navigational hazards or marks.
- > Anchor clearance.
- > Contingency plans.
- > Abort positions.
- > VTS and reporting points.
- > Air draft when passing under a bridge or object.
- > Alternative or contingency routing that may be required at short notice.
- > Specific guidance provided by local routing publications where applicable to the vessel and its route.
- > The process to verify that all navigational and environmental considerations have been included in the final passage plan.
- > The review and approval process for a passage plan including utilising the ECDIS route checking function.
- > The record keeping requirements as they relate to the progress of the passage and navigational events.
- > The actions to take to update the passage plan when circumstances change requiring the vessel to deviate from the originally agreed plan.
- > The requirement for preparing the passage plan on both paper charts and ECDIS where the vessel has a single ECDIS unit or has nominated paper charts as the primary means of navigation.
- > The requirement for a passage plan briefing including the Master, all navigation officers, and a representative from the engineering department.
- > A vessel should deviate from its planned passage to extend necessary to safely comply with the collision regulations. Therefore, it is important to carefully consider these factors when evaluating the passage and creating a plan to ensure sufficient sea room and suitable watch condition.

When using ECDIS for passage planning, the following factors should be considered:

- > Availability of and access to the required up-to date ENC and RNCs for the intended passage. This should include identification of areas where ECDIS may need to be in raster chart display system (RCD) mode and where paper charts might therefore be required;
- > If reusing a previous passage plan, the need to recheck the route to confirm that it remains safe and no changes are necessary;
- > An appropriate large scale ENC or RNC should be used when planning a route;
- > Making sure that any old or previous routes are removed from the display;
- > The need to select chart symbols (pick report) on ENCs to get additional detailed safety and navigational information;
- > Applying a maximum acceptable cross track distance (XTD) to each leg of a route. This should comply with any requirements in the SMS and be appropriate for the area;
- > Calculating safety depths and safety contour and setting them up in line with the under keel clearance (UKC) requirements in the SMS;
- > Setting estimated time of arrival (ETA) information manually or using route planning tools. If this is set incorrectly, it may affect tidal data and time dependent information for the route;
- > Applying current and tidal data, if integrated with ECDIS and up to date, to the route; and
- > Checking information about the vessel's characteristics and confirming it as correct. This includes details about draft (including any allowance for squat or additional safety margins), turn radius and vessel dimensions;
- > The passage plan should be saved, backed up and locked to prevent unauthorised editing.
- > The Master should check and approve the passage plan. The person responsible for the passage plan shall brief the bridge team. This must all be documented in accordance with Bridge Procedures Guide check list C.2.9.

The shallow contour value must be equal to or more than the lowest draft of the ship.

The officer of watch shall consider following when calculating the safety contour:

- > The ship's sailing draft and trim
- > Expected squat or allowance for squat
- > UKC as per company SMS
- > Expected height of tide

By setting a safety depth, spot soundings are highlighted in grey (deep waters) or black (shallow waters) when compared with the safety depth value entered by the OOW.

-In calculating the ship's safety depth, it is also important to consider the category zone of confidence (CATZOC) value of the chart in use.

(Bridge Procedures, 2022)

Route validation is a critical aspect of a passage plan. The route validation involves the following stages:

- > Visual checks
- > Manual and auto-validation features
- > Cross-checks by the bridge team
- > Final validation and authorisation by the Master
- > Re-validation along the route.

The Master should only authorise the plan once all stages of visual check and route validation have been completed.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

The route validation check of previous voyage should be documented and reviewed by inspector.

A list of ENC's used for the intended voyage should be part of passage plan. The parameters for the look-ahead zone should be planned so that the size of the zone is appropriate for the vessel's speed and manoeuvring characteristics. They should be set for each leg of the passage and should consider conditions such as proceeding from ocean to coastal waters, pilotage areas or speed. The look-ahead zone should be reassessed in CATZOC area that have reduced position accuracy (such as B, C, D, U) to ensure the vessel has a sufficient safety margin.

Amendment to the passage plan should be officially documented and specific changes recorded on the passage plan form, according to company SMS.

Alarm-setting parameters should be agreed by the Master and bridge team at the passage planning stage and captured in the relevant passage plan form.

(Recommendations on Usage of ECDIS and Preventing Incident, 2020)

The following marine environmental factors shall be taken into account during an appraisal of the passage plan:

- > Ballast water management
- > Emission Control Areas (ECA)
- > MARPOL Special Areas
- > National or regional requirements
- > Particularly Sensitive Sea area (PSSA)
- > Garbage disposal
- > Port reception facility

(Bridge Procedures Guide , 2022)

3.24

Was the ship's track monitored at sea and during pilotage, and were parallel indexing techniques used to monitor the passage in coastal and pilotage waters ,particularly during restricted visibility or nighttime conditions? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Radar overlays should be used for position verification at regular interval, as defined by company SMS requirements, and for various navigation conditions such as in open waters, confined waters, fairways/channels, or pilotage waters.

Position plotting should also be undertaken using traditional techniques, using lines of position to plot visual /radar fixes.

This will act as a cross check and will be recorded on the ECDIS data log. (Recommendations on Usage of ECDIS and Preventing Incident, 2020)

Compliance with the passage plan should be closely monitored by the OOW:

- > To check that the ship's position is maintained within an authorized XTD, including following alterations of course to avoid collision or following a planned course alteration;
- > By fixing the ship's position at the frequency based on existing conditions and the proximity of navigational hazards;
- > By cross checking the ship's position using all appropriate means including;
- > By visual and/or radar fixing techniques using ranges and bearing of charted objects;
- > By echo sounder to monitor charted depths and contours; and
- > By monitoring the integrity of information displayed on navigation equipment.

(Bridge Procedures Guide, 2022)

The following techniques should be used when monitoring the passage in coastal and pilotage waters, particularly in conditions of restricted visibility or at night:

- > Parallel indexing, which is recommended to ensure the ship's track is maintained.
- > Radar bearings; and
- > Radar ranges.

(Bridge Procedures Guide, 2022)

3.25 This question has been removed from the current version of the document.

3.26 Is the Global Navigation Satellite System (GNSS) set to the correct Geodetic Datum, and are officers aware of the errors and alarms associated with GNSS?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A GNSS is a satellite-based system that provides continuous worldwide position, time and speed(over ground) information. Two systems that give near global coverage are available to ships:

- > Global Positioning system (GPS) operated by the United States; and
- > Global Navigation Satellite System (GLONASS) operated by the Russian Federation.

Other satellite systems recognized as components of the World-Wide Radio Navigation System (WWRNS) are:

- > BeiDou Navigation Satellite System (BDS) operated by China; and
- > Galileo Global Satellite System (Galileo) operated by the EU.

GNSS generally have a based accuracy in the order of 15-25 meters. Differential GNSS receivers offers greater navigational accuracy by applying corrections received from ground based reference stations.

The OOW should be familiar with the GNSS system used on board.

The GNSS system should indicate its current operating status and any associated alarms or errors.

Some common errors are:

Dilution of precision error (DOP): This error occur when fewer satellites are available to the ship. It is common when sailing in area with high mountains such as Alaska or Norway.

Receiver autonomous integrity monitoring (RAIM): This error relates to the quality of the data being sent to the GNSS receiver. If the system detects a drop in quality, it will alert the user.

GNSS jamming or spoofing: This can happen in an area of increased military presence. The OOW may notice a position jump on ECDIS or, when cross- checking position, the GNSS position may be unreliable.

Multi path error: Similar to the DOP error. The GNSS receiver may be blocked or receiving a double signal. This could be due to interference by structure, mountains, etc.

-ENCs use WGS 84 as the geodetic datum. Many GNSS receivers have internal facilities to transform positions between different geodetic datum, eliminating the need to apply datum offsets manually.

(Bridge Procedures Guide, 2022)

3.27 Does the vessel utilise a weather routing service? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Weather routing allows the Master and the bridge team to follow a passage plan that avoids the worst weather in the interest of safety and fuel consumption efficiency.

Weather routing predicts the movement of weather systems associated with poor conditions and rough seas. The most favorable route is then planned, taking these systems into consideration.

The main benefits of weather routing are:

- > Increase safety;
- > Better conditions for cargo or passengers
- > Fuel and time saving; and
- > Reduced costs overall

Weather routing is an aid to navigation and the Master should always consider routing information as well as applying good seamanship.

The safety of the ship, its crew and its cargo or its passengers should always have priority over the ETA.

The bridge team should be familiar with dedicated software for weather routing on board.

(Bridge Procedures Guide ,2022)

3.28

Are there procedures in place to limit the use of cell phones, personal electronic devices, the internet, and email on the bridge?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A quiet bridge to allow VHF radio calls and sound signals to be heard.

The company should have a written policy requiring that mobile phones or other personal electronic devices should only be used on the bridge in circumstances approved by the Master. While on some occasions the use of mobile phones or personal electronic devices may be permitted, the company policy should minimise the distraction resulting from such devices by, in general, limiting their use to operationally necessary circumstances.

Where internet and email services are available on the bridge, the Company should have a policy to manage their use. Access to internet and email use by bridge watch keepers should generally be limited to those circumstances where it is necessary for the safe navigation of the ship, in order to minimise distraction that might be caused to the Bridge Team.

Internet access and email on the bridge should usually be restricted to:

- > Updates for nautical charts and publications, licences and permits.
- > Weather information.
- > Navigational warnings; and
- > Information relevant to the ship's operations and passage plan.

(Bridge Procedures Guide, 2022)

Section 4: ISM System

4.1 Has the vessel's manager established a documented system for personnel to effectively implement the ISM Code? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The latest revision of ISM manuals, procedure and instructions should be available. The inspector shall examine the compliance of the vessel with the procedures and instruction during the course of inspection

The company should define and document the responsibility, authority, and interrelation of all personnel who manager, perform and verify work relating to and affecting safety and pollution prevention.

The company should establish procedures, plans and instructions, including checklists as appropriate, for key shipboard operations concerning the safety of the personnel and, ship and protection of the environment. The various tasks should be defined and assigned to qualified personnel.

(ISM code and guidelines on the implementation of the ISM code, 2018)

The documents used to define and implement the SMS may be described as the Safety Management Manual. It may be more than one manual and may take the form that the company considers most appropriate. Policies, practices, and procedures are to be followed in order to ensure safe functioning of ships at sea.

A list of on-board publications shall be incorporated in SMS. A Finding should be filed by the inspector against ships not carrying publications in accordance with National requirements and the SMS.

Carrying solid bulk cargoes involves significant risks that must be carefully managed to protect the crew and the ship. These risks include reduced ship stability, and possibly capsize, as a result of cargo liquefaction; fire or explosion as a result of chemical hazards; and structural damage to ships as a result of incorrect loading procedures.

Carrying grain cargoes involves significant risks that must be carefully managed to protect the crew and the ship. These dangers include settling, shifting, contamination, dust explosions, and fire, as well as rotting caused by ingress of water and subsequent oxygen depletion or the evolution of toxic vapors.

General cargoes include industrial commodities, bagged cargoes, project cargoes, steel products, forest products, palletized cargoes, smaller break-bulk cargoes, automobiles, containers, heavy lifts, dangerous goods, and out-of-gauge cargoes.

Container cargo includes a number of risks for the crew, the ship, and the environment. Stowage, cargo securing, stability and stress calculations, dangerous goods segregation, mis-declared overweight containers, out of gauge containers, refrigerated containers, on deck and under deck stowage, half door or both door open containers, lashing strength, sailing conditions, visibility, and heavy weather all pose risks to the ship, crew, and environment if not addressed properly.

MSC-MEPC.2/Circ.2 of 1 June 2006, "IMO requirements on carriage of publications on board ships" provides the publications which are specifically required to be carried on board ships by IMO instruments.

IMO instruments such as the SOLAS, MARPOL, LL, COLREG and STCW Conventions deal with many operational aspects, inter alia, navigational responsibilities, safety-related training/drills on board, safe cargo handling, oil spill prevention, collision avoidance activities and watchkeeping standards.

Therefore, these publications, although not expressly required by IMO instruments, may need to be carried on board to improve the crew's knowledge and to enhance the implementation of IMO instruments. No Finding should be filed by the inspector against ships not carrying such publications on board unless otherwise required by the ships Safety Management System manual.

(IMO REQUIREMENTS ON CARRIAGE OF PUBLICATIONS ON BOARD SHIPS, 2006)

When a vessel is a gravity-based self-unloading and/or hybrid self-unloading type discharging bulk cargo either onshore or offshore, the following should be included in the SMS:

- > Starting procedure
- > How to avoid overloading the self-unloading system
- > Stopover procedure –Prevent forging material that may in the cargo hold hopper gates and damage the belt
- > Selecting a variable speed and discharge rate
- > Electric and hydraulic controls
- > Hold discharge gate, conveyors, and elevator
- > Boom conveyor
- > Energy isolation permits and isolation/lock
- > Gate problems including:
 - > Gate blockage
 - > Gate off-track
 - > Loss of gate power
 - > Foreign object in way of a gate
 - > Actuation of emergency stops
 - > Cleaning procedure for the gates

- 4.2 Has a safety officer been appointed and adequately trained, onboard safety meetings held regularly and acknowledged by the vessel's manager, with feedback provided where necessary, and are safety inspections of the main deck, machinery space, accommodation and forecandle store conducted at regular intervals in accordance with the company's procedures, and are health, safety, and environmental hazards effectively identified? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The safety officer is the safety adviser aboard the ship and shall provide valuable assistance to the company and to individual employers in meeting the statutory responsibilities for health and safety. Some training may be provided on board, but the safety officer should have attended a suitable safety officer's training course.

The safety officer should be familiar with the principles and practice of risk assessment and should be available to advise those preparing and reviewing risk assessments.

(Code of Safe Working Practices for Merchant Seafarer's 2024)

The Safety Officer training course shall adhere to the STCW Code 2010 Tables A-II/2 and A-III/2 and the IMO Model Course 3.11.

The safety committee is a forum for consultation between the master, safety officials and others of matters relating to health and safety. It may be used by individual employers for consultation with the company and seafarers. Its effectiveness will depend on the commitment of its members, in particular that of the master. Because of its broad membership, and because the master chairs it, the committee can take effective action in all matters it discusses other than those requiring the authorisation of the company and individual employers. Do not use safety committee meetings for instruction or training.

The frequency of meetings will be determined by circumstances, but the committee should meet regularly, considering the pattern of operation of the ship and the arrangement for manning, and frequently enough to ensure continuous improvement in safety. In particular, a meeting should also take place after any serious incident or accident on the ship, if the normal meeting is not due within a week.

The vessel manager should establish a procedure requiring regular safety inspections to identify hazards and potential risks to health, safety, and the environment across all accessible areas of the ship. This includes the main deck areas, forecandle, machinery space, and accommodation block. These inspections should be conducted at appropriate intervals by the designated Safety Officer. A specific checklist should be developed to facilitate these inspections.

- 4.3 Are the latest shipboard internal and external audits, as well as the superintendent inspection reports, available? Are corrective actions being taken in response to recorded non-conformances?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Internal audits should be held as required by the management system, at intervals not exceeding twelve months. Reports should be available on board. The interval of internal audit may be exceeded by not more than three months in exceptional circumstances. Internal audits should be conducted by personnel with a working knowledge of the vessel manager Safety Management System.

The vessel's manager is responsible for establishing an inspection program. This program should ensure that the vessel undergoes regular assessments by both Marine and Technical Superintendents. A systematic inspection of the vessel should be carried out, and the frequency of these visits should be clearly defined within the relevant procedures.

Inspections should be scheduled at six-month intervals, with a permissible tolerance of one month. However, the interval between inspections should not normally exceed seven months. Additionally, the interval between successive inspections conducted by either a Marine Superintendent or a Technical Superintendent should not exceed fourteen months.

The term systematic inspection refers to a structured process developed by the vessel's manager, resulting in an inspection report that records findings in a standardised format. This report should provide a comprehensive overview of the condition of all operational and accessible areas of the vessel, its equipment, and onboard management, thereby informing shore-based management.

The inspection report should:

- > Summarise any structural, machinery, or equipment defects identified during the inspection. These should be entered into the vessel's defect reporting system for follow-up and closure.
- > Highlight any procedural weaknesses, which should be addressed through the non-conformity process.
- > Be retained onboard as a complete record of each superintendent's visit.

An internal audit conducted by a Marine and/or Technical Superintendent may be counted as one of the superintendent's visits, provided a separate inspection report is prepared in addition to the documented audit report.

Remote Inspections

Where a vessel manager has implemented a formal remote vessel inspection program to cover vessels that cannot reasonably be visited by a company superintendent, such inspections may be accepted as qualifying visits, provided the following conditions are met:

- > The company has an established procedure for conducting remote vessel inspections, which clearly defines:
- > The circumstances under which a remote inspection may substitute for a physical inspection.
- > The processes used to verify the condition of all areas of the vessel under inspection.
- > The required content and format of the final inspection report.
- > Completed inspection reports, prepared in accordance with the company's procedure, are available for each remote superintendent inspection.

Visits by senior management, Electrical Superintendents, or Marine and Technical Superintendents that are conducted to address specific issues and do not result in a full inspection or a completed report are not considered qualifying inspections under this criterion.

The superintendent inspection should be conducted by personnel with working knowledge of the vessel's Safety Management System, and the use of a third-party company is not considered a qualifying inspection under these criteria.

4.4 Does the Master periodically review the effectiveness of the onboard Safety Management System, report the findings to shore based management and receive feedback from them? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Effective Master reviews should be carried out at least once every 12 months and evidence of the company's response to the Master's review should be available on board. The review should contain evidence of positive and negative feedback, rather than simply being a tick-box exercise with no substantial content. The review may also include input from the ship's management team.

4.5 Is the vessel provided with an enclosed space entry procedure, and is there documented evidence that it was followed, and is there evidence that the crew assigned to responsibilities requiring entry into enclosed spaces has attended a dedicated enclosed space entry course?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Rightship recommends that if the ballast treatment system is installed in an independent enclosed compartment, such compartment shall be identified as an enclosed space.

Persons entering enclosed spaces should be provided with calibrated and tested personal gas detection instrument or instruments that monitor the levels of oxygen, carbon dioxide, flammable gases or vapours, toxic gases (including carbon monoxide), and any other gases identified in the risk assessment. (IMO RESOLUTION MSC.581(110) 2025)

Record a finding if there is evidence that any person entered the enclosed space without a certified and calibrated personal gas detector.

Ships carrying solid bulk cargoes

On ships carrying solid bulk cargoes, dangerous atmospheres may develop within cargo spaces, connected and adjacent spaces. The hazards may include flammability, toxicity, oxygen depletion, carbon dioxide and/or carbon monoxide generation, or self-heating, as identified in the shipper's declaration and/or in the individual schedules in appendix 1 of the IMSBC Code. Solid bulk cargoes listed in the IMSBC Code should be carried in accordance with the provisions of the Code, including precautions, atmosphere testing, ventilation and other requirements specified. Solid bulk cargoes that are not listed in the IMSBC Code should be carried in accordance with section 1.3 of the Code, including the conditions for carriage and handling as determined by the relevant competent authorities. Grain cargoes and timber cargoes other than those listed in the IMSBC Code may also cause oxygen depletion and toxic gas emissions, primarily carbon dioxide, in cargo holds and connected and adjacent spaces.

Enclosed hold access trunks

In certain designs of hold access, a stair arrangement, sometimes referred to as the "Australian Ladder", substitutes for a vertical ladder. On some ships these are contained within enclosed protective structures that are open only at the top and bottom of the hold. When a cargo that presents the risk of a hazardous atmosphere is loaded, this "connected" space will quickly adopt atmospheric attributes similar to that of the source space. The enclosed Australian ladder access should not be used until it has been fully ventilated, tested and confirmed to be safe. It should be noted that such spaces are difficult to ventilate, unless cargo blocking the bottom of the ladder is removed.

When hatches are opened to ventilate the above cargo space at discharge, the hazardous atmosphere often gets trapped in the access trunk.

Where such type of access trunk is fitted on ships, the nature of the hazard should be identified outside the entrance to the space and should be listed in the Enclosed Space Register. When a cargo which may generate a hazardous atmosphere is loaded, the access door/hatch should be locked with a dedicated and unique locking arrangement, distinct from all other such arrangements, from the commencement of loading until the space is certified safe for entry by testing after completion of discharge.

Entry of personnel into a cargo hold using a hold vertical ladder should only be permitted when:

1. the atmosphere in the hold has been tested and found to be safe;
2. wearing a personal gas detector;
3. wearing a safety harness; and
4. an emergency response plan is in place.

Working spaces with connections to cargo holds

In certain designs of general cargo ships, bulk carriers and tankers, doors and ventilation trunks of cargo spaces, as well as pipework connecting to cargo spaces, are connected directly into working spaces such as forecastle workspaces, stores rooms, windlass hydraulics, bow thruster and other machinery rooms. When a certain cargo is stowed in the cargo space, there is a risk that gas or vapours from the cargo will penetrate into the connected working space.

These working spaces should be identified as "connected spaces" and consideration should be given to use of certified safe type of electrical equipment for an explosive atmosphere.

When the cargo space contains a hazardous atmosphere and, given the propensity for hazardous atmospheres to be trapped in those spaces, these connected spaces should continue to be considered hazardous until the atmosphere is ensured to be gas free by the test.

Action to be taken in the event of an emergency

The guidance contained in resolution A.1072(28) should be understood and form the basis of any emergency response plan. In the event of an emergency in an enclosed space the ship's crew should follow the ship-specific enclosed space emergency response plan. In an emergency the ship's crew, or any shore personnel, should NEVER perform rescue entering an enclosed space independently, but should always follow the agreed rescue plan.

The urge to enter an enclosed space where an accident has occurred is immense and should be always resisted. Many enclosed space accidents have been compounded by the good intentions of inadequately equipped ad hoc rescue attempts where the would-be rescuers have themselves become casualties.

If in port, a simple explanation of the enclosed space procedures placed at the entry to the ship together with an agreed emergency response plan will greatly assist the efficient rescue of any casualties following an enclosed space accident occurring on board the ship.

A clear understanding of the response required by ship and shore rescue teams will be most advantageous, even essential.

It is critical that the ship has an enclosed space emergency response plan, which is easily understood, regularly practised, verified as effective and followed precisely.

The emergency response plan should form a part of the company SMS.

Equipment should be provided for the ship's crew to utilize in the event of an enclosed space accident. Such equipment should fall into three main categories:

1. Equipment to test and verify the enclosed space atmospheric conditions and determine the hazards to life and the mitigations necessary prior to entry;
2. Equipment to ensure the safety of the rescue party such as self-contained breathing apparatus (SCBA), lifelines, harness, etc.; and
3. Equipment to facilitate the safe recovery of a casualty, such as recovery hoist, stretchers and resuscitation equipment.

The rescue of a casualty should be undertaken in a steady, controlled and methodical way. The aim is the safe rescue of the casualty without needlessly endangering the lives of those undertaking the rescue operation.

An example of an Enclosed Space Emergency Response Plan is contained in appendix of Resolution MSC.581(110) (IMO RESOLUTION MSC.581(110) 2025)

A dangerous space may not necessarily be enclosed on all sides. Some places may not be considered dangerous spaces but the atmosphere may become dangerous because of a change in the condition inside or in the degree of enclosure or confinement, which may occur intermittently, e.g. cargo space access ways.

(Code of Safe Working Practices for Merchant Seafarer's 2024)

The validity of the entry permit should be specified based on the risk assessment and should never be longer than eight hours. The company should ensure that single person entry into an enclosed space is not permitted.

(IMO RESOLUTION MSC.581(110) 2025)

It should also state in the entry permit the maximum permitted time between atmosphere-testing of spaces and when they are entered by personnel, as well as maximum time permitted between testing while the space is occupied. It is recommended this period should not exceed 30 minutes between testing and that records of the tests are maintained, and this should be defined in the operator's SMS.

(International safety guide for oil tankers & terminals, 2020).

Connected space means a space that is connected, by either permanent or temporary means (such as a door), to a source space that may contain a hazardous atmosphere.

For clarity, a space separated by a manual door, even if watertight, should be considered as "connected" as it is impossible to tell from outside the space whether it is open or closed or indeed properly sealed. A connected space should be treated as containing a hazardous atmosphere until testing proves otherwise. The nature of the connection may lead to a "trapped hazardous atmosphere".

Adjacent space means a space that shares a common boundary with a compartment that may contain a hazardous atmosphere. Such a space has no openings, temporary or permanent, into the hazardous compartment whatsoever and is designed to be a contiguous barrier. Such a space may only contain a hazardous atmosphere in the event of failure of that barrier. Precautions should relate to the possibility of such a failure.

Trapped Hazardous Atmosphere means a hazardous atmosphere that may be trapped in a connected space in a manner that causes that space's atmosphere to fill and/or to empty at a different rate to the source space. Such a space, while recognized as containing the same atmosphere, should be treated independently to the source space, and should be assumed to contain a hazardous atmosphere until proved otherwise by testing. For example, a trapped atmosphere may remain even after the cargo in the source space is discharged.

Competent person means a person with an operational level of competency to make an informed assessment of the likelihood of a dangerous atmosphere being present or subsequently arising in the space.

Responsible person means a person in a management level on board a ship (i.e. master, chief mate, chief engineer officer or second engineer officer) of competency and authorized by the shipping company to permit entry into an enclosed space.

Attendant means a person maintains a watch over those entering the enclosed space, to maintain communications with those inside the space and to initiate the emergency procedures in the event of an incident occurring.

(IMO RESOLUTION MSC.581(110) 2025)

Enclosed Space Register means a ship-specific register which lists all enclosed spaces on board the ship, along with their connected spaces and adjacent spaces, their hazards, associated risk mitigations if applicable, and how the atmosphere in these spaces may change depending upon the nature of cargo carried or the content of the space, and which forms a part of the safety management for enclosed spaces.

Identification of the hazards and assessment of risk

The company should ensure that a risk assessment is conducted to identify all enclosed spaces on board the ship and that the identified enclosed spaces are recorded in the Enclosed Space Register, which should be maintained on board the ship as well as ashore. This Enclosed Space Register and risk assessment should be kept up to date as appropriate to ensure its continued validity, particularly after loading and during the carriage of cargoes which may adversely affect the safety of the atmosphere within a space. A reassessment should also be made when the contents of the space change, such as in case a ship stores treated sewage or grey water temporarily in its ballast water tanks.

The company should consider the use of appropriate technology to assist in the hazard identification and mitigation. The Enclosed Space Register and risk assessment should form the basis of the development of the enclosed space emergency response plan (appendix 1 of IMO RESOLUTION MSC.581(110)).

It is recommended that the enclosed space emergency response plan be reviewed after each drill so that its effectiveness can be assessed and, if necessary, improvements made. As entries for enclosed spaces other than cargo spaces may be different, there is a need to record information for the benefit of personnel involved.

Every ship should have an Enclosed Space Register, the information within which may form the basis of a risk assessment. The Enclosed Space Register should contain:

1. physical layout of the space and access and egress points, including of connected spaces, if any;
2. physical hazards in the space, e.g. vertical ladders, unguarded openings, poor lighting, wet or slippery conditions, excessive heat;
3. connection to adjacent spaces;
4. specific hazards within the space, for example, the effect of ballast water treatment method on the atmosphere within ballast tanks;
5. if used, information related to additional technology, helping to determine enclosed space condition;
6. information related to fixed and portable ventilation systems including equipment and where the equipment is stored;
7. estimated time taken to achieve the air changes for safe entry, using forced or natural ventilation;
8. lighting and means for temporary lighting including intrinsically safe lighting where appropriate;
9. means for atmosphere testing;
10. any pertinent information that would assist the risk assessment process;
11. locking and "Safe to enter"/"Unsafe to enter" signage arrangements; and
12. the equipment necessary to facilitate emergency rescue from the space.

Training and awareness

Only trained and authorized personnel should be assigned the duties of entering, functioning as attendants or acting as members of rescue teams. Ships' crews with rescue and first aid duties should be drilled periodically as required by SOLAS regulation III/19.3.6 in rescue and first aid procedures.

Training should include as a minimum:

1. identification of the hazards likely to be faced during entry into enclosed spaces and whilst within the space, in particular the rapidity with which oxygen may be depleted in a space by corrosion or biological means especially in higher ambient temperatures. The pace at which oxygen gets depleted increases exponentially when ventilation is stopped (fan or other equipment in case of forced ventilation, and closing a vent or damper in case of natural ventilation) or other characteristics or properties of the contents or structure of the space. While higher ambient temperatures can increase the rate of oxygen depletion, it should be assumed that the rate of depletion will always be very rapid, regardless of the ambient temperature;
2. an explanation of how the pace at which oxygen is depleted will increase exponentially when ventilation is stopped (fan or other equipment in case of forced ventilation, and closing a vent or damper in case of natural ventilation), or when hatch covers are closed;
3. identification and use of the various sources of information on the hazards associated with individual solid bulk and liquid bulk cargoes, and the precautions to be adopted when entering spaces containing such cargoes, or their residues;
4. awareness of the fact that when a person or persons in an enclosed space shows signs of adverse health effects, that they should always assume that these effects are due to an oxygen-depleted or toxic atmosphere in the space, and that they should not enter it themselves;
5. recognition of the signs of adverse health effects caused by exposure to hazards during entry;
6. knowledge and experience in the use of personal protective equipment required for entry;
7. rescue, first aid, Cardio Pulmonary Resuscitation (CPR) techniques and evacuation procedures;
8. knowledge and experience in the use of communication equipment and procedures;
9. knowledge and experience in the use of instruments for measuring the atmosphere;
10. knowledge and experience in the use of rescue equipment and procedures;
11. knowledge of the IMO/WHO/ILO Medical First Aid Guide for Use in Accidents Involving Dangerous Goods (MFAG), where appropriate; and
12. knowledge of use of emergency and the first aid equipment for chemical tankers (IBC Code section 14.3) and for gas carriers (IGC Code section 14.3), where appropriate.

(IMO RESOLUTION MSC.581(110) 2025)

Drills:

Drills must be participated in by seafarers whose responsibilities include entry into or rescue from enclosed spaces.

Entrances to Enclosed Spaces

Entry doors or access hatches leading to enclosed spaces should at all times be secured against entry unless the spaces have been risk assessed, atmospherically tested as required and declared safe for entry. The ship may use a system of seals similar to those recommended in the ISPS Code. Suitable portable signage, which is easily understandable by the vessel's crew and also by shore personnel engaged on the ship at this time and in the port, indicating the hazards should be posted on entry doors or access hatches leading to an enclosed space. An example of such signage is contained within appendix 3. These signs should be updated when the space becomes safe for entry or when a safe space becomes unsafe.

A door or hatch cover which is opened to provide natural ventilation of an enclosed space may, wrongly, be taken to be an indication of a safe atmosphere and therefore it is recommended to station an attendant at the entrance or use a mechanical or physical barrier, such as a locked bar or chain positioned across the opening with an attached warning sign, to prevent accidental entry. It is recommended that operations such as these are included in the watch handover activity.

Safety of Shore Personnel

Before any personnel are authorized to enter any cargo space containing any cargo, the competent person should carry out a risk assessment of the cargo; identification of the physical characteristics of the cargo space(s) concerned; and the operations to be carried out, and the responsible person should:

1. where it is necessary for shore personnel to enter any such spaces, conduct a pre-operational risk assessment prior to commencement of cargo operations or arrival of shore personnel. When hazards are identified a joint risk assessment with terminal representatives responsible for operations on board the ship, or with other appropriate shore personnel, should be conducted; and
 2. identify and agree to the precautions required during entry and enclosed space entry permit arrangements to be used.
- (IMO RESOLUTION MSC.581(110) 2025)

Risk Assessment of an Enclosed Space

In order to ensure safety from the outset, entry and occupancy, a competent person should make an assessment of any potential hazards in the space to be entered, taking into account the characteristics of the previous and current cargo carried with Safety Data Sheets (SDS) and cargo information, ventilation of the space, coating of the space and other relevant factors. Opening of an enclosed space should be subject to a risk assessment, taking into consideration the potential for release of dangerous gases or potential to create an explosive atmosphere: the competent person's assessment should determine the potential for the presence of an oxygen-deficient, oxygen-enriched, flammable or toxic atmosphere, which includes carbon monoxide (CO) and carbon dioxide (CO₂) as well as other toxic or asphyxiant gases.

The competent person should bear in mind that the ventilation procedures for an adjacent or connected space may be different from those for the enclosed space itself. The details of the assessment should be recorded in a standard format, and maintained on board the ship.

The nature of the hazards that may be present should be fully understood by those on board the ship, both crew and shore-based personnel, and those ashore managing the ship.

The procedures to be followed for testing the atmosphere in the space and for entry should be decided on the basis of the assessment. These will depend on whether the assessment shows that:

1. there is minimal risk to the health or life of personnel entering the space; or
2. there is no immediate risk to health or life, but a risk could arise during the course of work in the space; or
3. a risk to health or life is identified.

Where the assessment indicates minimal risk to health or life or potential for a risk to arise during the course of work in the space, the precautions described in sections 5 to 8 and 10 of MSC581 (110) should be followed, as appropriate.

Where the assessment identifies a risk to life or health, if entry is to be made, the additional precautions specified in section 9 of MSC581 (110) should also be followed.

Throughout the assessment process, there should be an assumption that the space to be entered is considered as hazardous until positively proved to be safe for entry.

After completion of a suitable risk assessment targeted to the space to be entered steady readings of all of the following should be obtained:

1. 20.9% oxygen by volume;
 2. the level of carbon dioxide has been checked and is less than 0.5% by volume (5,000 ppm);
- Note: National requirements may differ when determining the safe atmosphere range for gases stated above.
3. less than 1% of lower flammable limit (LFL) on a suitably sensitive combustible gas indicator, where the assessment has determined that there is potential for flammable gases or vapours; and
 4. less than 50% of the occupational exposure limit (OEL) of any toxic vapours and gases.

(IMO RESOLUTION MSC.581(110) 2025)

Personal Protective Equipment (PPE)

When considering the planned work activity within an enclosed space where the atmosphere is known or suspected to be unsafe, suitably designed and constructed breathing apparatus of positive pressure type should always be worn, and only personnel trained in its use should be allowed to enter the space. Air-purifying respirators, dust masks and canister face masks do not provide a supply of clean air from a source independent of the atmosphere within the space and should not be used. Emergency Escape Breathing Devices (EEBDs) are not suitable to use for entry into enclosed spaces.

Notwithstanding paragraph 6.5.6 of MSC581(110), persons entering enclosed spaces that may contain a suspected atmospheric hazard should be provided with calibrated and tested personal gas detector suitable for the gas or gases assessed as likely to occur in the space.

Rescue harnesses should be worn and unless impractical, lifelines should also be used. A means to facilitate evacuation from the enclosed space should be available and ready for use, as per emergency response plan.

Appropriate protective clothing should be worn, particularly where there is any risk of toxic substances or chemicals coming into contact with the skin of those entering the space.

(IMO RESOLUTION MSC.581(110) 2025)

RightShip recommends that the enclosed space entry procedure comprise at a minimum the following:

- > Define enclosed space.
- > Dangers associated with enclosed space atmospheres
- > Identify a enclosed space
- > Precautions in general
- > Entry authorization
- > Requirements for entrance into enclosed space
- > Precautions to take before entering an enclosed space
- > Work in enclosed space
- > Education and awareness
- > An enclosed space's risk assessment
- > Drills
- > Requirement for personal protective equipment
- > Requirements for equipment and testing
- > Evacuation and rescue from enclosed spaces
- > Entering enclosed spaces with known or suspected hazardous atmospheres

4.6 Is entry into and rescue from enclosed space training undertaken and are regular drills conducted? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The company should ensure that rescue drills from enclosed space identified in their Enclosed Space Register are undertaken regularly as required by SOLAS regulation III/19.3.6 using the equipment provided to facilitate a rescue from an enclosed space. Such drills should focus on different aspects of operations involving enclosed spaces. SOLAS regulation III/19.3.6 requires:

1. checking and use of personal protective equipment required for entry;
2. checking and use of communication equipment and procedures;
3. checking and use of instruments for measuring the atmosphere in enclosed spaces;
4. checking and use of rescue equipment and procedures; and
5. instructions in first aid and resuscitation techniques.

(IMO RESOLUTION MSC.581(110) 2025)

Crew members with enclosed space entry or rescue responsibilities shall participate in an enclosed space entry and rescue drill to be held on board the ship at least once every two months.

The enclosed space entry/rescue drill should include a scenario involving the rescue of a person from the cargo hold, including access to and retrieval from the space.

4.7 Are procedures in place for the control of hot work, are they incorporated in the safety management system and is there documented evidence of compliance? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Permits for hot work should be specific regarding the exact risks associated with the specific hot work, location, timing, and hazards.

Permits to hot work should address dangers to all adjacent cargo or other flammable materials that may be exposed, as well as the necessity for additional protective covers.

Hot work means any work requiring the use of electric arc or gas welding equipment, cutting burner equipment or other forms of naked flame, as well as heating or spark generating tools, regardless of where it is carried out on board a ship. The safety management system (SMS) on board should include adequate guidance on control of hot work and should be robust enough to ensure compliance. Absence of guidance should be regarded as prohibition, rather than approval.

Whenever possible, a space such as a workshop where conditions are deemed safe, should be designated for hot work to be performed and first consideration given to performing any hot work in that space.

> Hot work performed outside that space should be subject to the following considerations.

Hot work outside the designated space:

- > The Master or designated safety officer should be responsible for deciding whether hot work is justified and whether it can be conducted safely.
 - > A permit-to-work system should be employed.
 - > Hot work procedures should take account of national laws or regulations or other national safety and health rules.
 - > A responsible officer, not involved in the hot work, should be designated to ensure that safe procedures are followed.
 - > A written plan for the operation should be agreed by all who will have responsibilities in connection with the hot work.
 - > The work area should be carefully prepared and isolated before hot work commences.
 - > Fire safety precautions should be reviewed, including fire equipment preparations, setting a fire watch in adjacent compartments and areas, and fire-extinguishing measures.
 - > Isolation of the work area and fire precautions should be continued until the risk of fire no longer exists.
- (Principles for Hot Work on Board all Types of Ships, 2003)

Hot work in places other than the workshop should be the subject of a permit to work.
(Code of Safe Working Practices for Merchant Seafarer's, 2024)

Hot Work Outside Designated Spaces

When hot work is required outside designated spaces, it is essential to follow strict safety protocols to ensure the operation is conducted safely and in compliance with company procedures.

A written plan for the hot work should be prepared and reviewed prior to commencement by a representative of the vessel's management company. Feedback should be provided, and formal agreement obtained to ensure the operation aligns with safety protocols and company procedures.

All personnel involved must be familiar with the permit-to-work system and adhere to the defined safety requirements throughout the operation.

4.8 Has a specific permit to work, Lock-Out/Tag-Out (LOTO) system, and Stop Work Authority policy and procedure been introduced and are they being used effectively? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The vessel's manager should identify the High-risk tasks on board and create a specific permit and risk assessment system for the ship.

The safety management system for individual ships will determine when permit to work systems should be used, and the form of the permit to work. (Code of Safe Working Practices for Merchant Seafarer's, 2024)

Wherever there is a high-risk job taking place, a written permit to work procedure should always be used. Jobs considered to be high risk should include:

- > Entry into enclosed or confined spaces.
- > Working on machinery or equipment which can start automatically or requires isolation.
- > Hot work including welding.
- > Working aloft or overside.
- > General electrical work (Under 1000 Volts).
- > Electrical high voltage work (Over 1000 Volts).
- > Working on elevator, and
- > Underwater Operation.

Additional Permits to Work may be required depending on the trade of the ship and the work carried out. Permits can be individual or cover a number of work types.

Working aloft or overside:

- > The ship's manager shall specify a height above a deck or tank top that is considered to be "working aloft or from height,
- > Define the meaning of working over or near the side.
- > Identify shipboard tasks that may require a seafarer to work from height or over the side and the need for the risk assessments for those tasks to identify and address the associated hazards.
- > Identify practical alternatives for completing routine-routine tasks without a seafarer needing to work from height or over the side;
- > Articulate the need for all seafarers to remain vigilant-vigilant and exercise care whenever they move about the ship.

Plant is a general name for equipment, machinery, appliances, tools and implements. Every year, seafarers at work are injured, sometimes fatally, when plant inadvertently activates or stored energy including electricity, heat, steam, coiled, tension, compression, pressure and fluids released during inspection, repair, maintenance, or cleaning. The vessel's manager shall implement an effective isolation procedure into the ship's SMS.

A procedure for working over the side to rig and recover accommodation ladders and combination pilot ladders should be incorporated into the company's SMS. The work permit and risk assessment forms should specially identify this task taking account of vessel's movement and weather conditions.

Lock Out/Tag Out system are used to prevent contact with a hazard while performing tasks that require the removal, by-passing, or deactivation of safeguarding devices, and the unintended release of hazardous energy (stored energy), or the unintended start-up or motion of machinery, equipment, or processes. Lock-Out/Tag-Out is a decommissioning/recommissioning work system. Decommission to make the work environment safe, and recommission to restore operational readiness.

Lock Out is the control of hazardous energy by the placement of a lock and tag on an energy-isolating device, indicating that the energy-isolating device is not to be operated until removal of the lock or tag. In practice, lockout is the isolation of energy from the system (a machine, equipment, or process) which physically locks the system in a safe mode. The energy-isolating device may be a manually operated disconnect switch, a circuit breaker, a line valve, or a block. Push buttons, selection switches and other circuit control switches are not considered energy-isolating devices.

Tag Out is a labelling process that is always used when lockout is required. The process of tagging out a system involves attaching or using a standardised label that includes the following information:

- > Why the lockout or tag out is required (repair, maintenance, etc.).
- > Time of Application of the lock or tag; and
- > The name of the authorised person who attached the tag and lock to the system

Only the authorised individual who placed the lock and tag onto the system is the one who is permitted to remove them. This procedure helps make sure the system cannot be started up without the authorised individual's knowledge. The following standards can be referred to for safe guidelines: AS/NZS 4836:2011, AS 4024.1603-2006.

Unless specifically addressed by ship-specific "Gantry Crane Isolation" procedures, the crane shall be subject to the company's general lockout/tagout procedures under the following circumstances:

- > When not in use, to prevent unauthorized use.
- > When being worked on, to ensure the safety of maintenance staff or operators.
- > When deemed unfit for use, either following an incident, inspection, or prior to commissioning.
- > When people are working near potential power sources, such as bus bars and electrical tracks.

Stop Work Authority (SWA) Policy:

The company should establish a Stop Work Authority (SWA) policy that empowers all personnel, regardless of role or rank, to halt any work activity that poses a risk to health, safety, the environment, or equipment.

This policy should clearly outline the following key requirements for implementation across all operations:

1. Universal Authority: Every individual has the right and responsibility to stop work when unsafe conditions or behaviours are observed.
2. No Retaliation: Personnel who exercise SWA in good faith should be protected from any form of disciplinary action or retaliation.
3. Immediate Response: Work should be stopped immediately upon identification of a hazard, and the issue should be reported without delay.
4. Assessment & Resolution: The hazard should be assessed, corrective actions taken, and the situation resolved before work resumes.
5. Training: All personnel should receive training on the SWA policy during induction and through ongoing safety briefings.
6. Leadership Support: Managers should actively support and promote the SWA policy to foster a strong and proactive safety culture.

Stop Work Authority (SWA)

The SMS should include a Stop Work Authority (SWA) policy and procedure. SWA recognizes the importance of encouraging any employee on board a vessel to express concern if they believe that an operation is being incorrectly undertaken or unsafe. SWA gives crewmembers the responsibility and obligation to intervene and stop work if they see something unsafe that may cause an accident.

A typical SWA is comprised of six steps:

Stop – When you or a colleague perceive condition(s) or behaviour(s) that pose imminent danger to person(s), equipment, or the environment, they must immediately initiate a stop work intervention with the person(s) potentially at risk.

Notify – Notify affected personnel and supervision of the stop work action.

Investigate – Affected personnel will discuss the situation and come to an agreement on the stop work action.

Correct – The affected area(s) will then be inspected by qualified experts to verify completeness of the modifications and ensure all safety issues have been properly resolved.

Resume – All affected personnel will be notified of what corrective actions were implemented, and the affected area(s) will be reopened for work by personnel with restart authority.

Follow-up – The Safety Manager will publish the incident details regarding the stop work action to all Operations Managers and employees, outlining the issue, corrective action, and lessons learned.

Toolbox talk

A toolbox talk before the work begins is key in ensuring that all workers involved in the work understand and are aware of any hazards and their associated risks.

A toolbox talk is another form of risk assessment carried out in support of a Task -Based-Risk assessment (TBRA). Its prime purpose is to talk through the procedures of the job in hand and the findings of the TBRA with the seafarers involved.

A task-based risk assessment (TBRA) is a systematic approach to identify, evaluate, and control hazards associated with specific tasks or activities in the workplace. It focuses on the individual steps or stages of a task to pinpoint potential hazards and implement targeted safety measures.

When carrying out a toolbox talk, it is important to actively involve workers doing the work and others who may be at risk; in other words, seafarers, sub-contractors and others on board ship who may be affected. Encourage full and active participation and discuss any questions or concerns. Once finished, confirm that everyone fully understands their role in the task and the precautions in place (this is known as 'closed-loop communication'). Record this along with details of any relevant risk assessment to which you have referred.

Give a toolbox talk before any work is carried out that involves more than one person and where there is significant risk to persons or assets.

(CPSWP 2025)

Integrating LOTO with the Permit to Work (PTW) System

To ensure the highest level of safety when working with hazardous energy sources, or within spaces containing such sources, it is essential to integrate Lockout/Tagout (LOTO) with the Permit to Work (PTW) system. When these systems operate independently, critical safety gaps can arise, such as work commencing before energy sources are fully isolated.

Key Benefits of Integration:

1. Ensures all hazardous energy sources are locked out before work begins.
2. Provides a single, unified workflow for managing high-risk tasks.
3. Enhances accountability, traceability, and regulatory compliance.
4. Reduces the risk of human error and miscommunication.

4.9

Is there a schedule of drills and exercises to address potential emergency shipboard situations and is it being conducted effectively? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

ISM requirement:

"The company should identify potential emergency shipboard situations and establish procedures to respond to them. The company should establish programs for drills and exercises to prepare for emergency actions".
(ISM Code and Guidelines on the Implementation of the ISM Code, 2010)

Emergency procedures should at least include collision, grounding, flooding, heavy weather damage, cargo damage, shift of cargo, loss of cargo, structural failure as per MSC Circ. 1143, fire (on deck and in cargo hold, the engine room and accommodation), damage to fixed and floating objects, explosion, pollution by harmful substances in packaged form, critical machinery failure, rescue from enclosed spaces, serious personal injury, emergency towing equipment, helicopter operations and pollution clean-up and emergency operation of hatch cover.

SOLAS requirement:

On-board training in the use of the ship's fire-extinguishing systems and appliances shall be planned and conducted in accordance with the provisions of regulation SOLAS III/19.4.1. 2.2.5

Fire drills shall be conducted and recorded in accordance with the provisions of regulations SOLAS III/19.3 and III/19.5.

Abandon ship drill: Each lifeboat shall be launched with its assigned operating crew aboard and manoeuvred in the water at least once every three months during an abandon ship drill.

Free fall lifeboat: In the case of a lifeboat arranged for free-fall launching, at least once every three months during an abandon ship drill the crew shall board the lifeboat, properly secure themselves in their seats and commence launch procedures up to but not including the actual release of the lifeboat (i.e., the release hook shall not be released). The lifeboat shall then either be free-fall launched with only the required operating crew on board or lowered into the water by means of the secondary means of launching with or without the operating crew on board. In both cases the lifeboat shall thereafter be manoeuvred in the water by the operating crew. At intervals of not more than six months, the lifeboat shall either be launched by free-fall with only the operating crew on board, or simulated launching shall be carried out in accordance with the guidelines developed by the Organization.

Muster lists and emergency instructions shall be exhibited in conspicuous places throughout the ship, including the navigation bridge, engine-room, and crew accommodation spaces.

The muster list shall specify details of the general emergency alarm and public address system, and action to be taken by crew and passengers when this alarm is sounded. The muster list shall also specify how the order to abandon ship will be given.

The muster list shall specify which officers are assigned to ensure that lifesaving and fire appliances are maintained in good condition and are ready for immediate use.

The muster list shall specify substitutes for key persons who may become disabled.

The muster list shall be prepared before the ship proceeds to sea.

The muster list shall show the duties assigned to the different members of the crew including:

- > Closing of the watertight doors, fire doors, valves, scuppers, side scuttles, skylights, portholes, and other similar openings in the ship
- > Equipping of the survival craft and other life-saving appliances
- > Preparation and launching of survival craft.
- > General preparations of other life-saving appliances
- > Muster of passengers
- > Use of communication equipment
- > Manning of fire parties assigned to deal with fires
- > Special duties assigned in respect to the use of fire-fighting equipment and installations.
- > Illustrations and instructions in appropriate languages shall be posted in cabins and be conspicuously displayed at muster stations and other spaces to inform crew of:
 - > Their muster station.
 - > The essential actions they must take in an emergency, and
 - > The method of donning lifejackets.

Rescue boat drill: As far as is reasonable and practicable, rescue boats other than lifeboats which are also rescue boats, shall be launched each month with their assigned crew aboard and manoeuvred in the water. In all cases this requirement shall be complied with at least once every three months.

On-board training in the use of davit-launched life rafts shall take place at intervals of not more than four months on every ship fitted with such appliances. Whenever practicable this shall include the inflation and lowering of a life raft. This life raft may be a special life raft intended for training purposes only, which is not part of the ship's life-saving equipment; such a special life raft shall be conspicuously marked.

Steering gear testing and drills: 1- Within 12 hours before departure, the ship's steering gear shall be checked and tested by the ship's crew. 2- All ships' officers concerned with the operation and/or maintenance of steering gear shall be familiar with the operation of the steering systems fitted on the ship and with the procedures for changing from one system to another. 3- Emergency steering drill shall take place at least every once every three months in order to practice emergency steering procedure. (SOLAS 74,2020)

Muster lists and emergency instructions shall be exhibited in conspicuous places throughout the ship, including the navigation bridge, engine-room, and crew accommodation spaces.

The muster list shall specify details of the general emergency alarm and public address system, and action to be taken by crew and passengers when this alarm is sounded. The muster list shall also specify how the order to abandon ship will be given.

The muster list shall specify which officers are assigned to ensure that lifesaving and fire appliances are maintained in good condition and are ready for immediate use.

The muster list shall specify substitutes for key persons who may become disabled.

The muster list shall be prepared before the ship proceeds to sea.

The muster list shall show the duties assigned to the different members of the crew including:

- > Closing of the watertight doors, fire doors, valves, scuppers, side scuttles, skylights, portholes, and other similar openings in the ship
- > Equipping of the survival craft and other life-saving appliances
- > Preparation and launching of survival craft.
- > General preparations of other life-saving appliances
- > Muster of passengers
- > Use of communication equipment
- > Manning of fire parties assigned to deal with fires
- > Special duties assigned in respect to the use of fire-fighting equipment and installations.
- > Illustrations and instructions in appropriate languages shall be posted in cabins and be conspicuously displayed at muster stations and other spaces to inform crew of:
- > Their muster station.
- > The essential actions they must take in an emergency, and
- > The method of donning lifejackets.

4.10

Are procedures for reporting, investigating, and closing out non-conformities, incidents, and hazardous situations available and being followed and, is an incident investigation report or a summarised lessons learned bulletin available for each reported incident? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Inspector shall verify the incident history of the vessel with the 24-month incident history provided by RightShip. Any incidents that are not documented in the RightShip record shall be recorded as a Finding. Record in the comments the number of near-miss reports submitted by the vessel over the past 12 months.

The SMS should include procedures ensuring that non-conformities, incidents, and hazardous situations are reported to the company, investigated and analysed with the objective of improving safety and pollution prevention. (ISM Code and Guidelines on the Implementation of the ISM code, 2010)

The inspector should review the records of reported incidents from the past 12 months and verify whether an incident investigation report or a lessons learned bulletin has been provided by the vessel's manager for each case where the manager has declared the investigation as completed.

Near-miss:

A near-miss is defined as a sequence of events and/or conditions that could have resulted in a loss, but where the outcome was avoided purely by chance or an unplanned interruption in the chain of events.

Potential losses may include personal injury, environmental harm, or adverse business impacts such as repair or replacement costs, scheduling delays, contractual breaches, or reputational damage.

Under the Hazardous Occurrences provisions of the ISM Code, companies are required to investigate near-misses as part of their regulatory obligations.

Vessel's managers should actively encourage the reporting of near-misses to support ongoing safety awareness and continuous improvement.

(MSC-MEPC.7/Circ.7)

4.11 Has a PPE Matrix for use of personal protective equipment been provided and is it being worn as required? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends that ship's crew use an inertia reel personal fall arrestor with a full body harness when working aloft or over side.

Record a Finding if the vessel is not equipped with a safety harness and fall protection device, or if wearing such personal protective equipment is not an obligatory requirement under the SMS when operating aloft or overside. Fall protection devices should be inspected every twelve months or after a fall. This inspection must be done in accordance with the manufacturer's instructions by a competent person. EN 365 specifies the minimum general requirements for instructions for use, maintenance, periodic examination, repair, marking, and packaging of PPE, which includes body holding devices and other equipment used in conjunction with a body holding device to prevent falls, for access, egress, and work positioning, to arrest falls, and for rescue.

"The company must ensure that seafarers are provided with suitable PPE where it is needed. The company should assess the equipment required to ensure that it is suitable and effective for the task in question and meets the appropriate standards of design and manufacture." All personnel who are working at height (i.e. in any position from which there is a risk of falling) shall wear a safety harness (or belt with shock absorber) attached to a lifeline.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

IMSBC requirements:

Many bulk cargoes are dusty. The effects of breathing dust can never be beneficial and are probably harmful in some cases at least. Where possible it is always best to avoid exposure to cargo dust and employers and their representatives have a duty to minimise dust. When exposure to hazardous solids, liquids or gases cannot be avoided respiratory protective equipment (RPE) and safety goggles must be worn.

The selection and use of the appropriate PPE is complex and extremely important. It should be part of the risk assessment process. For general shipboard use a simple respirator with a disposable filter where the wearer's lungs are used to draw air through the filter should be suitable for cargoes which are not stated to be hazardous.

Filters should be renewed according to manufacturers' instructions or, in the absence of instructions, when soiled.

When a chemical product and/or other specialist equipment is used during cargo hold cleaning process, full and correct PPE, suitable for the nature of the task must be available and worn at all times throughout the cargo hold cleaning.

(Guidance on Preparing Cargo Holds and Loading of Solid Bulk Cargoes, 2014)

The minimum PPE requirement when rigging a combination pilot ladder overside shall be incorporated into the guideline.

Employees working in areas where there are potential electrical hazards shall be provided with, and shall use, electrical protective equipment that is appropriate for the specific parts of the body to be protected and for the work to be performed. 29 CFR 1910.137 outlines the design, in-service care, and use regulations for electrical protection equipment, which includes rubber insulating gloves.

Various international standards specify requirements for safety helmets. While these standards may differ in certain details, they all share the common goal of ensuring adequate head protection for their intended applications. It is essential to follow the manufacturer's instructions and to replace helmets after any significant impact or when they no longer meet the required safety standards.

PPE for Ships Using Ammonia as Fuel

20.3 Protective equipment

20.3.1 Suitable protective equipment, including eye protection, to a recognized national or international standard, should be provided for the protection of crew members engaged in normal operations related to the ammonia fuel system.

20.3.2 Personal protective and safety equipment required in this chapter should be kept in suitable, clearly marked lockers located in readily accessible places.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

4.12 This question has been removed from the current version of the document.

4.13 Is a completed IMSBC/BLU Code ship/shore safety checklist for loading and unloading dry bulk carriers available and are the requirements of the checklist complied with? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Incomplete ship/shore safety checklist and/or non-compliance with the checklist should be recorded as a Finding.

The purpose of the ship/shore safety checklist is to improve working relationships between ship and terminal, and thereby to improve the safety of operations. Misunderstandings occur and mistakes can be made when ships' officers do not understand the intentions of the terminal personnel, and the same applies when terminal personnel do not understand what the ship can and cannot safely do.

(BLU Code, 2011)

For vessel fitted with gantry cranes following should be discussed during the meeting and documented in the ship/shore safety check list:

1. The time required for preparation before operations and securing after completion of cargo operation considered in the ship shore safety checks and information exchanged.
2. Operations limits of gantry as per makers manual established (including maximum weather conditions & visibility and ships list/trim) and this information made available to the operator.
3. Blind areas for operator, if any, identified and measures in place for supervision and signalling arranged. Special consideration when operating within shore structures / obstructions or in tandem with shore cranes.

When stevedores are working on board the vessel, the ship-shore safety checklist should be shared with them. Both the stevedore and the master's representative are responsible for ensuring full compliance with the checklist, including item 13 of the BLU (Bulk Loading/Unloading) ship-shore safety checklist. This is essential to maintain a safe atmosphere within the cargo holds and access points at all times while stevedores and other personnel are present in these compartments.

Access hatches to enclosed spaces should remain locked while stevedores are working on board. This measure is essential to prevent unauthorized and unsupervised entry into cargo hold access areas, thereby ensuring the safety of all personnel.

Effective access control to cargo holds and enclosed spaces—whether through physical barriers or robust entry procedures that restrict easy access—should be firmly established and maintained.

The ship-shore safety checklists should include a stevedore familiarisation section with emergency procedures.

The ship-shore checklist should verify compliance with cargo hold entry procedures at least once every watch.

4.14 Are Water Ingress Detector System(WIDS) and alarms maintained in good condition and are the records of tests being maintained? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the vessel is classified as a General Cargo Ship according to the Class certificate, the response to this question should be N/A.

- > Bulk carriers shall be fitted with water level detector
- > In each cargo hold, giving audible and visual alarms, one when the water level above the inner bottom in any hold reaches a height of 0.5m and another at a height not less than 15% of the depth of the cargo hold but not more than 2.0 m. On bulk carriers to which regulation 9.2 applies, only the latter alarm need be installed. The water level detectors shall be fitted in the aft end of the cargo holds. For cargo holds which are used for water ballast, an alarm overriding device may be installed. The visual alarms shall clearly discriminate between the two different level detectors in each hold.
- > In any ballast tank forward of the collision bulkhead required by regulation II-1/11, giving an audible and visual alarm when the liquid in the tank reaches a level not exceeding 10% of the tank capacity. An alarm overriding device may be installed to be activated when the tank is in use; and
- > In any dry or void space other than a chain cable locker, any part of which extends forward of the foremost cargo hold, giving an audible and visual alarm at a water level of 0.1 m above the deck. Such alarms need not be provided in enclosed spaces the volume of which does not exceed 0.1% of the ship's maximum displacement volume.
- > The audible and visual alarms specified in paragraph 1 shall be located on the navigation bridge.
- > The record of testing of alarm systems should be retained on board.
- > The electrical power supply should be from two separate sources, one should be the main source of electrical power and the other should be the emergency source, unless a continuously charged dedicated accumulator battery is fitted, having arrangement, location and endurance equivalent to that of the emergency source (18 hours). The battery supply may be an internal battery in the water level detector system.
- > The changeover arrangement of supply from one electrical source to another need not be integrated into the water level detector system.
- > Where batteries are used for the secondary power supply, failure alarms for both power supplies should be provided.

(SOLAS 74,2020)

(Resolution MSC. 188 (79)/Performance Standard for Water Level Detectors 2004)

4.15 Has a smoking policy been implemented; is it being followed and are designated smoking areas adequately identified? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Environmental Tobacco Smoke (ETS) is a substantial health risk to nonsmokers. There is no degree of exposure to ETS that is considered safe. Passive smoking entails the inhalation of ETS. ETS is a complex mixture of chemicals and particles (particulate matter) that is released into the air when a person smokes a cigarette, cigar, or pipe. Nonsmokers may face a variety of adverse health consequences as a result of ETS.

(Guidance Note on the Elimination of Environmental Tobacco Smoke in the Workplace, 2003)

Employers have a responsibility to ensure the health and safety of their employees and other persons on board ship, to the extent that this is reasonably practicable.

RightShip urges the vessel's manager to implement a smoking policy with the following objectives:

- > to promote the health and welfare of seafarers;
- > to provide and maintain a healthy shipboard environment;
- > to minimise the risks of tobacco smoke to non-smokers;
- > to educate seafarers about the harmful effects of smoking; and
- > to provide support and assistance to any seafarers who smoke and express a desire to quit.

Control methods that should be considered include the following:

- > Restriction of smoking locations to protect non-smoking personnel from ETS and to provide specified safe smoking areas for smokers during their off-duty hours.
- > Smoking should be permitted only in designated places, with visible instructions and prohibition notices.
- > Any room allocated for smoking should be utilised only for that purpose. It is not suggested that smoking be permitted in the cabins.
- > Where reasonably practicable, any room that is designated for smoking should be adequately ventilated and not ventilated into a smoke-free place.
- > Smoking should be prohibited in kitchens, galleys, pantries, storerooms or other places where food is stored, handled or prepared and notices to this effect should be displayed.
- > Careless disposal of burning matches and cigarette ends is dangerous: ashtrays, or other suitable containers, should be provided and used in locations where smoking is permitted.

4.16 Are portable gas detectors suitable for atmosphere testing of enclosed spaces provided; in good condition; calibrated in accordance with the manufacturer's instructions, and are officers trained and competent with their operation? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The SOLAS XI-1/7 regulation requires portable gas detectors to be dedicated 4-gas (oxygen, flammable gases, or vapours (% of LFL), carbon monoxide; and hydrogen sulphides), capable of 10 hours continuous operation, waterproof and dustproof to Ingress Protection rating IP67, and capable of remote detection (using a pump with a sample hose) suitable to test the atmosphere in an enclosed space before entry. Calibration is also a requirement, as prescribed by the manufacturers' instructions.

(SOLAS 74,2020)

The gas detection equipment should be appropriate for the cargo that the ship has carried and is carrying considering information provided in the shipper's declaration, the Safety Data Sheets (SDS), the IMDG Code, the IMSBC Code, International Bulk Chemical Code (IBC Code) and the IGC Code. The gas detection equipment, including devices for testing CO₂, should be capable of operating correctly even in oxygen-depleted atmosphere.

Appropriate testing of the atmosphere of a space should be carried out with properly calibrated equipment by persons trained in the use of the equipment. The manufacturers' instructions should be strictly followed. Testing of the atmosphere of a space should be carried out before any person enters the space and at regular intervals thereafter until all work is completed. Where appropriate, the testing of the atmosphere of a space should be carried out at as many different levels and areas as is necessary to account for gas stratification, and obtain a representative sample of the atmosphere in the space. In some cases, it may be difficult to test the atmosphere throughout the enclosed space without entering the space (e.g. the bottom landing of a stairway or complex areas of the structure within the space) and this should be taken into account.

All ships should carry at least two sets of gas detection equipment as required by SOLAS regulation XI-1/7, taking into account the Guidelines to facilitate the selection of portable atmosphere testing instruments for enclosed spaces as required by SOLAS regulation XI-1/7 (MSC.1/Circ.1477). Any ship which may carry a cargo capable of generating hazardous vapours and which requires regular entry into the cargo space for cleaning or inspection should carry two sets of gas detection equipment in addition to those required by SOLAS regulation XI-1/7 for assessing the risk to personnel entering the space. The use of flexible hoses or fixed sampling lines, which reach remote areas within the enclosed space, may allow for safe testing without the need to enter the space.

(IMO, Resolution MSC.581(110), 2025)

RightShip recommends that vessels carry a minimum of two portable gas detectors equipped with built-in sample pumps to ensure accurate and reliable atmosphere testing. Detectors fitted with manual aspiration pumps are acceptable only if supplied by the manufacturer and accompanied by clear, model-specific instructions stating the exact number of strokes required to fully purge the sampling line and deliver a representative sample to the sensor chamber. This is essential, as stroke volume varies significantly between models; insufficient strokes can result in incomplete sampling, inaccurate readings, and elevated safety risk to personnel. Bump test is a "qualitative function check where a challenge gas is passed over the sensor(s) at a concentration and exposure time sufficient to activate all alarm indicators to present at least their lower alarm setting. This is typically dependent on the response time of the sensor(s) or a minimum level of response achieved, such as 80% of gas concentration applied." This verifies that sensors and alarms are functioning properly; if they are not, a blockage may be present. In conclusion, bump testing examines function, not accuracy.

Calibration Check - A quantitative test utilizing a known traceable concentration of test gas to demonstrate that the sensor(s) and alarms respond to the gas within manufacturer's acceptable limits. This is typically $\pm 10\text{-}20\%$ of the test gas concentration applied unless otherwise specified by the manufacturer, internal company policy, or a regulatory agency.

Full calibration - The adjustment of the sensor(s) response to match the desired value compared to a known traceable concentration of test gas. This should be done in accordance with the manufacturer's instructions.

Personal gas monitoring equipment is designed for personal use only, to provide a warning against oxygen deficiency, toxic gases and explosive atmospheres whilst the wearer is in the space. This equipment should not be used as a means of determining whether a dangerous (enclosed) space is safe prior to entry, unless the specific equipment has the necessary certified/approved additional capability to conduct remote readings (i.e. pumping capability)

(The merchant shipping and fishing vessels (entry into enclosed spaces) regulations 2022, 2022)

4.17 Is the welding and gas-burning equipment, as well as the electric arc welding equipment, properly stored and maintained in good working order? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a finding if the annual check of the oxygen-acetylene welding equipment/installation on board has not been carried out by a competent person in accordance with the equipment manufacturer's guidelines, or if there is no recorded evidence of such a check. In the absence of the manufacturer's guidelines, such guidelines should be sought from the certified service providers engaged in this service.

Oxy-acetylene gas equipment typically consists of oxygen and acetylene cylinders, pressure regulators, safety devices such as non-return valves and flame arrestors, and a flexible hose assembly capable of supplying a mixed gas output via a blowpipe.

A pressure regulator should be selected based on the compatibility of the gas. Regulators are designed for a certain gas and must not be used with any other gas once installed in that gas service. Oxygen regulators are intended for use in oxygen service only and acetylene regulators are intended for use in acetylene service only. The use of LPG / Propane in gas cutting and welding systems are prohibited.

The pressure regulator shall be replaced every five years or as indicated by the manufacturer.

Hose tail refers to the end of a coupling device (such as a nipple/nozzle) that is intended to be placed into a hose.

The hose shall be crimped to the hose tail using an appropriate crimping instrument that provides reproducible crimping performance. To secure the hose to the hose tail, no worm screw drive or similar detachable clips or clamps shall be utilized.

Hoses shall be constructed from a material compatible with the gas being used. Hoses shall not be utilized for gases or at pressures greater than those for which they were intended.

- > Hose assemblies shall be permanently labelled with the following information: The manufacturer's name or trademark; and
- > A reference to a recognized international standard, for example EN 1256.

This information shall be marked on the mounting device for the hose or on a separate band attached to the hose adjacent to the coupling or on the mounting device.

When a hose's general condition deteriorates, it must be discarded.
Hoses used with welding equipment are color-coded. Hoses for oxygen should be blue, whereas hoses for acetylene should be red.

Blowpipes have been referred to as torches, lamps, blowtorches, guns, shanks, burners, and handles.

Every five years from the date of manufacture, or as recommended by the manufacturer, the flame arrestor shall be renewed.

Blowpipes shall be maintained in accordance with the manufacturer's / supplier's recommendations.

Prior to each blowpipe inlet connection, a non-return valve (hose check valve) shall be inserted into the assembled hose. The non-return valve should be replaced according to the manufacturer's recommendations.

Copper pipe or fittings shall not be used to connect acetylene hoses.
The pipeline and fittings for oxygen distribution must be made of seamless steel or copper.

The hard pipe for oxygen and acetylene must be color coded, with blue piping for oxygen and red piping for acetylene.

(BS EN 1256:2006 Gas welding equipment. Specification for hose assemblies for equipment for welding, cutting and allied processes, 2006)

(Code of Practice 7 The Safe Use of Oxy-Fuel Gas Equipment (individual Portable or Mobile Cylinder Supply), 2018)

(BS EN 1256:2006 Gas welding equipment. Specification for hose assemblies for equipment for welding, cutting and allied processes, 2006)

Regular inspection, thorough examination, and testing of all components to ensure that all the oxy/ acetylene equipment in use on board is in a safe operational condition. This should be performed at least annually by a competent person in accordance with the equipment manufacturer's guidelines. Objective evidence of such tests should be available on board.

Pressure testing of the on-board piping system should be performed by a competent person in accordance with applicable CLASS/ FLAG rules. An appropriate certificate or service record should be issued.
(UK P&I club, Technical Bulletin-Oxy/Acetylene equipment, 2008)

Electric Arc Welding

Welding Electrocutation Risk

Electric arc welding can cause electrocution due to three main factors:

- > High Open Circuit voltage(OCV): Open Circuit Voltage (OCV), No-Load Voltage, Safety Voltage, etc., indicate the expected welding output voltage when the machine is ON but not actively welding. A high OCV increases the chance of electric shock.
- > Faulty or non-insulated electrode holder: A damaged or non-insulated electrode holder exposes the welder to electrocution risk.
- > Return clamp connected to hull structure: If the welder connects the return clamp to the ship's structure instead of the workpiece, it increases the risk of getting electrocuted."

(Code of Safe Working Practices for Merchant Seafarers , 2024)

Welding Personal Protective Equipment (PPE) Requirements

24.4.2 The operator should normally wear:

- > welding shields or welding goggles with appropriate shade of filter lens conforming to EN 169 or to a recognised international standard. (goggles are only recommended for gas welding and flame cutting);
- > leather gauntlets;
- > leather apron (in appropriate circumstances);
- > long-sleeved natural-fibre boiler suit or other approved protective clothing.

24.4.3 Clothing should be free of grease and oil and other flammable substances.

Welding Power Source Requirements

24.6.1 To minimise personal harm from electric shock, electric welding power sources for shipboard use should have a direct current (DC) output not exceeding 70 volts, with a minimum ripple. See section 24.6.13 for further information on DC power sources.

24.6.2 When DC equipment is not available, AC output power sources may be used providing:

- > they have an integral voltage-limiting device to ensure that the idling voltage (the voltage between electrode and workpiece before an arc is struck between them) does not exceed 25 V rms the proper function of the device (which may be affected by dust or humidity) is checked each time a welding set is used.

24.6.3 Some voltage-limiting devices are affected by their angle of tilt from the vertical. Therefore mount and use them in the position specified by the manufacturers. This requirement can be affected by adverse sea conditions.

24.6.12 Provide a local switching arrangement or other suitable means to rapidly cut off current from the electrode if the operator gets into difficulties, and to isolate the holder when changing electrodes.

24.7.1 In addition to the protective clothing specified in section 24.4.1. the welding operator should wear non-conducting safety footwear complying with BS EN ISO 20345:2012/BS EN 50321-1:2018. or to a recognised international standard. Clothing should be kept as dry as possible as some protection against electric shock; it is particularly important that gloves should be dry because wet leather is a good conductor.

4.18

Are the lifeboats, rescue boat and davit-launched life raft their equipment and launching arrangements serviced periodically in good condition, and are the crew familiar with the launching procedure and operation? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When the vessel is equipped with a davit-launched life raft, the inspector shall check the crew's familiarity with the operation of the davit and test its operation.

Wire rope slings and pendants are occasionally sheathed in plastic by the manufacturer. This sheathing allows saltwater ingress, which aids corrosion by retaining moisture and prevents external inspection and the application of lubricant or other corrosion-inhibiting substances.

The vessel's managers should establish a procedure for the regular inspection and maintenance of wire rope slings and pendants sheathed in plastic. Consideration should be given to the use of wire rope slings and pendants without plastic sheathing, or an alternative method of encasing the sheath that allows for removal during inspection.

The vessel's managers should ensure that wire rope slings and pendants are renewed regularly, following the same maintenance principles as lifeboat falls, as stipulated in SOLAS Chapter III, Regulation 20.4.

Wire-rope grips for the falls of life-saving appliances, such as bulldog grips, are not acceptable for any primary load-bearing terminations. This includes falls for lifeboats, rescue boats and davit-launched life rafts, as well as hanging off pendants and recovery stops. Wire rope terminations are to be manufactured and the termination process is to be in accordance with the requirements of the manufacturer and/or a National or International Standard. Wire rope grips are not to be used as the primary load bearing termination.

A davit-launched rescue boat is typically deployed using standard electric or hydraulic power. In emergency situations, alternative methods such as hydraulic accumulator power, compressed air, manual rotation, or gravity/crank mechanisms may be used. Inspectors should confirm that officers and crew are familiar with the emergency operation of the davit, and verify that the rescue boat davit is fully operational under both normal and emergency power sources.

Each survival craft shall be stowed in a state of continuous readiness so that two crew members can carry out preparations for embarkation and launching in less than five minutes.

(SOLAS 74,2020)

Falls used in launching shall be inspected periodically with special regard for areas passing through sheaves and renewed when necessary due to deterioration of the falls or at intervals of not more than five years – whichever is the earlier.

(Measures to Prevent Accidents with Lifeboats, 2006)

Each free-fall lifeboat shall be fitted with a release system which shall be designed to test the release system without launching the lifeboat.

Each lifeboat shall be clearly marked with the number of persons for which the lifeboat is approved and the name and port of registry. Means of identifying the ship to which the lifeboat belongs, and the number of the lifeboat shall be marked in such a way that they are visible from above.

(Life-saving appliances including LSA Code, 2017)

The release system of lifeboats, rescue boats, free-fall lifeboat including davit-launched life rafts shall be:

- > Maintained in accordance with instructions for on-board maintenance as required by regulation 36.
- > Subjected to a thorough examination and operational test during the annual surveys required by regulations I/7 and I/8 by properly trained personnel familiar with the system; and
- > Operationally tested under a load of 1.1 times the total mass of the lifeboat when loaded with its full complement of person and equipment whenever the release gear is overhauled. Such over-hauling and testing shall be carried out at least once every five years.

(SOLAS 74,2020)

Davit-launched life raft automatic release hooks shall be:

- > Maintained in accordance with instructions for on-board maintenance as required by regulation 36.
- > Subjected to a thorough examination and operational test during the annual surveys required by regulations I/7 and I/8 by properly trained personnel familiar with the system; and
- > Operationally tested under a load of 1.1 times the total mass of the lifeboat when loaded with its full complement of person and equipment whenever the release gear is overhauled. Such over-hauling and test shall be carried out at least once every five years.

Note: Of particular importance in the checking of lifeboats is the on-load release system fitted to enclosed lifeboats and the maintenance routines for them. A high percentage of accidents at sea are attributed to lifeboats and their release systems.

(Measure to Prevent Accident with Lifeboats, 2006)

SOLAS Regulation III/1.5 requires all ships subject to SOLAS, regardless of build date, to identify existing on-load release mechanisms that do not comply with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the International Life-Saving Appliance (LSA) Code, as amended by IMO Resolution MSC.320 (89); and replace them with compliant release mechanisms no later than the next scheduled dry-docking after July 1, 2014 (but in any case, before July 1, 2019). SOLAS Regulation III/1.5 does not apply to the release mechanisms on free-fall lifeboats.

(IMO Circular MSC.1/Circ.1392, Guidelines for Evaluation and Replacement of Lifeboat Release and Retrieval Systems)

Wire rope grips can be one of the most dangerous fitting if not used correctly. At least three wire rope grips should be used, with the saddles on the live part of the rope, and the U-bolt pressing on the less heavily loaded tail of the rope. They should be spaced at least six wire diameters apart.

The operating instructions for the lifeboat and life raft shall be displayed.
On or near survival craft and their launching controls, posters or signs shall:

1. Clearly indicate the purpose of the controls and the procedures for operating the appliance, as well as any pertinent instructions or warnings.
2. Be easily visible under emergency lighting conditions; and
3. Use symbols consistent with resolution A.760, as amended by MSC.82.

(SOLAS 74,2020)

Some vessels may be equipped with a common davit to launch the rescue boat and the liferafts. These davits have two separate hooks for lifting the rescue boat (on-load release) and the liferafts (off-load release). The hooks must be properly marked and their use clearly understood by all personnel to avoid the wrong hook being used with the wrong type of equipment.

(Survival Craft A Seafarer's Guide 2008)

4.19 Are life rafts in good order and are hydrostatic release units maintained and installed correctly? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Cargo ships shall carry one or more inflatable or rigid liferafts, stowed in a position providing for easy side-to-side transfer at a single open deck level and of such aggregate capacity as will accommodate the total number of persons on board. If not stowed in a position providing for easy side-to-side transfer at a single open deck level, the total capacity available on each side shall be sufficient to accommodate the total number of persons on board.

If a free-fall lifeboat is fitted, cargo ships shall have one or more inflatable or rigid liferafts, on each side of the ship, of such aggregate capacity as will accommodate the total number of persons on board. The liferafts on at least one side of the ship shall be served by launching appliances.

(SOLAS 74,2020)

For davit launched liferafts, the launching appliance shall include an automatic release hook arranged so as to prevent premature release during lowering and shall release the liferaft when waterborne. The release hook shall include a capability to release the hook under load. The on-load release control shall:

- > Be clearly differentiated from the control which activates the automatic release function;
- > Require at least two separate actions to operate;
- > Be designed such that crew members on deck can clearly observe when the release mechanism is properly and completely set.

(Life-saving appliances including LSA Code, 2017)

Every liferaft shall be stowed with its painter permanently attached to the ship.

Each liferaft or group of liferafts shall be stowed with a float-free arrangement so that each floats free and if inflatable, inflates automatically when the ship sinks.

Liferafts shall be so stowed as to permit manual release of one raft or container at a time from their securing arrangements.

(SOLAS 74,2020)

Some hydrostatic release manufacturers recommend that each liferaft is fitted with its own individual hydrostatic release unit (HRU), to prevent the possibility, where more than one liferaft is utilising the same release, of one of the liferafts breaking the weak link before the second or subsequent liferafts have inflated.

When multiple liferafts are connected to a single HRU, each raft must be equipped with its own weak link. A HRU is not required for liferafts stored in the forward part of a vessel.

4.20 Are life jackets in good condition, allocated as per the plan and donning instructions clearly displayed? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The vessel must carry: (a) a lifejacket for each person that the vessel is certified to carry, including a suitable lifejacket for each person aboard the vessel who weighs less than 32 kg; and (b) a sufficient number of lifejackets stowed in working spaces for the use of seafarers who may be required to remain on duty in those spaces. (2) A lifejacket for an adult must: (a) be designed to fit a person weighing up to 140 kg and with a chest measurement of at least 1 750 mm; or (b) have available suitable accessories to enable its use by that person.

Lifejackets selected for free-fall lifeboats and the manner in which they are carried or worn, shall not interfere with entry into the lifeboat, occupant safety or operation of the lifeboat.

(SOLAS 74, 2020)

For ships having keel laying on or after 01 July 2010, the method of securing the lifejacket to the wearer has quick and positive means of closure that do not require tying of knots.

4.21 Are immersion suits in good condition, allocated as per the fire and safety plan and donning instructions clearly displayed? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Guidelines for monthly shipboard inspection of immersion suits and anti-exposure suits (MSC/Circ.1047) are very helpful in identifying obvious problems with a suit, but do not adequately address deterioration of seams and closures (zippers, etc.) which may not be readily apparent by visual inspection. Such deterioration can be detected by pressurisation of the suit with air and testing the seams and closures for leaks with a soapy water solution.

To ensure the maintenance of adequate strength and watertightness of seams and closures of immersion suits and anti-exposure suits with age, it is recommended that each suit be subjected to an air pressure test such as the following, at intervals not exceeding three years, or more frequently for suits over ten years of age:

- > A suitable head piece, fitted with a means to inject air into the suit, should be inserted into the face orifice of the suit and secured so as to minimize leakage around the face seal. A low-pressure monitoring device, either integral to the fitting for air injection or as a separate device, should also be inserted. If the suit is fitted with detachable gloves and/or boots, the wrists and/or cuffs should be sealed by inserting a short length of suitable diameter plastic pipe and securing the gloves and/or boots with suitable wire ties or hose clamps. The zipper should be fully zipped, and any face flap closed. The suit should then be inflated to a pressure of 0.7 to 1.4 kPa (0.1 to 0.2 psi). If an auxiliary inflatable means of buoyancy is provided, it should be inflated through the oral valve to a pressure of 0.7 kPa (0.1 psi) or until firm to the touch.
- > Each seam and closure of the suit and each seam, oral tube and attachment points and joint or valve of any auxiliary inflatable means of buoyancy should then be covered with a soapy water solution containing enough soap to produce bubbles (if leakage is noted at a foot valve to the extent that air pressure cannot be maintained, the valves should be sealed for the test).
- > If leaks are revealed by the propagation of bubbles at seams or closures, the leaking areas should be marked and, after cleaning the suit thoroughly with fresh water and drying it, repaired in accordance with the suit manufacturer's recommendations.

(MSC/Circ.1114 –Guidelines for Periodic Testing of Immersion Suit and Anti-Exposure Suit Seams and Closures, 2004)

4.22 This question has been removed from the current version of the document.

4.23 Has a sample of foam compound, applicable to both fixed and portable systems, been sent for regular testing and is evidence of satisfactory results available? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Protein-based alcohol-resistant foam concentrates should be subjected to a stability test with acetone. The first periodical control of foam concentrates should be performed not more than three years after being supplied to the ship, and after that, every year. The tests should be performed prior to delivery to the ship and annually thereafter

(MSC.1/Circ.1312).

Portable foam applicators:

- > Portable containers or portable tanks containing foam concentrate, excluding protein-based concentrates, less than 10 years old, that remain factory sealed can normally be accepted without the periodical foam control tests required in MSC.1/Circ.1312 being carried out.
- > Protein based foam concentrate portable containers and portable tanks should be thoroughly checked and, if more than five years old, the foam concentrate should be subjected to the periodical foam control tests required in MSC.1/Circ.1312, or renewed; and
- > The foam concentrates of any non-sealed portable containers and portable tanks, and portable containers and portable tanks where production data is not documented, should be subjected to the periodical foam control tests required in MSC.1/Circ.1312.

(MSC.1/Circ.1432 -Revised guidelines for the maintenance and inspection of fire protection system and appliances 2012)

Perfluoro-Octane Sulfonic Acid (hereinafter referred to as PFOS) is known to be harmful to humans, other creatures and the environment. Where fire-extinguishing media used or stored on ships are confirmed to contain PFOS, they need to be disposed to appropriate shore-based reception facilities no later than the first survey on or after 1 January 2026.

(Amendments to SOLAS regulations II-2/1.2.10 and II-2/10.11 as adopted by resolution MSC.532(107 2023)

4.24 Are fire mains and associated isolation valves, fire boxes, hoses, nozzles, applicators, and spanners regularly inspected and maintained and found to be in a satisfactory operating condition? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

4.25 Are the International Shore Connection fitting arrangements clearly marked and well maintained and are the crew aware of their location? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

There should be at least one shore connection for ship greater than 500GRT.

The international shore connection is a standard sized flange with nuts, bolts and washers and a coupling for ship's fittings. The fitting and joining must be suitable for a working pressure of 10.5 bar. Four bolts are required of 16mm diameter and 50mm length, also eight washers and a gasket of any suitable material.

(SOLAS 74, 2020)

4.26 Are records available to show that fixed fire detection and fire alarm systems have been tested at regular intervals and maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If a fire detection and alarm system is found to be malfunctioning, the machinery space shall not be operated in UMS mode until the system is repaired.

Periodically, fixed fire detectors and fire alarm systems shall be tested in accordance with MSC.1/Circ.1432 using equipment suited to the types of fires to which the detector is designed to respond.

If indicated by the manufacturer, the test protocol and specialised test equipment should be followed and used.

Spaces not covered by a fire detection system shall be covered by regular fire patrols.

4.27 Are the fixed fire extinguishing systems (where fitted) inspected, tested and in good order? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector is not required to witness a physical test of the head of water mist systems unless there are visible potential defects warranting such a test.

Fixed hold fire extinguishing systems, such as CO₂ lines, should be blown through with compressed air and checked to ensure they are free of dust and debris.

Paint lockers shall be protected by:

- > A carbon dioxide system, designed to give a minimum volume of free gas equal to 40 % of the gross volume of the protected space; or
- > A dry-powder system, designed for at least 0.5 kg powder/m³; or
- > A water-spraying or sprinkler system, designed for 5 l/m² min. Water spraying systems may be connected to the fire main of the ship; or
- > A system providing equivalent protection, as determined by the Administration.

In any case, the system shall be operable from outside the protected space.

Flammable liquid lockers shall be protected by an appropriate fire-extinguishing arrangement approved by the Administration.

For lockers of a deck area of less than 4 m², which do not give access to accommodation spaces, a carbon dioxide portable fire extinguisher sized to provide a minimum volume of free gas equal to 40 % of the gross volume of the space may be accepted in lieu of a fixed system. A discharge port shall be arranged in the locker to allow the discharge of the extinguisher without having to enter into the protected space. The required portable fire extinguisher shall be stowed adjacent to the port. Alternatively, a port or hose connection may be provided to facilitate the use of fire main water.

(SOLAS 74, 2020)

The foam pump shall be maintained as per manufacturer recommendation and the inspector shall witness the free movement of the foam pump when possible.

Foam proportioners and other foam mixing devices should be tested to ensure they are operating within the correct mixing ratio tolerance at least every five-yearly servicing.

4.28 Is the emergency fire pump being regularly tested, in good operational condition and are starting instructions clearly posted? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When agreeable by the Chief Engineer and safe to do so, the inspector shall witness the starting and operation of the emergency fire pump and check the following:

- > Fire pumps should operate satisfactorily and be able to maintain proper pressure
- > Pressure gauges should be in good order
- > The operating condition of the priming system and/or non-return valve should be in good order
- > The operating condition of the isolating valves and cocks should be in good order
- > Witness delivered flow from hose with nozzle.

All fire pumps, including emergency fire pumps, must be flow tested once a year to ensure proper pressure and capacity (reference: MSC.1/Circ.1432).

When possible and with the Master's agreement, the inspector shall witness the test of the emergency fire pump's capacity and pressure by deploying a fire hose on the bridge wing and another on the forecastle and observing the water flow.

4.29 Are portable fire extinguishers being maintained in good order, and ready for immediate use in an emergency? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Although there is no clear requirement concerning the validity date of powder in the FSS Code, it is generally considered necessary to refill the powder every 5 or 6 years, in principle. It is highly recommended that spare charges should also be replaced at the same time considering age deterioration, even though the expiration date of the charges is not specified by the manufacturer. In light of the above situation, crew shall consider replacing the spare charges of powder fire extinguishers at the same intervals as the refilling intervals specified by the manufacturer (Class NK Bulletin 20, 2020).

Periodic inspections and maintenance of portable fire extinguisher:

Extinguishers should be subject to periodical inspections in accordance with the manufacturer's instructions and serviced at intervals not exceeding one year.

- > At least one extinguisher of each type manufactured in the same year and kept on board a ship should be test discharged at five yearly intervals (as part of a fire drill).
- > All extinguishers together with propellant cartridges should be hydraulically tested in accordance with the recognized standard or the manufacturer's instruction at intervals not exceeding ten years.
- > Service and inspection should only be undertaken by, or under the supervision of, a person with demonstrable competence, based on the inspection guide in table 9.1.3 in Resolution A.951 (23).
- > Records of inspections should be maintained. The records should show the date of inspection, the type of maintenance carried out and whether a pressure test was performed.
- > Extinguishers should be provided with a visual indication of discharge.
- > Instructions for recharging extinguishers should be supplied by the manufacturer and be available for use on board.

(Resolution A.951 (23)/Improved Guidelines for Marine Portable Fire Extinguishers, 2004)

Spare charges shall be provided for 100% of the first ten extinguishers and 50% of the remaining fire extinguishers capable of being recharged on board. Not more than sixty total spare charges are required.

For fire extinguishers which cannot be recharged on board additional portable fire extinguishers of the same quantity, type, capacity, and number as determined in paragraph above shall be provided in lieu of spare charges.

(SOLAS 74, 2020)

4.30

Are firemen's outfits including their equipment, two-way portable radiotelephone apparatus for fire-fighter's communication and breathing apparatus in good condition, fit for purpose and available for instant use? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Each firefighting team shall be equipped with portable two-way radiotelephones for the purposes of communication. The efficient use of these radios shall not be inhibited by the use of any firefighting equipment, including the firefighter's outfit. To be fit for purpose a firefighter wearing a fireman's outfit and Breathing Apparatus must be able to communicate clearly and without restriction with the Command and Control team.

When possible, the inspector should evaluate the effectiveness of the two-way radio communication equipment provided. Record a Finding if there are insufficient intrinsically safe two-way portable radios for the number of fire teams on the muster list, or if wearing a fireman's outfit and Breathing Apparatus hinders the firefighter's ability to operate the radios while fighting a fire.

One complete fireman's outfit should include following items:

- > SCBA (self-contained breathing apparatus)
- > Fireman's suit
- > Fireman's helmet
- > Fireman's rubber boots
- > Explosion proof light

Fireproof lifeline combination rope of wire and nylon or hemp rope, complete with safety snap hook. Available in 30, 40 and 50 metre lengths.

- > Fire axe
- > Fireman safety belt

"Compressed air breathing apparatus shall be fitted with an audible alarm and a visual or other device which will alert the user before the volume of the air in the cylinder has been reduced to no less than 200 litres." This applies to ship's constructed (keel laid) on or after 1 July 2014. Ships constructed (keel laid) before 1 July 2014 must comply no later than 1 July 2019.

"An on-board means of recharging breathing apparatus cylinders used during drills shall be provided or a suitable number of spare cylinders shall be carried on board to replace those used." This applies to all ships on or after 1 July 2014.

"For ships constructed on or after 1 July 2014, a minimum of two two-way portable radiotelephone apparatus for each fire party for fire-fighter's communication shall be carried on board. These two-way portable radiotelephone apparatuses shall be of an explosion-proof type or intrinsically safe. Ships constructed before 1 July 2014 shall comply with the requirements of this paragraph not later than the first safety equipment survey after 1 July 2018."

(SOLAS 74, 2020)

The two-way portable radio should comply with the IACS UI SC291.

The purpose of these fire-fighter radios is to provide a dedicated means of communication between the team of fire fighters entering the space and the crew member located outside the space who is assigned to control this team.

The two-way portable radio, intended for use by the fire fighters in the Fire Party, should be capable of being used by crew members wearing full fire fighter suits and full breathing apparatus/smoke diver equipment, as well as by fire fighters not involved in smoke diving who are wearing fire-fighter outfits and helmets.

The engineering specification of the firefighter's outfit in the FSS Code (Chapter 3, Section 2.1.1) requires that the personal equipment shall consist of protective clothing made of material designed to protect the skin from heat radiating from the fire and from burns and scalding by steam.

4.31 Is the operation and maintenance of the breathing apparatus air recharging system (where fitted) incorporated in the ship's safety management manual, and has the annual air quality check for breathing apparatus air recharging systems been carried out? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Annual testing should be carried out to ensure the air quality of breathing apparatus air recharging systems.

(MSC/Circ.1432, Revised Guidelines for the Maintenance and Inspection of Fire-Protection Systems and Appliances, 2012)

4.32 Are records available to show that Emergency Escape Breathing Devices (EEBDs) in the accommodation and engine room are being inspected, in good condition and are available for instant use? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The minimum number of EEBDs to be kept within accommodation spaces should be:

For cargo ships: two (2) EEBDs and one (1) spare EEBD.

In machinery spaces for category A containing internal combustion machinery used for main propulsion, EEBDs should be positioned as follows:

1. One (1) EEBD in the engine control room, if located within the machinery space.
2. One (1) EEBD in workshop areas. If there is, however, a direct access to an escape way from the workshop, an EEBD is not required; and
3. one (1) EEBD on each deck or platform level near the escape ladder constituting the second means of escape from the machinery space (the other means being an enclosed escape trunk or watertight door at the lower level of the space).

Alternatively, different number or location may be determined by the Administration taking into consideration the layout and dimensions or the normal manning of the space.

For machinery spaces of category A other than those containing internal combustion machinery used for main propulsion, one (1) EEBD should, as a minimum, be provided on each deck or platform level near the escape ladder constituting the second means of escape from the space (the other means being an enclosed escape trunk or watertight door at the lower level of the space).

For other machinery spaces, the number and location of EEBDs are to be determined by the Administration.

(MSC/Circ.1081, Unified Interpretations of SOLAS Regulations II-2/13.3.4 and II-1/13.4.3, 2003)

The EEBD should be maintained in accordance with the manufacturer's instructions.

Spare EEBDs should be kept on board.

Maintenance requirements, manufacturer's trademark and serial number, shelf life with accompanying manufacture date and name of approving authority should be printed on each EEBD.

Training in the use of the EEBD should be considered as a part of basic safety training.

All EEBD training units should be clearly marked.

Personnel should be trained to immediately don an EEBD prior to exiting a space when the atmosphere becomes life threatening. This is necessary due to the possibility of encountering smoke during escape. Such training should be accomplished by scheduling routine escape drills for crew members working in the engineering or machinery spaces.

(MSC/Circ.849, Guidelines for The Performance, Location, Use and Care of Emergency Escape Breathing Devices (EEBDs), 1998)

4.33 Are ventilation fire dampers clearly marked with open/close positions and space served and is there evidence of regular testing and maintenance? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

4.34 Are Material Safety Data Sheets (MSDS) for all bunkers, chemicals, paint, corrosive, and toxic materials available, and are all crew familiar with their contents? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

It is essential before use of any hazardous substance that the manufacturer's safety data sheet (SDS) is referred to, to select appropriate personal protective equipment (PPE) and working methods.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

All stores on board where hazardous or toxic substances are kept, such as paint and chemical stores, shall have readily accessible MSDS.

4.35 Is a safe means of access to the vessel being provided? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Accommodation Ladders:

In accordance with their design, Accommodation ladders should, if reasonably practicable be firmly landed on the wharf.

The use of an accommodation ladder in a suspended or "floating" condition poses elevated safety risks to personnel and mitigating measures must be taken to reduce these to as low as reasonably practicable.

Landed Accommodation Ladders:

If a suitable surface is available, the accommodation ladder should be firmly landed. The following requirements apply:

- > Risk Assessment: Conduct a specific risk assessment to identify hazards and establish appropriate controls.
- > Location: The means of embarkation and disembarkation should be sited clear of obstructions and working areas and should not be placed where cargo or other suspended loads may pass overhead.
- > Supervised Use: Ensure the means of access arrangement is attended whenever personnel are using it.
- > Emergency Equipment: Position a lifebuoy with a buoyant line and self-igniting light in the vicinity of the embarkation and disembarkation arrangement for immediate deployment in case of emergency.
- > Lighting: Provide clear illumination of the ladder arrangement and immediate approaches during hours of darkness.
- > Marking: The accommodation ladder should be clearly marked at each end with a plate showing the restrictions on the safe operation and loading, including the maximum and minimum permitted design angles of inclination, design load, maximum load on bottom end plate, etc.
- > Records of Testing: Records must be available to confirm static load and operational testing have taken place on the winch and accommodation ladder.
- > Maintenance: The accommodation ladder, winch, wires and connection to the deck shall be inspected and maintained in suitable condition for their intended purpose, taking into account any restrictions related to safe loading. All wires used to support the means of embarkation and disembarkation shall be maintained as specified in SOLAS III/20.4.
- > Angle: Accommodation ladders should not be used at an angle greater than 55° from the horizontal, unless designed and constructed for use at angles greater than these and marked as such.
- > Wires: Accommodation ladder wires are to be inspected periodically, and renewed when necessary due to deterioration of the wires or at intervals of not more than 5 years, whichever is the earlier; wire rope certificates should be kept available. When the accommodation ladder is landed on the wharf, the wires should not obstruct use of the walking surface.
- > Side rails, stanchions and ropes: During use, the side rails, stanchions and ropes should be maintained fully rigged with side ropes taut.
- > Safety Net: A safety net should be rigged in way of the accommodation ladder, from the top outboard rail, under the ladder, to either the top inboard rail or preferably to the ship's side rail covering the gap between the ladder and the ship's side or where impracticable to the top inboard rail of the ladder, preferably to the ship's side covering the gap between the ladder and the ship side or where impracticable to the top inboard rail of the ladder. In either case, the net should be sufficiently taut so as to provide cushioning to any person or object falling on to it.
- > Bottom Platform: The bottom platform should be maintained in a horizontal condition.

Suspended/Floated Accommodation Ladders:

Where the berth design and/or terminal/berth regulations prevent the ladder from being landed, this limitation should be communicated to the Master prior to berthing. In such cases, a brow/gangway may be rigged in conjunction with the accommodation ladder, between the ladder's bottom platform and the wharf surface; where this cannot be safely achieved, use of a suspended offshore accommodation ladder in conjunction with launch access should be considered.

In addition to the above requirements, the following risk mitigation measures should be implemented:

- > Risk Assessment: Conduct a specific risk assessment to identify hazards and establish appropriate controls, noting the suspended/floated status of the accommodation ladder and brow.
- > Signage: Display a placard warning of the risks associated with the suspended/floated accommodation ladder and brow and instructing users to proceed one at a time.
- > Inspection of Ladder Pins: Prior to fixing a brow, the pins on the lower platform of the accommodation ladder are to be inspected to confirm they are secure and in good condition.
- > Certification: A brow must be inspected and found fit for purpose, and the Master must hold the relevant approval certificate for ship's equipment, or have sighted the relevant approval certificate if the brow is shore-supplied.
- > Brow Rigging: A brow must be rigged such that one end is firmly located and secured on the horizontal lower platform of the accommodation ladder, and the other end is firmly located on the wharf surface. Any angle of inclination limits are to be complied with.
- > Side rails, stanchions and ropes: Where not fixed the side rails, stanchions and ropes for a brow and an accommodation ladder should be maintained fully rigged with side ropes taut.
- > Safety Net: Rigging of the safety net should incorporate any brow arrangement.
- > Emergency Equipment: Where offshore boarding by launch is utilised a lifejacket must be worn.
- > Compliance: A suspended/floated accommodation ladder or brow are not to be used if flag State, port State, local or other regulations require otherwise.

Masters' attention is drawn to MSC.1/Circ.1331, SOLAS II-1/3-9, and SOLAS III/20.4.

Inspectors should not record a finding when an accommodation ladder complies with the above specified provisions and is landed or when a vessel uses a suspended or floating accommodation ladder, provided that all relevant mitigation measures have been properly implemented and verified.

The RightShip best practice & lessons learned for Safe Means of Access must be taken in to account when rigging a safe means of access. Please [click here](#)

4.36 Are accommodation ladders and gangways maintained in good condition, marked clearly, and inspected regularly (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding If the maintenance and inspection of the accommodation ladder turn table and its pin were not included in the PMS

Accommodation ladders and gangways, including associate winches and fittings, should be properly maintained, and inspected at appropriate intervals as required by SOLAS regulation III/20.7.2, in accordance with manufacturers' instructions. Additional checks should be made each time an accommodation ladder or gangway is rigged, looking out for signs of distortion, cracks, and corrosion. Close examination for possible corrosion should be carried out, especially when an aluminium accommodation ladder/gangway has fittings made of mild steel.

Bent stanchions should be replaced or repaired and guard ropes should be inspected for wear and renewed where necessary.

Moving parts should be free to turn and should be greased as appropriate.

The lifting equipment should be inspected, tested, and maintained paying careful attention to the condition of the hoist wire. The wires used to support the means of embarkation and disembarkation should be renewed when necessary, as required by SOLAS regulation II-1/3-9.

Arrangements should also be made to examine the underside of gangways and accommodation ladders at regular intervals.

All inspections, maintenance work and repairs of accommodation ladders and gangways should be recorded to provide an accurate history for each appliance. The information to be recorded appropriately on board should include the date of the most recent inspection, the name of the person or body who carried out that inspection, the due date for the next inspection and the dates of renewal of wires used to support the embarkation and disembarkation arrangement.

Winch:

During annual surveys required by SOLAS regulations I/7 and I/8, the following items should be examined for satisfactory condition:

- > Brake mechanism including condition of brake pads and band brake, if fitted.
- > Remote control system; and
- > Power supply system (motor).

At every five-yearly survey, the winch should be operationally tested with the specified maximum operational load of the accommodation ladder.

Marking:

Each accommodation ladder or gangway should be clearly marked at each end with a plate showing the restrictions on the safe operation and loading, including the maximum and minimum permitted design angles of inclination, design load, maximum load on bottom end plate, etc. Where the maximum operational load is less than the design load, it should also be shown on the marking plate.

(MSC.1/Circ.1331, Guidelines for Construction, Installation, Maintenance and Inspection/Survey of Means of Embarkation and Disembarkation, 2009)

The certificate (s) for a five-year load test of an accommodation ladder should be kept on board, and the load test should be performed within the stipulated time period. A manufacturer's certificate for a fall wire in service should be provided, as well as confirmation that accommodation ladder fall wires have been replaced within the last five years. The date of fall wire renewal should be stencilled in the vicinity of the fall wire winch.

All wires used to support the means of embarkation and disembarkation shall be maintained and inspected with special regard to the areas passing through sheaves. The falls should be 'renewed when necessary due to the deterioration of the falls or at intervals of not more than five years, whichever is the earlier as per SOLAS III/20.4, SOLAS II-1/3- and MSC.1/Circ.1206.

4.37

Are pilot ladders used for pilot transfer in good condition, and inspected regularly, clearly identified with tags or with permanent marking and are maintenance records available and, are crew members capable of demonstrating proper rigging of the pilot-ladder arrangement?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Counterfeit pilot ladders and certificates pose a serious safety risk and remain a significant industry concern. To mitigate this risk, ship managers should adopt one or both of the following measures:

Maintain Accurate Records:

Ensure that the Record of Approved Safety Equipment, where it includes pilot transfer arrangements, is updated to accurately reflect the details specified in the relevant form.

Verify Certificate Authenticity:

Ship managers should establish and implement a formal procurement procedure that requires direct verification of the pilot ladder certificate's authenticity with the issuing authority (e.g., manufacturer, classification society) prior to accepting delivery. Objective evidence of this verification, such as email confirmation, or an verifiable digital certificate, should be retained onboard and kept readily available for audit or future compliance checks.

Comments should also specify whether the ladder steps are made of hardwood, aluminum, or composite materials.

If a pilot ladder is made of a material other than hardwood, it should possess equivalent strength, stiffness, and durability to the satisfaction of the Administration. (IMO Res 10459(27))

When plastic materials are used in the construction of a pilot ladder, ISO 877 provides guidance with regards to testing for both original tensile and original impact strength.

(Pilot ladder manual 2024)

Expected service life of the pilot ladder, which may be less than 30 months, especially on ladders with mechanically placed metal clamps that prevent inspection of the side ropes. If a pilot ladder's service life exceeds 30 months, the test specified in section 6.6 of ISO standard 799-2021 should be conducted.

Personnel responsible for rigging and inspecting pilot ladders shall receive periodic training in the inspection requirements, regulations and standards associated with pilot ladders and their use.

Where other arrangements or equipment are essential for use in conjunction with a pilot ladder, those arrangements or equipment shall be included in such training. This requirement may be fulfilled by the use of an onboard computer-based training module.

(BS ISO 799-2:2021 Ships and marine technology. Pilot ladders. Maintenance, use, survey, and inspection, 2021)

The vessel's manager should follow the international standards for inspecting the pilot ladder, such as ISO799-2:2021 Annex A. This standard provides checklists for pre-use, post-use, three-monthly and annual inspections. RightShip recommends this practice to ensure the safety and quality of the pilot ladder.

All pilot ladders used for pilot transfer shall be clearly identified with tags or other permanent marking so as to enable identification of each appliance for the purposes of survey, inspection and record keeping. A record shall be kept on the ship as to the date the identified ladder is placed into service and any repairs effected.

(SOLAS 74, 2020)

The top of the pilot ladder should be secured to the approved fixing point and not to handrails. Ladder steps or spacers should not be rigged in a position in which they are taking the weight of the ladder.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

When utilising a combination pilot ladder arrangement, it is necessary to secure both the pilot ladder and accommodation ladder to the ship's side. [CLICK HERE](#) to access the IMO Circular MSC.1/Circ. 1428, which illustrates the pilot transfer arrangements required by SOLAS V/23.

SOLAS Chapter V, Regulation 23, necessitates securing the lower platform of the accommodation ladder to the ship's side. This is to ensure that the lower end of the accommodation ladder and the lower platform are held firmly against the ship's side.

If the accommodation ladder is not secured to the ship's structure, it is possible for the combination ladder to swing violently and uncontrollably with a pendulum motion.

When a combination arrangement is used for pilot access, means shall be provided to secure the pilot ladder and manropes to the ship's side at a point of nominally 1.5 m above the bottom platform of the accommodation ladder.

(SOLAS 2020)

(The Pilot ladder Manual 2024)

There are several methods available to ensure multiple securing points:

- > Fastening to fixed lashing points built into the ship's hull.
- > Use of movable magnetic or pneumatic fixing equipment.

The easiest way to secure the ladder is the use of two strong (at least 2 x 24 kN) manila ropes directly attached to each side rope of the pilot ladder, by means of a rolling hitch knot.

Any pilot ladders not in use should be clearly identified and tagged.

The requirements in SOLAS V/23, deal with the standards for equipment installed and arrangements for pilot transfers on ships on or after 1 July 2012. The standards adopted by the IMO can be found in IMO Resolution A.1045(27) "Pilot transfer arrangements". SOLAS V/23.2.3 additionally states that a pilot ladder shall be certified by the manufacturer as complying with V/23 or "with an international standard acceptable to the Organization" and ref and refers to ISO 799:2004 Ships and marine technology.

Counterfeit pilot ladders and certificates:

A counterfeit pilot ladder is a ladder that purports to be a genuine, approved product from a manufacturer certified by an approved body in the manufacture of pilot ladders to a certified standard, but is not.

At a minimum, a pilot ladder certificate should have:

1. The name and address of the manufacturer
2. The serial number
3. The model and length of the ladder
4. The date of manufacture
5. The standards, resolutions and regulations to which the ladder complies (eg. SOLAS 74 Reg. V/23 & SOLAS 74 Reg. X/3, IMO Res. A.1045(27), IMO Res. MSC/Circ. 1428, ISO 799:1 2019)
6. The certifying authority

RightShip strongly encourages the vessel's manager to verify the authenticity of the pilot ladder certificate by contacting the relevant certification body. It is important to keep a record of such verification on board for future reference.

4.38

Has the vessel been provided with ship-specific fire safety and SOLAS training manuals and operational booklets? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Fire Safety training manual: A training manual shall be written in the working language of the ship and shall be provided in each crew mess room and recreation room or in each crew cabin. The manual shall contain the instructions and information required in regulation II-2/15.2.3.4. Part of such information may be provided in the form of audio-visual aids in lieu of the manual. (SOLAS 74, 2020)

SOLAS training manual: Specific training manuals shall be provided in each crew mess room and recreation room, or in each cabin and shall contain instructions and information on the life-saving appliances provided in the ship. It shall also contain information on the best methods of survival. The material in the manual shall be in easily understood terms and illustrated where appropriate.

The following when applicable, shall be incorporated into the manual and explained in detail:

- > Donning of lifejackets, immersion suits and anti-exposure suits, as appropriate.
- > Muster at the assigned stations.
- > Boarding, launching, and clearing the survival craft, rescue boats, fast rescue boats, free-fall boats and inflated boats.
- > Method of launching from within the survival craft.
- > Release from launching appliances.
- > Methods and use of devices for protection in launching areas.
- > Illumination in launching areas.
- > Use of all survival equipment.
- > Use of all detection equipment.
- > With the assistance of illustrations, the use of radio lifesaving appliances.
- > Use of sea anchors.
- > Use of engine and accessories.
- > Recovery of the survival craft, rescue boats, fast rescue boats, free-fall boats and inflated boats including stowage and securing.
- > Hazards of exposure and the need for warm clothing.
- > Best use of the survival craft facilities in order to survive.
- > Methods of retrieval, including the use of helicopter rescue gear (slings, baskets, stretchers), breeches-buoy and shore life-saving apparatus and ship's line-throwing apparatus; all other functions contained in the muster list and emergency instructions.

The fire safety operational booklet shall contain the necessary information and instructions for the safe operation of the ship and cargo handling operations in relation to fire safety. The booklet shall be written in the working language of the ship and be provided in each crew mess room and recreation room or in each crew cabin. The booklet may be combined with the fire safety training manuals required in regulation II-2/15.2.3. The booklet may be combined with the fire training manual. (SOLAS 74, 2020)

4.39

If the vessel is provided with a helicopter operating area, does the area comply with the requirements of ICS guidelines, and is there a safe access from the hatch cover to deck? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if a strength confirmation letter from a classification society is not available on board. The inspector shall obtain a copy of the strength letter and attach it to the RightShip inspection report under Question 4.37. Additionally, the inspector should provide 360-degree photographs of the landing site and access, when practicable.

If the helicopter operation area is designated as "Winch Only," the requirement for a strength confirmation letter does not apply to the vessel.

The construction and usage of inclined ladder(s) for accessing the helicopter operation area may not be possible on a number of cargo vessels, specially smaller cargo ships. This is due to factors such as the structural design, location of fittings, and limited deck space. If that is the situation, the inspector should answer Yes to the question and provide an explanation in the comment area regarding why the vessel was unable to use the inclined ladder(s).

The strength of the HLS for the vessel is to be confirmed by the following documents:

1. Maximum designed load for helicopter landing on HLS as described in the structural drawings such as hatch cover, upper deck, etc.
 2. Strength confirmation letter by HLS designer (shipyard, hatch cover maker, etc.) or the Flag State.
 3. Strength confirmation letter issued by classification society in cases where neither items 1 nor 2 above are available.
- In order to issue the letter under this paragraph, the structural drawing of the HLS is to be submitted to the classification society. In addition, fire-fighting equipment for helicopter facilities are required to comply with the "Guide to Helicopter/Ship Operations, 5th Edition issued by the International Chamber of Shipping.

The access from hatch cover to deck should be a fixed height inclined ladder with fixed handrails and front platform with operational load for two persons (150 KG). The steps and platforms should be made of non-slip materials.

The Master of a vessel must ensure that any obstacle within the landing or winching area is clearly marked if it does comply with the recommendations for obstacles in the ICS Guide.

(AMSA Marine Orders Part 57)

In addition to the marking arrangements described, the vessel's manager should ensure that, if possible, a minimum of two access/egress routes to and from the landing area available to ensure that, in the event of an incident on the landing area, helicopter passengers and crew can escape upwind of the incident.

Handrails exceeding the height limitation set out in section 4.1.2 of the Guide to helicopter/ship operation shall be retractable, collapsible or removed. Such handrails should be painted in a contrasting colour scheme and procedure should be in place to retract, collapse or remove them prior to the arrival of the helicopter.

Red and white strips should be used for marking the position of notifiable objects within either the manoeuvring zone or clearing zone that exceed the height limits for those zone(refer figure 4.1 of the ICS Guide to Helicopter/Ship Operations Edition 5):

- > Object within the clear zone of height exceeding 2.5 cm
- > Objects outside the clear zone but within the manoeuvring zone of height exceeding 25 cm.

Yellow should be used for marking the position of objects beyond the manoeuvring zone to which it is considered appropriate to draw the attention of the helicopter pilot. Yellow may also be used to mark objects within manoeuvring zone and clear zone below the height limits for either the clear zone(2.5 cm) or the manoeuvring zone(25 cm) and to which it is considered appropriate to draw the attention of the helicopter pilot.

(ICS Guide to Helicopter/Ship Operations Edition 5)

4.40 This question has been removed from the current version of the document.

4.41 Are the crew familiar with the helicopter operation at sea, and are records available to show that the proper communication, shipboard helicopter safety checklist and specific risk assessment conducted prior to helicopter operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inventory of helicopter equipment shall be in compliance with section 9.3 of the record of approved Ship Safety Equipment.

Evidence of communication between helicopter and vessel's bridge team, shipboard safety checklist for helicopter operations, specific risk assessment (as per appendix B of Guide to Helicopter/Ship operation) should be available and reviewed by inspector.

Helicopter operations are commonly used for embarkation and disembarkation of Pilot and medical evacuation in emergency situations. The helicopter operation is a complicated, high-risk operation. This operation demands accuracy, training, and clearly established procedures. The officers and crew members associated with these operations should show a high level of situational awareness and good seamanship.

For additional information, reference should be made to the Rightship best practice & lessons learned for Helicopter Operations High Potential Near Miss Incidents. Please download the document [via this link](#).

4.42 Are the lifebuoys, related equipment, and pyrotechnics in good working order?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Cargo ships shall carry not less than the following numbers of lifebuoys:

- > Under 100 metres in length - 8
- > Between 100 metres and under 150 metres - 10
- > Between 150 metres and under 200 metres - 12
- > 200 metres and over - 14

At least one lifebuoy on each side of the ship shall be fitted with a buoyant line, equal in length to not less than twice the height at which it is stowed above the waterline in the lightest seagoing condition, or 30 metres, whichever is the greater.

Not less than one half of the total number of lifebuoys shall be provided with self-igniting lights; Not less than two of these shall also be provided with lifebuoy self-activating smoke signals capable of quick release from the navigating bridge; Lifebuoys with lights and those with lights and smoke signals shall be distributed equally on both sides of the ship and shall not be the lifebuoys provided with lifelines.

12 rocket parachute flares must be carried and stored on or near the navigation bridge. There shall be a line throwing appliance that complies with the requirements of Section 7.1 of the Code. The officer of the watch shall have readily available an illustrated table describing the life-saving signals. The line throwing apparatus should be immediately available for use. The line and rockets should not be separated when stowed. Self-contained rocket line throwing appliances are frequently dismantled prior to transport to the vessel. Inspectors should verify that all equipment has been reassembled and rockets are properly positioned for immediate use. Containers, brackets, racks, and other similar stowage locations for life-saving equipment shall be labelled in accordance with IMO Resolution A.760(18) with symbols indicating the devices stowed in that location for that purpose. If that location is used to store more than one device, the number of devices must also be indicated.

(SOLAS 74, 2020)

4.43 Are adequate familiarization procedures in place that cover the personal safety and professional obligations of new personnel and personnel transferred to a new assignment, and are documents available to confirm effective familiarization?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The company should establish procedures to ensure that new personnel and personnel transferred to new assignments related to safety and protection of the environment are given proper familiarization with their duties. Instructions which are essential to be provided prior to sailing should be identified, documented and given.

(ISM code and guidelines on the implementation of the ISM code, 2018)

Deck officer familiarization with navigation equipment should be delivered one-on-one using a common language using the Bridge Procedures Guide's (BPG) checklist C2.3 and C2.4. Familiarization should include all bridge equipment and procedures relevant to the roles and responsibilities of each bridge team member.

(Bridge Procedures Guide, 2022)

4.44 Does the vessel manager have documented procedures in place to control stevedore activities onboard by identifying potential hazards and implementing appropriate mitigation measures?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

> **Stevedore Safety in Cargo Operations**

- >
- > Stevedores engaged in cargo operations are exposed to a variety of hazards, particularly when adequate supervision is not maintained onboard. These risks may arise from the nature of the cargo, unfamiliarity with shipboard safety procedures, or the working environment itself.
- > Common hazards include:
 - > Entry into confined or enclosed spaces
 - > Slips, trips, and falls
 - > Machinery and equipment-related risks
 - > Exposure to hazardous chemicals
 - > Fire hazards

- > Without adequate training and awareness, stevedores may inadvertently engage in unsafe practices, significantly increasing the risk of serious incidents or fatalities.
- > The safety of stevedores during cargo operations primarily rests with the stevedore company which must ensure its personnel are :
 - Properly trained and competent in their roles
 - Familiar with shipboard procedures and safety protocols
 - Equipped with appropriate personal protective equipment (PPE)
 - Fully aware of operational hazards and risk controls
 - A designated company representative must be present onboard to supervise stevedores throughout the operation.

This includes actively monitoring their activities, identifying unsafe behaviours, and intervening immediately when necessary.

Additionally, vessel managers are responsible for ensuring the designated stevedore representative receives a pre-operation safety briefing who is responsible to brief stevedores in their working language at each shift. The ship-shore safety checklist in accordance with BLU Code including item 13 to be shared with the stevedore representative.

Pre-Work Briefing / Toolbox Meeting

A concise pre-work briefing or toolbox meeting should be conducted between the Master (or their designated representative) and the stevedore designated representative . This meeting should:

- > Outline relevant safety procedures
- > Emphasise the consequences of non-compliance
- > Reinforce the vessel's Stop-Work Authority, including the right to suspend operations if unsafe conditions arise
- > The briefing should be documented in the cargo logbook or, where appropriate, included in the ship-shore safety checklist.

Access Control & Enclosed Space Management

Strict access control to cargo holds and enclosed spaces is critical. The vessel should:

- > Lock and secure all access points to confined spaces, including cargo holds not in use
- > Implement controlled entry procedures, including:
 - Physical barricading
 - Clear signage
 - Padlocking of restricted areas

The Master retains ultimate authority over access approvals and should ensure full compliance with enclosed space entry protocols throughout cargo operations.

All enclosed space entries by shore personnel should include:

- > Shore personnel should be informed of the hazards associated with enclosed spaces that may exist on board vessels. Both the general and specific requirements outlined in MSC.1/Circ.581 (110) should be carefully considered and adhered to.
- > Verified atmosphere testing
- > Permit-to-enter signed by a ship's officer
- > Appropriate caution signage and physical barriers

Stop-Work Authority

The vessel manager should reinforce the crew's authority to stop work immediately if unsafe conditions, equipment, or behaviours are observed. This includes:

- > Immediate suspension of cargo operations upon detection of unsafe practices
- > Documentation of all unsafe conditions and behaviours
- > Resumption of operations only after corrective actions have been implemented by the stevedores

The vessel manager should clearly define:

- > Conditions that warrant activation of stop-work authority
- > Criteria for safely resuming operations

Supervision and Risk Assessment

Stevedore operations should be monitored by shipboard personnel and the stevedore company to ensure safety and compliance. Any unauthorized access or entry into restricted areas should be immediately challenged and reported through the appropriate channels.

Risk assessments should be:

- > Specific to the vessel and its operations
- > Developed by both the ship and the stevedore company and exchanged prior to commencement of cargo operations
- > Discussed during pre-work briefings or toolbox meetings

These assessments should address operational hazards associated with cargo handling methods, equipment usage, and stevedore work practices, and should be reviewed and discussed during the Pre-Work Briefing or Toolbox Meeting.

Section 5: Pollution Prevention and Control

5.1 Is the Oil Record Book (Part 1) completed correctly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Flag Administration may permit the use of an electronic oil record book as an alternative substitute of the traditional paper ORB. However, Flag approval should be available on board and verified by the inspector.

Non-automatic starting of discharge overboard via 15 ppm equipment, transfer, or disposal otherwise of bilge water which has accumulated in machinery spaces should be recorded in section D.

Pumping of bilge water from engine-room bilge wells to a tank listed under item 3.3 in the Supplement to the IOPPC should be recorded in section D 15.3.

Automatic starting of discharge overboard, transfer or disposal otherwise of bilge water which has accumulated in machinery spaces should be recorded in section E. The automatic starting systems will be activated by float switches in bilge wells or bilge holding tanks. This system is rarely installed in the machinery space of dry cargo vessels.

The condition of oil filtering equipment and oil content meter or stopping device, including the alarm and automatic stopping devices when defective should be recorded in section F. A code 'I' entry should also be made indicating that the overboard valve was sealed shut due to non-working oil filtering equipment or oil content meter.

On the date when the system is functional again, a new entry, using code F should be made. A code 'I' entry should also be made indicating that the overboard valve was unsealed since the operation of the oil filtering equipment or oil content meter has been restored.

Accidental or other exceptional discharges of oil should be recorded in section G.

Bunkering of fuel or bulk lubricating oil should be recorded in section H. Separate entries are required for each grade of fuel oil and lubricating oil respectively to ensure transparency. This entry is not required if lubricating oil are delivered on board in packaged form (55-gallon drum, etc.).

Voluntary declaration of quantities retained in bilge water holding tanks (ref MEPC.1/Circ.640) should be record weekly in section I.

(MEPC.1/Circ.736/Rev.2, Guidance for the Recording of Operations in the Oil Record Book Part I- Machinery Space Operations (All Ships), 2011)

"When disposal of engine-room oil water or sludge to a shore reception facility has taken place, the entry in the Oil record Book shall be made accurately and in consistency with the shore reception facility receipt."

5.2 Is an approved MARPOL Shipboard Oil Pollution Emergency Plan (SOPEP) available, and up to date and are ship's personnel familiar with their duties? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Every ship other than an oil tanker of 400 gross tonnage and above shall carry on board a shipboard oil-pollution emergency plan approved by the Administration.

Such a plan shall be prepared based on guidelines developed by the Organisation and written in the working language of the Master and officers. The plan shall consist at least of:

- > The procedure to be followed by the Master or other persons having charge of the ship to report an oil pollution incident
- > The list of authorities or persons to be contacted in the event of an oil pollution incident
- > A detailed description of the action to be taken immediately by persons on board to reduce or control the discharge of oil following the incident
- > The procedures and point of contact on the ship for coordinating shipboard action in combating the pollution with national and local authorities
- > Description of equipment, its location, a plan for deployment and specific crewmember duties for handling small spills, and
- > An up-to-date IMO Coastal Contact List.

The SOPEP must be re-approved after a change of management. The list of national operational contact points is issued electronically on a quarterly basis on the 31 January, 30 April, 31 July and 31 October at www.imo.org.

5.3 Are the ship's personnel aware of the requirements of MARPOL Annex V with respect to the disposal of operational waste and cargo residues from ships? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Operational waste means all solid waste (including slurries) not covered by other Annexes that are collected on board during normal maintenance or operation of a ship or used for cargo stowage and handling.

Operational waste also includes cleaning agents and additives contained in cargo holds and external wash water.

Operational waste does not include grey water, bilge water, or other similar discharge essential to the operation of a ship, taking into account the guidelines developed by the Organisation.

Cargo residues means the remnants of any cargo which are not covered by other Annexes and which remain on the deck or in holds following loading and unloading; including loading and unloading excess or spillage, whether in wet or dry condition or entrained in wash water; but does not include cargo dust remaining on the deck after sweeping or dust on the external surface of the ship.
(MARPOL, 2017)

The SKULD P&I club information paper "Guidance on disposal of cargo residues in line with MARPOL Annex V" provides further information.

Deck scuppers should be plugged during cargo operations involving residues that are harmful to the marine environment. Vessel managers should establish procedures to ensure the integrity of deck containment is maintained. This includes the proper use of drain valves where fitted and ensuring controlled drainage of accumulated water from the deck during cargo operations.

5.4 This question has been removed from the current version of the document.

5.5 Is the vessel free from any visible bulkhead leakage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The examples of bulkheads are:

- > The engine room forward bulkhead at its intersection with the topside tank structure in the aftermost cargo hold.
- > The side shell plating of the cargo hold side structure.
- > Side shell plating in the foremost cargo hold.
- > The stool shelf plates of the transverse bulkheads in the cargo hold.
- > The transverse bulkheads at the topside tank connection, in the cargo hold.
- > The vertical corrugations of transverse bulkheads in the cargo hold.
- > The corrugated bulkheads at the intersection of the shredder plates in the cargo holds.
- > Any fuel tank bulkheads within the machinery space

5.6 Are the cargo hold bilge pumping systems and bilge arrangements appropriately set, in good order and tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Bilge wells, including bilge covers, strum boxes; and bilge well valves, including non-return valves, should be in a clean and sound condition.

Non-return valves must be checked to ensure they are fully operational. Overhaul of non-return valves at regular intervals should be incorporated into the planned maintenance system. Inspection and testing of these non-return valves should be incorporated in the pre-loading checks of the holds. The presence of previous cargo residues and/or scale around the valve's seat may prevent the correct operation of the non-return valve.

Bilge lines should be blown back to confirm the effectiveness of the valves regularly.

Bilge high-level alarms should be tested regularly. Records of testing of alarm systems should be retained on board.
(Bulk Cargoes-Hold Preparation and Cleaning, 2011)

When the cargo hold bilge system is not in use, all valves should be effectively shut and measures should in place to ensure that they remain shut i.e. visible signs.

5.7 Is the sounding of cargo hold bilge, ballast tanks, chain lockers, pipe ducts and other void spaces regularly performed for accumulations of water, or alternative evidence of regular monitoring? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

5.8 Are suitable containment arrangements in place around the hydraulic components of deck machinery? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The hydraulic component can include hatch cover rams and remote-control stand, cranes, winches, windlass, piping, and hoses.

5.9 Are the emergency arrangements to pump out the spaces forward of the collision bulkhead in the event of flooding in good order, and are these arrangements prominently marked? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Water level detection and alarm system is to be located on the navigation bridge and be capable of detecting water ingress at all cargo holds and spaces and ballast tank forward of collision bulkhead. In general, Fore Peak Tank, Bosun's Store, Forecastle Space excluding chain lockers are considered as these spaces. The systems is to be a type being approved and to be supplied with electrical power from two independent supplies and failure of the primary electrical power supply is identified by an alarm.

(SOLAS 74 2020)

The vessel manager should develop vessel-specific procedures to pump out the spaces forward of the collision bulkhead in the event of flooding, including guidance on:

- > The use of the pumps/eductors connected to the systems, their direct suctions, and overboard valves.
- > The marking of system valves and controls to ensure correct operation and avoid accidental opening.
- > The vessel-specific instructions for pumping out the spaces forward of the collision bulkhead in an emergency may be included as part of the vessel emergency response plan for forecastle space flooding.

5.10 Are the master and officers familiar with the approved Ballast Water Management Plan, was the ballast water treatment system maintained as per the manufacturer's recommendations, and is the record-keeping of ballast water operations in the ballast water record book completed correctly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Ballast Water Management Plan (BWMP) is the document that details the procedure for the discharge of ballast water in accordance with regulation D-1 (exchange), and/or regulation D-2 (treatment). Conducting ballast water discharge in accordance with the BWMP ensures compliance with regulations D-1 or D-2.

By September 2024, all ships must confirm conform to the D-2 standard.

All ships are required to carry:

- > A ship specific BWMP
- > A Ballast Water Record Book (BWRB)
- > An International Ballast Water Management Certificate (IBWMC)

The BWMP is approved by the ship's flag state and should set out a particular and realistic set of measures to be followed by the crew. These should include the following procedures:

- > Best practices are in place to ensure the BWMS is not disabled or malfunctioning due to mechanical or electrical faults that are easily preventable;
- > Verify the next port of discharge and place into IMO or United States Coast Guard (USCG) mode accordingly;
- > Ensure that all valve position are verified before the start of each ballast operation;
- > All maintenance work is performed inline with the PMS;
- > All consumables (reagents, disinfectants, chemicals, etc.) and critical spare parts are readily available; and
- > Ensure that all troubleshooting and mitigation measures are followed as outlined in the original equipment manufacturer's instructions.

The ship-specific and crew familiarisation program should cover:

- > Maintenance of the system, including checklists and schedules for maintaining the system's optimal condition, including:
 - Inspection, cleaning, and calibration; and
 - Maintaining adequate spare parts on board, as well as active substances and neutralisers on board, if applicable.
- > Troubleshooting and mitigation where best practices should be established for investigating and resolving common/critical BWMS technical malfunctions that may arise;
- > Record keeping; and
- > The procedure for notifying the port state that they will receive ballast water discharge impacted by failure of malfunction of the BWMS.

For record keeping, the following actions requires a note to be made in the BWRB:

- > When ballast water is taken onboard;
- > When ballast water is discharged into the sea
- > Whenever ballast water is exchanged into the sea
- > Whenever ballast water is exchanged, circulated and treated for ballast water management purposes;
- > Uptake or discharge of ballast water from/to a port-based or reception facility
- > Accidental discharge/ingress or other exceptional uptake or discharge of ballast water;
- > Failure and malfunction of the BWMS; and
- > Ballast tank cleaning/flushing, removal and disposal of sediments

The duties of the officer in charge of ballast water operations must be specified in the ship's BWMP.

The officer in charge must fill in and sign every BMRB entry and the master should sign every full page.

Should the BWMS fail, the shipowner/manager should be contacted before bypassing the ballast water treatment system.

The ship will need to demonstrate that any bypass of the system will not be a risk to the environment and should only be used as a last resort.

(Engine Room Procedures Guide 2024)

Ballast Water Biosecurity Risks

An Active Substance (AS) is defined as 'a substance or organism, including a virus or a fungus, that has a general or specific action on or against harmful aquatic organisms and pathogens'.

BWMS that use Active Substances (electro-chlorination & chemical-based)

Repeated 'low TRO' alarms during uptake: The Total Residual Oxidant (TRO) sensor is a critical component of a properly installed BWMS as it measures if ballast water is treated within the system design limits. Repeated 'low TRO' or similar alarms during uptake may indicate that the ballast water has not been effectively treated due to an insufficient dose or generation of Active Substance (AS). If ballast water is treated with an AS at a concentration below the lower alarm limit of the system, an increased level of inspection may apply.

'High TRO' alarms during discharge: This alarm could indicate a problem with neutralisation. If the AS in the ballast water is not properly neutralised, the ballast water may cause marine pollution. Commonly encountered issues include seized metering pumps, insufficient or incorrectly mixed neutraliser in the neutraliser tank, blocked TRO sensor pipes, expired TRO reagents, and incorrectly arranged valves.

'TRO Communication' alarm: During uptake, TRO production by the electrolyser or dosage from the chemical tank is linked to the values obtained by the TRO sensor. If the sensor is not properly reporting the AS concentration to the BWMS, then the treatment may not be effective. Further analysis of the error code on the TRO unit could be required. Common problems include intake valves not opened, intake lines blocked, dirty cuvette, expired reagents, or seized TRO pumps.

BWMS that use ultra-violet (UV) light

These systems rely on UV light to inactivate or kill organisms in the ballast water and often treat on uptake and discharge. The systems use multiple sensors for UV intensity, temperature, water turbidity, and water flow to maintain effective treatment.

Alarms indicate that the system may not be treating the ballast water effectively. Continually operating when there are repeated alarms for low light intensity, high filter differential pressure, low water flow, or high temperature may result in a detailed inspection and a direction to stop discharge.

(AUS Gov- Department of Agriculture,Fisheries and Forestry 2025)

The TRO sensor measures the amount of free chlorine or chlorine compounds present in the ballast water after it has been treated by the Ballast Water Treatment System . The TRO sensor is used to evaluate the efficiency of the treatment process. The TRO level indicates how effective the treatment has been in killing harmful organisms and pathogens. If the TRO is operating outside the permitted tolerance levels, it may indicate that the treatment has not been effective. The result could mean that the ballast water may still contain harmful organisms or there is too much chlorine (Cl₂) in the ballast water.

The TRO sensor should be maintained and calibrated as per the manufacturer's recommendations.

All ships with an approved BWMP should maintain record-keeping of ballast water operations in the BWRB in accordance with guidance BWM.2/Circ.80. They should also apply the codes A to H, together with the specific item number for different ballast water operations specified in MEPC.369(80).

If the approved electronic record books are provided and the hard copy record book is replaced by an electronic one, a ship-specific declaration must be carried on board. This declaration should confirm that the installation of the electronic system designed to record entries in accordance with the BWM Convention meets the requirements of the IMO guidelines.

5.11 This question has been removed from the current version of the document.

5.12 If ballast tanks are located adjacent to fuel oil tanks, or there is a possibility of contamination by hydraulic oil, are ballast tank contents being checked to ensure there has been no contamination of the water by oil prior to discharge? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

There have been cases where fuel has leaked into ballast water tanks due to fractures, pitting or corrosion in the shared bulkhead between the tanks. This can lead to substantial financial losses and penalties if the contaminated ballast water is discharged at sea.

Ballast tanks adjacent to bunker tanks, or those with bunker lines running through them, need to be monitored for potential leaks from nearby bunker tanks or bunker lines. It's crucial that ballast water, if it contains an oil sheen on the surface, is not discharged. Monitoring these ballast tanks may involve one of several options. These could include checking the quality of the water inside the ballast tanks by visually inspecting the surface, drawing samples from the tanks, or monitoring the atmosphere within the tanks by installing a gas sampling system.

5.13 Are the emergency bilge suction and emergency overboard discharge valves in the engine room in good order and clearly identified with a notice warning against accidental opening and, is the area around the bilge injection suction bellmouth clear of debris and clean? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall review the test procedure of emergency suction valves.

Emergency bilge discharge valves and other overboard discharge valves of a similar nature that are normally closed are sealed in the closed position with numbered seals. The SMS should implement a suitable method, either manual or electronic, for recording the changes in the process, including removal and replacement of numbered seal tags, testing of valves, maintenance, and other operational requirements. In accordance with MSC-MEPC.4/Circ.3, the sealing of valves of an emergency nature shall not be construed as a requirement for the valve to be blanked or physically locked. It shall be ensured that such valves always remain available for use in case of an emergency, and valve sealing may be accomplished through use of a breakable seal, electronic tracking, or similar method.

5.14 Are arrangements for sludge collecting pumps free from any connection to a direct overboard discharge? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Sludge collecting pumps are pumps capable of taking suction from any oil residue (sludge) producing equipment or tank, other than an oil residue (sludge) tank(s) and discharging only to oil residue (sludge) tank(s).

(MEPC.1/Circ.642, Revised Guidelines for Systems for Handling Oily Wastes in Machinery

Spaces of Ships Incorporating Guidance Notes for an Integrated Bilge Water Treatment System (IBTS), 2008)

5.15

Are the engineering officers familiar with the company procedures for operating the oil filtering equipment, and was the equipment in satisfactory condition and used in compliance with company procedures, the manufacturer's instructions, and MARPOL Annex I? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Oil filtering equipment installed on a ship on or after 1 January 2005 must be approved to meet Resolution MEPC.107(49) to comply with MARPOL Annex I Regulation 14.

The requirements include;

- > a truly representative sample of the effluent with adequate pressure and flow is supplied to the 15ppm bilge alarm (6.2.2).
- > provision of fail-safe arrangements to avoid any discharge in case of malfunction of the 15ppm bilge alarm (4.1.3).
- > 15ppm bilge alarm is fitted with an electronic device that is pre-set to activate when the effluent exceeds 15ppm and operates automatically if at any time the 15ppm bilge alarm should fail to function (4.2.7).
- > the response time of the 15ppm bilge alarm. This is the time that elapses between an alteration in the sample being supplied to the 15ppm bilge alarm and the ppm display showing the correct response. Response time should not exceed 5 seconds (4.2.6).
- > the setup of the installation must minimise the time it takes for the system to respond when the discharge from the 15ppm bilge separator exceeds 15ppm. This includes the time it takes for the automatic stopping device to activate and prevent discharge into the water. This should take less than 20 seconds. It must not exceed that limit under any circumstances (as per section 6.2.1).

The inspector should inspect the condition, and operation, of the oily water separator, filtering equipment and alarm and stopping or monitoring arrangements.

Operational testing of oil filtering equipment will require the equipment to be configured to circulate liquid from bilge tank to bilge tank (recirculating facility) and provide an effluent sample to the 15ppm bilge alarm – simulating the discharge of 15ppm bilge separator effluent overboard. When a simulation of effluent sample greater than 15ppm is applied, the inspector shall confirm that the alarm is activated, and that the automatic stopping device (3-way valve) stops effluent discharge overboard. This indicates compliant operation of the system.

The inspector should confirm that there is a flow of effluent sample from the 15ppm bilge separator that is truly representative, with adequate pressure and flow, to the 15ppm bilge alarm while effluent is being simulated to flow overboard. In cases where the flow of effluent sample is not a representative sample, including blockage of the sample line or incorrect operation of valves, it is expected that, in accordance with MEPC.107(49) requirements, the fail-safe arrangement will activate the automatic stopping device (3-way valve) and stop effluent discharge overboard.

RightShip's view aligns with AMSA's interpretation that the failure of the 15ppm bilge alarm to activate the automatic stopping device, in the absence of a representative sample of the effluent, constitutes noncompliance with Resolution MEPC.107(49). This indicates that there is no fail-safe arrangement as required by technical specification 4.1.3.

MEPC 107(49) does not specifically require the fitting of flow or pressure sensors. However, classification societies advocate for the installation of "flow sensors" in the 15ppm bilge alarm sample line. The flow or pressure sensors activate an alarm and operate the automatic stopping arrangements when a truly representative sample, with adequate pressure and flow, is not present at the 15ppm bilge alarm. They also recommend the sealing of all valves installed in the effluent sample pipes so that the valves are locked and sealed in their normal operating position to ensure adequate effluent sampling.

If a manual valve on the sampling line of the MEPC.107(49)-approved oily water separator is fitted and is not sealed in the open position by Class, the inspector should record a finding.

If the manual valve located on the sampling line of the MEPC.107(49)-approved oily water separator is not sealed by Class, the inspector should witness the operational testing of the equipment by shutting the sampling valve and stopping the sample water flow to the 15ppm bilge alarm. If the 15ppm bilge alarm does not activate when the effluent sample flow is stopped for more than five seconds, and the automatic stopping device is not triggered within twenty seconds, this is considered a failure of the oily discharge monitoring and control system and the 15ppm alarm arrangements. The inspector should record a finding.

Inspectors shall, when agreed by the Chief Engineer and in full compliance with MARPOL and local environmental requirements, conduct the following operational tests of the oily water separator:

1. The equipment should be configured to circulate liquid from bilge tank to bilge tank (recirculating facility) and provide an effluent sample to the 15ppm bilge alarm, simulating the discharge of 15ppm bilge separator effluent overboard. When a simulation of an effluent sample greater than 15ppm is applied, the inspector shall witness that the alarm is activated, and that the automatic stopping device (3-way valve) stops effluent discharge overboard.

2. If a sampling valve is provided and is not sealed by Class, the inspector shall request the accompanying engineer officer to close the sample valve, thereby interrupting the effluent flow to the 15ppm bilge alarm. The 15ppm bilge alarm must activate within 5 seconds of the effluent sample flow being stopped, and the automatic stopping device must activate within 20 seconds. If not, this is considered a failure of the oily discharge monitoring and control system and the 15ppm alarm arrangements.

Inspectors should document any evidence of operational failure of system or failure to respond to an interruption of the effluent sample. In the latter instance, the inspector must clearly state the time at which the system responded to the interruption of the effluent sample, triggering the 15ppm bilge alarm, and the time taken for the automatic stopping device to activate.

According to the MEPC 107(49) Chapter 6.1.1, Installation Requirements, it states that "For future inspection purposes on board ship, a sampling point should be provided in a vertical section of the water effluent piping as close as is practicable to the 15ppm bilge separator outlet."

The requirement to install the sampling point in a vertical section of the effluent pipe is to ensure that the sample is representative and homogeneous. If the sampling point is installed in a horizontal section of the effluent pipe, there is a risk that the oil is floating at the top and will not be part of the sample. Although it may be demonstrated that sampling from a horizontal section of the water effluent piping can also result in a representative sample, it doesn't comply with the explicit requirement in MEPC.107(49) to have a sampling point in a vertical section.

Specific warning signs should be posted at the overboard discharge valve of the Oily Water Separator. Effective sealing arrangements must be in place to prevent accidental opening, and the system should be engineered to provide protection against unauthorized access or unintentional operation of the valves.

5.16 This question has been removed from the current version of the document.

5.17 This question has been removed from the current version of the document.

5.18 Has a declaration been provided by the shipper as to whether the cargo is harmful to the marine environment (HME)? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Solid bulk cargoes shall be classified in accordance with the criteria specified in the 2012 Guidelines for the implementation of MARPOL Annex V MEPC. 219(63) and a declaration provided by the shipper as to whether or not they are harmful to the marine environment.

Cargo residues classified as harmful to the marine environment (HME), which cannot be recovered using commonly available methods for unloading, cannot be discharged into the sea. This waste must be discharged to an onshore waste reception facility. MARPOL, 2017.

(Resolution MEPC.219 (63), Guidelines for the Implementation of Marpol Annex V, 2012)
(Resolution MEPC.278 (70) Amendments to the Annex of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the protocol of 1978 relating thereto, 2016)

5.19 Has a Garbage Management Plan been provided and is the Garbage Record Book (GRB) being correctly maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Every ship of 100 gross tonnage and above and every ship which is certified to carry 15 persons or more shall carry a garbage management plan which the crew shall follow.

Every ship of 400 gross tonnage and above and every ship which is certified to carry 15 or more persons engaged in voyages to ports or offshore terminals under the jurisdiction of another Party to the Convention and every fixed or floating platform shall be provided with a Garbage Record Book. The Garbage Record Book, whether as part of the ship's official logbook or otherwise, shall be in the form specified in the appendix to this Annex:

1. Each discharge into the sea or to a reception facility, or a completed incineration, shall be promptly recorded in the Garbage Record Book and signed for on the date of discharge or incineration by the officer in charge. Each completed page of the Garbage Record Book shall be signed by the master of the ship. The entries in the Garbage Record Book shall be at least in English, French or Spanish. Where the entries are also made in an official language of the State whose flag the ship is entitled to fly, the entries in that language shall prevail in case of a dispute or discrepancy.

2. The entry for each discharge or incineration shall include date and time, position of ship, category of the garbage and the estimate amount discharged or incinerated.

3. The Garbage Record Book shall be kept on board the ship or the fixed or floating platform, and in such a place as to be readily available for inspection at all reasonable times. This document shall be preserved for a period of at least two years from the date of the last entry made in it.

4. in the event of any discharge or accidental loss referred to in regulation 7 of this Annex and entry shall be made in the Garbage Record Book, or in the case of any ship of less than 400 gross tonnage, an entry shall be made in the ship's official logbook, of the location, circumstances of, and the reasons for the discharge or loss, details of the items discharged or lost, and the reasonable precautions taken to prevent or minimize such discharge or accidental loss.

4.1 Garbage is to be grouped into categories from the purposes of the Garbage Record Book (or ship's official logbook) as follows:

- A- Plastics
- B- Food Wastes
- C- Domestic Wastes
- D- Cooking Oil
- E- Incinerator Ashes
- F- Operational Wastes
- G- Animal carcasses
- H- Fishing gear
- I- E-waste

4.2 Amount of garbage

The amount of garbage on board should be estimated in cubic metres, if possible separately according to category.

The Garbage Record Book contains many references to estimated amount of garbage. It is recognized that the accuracy of estimating amounts of garbage is left to interpretation. Volume estimates will differ before and after processing. Some processing procedures may not allow for a usable estimate of volume, e.g. the continuous processing of food waste. Such factors should be taken into consideration when making and interpreting entries made in a record.

(MARPOL, 2017)

(Resolution MEPC.220 (63), Guidelines for the Development of Garbage Management Plans, 2012)

(Resolution MEPC.295 (71), Guidelines for the Implementation of MARPOL Annex V, 2017)

Garbage collected throughout the ship should be delivered to designated processing or storage locations. Cleaning and disinfecting of garbage storage location are both preventative and remedial pest control methods that should be applied regularly in garbage storage areas.

(GUIDELINES FOR THE IMPLEMENTATION OF MARPOL ANNEX V, 2017)

5.20 This question has been removed from the current version of the document.

5.21 Has a ship-specific Energy Efficiency Management Plan been provided to the vessel?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

To support ships' energy performance and efficiency objectives, the IMO has developed the SEEMP. This three-part operational measure establishes a cost-effective mechanism for improving ship energy efficiency over time.

SEEMP Part I:

Came into force on January 1, 2013.

Required all ships over 400 GT making international voyages to have SEEMP documentation on-board.

SEEMP PART II

Came into effect on January 1, 2019, as part of the IMO Data Collection System (DCS).

Required every ship over 5,000 GT to collect data and report on their fuel oil consumption.

SEEMP Part II must be verified by the relevant flag administration, or any organization duly authorized by it.

SEEMP PART III

Came into effect on January 1, 2023, and concerns ships' Carbon Intensity Indicator (CII) and ratings.

Requires ships over 5,000 GT (that fall into one of the categories listed in MARPOL Annex VI, regulation 26) to describe and support the ship's carbon intensity objectives, using data from the IMO DCS to assess their performance.

Contains a ship's CII calculation methodology, the required CII values for the next three years, a three-year implementation plan, and self-evaluation and improvement procedures.

Must be verified by the relevant flag administration or any organization duly authorized by it.

The SEEMP Part III is designed to assist companies in achieving the required Carbon Intensity Indicator (CII). In relation to this annual rating, the SEEMP Part III is a mandatory, ship-specific document that outlines the plan to improve the CII, and consequently, the vessel's operational energy efficiency, over the next three years.

The SEEMP Part III is a dynamic document that is subject to regular updates and revisions, reflecting changes in performance and required measures. It must be verified and kept on board the respective vessel from January 1, 2023, along with the Confirmation of Compliance (CoC)

Connection between DCS, CII and SEEMP Part III

Starting in 2024, the CII must be calculated and reported to the DCS verifier, along with the aggregated DCS data for the previous year, including any correction factors and voyage adjustments. The attained CII and the environmental rating (A to E) will be noted on the DCS Statement of Compliance (SoC), which is required to be kept on board for five years.

In case of a D rating for three consecutive years or one E rating, the SEEMP Part III must be updated with a Corrective Action Plan and verified before the DCS SoC can be issued. The Corrective Action Plan should consist of an analysis of why the required CII was not achieved and include a revised implementation plan.

(MEPC.346(78) 2022)

SEEMP Part III is a dynamic document that needs to be updated regularly with prevailing fuel consumption data. This allows the vessel's manager to tangibly track the CII performance and the effectiveness of the ship's energy efficiency measures. The SEEMP Part III plan provides the information needed to assess the operational carbon intensity (CII) of the ship. Regulation 26.3.3 of Annex VI requires SEEMP Part III to be subject to periodical company audits, which should be undertaken at least every three years.

The scope for such company audits is outlined in Resolution MEPC.347(78) – Guidelines for the Verification and Company Audits by the Administration of Part III of the Ship Energy Efficiency Management Plan (SEEMP).

Company audits are to be undertaken within a period of six months after the issue of the Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating, as required by MARPOL Annex VI, Regulation 6.6.4.

Classification Societies can act on behalf of the flag Administration to perform company audits. The Class should issue a statement confirming that the periodical company audit has been conducted to demonstrate compliance with this requirement.

An additional verification shall be carried to a ship rated as D for three consecutive years, to ensure that a plan of corrective actions has been established. A company audit should take place during the third year, after 2 consecutive years, rated as D.

An additional verification shall be carried to a ship rated as E, to ensure that a plan of corrective actions has been established. A company audit, in conjunction with a shipboard audit, should take place during the immediate year after, being rated as E.

5.22 This question has been removed from the current version of the document.

5.23 If the vessel has an Exhaust Gas Cleaning System (scrubber system), is it in good working order, are the engineers familiar with its safe operation, and have procedures been incorporated into the Safety Management System?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

There are three types of exhaust gas cleaning systems at present – open loop system, closed loop system and Hybrid system. Record the type of exhaust gas cleaning system in comments.

Record Finding if there was soot on the water surface, which have been traced to the wash water discharge from vessels.

Corrosion of overboard distance piece attached to the hull on SOX scrubber discharge water line can result in water ingress into areas such as the engine room, ballast tanks and cargo holds. Absence of or poor application of protective coatings on the inside of the pipe and at the welds, along with poor application of paint on hull plating near the wash water discharge were identified as the causes of accelerated corrosion. Rightship recommends that the vessel manager implement a method of monitoring the internal condition of distance piece in the PMS.

Crew should be fully trained to handle the scrubber system and hazardous chemicals used for the process. They should also be trained to deal with medical emergencies. Hazardous chemicals are used in a number of Exhaust Gas Cleaning System (EGCS) and adequate controls should be put in place to protect the ship's staff.

There is also a possibility of further hazardous chemicals and compounds (such as ammonium bisulphate in selective catalytic reduction (SCR) systems) being generated. These will require robust procedures and crew training, as well as adequate signage and personal protective equipment (PPE). Crew training should cover the normal operation of the scrubber system, including bunkering of any chemicals (consumables), calibration of sensors and routine maintenance, as well as the procedures to be followed in case of system failure and deviation from normal operation.

(Your options for emissions compliance Guidance for shipowners and operators on the Annex VI SOx and NOx regulations, 2015)

The maintenance, calibration, cleaning, and chemical handling of the exhaust gas cleaning system shall be incorporated in the plan maintenance system. The system shall be approved by the classification society.

EGCS malfunction:

Any EGCS malfunction should be noted in the EGCS record book and engine room logbook, including the date and time of malfunction, if and how it was resolved, and any follow-up action. The ship manager should be informed, as should the flag state administration, as necessary.

Emissions may be exceeded in the short term when a sudden change in the exhaust gas flow rate changes the EGCS dynamic response. The ECS technical manual will specify the typical operating conditions that may result when emissions are exceeded for a short time.

Repetitive malfunctions and any malfunctions that last more than an hour should be reported to the flag and port state, along with an explanation of the steps the operator is taking to address the failure. At their discretion, the flag or port state may take this information and other relevant circumstances into account when deciding what action to take.

If non-compliance cannot be settled resolved within an hour, changeover to compliant fuel should be considered and the EGCS stopped. Flag and or port state should be consulted. All relevant entries should be made in engine room/deck logbooks.
(Engine Room Procedure Guide 2024)

Malfunctions of EGCS lasting over an hour should be reported to the flag and port state administrations. Sometimes, malfunctions go undetected due to poor maintenance, as monitoring equipment may not register increased SO2 levels.

Zero or negative SO2 readings don't always indicate analyzer failure; they can be within calibration tolerance. Low sulphur fuel, low engine load, and high wash water flow typically result in low SO2 levels.

However, zero or negative readings should be scrutinized. Indicators of false readings include irritating exhaust odors, stable readings despite wash water adjustments, and CO2 levels below 5%. High sulphur fuel should show noticeable SO2 levels. Regular self-checks and error message reviews are crucial.

For suspected false readings, follow the manufacturer's corrective actions. Check for leaks in the measuring system, ensure proper calibration, and avoid systematic errors.

Recommendations:

- > Be alert to prolonged zero or negative readings and perform plausibility checks.
- > Perform a validation test: Validate by increasing engine load and decreasing wash water flow; if SO2 doesn't increase, service the analyzer.
- > Follow strict service, maintenance, and calibration procedures to minimize malfunctions.

(DNV Technical and Regulatory News No.25/2024)

5.24 This question has been removed from the current version of the document.

5.25 Are the ballast pumping systems, including their associated instruments, controls, valves, pipework, and the control panel (including pressure gauges, draft gauges, and the remote control system), in good working order and is there recorded evidence of regular inspection?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The maintenance and testing of the ballast valves should be incorporated into the PMS.

Valve Remote Control System: This refers to devices that utilize hydraulic oil pressure, pneumatic pressure, and/or electricity as power sources. These devices enable remote centralized control and position display of valves within the ship's piping systems, all from the control console.

The controls and indicators of the Ballast System should be accurate and fully operational, including, where fitted:

- > Monitoring of draft, trim, and list
- > Control and monitoring of ballast pumps and associated plant
- > Control of ballast system valves and indication of their status
- > Information mimics/displays of the ballast system

All pressure, temperature, and level sensors of the ballast system should be periodically verified for accuracy.

Incorrect Valve Position Indication: The primary function of the hydraulically controlled valve is to enable remote operation. Ballast control panels for such valves are equipped with position indicators that display the current position of the valve (Open or Shut). An incorrect position displayed by the indicator can lead to confusion. This may cause the officer in charge of cargo operation to proceed with ballast operation without knowing the actual position of the valve.

Common Causes for Incorrect Valve Position Indication:

The following are the most common causes for incorrect position indication:

- > Defective flow meter
- > Defective pressure switch
- > Problems with the indicator panel
- > Slow valve movement

Once the signal is given to operate a hydraulic valve from the ballast control panel, the valve should fully open or shut within a given time. A delay in the operating timing of the valve can be caused by:

- > Defective control valve operation
- > Clogged flow reducer
- > Clogged filters
- > Low oil pressure
- > Low oil temperature
- > Damaged valve actuator

Valve Opening/Closing Time: This refers to the duration the actuator of the Valve Remote Control System takes to drive the valves into an open or closed position.

It is recommended that all shipside valves and main valves be opened out and overhauled, repaired and surveyed at dry dock to ensure their correct operation and tightness.

Operating time of the power operated ballast valves should be checked regularly, and the manufacturers guidance should be followed for optimum opening and closing times of the ballast valves.

5.26 Are bunker and ballast tank manholes maintained in good condition?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The gaskets and fastening bolts should be fitted in their original condition and maintained in good condition.

5.27 Are the crew members familiar with the operation and maintenance of the sewage treatment plant, and is the sewage treatment plant, including its associated piping, discharge pumps, and air blowers, in good condition?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The sewage regulations prohibit the discharge of sewage into the sea, except when the ship has in operation an approved sewage treatment plant; is discharging comminuted (or macerated) and disinfected sewage at a distance of more than three nautical miles from the nearest land; or is discharging untreated sewage from a holding tank at a prescribed rate and at a distance of more than 12 nautical miles from the nearest land. The sewage regulations also include standards for discharge connections to facilitate the disposal of sewage from ship to shore.

Engineer officers should be familiar with the operation and maintenance of the sewage treatment plant, following the manufacturer's guidelines. The effectiveness of the plant's aeration and air blowers can be verified by observing fine air bubbles in the sludge return line.

The final stage of water treatment incorporates a chlorinator before the water is pumped overboard. The chlorinator could be either a tablet dosing type or a chemical injection type.

In the tablet-based chlorinator, clean water directly contacts the chlorine tablets, forming a chlorine solution. The chlorinator is equipped with cylinders for tablet insertion. A sufficient number of tablets of adequate dimensions that can be inserted without breaking and as recommended by the manufacturer, should be used.

In the chemical pump type, a predetermined quantity of Sodium Hypochlorite (NaOCl) is injected into the sterilisation/chlorination tank using a diaphragm-type reciprocating pump.

Section 6: Ship's Structure

6.1 Is the vessel free of any hull repairs unreported to class? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if documents or visual evidence indicated that unauthorised hull repairs have been carried out.

The vessel's manager may engage a "Riding Crew" consisting of a qualified welder and fitters who carry out repairs and steel renewal at sea. The Riding Crew may be involved in welding fractures and replacing deck plate and hatch coaming in various locations, including cargo holds and ballast tanks. Many repairs carried out on voyages are not brought to the attention of the class society and are not always carried out in a professional way.

It is the responsibility of the shipowner to maintain and repair the vessel in periods between regular surveys. Moreover, the shipowner is required to inform the corresponding Classification Society as soon as any damage or defect which may affect conformance with Classification rules is discovered. There is no precise definition of what deficiencies are relevant in this respect. In general, these would be defects which diminish the structural capability of the hull, breach the watertight integrity of tanks or the hull, or impair redundancy or normal operation of a vessel's propulsion, steering, power generation, auxiliary machinery, and associated systems. In case of doubt as to whether a particular deficiency warrants Class attention, shipowners should contact their Classification Society for clarification.

(Onboard Repairs - Compliance with Class and Statutory Requirements - A P&I Perspective, 2017)

6.2 Does the SMS include procedures and instructions for regular inspection of cargo holds, ballast tanks, void spaces, trunks, duct keel and cofferdams by the ship's personnel and are records maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record of inspection, photo and/or video evidence of such inspection shall be available. Record a Finding:

1. When the inspection report is not available, or
2. When inspection report was not supported with photos and /or video evidence.

After every discharge and each cleaning, holds should be formally inspected by the Master or Chief Officer. The ballast tanks, void spaces, cofferdams, and duct keel should be inspected at least annually. Ballast tanks and void spaces adjacent to grab or bulldozer's damage shall be inspected after completion of the discharge. This inspection should be recorded with photographs.

The inspection plan and records should at least cover the following:

- > Framing of the holds – damaged and 'tripped' brackets
- > Condition of bulkhead coatings on the holds
- > Condition of hatch covers, trackways, compression bars, channel drainage, hatch rubbers, cross, hatch drain valve and side cleats
- > Hatch and hold vents and watertight lids, including access hatch lids, rubber packing and closing cleats and dogs
- > Tank top, any damage
- > Condition of coating in ballast tanks, void space, cofferdam, and duct keel
- > Condition of tank top double bottom or side tank access lid, condition and the fitting of the gaskets, condition of nuts
- > Condition of hold ladders, platforms and handrails
- > Condition of hold piping, air vent and water ballast sounding lines, and piping protection brackets
- > Condition of bilge wells, including bilge covers, strum boxes, and bilge well valves, including non-return valves
- > Condition of bilge high-level alarms
- > Condition of lights and light fittings.

Self-unloading vessels that handle bulk materials may utilize linings specifically designed to reduce friction between the cargo and the beds or chutes. This design facilitates and enhances the mass flow of materials, increasing the rate of flow by preventing blockages, build-ups, and funnel flow. The frequency of lining replacement can depend on various factors, including the type of cargo, the frequency of operations, and environmental conditions. It is recommended that the lining be maintained in accordance with the manufacturer's guidelines. If linings are provided, they should be regularly inspected and maintained as an integral part of routine vessel maintenance.

6.3 Is the Enhanced Survey report file adequately maintained and does the condition evaluation report confirm the fitness of the ship for its intended service for the next five years? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record the tank coating condition in comment. This question must be answered N/A if the Class certificate indicates the vessel as a General Cargo Ship.

Bulk carriers and oil tankers shall have a survey report file and supporting documents complying with paragraphs 6.2 and 6.3 of annex A and annex B of resolution A.744(18) – Guidelines on the enhanced programme of inspections during surveys of bulk carriers and oil tankers.

Note: refer to the requirements of survey report file and supporting documents for bulk carriers and oil tankers as referred to in paragraphs 6.2 and 6.3 of annex A/annex B, part A/part B, 2011 ESP Code.

(SOLAS 1974, 2020)

6.4 Are the access points to cargo holds, ballast tanks, and void spaces including vertical ladders, spiral ladders, rungs, stations, and platforms being maintained and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Corrosion wastage affects the structural safety. Record a Finding if vertical ladders, spiral ladders, rungs, station, and platform are found damaged or corroded.

6.5 Are the air pipes and sounding pipes in the cargo holds and void spaces in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Deck sounding pipes pass through the weather deck and are fitted with screw-down caps. Sounding pipes for engine room double-bottom tanks are fitted with counterweight self-closing cocks. It is imperative that sounding pipe caps or cocks be kept shut and well maintained.

Pipes passing through a dry cargo space must be inspected for physical damage after the completion of discharging operations. It is advisable to open and inspect air pipe headers on the exposed weather deck once every five years, following the first special survey. This is necessary because corrosion on the inside of an air pipe header will not be noticeable externally. To extend the life of air pipe headers, they should be galvanised.

Screw-down caps are fitted to the top of sounding pipes. These caps should never be mislaid or replaced with wooden plugs. The self-closing cocks on engine room sounding pipes should never be tied open.

(A Master's Guide to Ship's Piping, 2012)

6.6 If the vessel has a duct keel, is the access, mechanical ventilator, and lighting adequate and is it free of water? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

6.7 Is the vessel free of any apparent structural defects? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall specify the structural component(s) inspected.

6.8

Are cargo hold ventilation systems being maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V**Guide to Inspection**

Every ventilator should have a positive means of closing. The closing mechanism could be in the form of a weathertight door or a ventilator flap (or damper) set within the vent trunk and operated by an external lever, or it may consist of a cowl which can be screwed down into a closed position by the operation of a valve wheel. It is essential that the closing devices are maintained and in a good condition, which includes being greased as needed and inspecting the gaskets to ensure an effective seal, especially in the case of a fire or shipping spray in the vicinity of ventilator intakes.

Ventilation ports and fan spaces must be checked for possible loose rust or paint chips that might fall onto the cargo, causing contamination. Prior to any loading operation, the fans for mechanical ventilation should be checked to ensure they are in operation.

It is recommended that the mechanical ventilators are prominently and permanently marked with the space (that is being serviced by the vent) and that it is indicated whether the shut-off is open or closed with the direction of the damper mechanism.

For various cargoes, wire mesh guards shall be fitted over the fan openings on deck. The wire mesh guards shall have a mesh size not exceeding 13 x 13 mm and shall prevent foreign objects entering the fan casing which could produce sparks with the rotating impeller (SOLAS Regulation II-2/19.3.4.2 and MSC/Circ.1120). In addition, for the carriage of SEED CAKE UN 1386 (b), SEED CAKE UN 2217 and SULPHUR UN 1350, all ventilation openings on the deck shall be fitted with spark-arresting screens (IMSBC Code Appendix 1).

These screens have a much finer mesh size than wire mesh guards. A definition of the term is not included in the IMSBC Code. However, the U.S. Coast Guard (46 CFR §151.03-25), for example, defines the mesh size as follows: single screen with at least 30 x 30 threads per square inch or two screens with 20 x 20 threads per square inch fitted in series not less than half an inch or more than one and a half inches apart.

(CARGO AND CARGO HOLD VENTILATION, 2020)

Section 7A: Fuel Management (Oil Fuel)

Note: Oil Fuel means any oil used as fuel in connection with the propulsion and auxiliary machinery of the ship in which such as oil is carried.

7.1 Is adequate manifold spill containment provided under the bunker manifolds, and are they clean and empty? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

7.2 Are bunker transfer systems hydrostatically tested to their Maximum Allowable Working Pressure (MAWP) on an annual basis and to 1.5 times their MAWP at least twice within any five years period?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Bunker pipelines are defined as any pipeline used for taking on, discharging or internally transferring any fuel for consumption on board.

A vessel's 'Bunker Transfer System' should be tested to 100% of their rated working pressure (Sometimes referred to as Maximum Allowable Working Pressure - MAWP) at least annually. 'Bunker Transfer Systems' should be tested to 1.5 times their rated working pressure at least twice within any five-year period.

Pipelines should be marked with the date of test and the test pressure. A vessel's 'Bunker Transfer System' includes the discharge pump and piping between the pump and the vessel's manifold, excluding any non-metallic hoses. In this case the MAWP can be assumed to be either the pressure at which the transfer piping relief valve is set or, where no relief valve(s) are fitted, the maximum discharge pressure that can be developed by the vessel's pump.

Pressure testing should be a hydrostatic test, pressure testing using compressed air is not acceptable.

7.3 Are the drains, vents, pressure gauges, and thermometers on the bunker manifolds in good working order, and are blanks fitted when they are not in use? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

7.4 Are save-alls fitted around all fuel, diesel, and lubricating oil tank vents; are they clean and empty, and is the drain plug secured with a strap chain to a save-all? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The height of any save-alls surrounding fuel, diesel, and lubricating oil tank vents should be lower than the vent heads themselves, including any extension pieces fitted to them. This precaution is necessary to prevent the ingress of water or the formation of a water seal, which could obstruct the release of vacuum pressure within the tank during adverse weather conditions, should the save-alls become filled with water.

The vent heads should be clearly labelled to indicate the space that they serve. Containers should be clean, empty of water and free of oil. Drain plugs should be in place in port.

7.5 Are there procedures for the analysis of fuel, lubricating and hydraulic oils, and are oil sampling requirements aligned with equipment manufacturer's recommendations? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall review the test results and recommendations from the last analysis/report. A Finding should be recorded only if the status of the analysis report is marked as "Critical" and no objective evidence is available to confirm that corrective actions have been taken. Record in comments any actions that have been taken if the status of the analysis report was marked "Critical."

Record a Finding when the instructions from the engine manufacturer as to how often oil samples should be drawn for testing was not followed.

RightShip recommends that the vessel's manager subscribe to a fuel bunker analysis and advisory service.

Fuel oil means any fuel delivered to and intended for combustion purposes for propulsion or operation on board a ship, including gas, distillate, and residual fuel.
(Regulation 2.1.14 of MARPOL Annex VI)

Lubricating oil analysis by approved organisations is an important tool for monitoring the condition of machinery and components. It is highly recommended that laboratory analysis programmes for lubricating oil and hydraulic oil are implemented on board and are closely monitored by the shore-based technical management.

The instructions and procedures for the analysis of fuel shall be incorporated in the safety management system.

In the absence of clear instructions from the engine manufacturer as to how often oil samples should be drawn for testing, the lubricant manufacturer should be contacted for advice.

Appendix V of MARPOL Annex VI details the information that should be displayed on the Bunker Delivery Note (BDN). From 1st May 2024, the BDN should state a specific flashpoint value of the fuel if the flashpoint is below 70°C. Otherwise, it should include a statement that the flashpoint has been measured at or above 70°C. The BDN should also stipulate the relevant test method for the flashpoint, which is ISO 2719:2016, 'Determination of Flashpoint – Pensky-Martens Closed Cup Method.' This could be either Procedure A (for distillate fuels) or Procedure B (for residual fuels).

7.6 This question has been removed from the current version of the document.

7.7 **Are bunkering and oil fuel transfer procedures carefully planned and executed in accordance with industry standards, are the details of the last operation in accordance with industry standards, is the vessel equipped with a procedure for sampling the oil fuel used on board, and are bunker samples stored in a sheltered location? (V)**

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Planning of bunkering operations should include the following:

- > Deck scuppers must be securely plugged during all bunkering operations to prevent the accidental discharge of fuel into the sea.
- > An accurate summary of the different quantities and grades of fuel to be supplied.
- > A plan of which bunker tanks are to be filled, which must include the type and quantity assigned to each tank and the maximum filling volumes.
- > A schematic diagram of the bunker system and proper valve line-up.
- > The filling sequence and the required pumping rate, including initial, maximum, and topping off rate.
- > An indication of the safety margin or "slack" space to be left in each tank. For example, no tank is to be more than 90% full.
- > Soundings of each tank prior to commencement of bunkering and the expected soundings/ullages on completion.
- > The method of sounding and/or ullaging, which can be stipulated to avoid confusion.
- > Details of who is in overall charge of the operation; this is usually the Chief Engineer, and the plan should also indicate who else is involved and their respective duties.
- > Emergency procedures and contacts
- > Procedure of line draining and blowing after completion of bunkering
- > If a common line is used for multiple grades, then the line flushing volumes and procedures
- > Testing of high-level alarms setting in the fuel oil tanks or a substitute means in case alarms are not provided.
- > Proper identification and markings of the valves on the bunker lines.
- > Procedure for changing over tanks during the bunkering
- > Vessel stability drafts, trim, and list during the various stages of bunkering
- > Manning requirements to execute the operation safely.

(Safe Bunkering Practices, 2013)

The procedure should specify the locations of fuel oil sampling points and the sampling procedures to be used to confirm the fuel oil's Sulphur level.

- > The MARPOL sample of the bunker fuel delivered to the ship during the bunkering operation must be taken in accordance with MSC-MEPC.2/Circ.18,
- > Commercial samples' taken during bunker operation for the purpose of verifying physical and chemical properties should be in accordance with ISO 8217;
- > For an in-use sample of a ship's fuel oil (MEPC.1/Circ.864/Rev.1)
- > A sample of the fuel oil to be used or carried in tanks for usage on board (MEPC.1/Circ.889)

It is critical to distinguish between a MARPOL sample and one obtained commercially. The sampling location of the MARPOL sample is regulated, whereas for the commercial sample it is not. As a result, the location of commercial samples is left to the parties.

The retained sample should be stored in a sheltered location where it will not be subject to elevated temperatures, preferably at a cool/ambient temperature, and where it will not be exposed to direct sunlight.

Pursuant to regulation 18(6) of Annex VI of MARPOL 73/78, the retained sample should be retained under the ship's control until the fuel oil is substantially consumed, but in any case, for a period of not less than 12 months from the time of delivery.

(2009 Guidelines for the Sampling of Fuel Oil for Determination of Compliance with the Revised Marpol Annex VI, 2009)

Rightship urges the vessel's manager to establish a procedure for bunker sampling management, including the safe disposal of the sample when no longer required. Consideration should be given to storing samples in a location protected by a localised fixed fire fighting system, such as the paint store.

7.8 This question has been removed from the current version of the document.

7.9 Are ship-specific procedures to control the change from residual to low-sulphur / distillate fuels and vice versa provided, and is the fuel oil change over logbook and data collection system being maintained correctly? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The use of a fuel change over calculator is recommended.

Switching from one type of fuel to another is an operation that does have risks. Vessels trading between areas with different sulphur limitations are required to have specific and detailed change-over procedures. The crew needs to be well trained and aware of any risks associated with the change-over – otherwise they risk engine failure, power loss or even blackout. A full risk assessment should be conducted by all involved in the procedure.

(Emission Control Areas – Ultra Low Sulphur Fuel Oil Change-over Procedures, 2014)

According to MARPOL Annex VI, vessels that use both high and low sulphur fuel oils should have a written change-over procedure. The method should detail how the change-over will be carried out, as well as the time required to flush high sulphur fuel out of the system following the changeover of service tanks, as well as the number of hours required before entering the ECA to begin the change-over.

Before entering an ECA, the ship should switch from high sulphur fuel to low sulphur fuel with a sulphur content of less than 0.10 percent by mass. This operation begins by shutting down consumption from the high sulphur service tank using the three-way valve and replacing it with fuel from the low sulphur service tank. Throughout the changeover procedure, the low sulphur fuel will continually dilute the fuel in the service system. The time required to reach the 0.10 percent sulphur level varies according to the amount of machinery fuel oil consumed, the volume of the service system, and the sulphur content of the fuel.

Throughout the changeover procedure, all steps must be recorded in the engine logbook and Marine sulphur record book. All entries must accurately reflect the quantities, the time of changeover, and the ship's position. Additionally, what time did the ship enter and exit the ECA/SECA should be documented along with the vessel's position. When the changeover process is complete and the vessel is operating on low sulphur fuel oil, the Chief Engineer shall notify the Master.

(Fuel change-over procedure, 2022)

From 01 May 2025, amendments to MARPOL Annex VI Regulation 14.3.5, as amended by IMO Resolution MEPC.361(79), prohibited ships operating within the Mediterranean Sea ECA from using fuel oils with a sulphur content exceeding 0.10% m/m unless an approved equivalent arrangement is used such as Exhaust Gas Cleaning Systems.

From 01 March 2026, in accordance with MARPOL Annex VI Regulations 13, as amended by IMO Resolution MEPC.392(82), ships operating in either the Canadian Arctic ECA or Norwegian Sea ECA with a marine diesel engine with power output of more than 130kW are required to be certified to the NOx Technical Code 2008 to meet the NOx Tier III standard, as follows:

For the Canadian Arctic ECA, ships with keels laid or at a similar stage of construction on or after 1 January 2025.

For the Norwegian Sea ECA:

- > Ships with a building contract placed on or after 1 March 2026
- > In absence of a building contract, ships with keels laid or at a similar stage of construction on or after 1 September 2026; or
- > The delivery is on or after 1 March 2030.

7.10 Are the Quick Closing Valves serving fuel and lubricating oil systems being regularly tested and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Quick Closing Valves are fitted to the outlets of lubricating and fuel oil storage, and settling and service tanks within the machinery space, boiler room and the emergency generator room. These spring-loaded valves may be operated locally or remotely by pull wires, hydraulics, or compressed air. Quick Closing Valves are essential safety devices. They should be properly maintained.

(Quick Closing and Self Closing Valves, 2011)

In multi-engine installations which are supplied from the same fuel source, means of isolating the fuel supply and spill piping to individual engines, shall be provided. The means of isolation shall not affect the operation of the other engines and shall be operable from a position not rendered inaccessible by a fire on any of the engines. (SOLAS 74 2020)

AMSA Marine Notice 2024/05, "Fuel Isolation for Multi-Engine Installations," states that one common solution is to install a quick-closing valve on the fuel supply line to the engine, between the fuel changeover valves and the engine, with a remote actuator in another part of the machinery space.

For ships built after June 2009, where possible, the valves or the actuator for the remote closing valves should be at least 5 metres from the engine in any direction. Where this is not possible, protection by obstructions may be implemented.

For further inspection references, please refer to AMSA Marine Notice 2024/05 and pages 24, 25, 26, and 27 of MSC.1/Circ.1321, dated 11 June 2009.

7.11 Are high pressure fuel delivery pipes of diesel engines protected with a jacketed piping and alarm system, and is the alarm system being tested regularly and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

External high pressure fuel delivery lines between the high-pressure fuel pumps and fuel injectors shall be protected with a jacketed piping system capable of containing fuel from a high-pressure line failure. A jacketed pipe incorporates an outer pipe into which the high-pressure fuel pipe is placed, forming a permanent assembly. The jacketed piping system shall include a means for the collection of leakages and arrangements shall be provided for an alarm to be given of a fuel line failure.

(SOLAS 74, 2020)

7.12 Are purifier rooms and fuel and lubricating oil handling areas ventilated, free of oil leaks and clean? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

7.13 Is the reserve fuel tank of the emergency generator filled with sufficient fuel of a suitable type for at least 18 hours operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the vessel is trading in area with sub-zero temperature, the fuel tank of the emergency generator should be charged with fuel designed for use in sub-zero temperatures.

- > The generator should be capable of providing full load requirements for at least 18 hours.
- > Every oil fuel pipe (which, if damaged, would allow oil to escape from a storage, settling or daily service tank situated above the double bottom) shall be fitted with a cock or valve directly on the tank capable of being closed from a safe position, outside the space concerned, in the event of a fire occurring in the space in which such tanks are situated.
- > Oil fuel pipes (which, if damaged, would allow oil to escape from a storage, settling or daily service tank having a capacity of 500 litres and above situated above the double bottom) shall be fitted with a cock or valve directly on the tank capable of being closed from a safe position, outside the space concerned, in the event of a fire occurring in the space in which such the tanks are situated.
- > The controls for the remote operation of the valve for the emergency generator fuel tank shall be in a separate location from the controls for the remote operation of other valves for tanks located in machinery spaces.

(SOLAS 74, 2020)

Section 7B: Fuel Management (Alternative Fuel- LNG)

Note: The IGF Code applies to ships using low-flashpoint fuels for which the building contract is placed on or after 1 January 2017; the keels of which are laid, or which are at a similar stage of construction on or after 1 July 2017 (in the absence of a building contract); or the delivery of which is on or after 1 January 2021. Ships which commence a conversion to use low-flashpoint fuels (or use additional or different low-flashpoint fuels other than those for which the ship was originally certified) on or after 1 January 2017 will also be required to comply with the IGF Code (see SOLAS regulation II-1/56).

Low-flashpoint fuel means gaseous or liquid fuel having a flashpoint lower than otherwise permitted under paragraph 2.1.1 of SOLAS regulation II-2/4.

This section shall be completed when equipment, machinery, or a system on board the ship is using LNG fuel.

7.1 Is there an approved LNG Fuel Handling and Emergency Procedure Manual, and are crew familiar with the bunkering and emergency procedures such as leakage, fire or potential fuel stratification resulting in rollover? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

LNG bunker operations shall be conducted in accordance with the detailed fuel handling manual and the emergency procedures specified in 18.2.3 of the IGF Code that have been approved for the vessel or vessels by their flag State, recognized organization or classification society that has classed the vessel and complies with the applicable uniform interpretations and requirements posted by the IACS.

Transfers from terminals or mobile facilities shall be conducted in accordance with approved terminal or mobile facility transfer procedures.

The LNG Fuel Handling and Emergency Procedure Manual requires a level of staffing during bunkering operations; however, it does not relieve vessel captains or facility operators from their responsibilities.

(ISO 20519:2017, 2017)

A low flashpoint bunkering manual should be established with involved parties agreeing technically and commercially on methodology, flow rate, temperature, pressure of the delivery of low flashpoint fuels and receiving tanks. This manual shall gather all the information, certificates, procedures, and checklist(s) necessary for an effective and safe low flashpoint bunkering operation.

The documented operational procedures shall cover the loading, storage, operation, maintenance and inspection of systems and emergency procedures.

"Guidelines for Gas Fuelled Ships" and "Guidelines for Ships Using Low Flashpoint Fuels" from ClassNK provides further guidance on how to prepare such a manual.

7.2 Do the Master, engineering officers and all personnel with immediate responsibility for the care and use of fuels and fuel systems on ship, hold a certificate in advanced training for service on the ships subject to the IGF Code? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Masters, engineering officers and all personnel with immediate responsibility for the care and use of fuels and fuel systems on ships subject to the IGF Code shall hold a certificate in advanced training for service on ships subject to the IGF Code.

Every candidate for certification in advanced training for service on ships subject to the IGF Code shall:

1. Have successfully completed the approved advanced training required by regulation V/3, paragraph 7 in accordance with their capacity, duties and responsibilities as set out in STCW Code table A-V/3-2; and
2. Provide evidence that the required standard of competence has been achieved in accordance with the methods and the criteria for evaluating competence tabulated in columns 3 and 4 of STCW Code table A-V/3-2; or
3. Have received appropriate training and certification according to the requirements for service on liquefied gas tankers as set out in STCW Code regulation V/3, paragraph 8.

Seafarers holding the advanced training for service on ships subject to the IGF Code shall, at intervals not exceeding five years, undertake appropriate refresher training or be required to provide evidence of having achieved the required standard of competence within the previous five years.

(Regulation V3, STCW 2010)

7.3 Do seafarers responsible for designated safety duties associated with the care, use or emergency response to the fuel onboard the ship, hold a certificate in basic training for service on ships subject to the IGF Code? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Every candidate for certification in basic training for service on ships subject to the IGF Code shall:

1. Have successfully completed the approved basic training required by regulation V/3, paragraph 5, in accordance with their capacity, duties and responsibilities as set out in STCW Code table A-V/3-1; and
2. Be required to provide evidence that the required standard of competence has been achieved in accordance with the methods and the criteria for evaluating competence tabulated in columns 3 and 4 of STCW Code table A-V/3-1.

Seafarers holding the basic training for service on ships subject to the IGF Code shall, at intervals not exceeding five years, undertake appropriate refresher training or be required to provide evidence of having achieved the required standard of competence within the previous five years.

(Regulation V3, STCW 2010)

7.4 Does the schedule of drills and exercises related to LNG fuels address potential emergency shipboard situations and has it been conducted effectively? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Exercises related to low-flashpoint fuels should include at least the following:

- > A desktop exercises.
- > Review of fuelling procedures based on the ISM approved Company Operations Procedures Manual.
- > Responses to identified hazardous contingencies.
- > •Tests of equipment intended for contingency response.
- > Reviews to confirm that assigned seafarers are trained to perform assigned duties during fuelling and contingency responses.

Gas related exercises may be incorporated into periodical drills required by SOLAS. The response and safety system for hazards and accident control shall be reviewed and tested.

(RESOLUTION MSC.391(95), ADOPTION OF THE INTERNATIONAL CODE OF SAFETY FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE), 2015)

7.5 Are hazardous areas marked with clearly visible warning signage and are the crew familiar with the special precautions and the risks for those areas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

During the transfer operation, personnel in the bunkering manifold area shall be limited to essential staff only. All staff engaged in duties or working in the vicinity of the operations should wear appropriate personal protective equipment (PPE) and every individual shall wear portable gas detector.

Hazardous area means an area in which an explosive gas atmosphere or a flammable gas or vapour is or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of electrical apparatus or any other equipment that may provide potential sources of ignition. Hazardous areas are divided into zone 0, 1 and 2.

For additional information, refer to the IGF Code for the definitions of Zone 0,1 and 2.

Hazardous Area Zone 0

This zone includes but is not limited to the interiors of fuel tanks, any pipework for pressure relief or other venting systems for fuel tanks, pipes and equipment containing fuel.

Hazardous Area Zone 1

This zone includes, but is not limited to:

1. Tank connection spaces, fuel storage hold spaces and intercarrier spaces;
2. Fuel preparation rooms arranged with ventilation.
3. Areas on open deck, or semi-enclosed spaces on deck, within 3m of any fuel tank outlet, gas or vapour outlet, bunker manifold valve other fuel valve, fuel pipe flange, fuel preparation room ventilation outlets and fuel tank openings for pressure release provided to permit the flow of small volumes of gas or vapour mixtures caused by thermal variation.
4. Areas on open deck or semi-enclosed spaces on deck, within 1.5 m of fuel preparation room entrances, fuel preparation room ventilation inlets and other openings into zone1 spaces.
5. Areas on the open deck within spillage coamings surrounding gas bunker manifold valves and 3m beyond these, up to a height of 2.4 m above the deck.
6. Enclosed or semi-enclosed spaces in which pipes containing fuel are located, e.g. ducts around fuel pipes, semi-enclosed bunkering stations;
7. The emergency shutdown (ESD)-protected machinery space is considered a non-hazardous area during normal operation, but any equipment which will need to be operated there following detection of gas leakage must be certified as suitable for zone 1;
8. A space protected by an airlock is considered as a non-hazardous area during normal operation but any equipment which will need to be operated there following detection of gas leakage must be certified as suitable for zone1;
9. Except for type C tanks, an area within 2.4 m of the outer surface of a fuel containment system where such surface is exposed to weather.

Hazardous Area Zone 2

This zone includes but is not limited to areas within 1.5 m surrounding open or semi-enclosed spaces of zone 1. Spaces containing a bolted hatch to a tank connection space.

(IGF Code 2016)

7.6 Are staff responsible for LNG bunkering aware of their responsibilities and actions to be taken in case of malfunction or emergency and are instructions and warning signs clearly posted on site for safe LNG bunkering operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

There shall be written detailed instructions for the bunkering process on both ships regarding responsibility and actions required in case of malfunction or emergency. The instructions are to be quickly available at all times and all personnel involved in bunkering operations are to be familiar with the content and location of the instructions. The instructions should cover the following areas:

- > Loss of communication or control system (ESD)
- > Loss of power
- > Safe break-away of ships in case of fire
- > Handling of cryogenic and petroleum products including use of personal protection equipment, ice formation and awareness of sharp edges.
- > Waves and weather conditions

There shall be warning, and instruction signs posted around hazardous area on both ships. The signs are to be clearly visible and placed according to an accepted guideline for placement of warning signs. The warning signs are to cover the risks of handling cryogenic liquid, fire and safety issues and show restricted areas.

(LNG bunkering ship to ship procedure, 2020)

7.7 Is the safety zone clearly marked and, have restrictions within the safety zone been enforced and followed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Since the receiving ship normally is larger than the bunker ship it is important to have a safety zone above the bunker station during bunkering. The extent of the safety zone should be 10 metres on each side of the bunker station manifold.

This safety zone shall be clearly marked and have the following restrictions:

- > No unauthorised persons to be able to access open deck areas directly above the bunker area
- > Warning signs to be posted around the area
- > Access doors to be locked and only to be opened by trained and authorised personnel
- > No overhead crane lifting in this area during bunkering
- > No maintenance work in the area during bunkering
- > No manoeuvring of ship equipment in the area during bunkering
- > Ventilation inlets in the area to be closed during bunkering

(LNG bunkering ship to ship procedure, 2020)

7.8 Are the self-igniting lights of lifebuoys located in the hazardous area intrinsically safe? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Lifebuoy self-igniting lights do not need to be intrinsically safe if located outside of the hazardous area. However, there must be strict controls in place to avoid those non-intrinsically safe lights being misplaced into the hazardous zone. This may include highlighting/markings of those lights or other appropriate means.

7.9 Is the LNG bunkering operator control panel fitted with an earth indicator light to indicate the faulty circuits and is the control panel free of any faulty earth indication during LNG bunkering? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The main switchboard on the bunker ship and the control panel on the receiving ship are to have earth indicator lights to indicate faulty circuits.

Any indications of faulty circuits are to be immediately traced and isolated to avoid arcing around the bunker area. The bunkering operation is to be suspended in case of faulty earth indication during ongoing transfer.

(LNG bunkering ship to ship procedure, 2020)

7.10 Is the main radio aerial earthed and are portable two-way UHF radios approved for use in hazardous areas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ships' main radio transmissions may cause electrical resonance in insulated parts of some ship fittings such as mast stays, and this can cause arcing across deck fittings. Radio aerials should be earthed but can induce arcing if insulators are coated with salt, dirt, or water. The use of ships main radio equipment during transfer operations can be dangerous and should be restricted during the process. The equipment is not to be used if there is a possibility of flammable gas in the vicinity of the antenna.

Satellite communication equipment normally operates at low power levels and is considered to be a low ignition hazard. The equipment is not to be used if there is a possibility of flammable gas in the vicinity of the antenna. VHF and UHF communications are low voltage operated and are considered to be safe to use. Hand-held VHF or UHF radios are considered to be intrinsically safe.

Portable electronic devices such as mobile phones, cameras etc using batteries are not allowed in hazardous areas unless they are intrinsically safe. It is especially important for personnel working in or visiting such areas to be aware of this. Warning/notification signs are to be posted around these areas.

(LNG bunkering ship to ship procedure, 2020)

Rightship recommends that only intrinsically safe torches and portable two-way UHF radios should be available and used on board every ship covered by the IGF Code.

7.11 Has a pre-compatibility assessment and study of the weather and current forecast been carried out prior to confirming the bunkering operation and is there documented evidence of such assessment and study? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Weather and current forecast for the area are to be studied before commencing bunkering operations. Each Master is responsible for his own ship and bunkering is only allowed when both Masters agree that ambient conditions (like wind and weather) are acceptable. Each Master is also responsible for determining restrictions and taking action in case of a sudden change of ambient conditions during a started bunker transfer.

(LNG bunkering ship to ship procedure, 2020)

A compatibility assessment of the bunkering facility and receiving ship should be undertaken prior to confirming the bunkering operation to identify any aspects that require particular management. The compatibility assessment should be undertaken with the assistance of an appropriate checklist to be completed and agreed by Master(s) and Person in Charge (PIC) prior to engaging in the bunkering operation.

Where applicable, as a minimum, compatibility of the following equipment and installations should be checked prior to engaging further in any low flashpoint bunkering operation:

- > Communication system (hardware, software if any and language) between the PIC, ship's crew and Bunkering Facility Organisation (BFO) personnel
- > ESD system
- > Bunker connection
- > Emergency release system (ERS) or coupling (ERC)
- > Vapour return line
- > Nitrogen lines' availability and connection
- > Mooring equipment
- > Bunker station location
- > Transfer system sizing and loading on manifold
- > Location of ERS
- > Closure speed of valves
- > Hazard Operability Analysis (HAZOP) results as applicable

7.12 Is all lighting around the bunker area Ex-rated and does it appear adequate to illuminate the bunker area? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The bunkering operation is preferably to be conducted during daylight hours. It is necessary to have adequate lighting in case of mooring and bunkering operations after daylight hours. The minimum lighting requirements are the bunker ship deck, the receiving ship bunker station, and the mooring bollards. Normal deck-lighting should in most cases be sufficient, but portable spotlights or bridge wing spotlights may be useful for night operations. Note that all lights around the bunker area are to be of Ex- rated.

(LNG bunkering ship to ship procedure, 2020)

7.13 Have the key components of the LNG bunkering system been identified, included within the PMS, maintained and where applicable, calibrated as per the manufacturer's recommendation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Key components of the ship's systems are to be identified with emphasis on safety to avoid leakage and ignition sources in and around the bunker areas. These components should have a maintenance and replacement schedule where inspections and actions are documented and stored on board. These components shall have redundancy back-up which can start up within a short period of time.

(LNG bunkering ship to ship procedure, 2020)

7.14 Is there a procedure for communication failure during LNG bunkering operation and are crew familiar with such a procedure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Communication failure during approach: Abort approach and re-establish contact before attempting a new approach. Communication failure during bunker operations: Sound the emergency signal and suspend all operations in progress immediately. Operations shall not be resumed before communication has been re-established.

(LNG bunkering ship to ship procedure, 2020)

7.15 Is there an agreed method of tank pressure and temperature control between the delivering and receiving vessels and is there recorded evidence to show that both ships' combined temperature and pressure range are within the safety limits before commencing LNG bunkering? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Both ships must check the LNG tanks regarding temperature and pressure prior to bunkering and note this on the pre-transfer bunker checklist. If the temperature of the receiving tank is significantly higher than the bunker tank, there will be an initial vaporisation when starting to transfer the LNG. This will increase the tank pressure and can trigger the pressure-relief valve to open if the pressure exceeds the set limit. The pressure of both tanks must be reduced prior to the bunkering in case of a high receiving-tank temperature.

The bunker ship Master is to confirm that both ships combined temperature and pressure range are within the safety limits before commencing transfer.

(LNG bunkering ship to ship procedure, 2020)

7.16 Is there evidence to show that a detailed mooring plan was exchanged between the delivering and receiving vessels and has the Master of the receiving vessel reviewed the type and size of fenders of the delivering vessel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The receiving ship should be able to supply, if requested, a sketch with information about placement and number of fairleads and mooring bits and their relative distances to the bunker station. A mooring plan, showing number of lines and fenders and their locations should be agreed upon before making berth.

It is recommended to use pneumatic type main fenders with a diameter of approx. 1 metre. Size and type of secondary fenders to be determined due to the design of the bunker ship. All fenders to be approved by class.

(LNG bunkering ship to ship procedure, 2020)

7.17 Has the LNG hose handling operation been carried out and supervised by trained personnel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Dedicated bunker ships may be fitted with specialised hose-handling equipment, but a rather common method is to use a hose crane to deliver bunker hoses from the bunker ship to the receiving ship.

The hoses are to be supported to the receiving ship, disconnected from the hose crane, and connected to the manifold, by trained personnel from the receiving ship, before the operation commences.

Each manifold is to be earthed and the receiving ship shall be equipped with an insulating flange near the coupling to prevent a possible ignition source due to electrostatic build-up. The hoses with couplings should not touch any un-earthed part before connection to avoid possible electrical arcing.

(LNG bunkering ship to ship procedure, 2020)

7.18 Is a water curtain system provided for the ship's sides in way of manifold and is the manifold tray arrangement adequate and free of any sharp edges? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

There shall not be any sharp edges in the hose handling area. If the receiving vessel has on-board traffic in the vicinity of the bunker station, there should be reinforcements built-in to protect the equipment from traffic impact.

Both ships must have insulated stainless steel trays, below the LNG and vapour-return manifolds, to prevent damage to the steel hull in case of leakage. The cold LNG liquid causes brittle fractions contacting mild steel. Each tray should have an outlet overboard which can be a temporary fitted pipe or hose to lead a possible spill to the water without contact to the hull.

(LNG bunkering Ship to Ship procedure, 2020)

For cargo temperatures below -110°C, a water distribution system shall be fitted in way of the hull under the shore connections to provide a low-pressure water curtain for additional protection of the hull steel and the ship's side structure. This system is in addition to the requirements of 11.3.1.4 and shall be operated when cargo transfer is in progress.

(International code for the construction and equipment of ships carrying liquefied gases in bulk, 2016)

7.19 Has the LNG fuels bunkering checklist been correctly completed and is there evidence to show that they are effectively managing their obligations as accepted in the checklist? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The pre-transfer bunker checklist is to be filled out, signed by the responsible operator on the receiving ship and returned to the bunker ship before starting any transfer. The signed checklist is to be kept on board the bunker vessel for 3 months. No bunker operation is to begin until this checklist is signed and returned to the bunker ship.

(LNG bunkering ship to ship procedure, 2020)

ISO 20519:2017(E) and the International Association of Ports and Harbors (IAPH) have developed three bunkering checklists for LNG bunkering.

For additional information, refer to the IAPH website ([Click Here](#)).

7.20 Are LNG bunker lines being inerted immediately after completion of LNG bunkering and disconnection of hoses from the manifolds and before departure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The receiving ship must inert the bunker lines before departure, which means that the inerting sequence is to start as soon as the hoses are disconnected from the manifold and run until lines are gas free.

There shall be a system for gaseous nitrogen onboard the ships. This is needed for purging the piping system from LNG and natural gas after bunkering (inerting). The manually operated valves, which will introduce nitrogen to the LNG system, are located in the bunker stations.

If liquefied gas is trapped in a pipe between two valves or a tank without an exit, the pressure in the tank or pipe will rise until the pipe or tank bursts. The consequence can be severe injury to personnel. All pipe sections and tanks must therefore be secured with thermal relief valves.

(LNG bunkering ship to ship procedure, 2020)

7.21 Are system safety valves in good order and officers aware of the requirements? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The system includes valves regarded as ordinary safety relief valves as well as so called thermal relief valves. The tanks' main safety valves are designed to meet the requirements for a LNG tank. Thermal relief valves are designed to meet capacities in a trapped volume in pipes. A safety valve exhausts/vents to a vent mast.

To ensure that both safety valves to the LNG tanks are not out of operation at the same time, the safety valve system incorporates an "interlock system". The interlock system consists of lockable valves and a set of keys that permits only one of the safety valves on each tank to be closed.

It will be possible to divert gas from different parts of the system to the atmosphere through a vent mast. The LNG tanks' safety valves will also exhaust to this vent mast, as well as the nitrogen used for purging.

(LNG bunkering ship to ship procedure, 2020)

7.22 Is the receiving vessel in a high state of readiness at all times during LNG bunkering operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Both ships are to be in a high state of readiness at all times during bunkering operations. The following arrangements should be made on both ships:

- > ESD-system tested and in operation mode
- > Emergency stop box (or Link) led from bunker ship to receiving ship
- > Fire-fighting equipment made ready for immediate use
- > Ships prepared to disconnect hoses at short notice
- > Axes placed at bunker ship mooring stations for quick release of mooring lines
- > Soft rope mooring lines (or tails) are being used for easier emergency cutting
- > Ships have main engines ready for immediate use
- > Outlet from LNG spill trays are led overboard and away from hull.

It is possible to have a water curtain system which, in an emergency, sprays water over the ship's sides around the bunker stations to protect the hulls from direct LNG contact, if in place, it should be ready to use.

(LNG bunkering ship to ship procedure, 2020)

7.23 Was the vessel provided with contingency plans for dealing with emergencies? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Due to the risk of accidents and the potential consequences, it is required that each ship has contingency plans for dealing with emergencies. A contingency plan is a summary of individual emergency procedures and shows emergency duties for all ship personnel and plans for taking care of passengers. The contingency plans should be integrated with port and local authorities and agreed upon between both ships prior to commencing operations.

The following emergencies are example of sections in the contingency plan:

- > Fire on either ship
- > LNG leakage
- > Hose failure
- > Hose quick release arrangements
- > Mooring line failure
- > Communication failure
- > Personnel injuries (frost burns, suffocation etc.)
- > Emergency departure procedure
- > Oil pollution from additional petroleum bunkering
- > Fender burst
- > These potential emergencies are to be evaluated to see if some of the risk scenarios are more likely to occur; if so, they should be included in the contingency plan.

(LNG bunkering ship to ship procedure, 2020)

7.24 Is the emergency shutdown system in good order and is there recorded evidence of regular testing? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The bunker ship should preferably provide an emergency stop to the receiving ship in order for both ships to be able to stop the pumps. (LNG bunkering ship to ship procedure, 2020)

The bunkering facility and receiving ship should both test their emergency shutdown (ESD) systems not more than 24 hours before bunkering operations commence. These tests should be documented in accordance with the bunkering procedure manual. The time taken for emergency shutdown valves to move from open to closed, and from closed to open, should be checked regularly and documented.

7.25 Are tank domes, domes' insulation, vapour and filling pipes' insulation, manhole cover insulation and associated fittings in good order, free from leaks and corrosion? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The main purposes of the insulation system are to:

- > Minimize loss of boil-off rate of fuel gas (LNG) by restricting heat ingress.
- > Protect the hull structure against harmful temperature fluctuations and absolute temperatures.
- > Minimize condensation or forming of ice on the cold surfaces and thereby reduce accumulation of water and moisture in the cargo containment system.

7.26 Are LNG fuel tanks protected by an independent LNG tank level alarm device and is there recorded evidence to show that the device has been tested regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

7.27 Is there recorded evidence of regular calibration of thermometers, pressure gauges, the gas detection system and tank level gauges? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Instruments shall be tested to ensure reliability under working conditions and recalibrated at regular intervals. Test procedures for instruments and the intervals between recalibration shall be in accordance with manufacturers' recommendations.

(International code for the construction and equipment of ships carrying liquefied gases in bulk, 2016)

7.28 Is prevention of over-pressurization of the LNG transfer system in the event of activation of the ERS or the ESD documented in the LNG fuel-handling manual? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Select N/A if the transfer system is designed to consider over pressurization due to surge pressure in the event the ERS or the ESD is activated.

7.29

Are precautions to prevent electrostatic charge in the LNG bunker hose being taken and, have the minimum and maximum hose lengths and diameters that the hose support loading arm and/or hose saddles can support been documented in the LNG fuel-handling manual? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

To prevent arcing between the ships the manifolds on both ships are to be earthed, all hoses are to be electrically continuous, and each hose string shall be fitted with an insulating flange on the bunker ship manifold. It is important that the insulating flange is only fitted to one ship; otherwise, there may be an electrostatic build-up in the hose between the insulating flanges which can result in arcing.

Electrical Arcing

Other places (besides hose connections) where arcing can occur are:

- > Mooring lines (should be insulated)
- > Ladders or gangways between ships (should be insulated)
- > Crane wire runners and hooks (operate carefully)
- > Bare wires and chains for fender support (should be insulated)

(LNG bunkering Ship to Ship procedure, 2020)

Hose support loading arm and hose supports (saddles), if used, shall conform to ISO 16904 or EN 1474-3 and be designed to safely support the loads (static and dynamic) imposed by the LNG transfer operations during hose connection, transfer operations and when the hose is disconnected under emergency conditions. They shall provide the necessary support so that the hose bending radius is not below recommended minimum bending radius specified by the hose manufacturer.

The minimum and maximum hose lengths and diameters that the hose-support loading arm and/or hose saddles can support shall be documented in the LNG bunkering procedures manual.

(ISO 20519:2017, 2017)

7.30

Are the officers aware of any LNG bunker loading limitations for the vessel and are these limitations, if applicable, clearly posted at the LNG bunker operation panel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A document shall be provided to the ship, specifying the maximum allowable loading limits for each cargo tank and product, at each applicable loading temperature and maximum reference temperature. The information in this document shall be approved by the administration or recognised organisation acting on its behalf. The pressure at which the pressure relief valves (PRVs) have been set shall also be stated in the document. A copy of the document shall be permanently kept on board by the Master.

(International code for the construction and equipment of ships carrying liquefied gases in bulk, 2016)

Section 7C: Fuel Management (Alternative Fuel- Methanol)

This section is applicable when any equipment, machinery, or system aboard the ship operates using these alternative fuels

7.1

Has a Certificate or Statement of Compliance been issued by Class to confirm the safe use of Methyl/Ethyl Alcohol as fuel by the vessel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

According to the SOLAS Convention, specifically Chapter II-1, Regulations 56 and 57, ships using methanol as fuel shall adhere to the 'International Code of Safety for Ships Using Gases or Other Low-Flashpoint Fuels', also referred to as the IGF Code. It's important to note that the IGF Code does not provide specific safety requirements for the use of methanol as a fuel. At present, the requirements are confined to the 'Interim Guideline for the Safety of Ships Using Methyl/Ethyl Alcohol as Fuel' (MSC.1/Circ. 1621), which was issued by the IMO. The mandatory safety requirements for this scenario have yet to be defined by international conventions. Please [CLICK HERE](#) to access MSC.1/Circ. 1621.

7.2

Was the vessel provided with a procedure for bunkering Methyl/Ethyl Alcohol and were the Master and officers familiar with its contents?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A procedure outlining methanol bunkering procedures which may be incorporated in the ship's fuel handling manual, must be available. It should include:

- > A list of personnel who are currently qualified to conduct methanol bunkering operations, taking into account the specific hazards of methyl/ethyl alcohol used as fuel (colourless flame, low flash point, high flammability range).
- > A description of the duties of all personnel involved.
- > A description of the bunkering parameters for which the system has been designed.
- > De-bunkering procedures
- > Identification of safety zone and Simultaneous Operations (SIMOPS) assessment (when applicable)
- > Nitrogen padding of bunker tanks
- > Pressure test and purging of bunker hose (liquid and vapour) with Nitrogen
- > Vapour recovery procedures have been established to prevent operational venting
- > A list of any limitations on bunkering operations identified in the risk assessment or imposed by authorities, such as monitoring and early detection of flame and lightning risk.
- > List of personal protective equipment, firefighting equipment's, methanol gas detector and thermal cameras.
- > Use and location of Ship Shore Link (SSL) for automatic Emergency Stop Device (ESD) communication with the receiving ship and the delivering facility ESD system
- > Consider the use of Quick Connect and Disconnect Coupling (QCDC) at the bunkering manifold.
- > Emergency contact information.
- > This procedure can be incorporated into an existing ISM manual, provided it is easily accessible to all personnel involved at each transfer site.

7.3

Were the Master, officers, and ratings suitably trained and qualified to serve on board a ship using methylene/ethyl alcohol fuels?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The company should ensure that seafarers on board ships using methyl/ethyl alcohol fuels have completed specific training to attain the abilities that are appropriate to the capacity to be filled, and duties and responsibilities to be taken up.

The master, officers, ratings and other personnel on ships using methyl/ethyl alcohol fuels should be trained and qualified in accordance with regulation V/3 of the STCW Convention and section A-V/3 of the STCW Code, taking into account the specific hazards of methyl/ethyl alcohol used as fuel.

(MSC.1/Circ.1621 2020)

Methanol and Methyl/ethyl alcohol fuel-related drills and exercises should be incorporated into the company's SMS including schedule for periodical drills.

7.4

Is a suitable firefighting system in place for using methyl/ethyl alcohol as fuel, and were the crew members familiar with the location, purpose, and its operation?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Where fuel tanks were located on open deck, there should be a fixed fire-fighting system of alcohol-resistant foam type, as set out in chapter 17 of the IBC Code and, where appropriate, chapter 14 of the FSS Code.

The alcohol-resistant foam type fire-fighting system should cover the area below the fuel tank where a spill of fuel could be expected to spread.

The bunker station should have a fixed fire-extinguishing system of alcohol resistant foam type and a portable dry chemical powder extinguisher or an equivalent extinguisher, located near the entrance of the bunkering station.

Machinery space and fuel preparation space where methyl/ethyl alcohol-fueled engines or fuel pumps are arranged should be protected by an approved fixed fire-extinguishing system in accordance with SOLAS regulation II-2/10 and the FSS Code. In addition, the fire-extinguishing medium used should be suitable for the extinguishing of methyl/ethyl alcohol fires.

An approved alcohol-resistant foam system covering the tank top and bilge area under the floor plates should be arranged for machinery space category A and fuel preparation space containing methyl/ethyl alcohol.

(MSC.1/Circ.1621 2020)

The ship's fuel handling manual should incorporate the fire-fighting and emergency procedures: operation and maintenance of fire-fighting systems and use of extinguishing agents.

Flammable vapors burn over a methanol pool, and the liquid evaporates due to the heat, contributing to the burn. Therefore, the most effective ways of fighting a methanol fire are to smother the vapors or to dilute the flammable substances below their lower flammable limit.

Portable dry chemical or CO2 extinguishers can be used for small methanol fires where there is less risk of methanol pool evaporation. For larger volumes of methanol, water extinguishers may be used, if the volume of water is at least four times the size of the methanol pool. Alcohol Resistant Film Forming Foam (AR-FFF) extinguishers with foam water proportioning equipment are a highly recommended method for large methanol pool fires, such as a potential fire below methanol fuel tanks.

(Sustainability whitepaper Methanol as marine fuel 2021)

7.5 Is the ventilation system in the fuel preparation spaces and bunkering station, when not located on the open deck, adequate, and were the Master, officers, and crew familiar with the ventilation system's location, purpose, and the procedures for stopping the power ventilation systems from an external location?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Fuel preparation spaces should be provided with an effective mechanical forced ventilation system of extraction type. During normal operation the ventilation should be at least 30 air changes per hour.

The number and power of the ventilation fans should be such that the capacity is not reduced by more than 50% if a fan with a separate circuit from the main switchboard or emergency switchboard or a group of fans with common circuit from the main switchboard or emergency switchboard is inoperable.

Ventilation systems for fuel preparation spaces and other fuel handling spaces should be in operation when pumps or other fuel treatment equipment are working.

Bunkering stations that are not located on open deck should be suitably ventilated to ensure that any vapor being released during bunkering operations will be removed outside. If the natural ventilation is not sufficient, the bunkering stations should be subject to special consideration with respect to provisions for mechanical ventilation. The Administration may require special risk assessment.

Any loss of the required ventilating capacity should give an audible and visual alarm on the navigation bridge, and in a continuously manned central control station or safety center as well as locally.

(MSC.1/Circ.1621 2020)

7.6 Were the fuel tank level indicators and their overflow control system in good working order, and was there recorded evidence of regular testing?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Each fuel tank should be fitted with closed level gauging devices, arranged to ensure a level reading is always obtainable and unless any necessary maintenance can be carried out while the fuel tank is in service, two devices should be installed.

Each fuel tank should be fitted with a visual and audible high-level alarm. This should be able to be function tested from the outside of the tank and can be common with the level gauging system (configured as an alarm on the gauging transmitter), but should be independent of the high-high-level alarm.

An additional sensor (high-high-level) operating independently of the high liquid level alarm should automatically actuate a shut-off valve to avoid excessive liquid pressure in the bunkering line and prevent the tank from becoming liquid full.

The high and high-high-level alarm for the fuel tanks should be visual and audible at the location at which gas freeing by water filling of the fuel tanks is controlled, given that water filling is the preferred method for gas freeing.

7.7 Was the remote control and monitoring system in the bunkering remote control station in good working condition, and was there documented evidence to confirm that regular testing has been conducted?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Bunkering control should be from a safe remote location. At this safe remote location:

1. tank level should be capable of being monitored;
2. The remote-control valves, which are integral to the bunkering Emergency Shutdown (ESD) system, should be operable from this location; closing of the bunkering shutdown valve should be possible from the control location for bunkering and from another safe location; and
3. overfill alarms and automatic shutdown should also be indicated at this location.

If the ventilation in the ducting enclosure or annular spaces of the double walled bunkering lines stops, an audible and visual alarm should be activated at the bunkering control location.

If fuel leakage is detected in ducting enclosure or the annular spaces of the double walled bunkering lines, an audible and visual alarm and emergency shutdown of the bunkering valve should automatically be activated.

Fire detection in machinery space containing methyl/ethyl alcohol engines and fuel storage hold spaces should give audible and visual alarms on the navigation bridge and in a continuously manned central control station or safety center as well as locally.

(MSC.1/Circ.1621 2020)

7.8 Were safety notices, caution placards and signs prominently displayed and customised for each compartment to underline the safety instructions?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the fuel supply is shut off due to activation of an automatic valve, the fuel supply should not be opened until the reason for the disconnection is ascertained and the necessary precautions taken. A readily visible notice giving instructions to this effect should be placed at the operating station for the shut-off valves in the fuel supply lines.

A caution placard or signboard should be permanently fitted in the machinery space containing methyl/ethyl-fueled engines stating that heavy lifting, implying danger of damage to the fuel pipes, should not be done when the engine(s) is running on methyl/ethyl.

(MSC.1/Circ.1621 2020)

7.9 Were drills and emergency exercises, related to methyl/ethyl alcohol fuels, integrated into the company's defined emergency drills, and were there available records to show that company-defined emergency drills have been completed and documented as required by company procedures?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Drills and emergency exercises on board are to be conducted at regular intervals. Exercises related to low flash point fuels, such as methanol are to at least include following:

1. tabletop exercise.
2. review of fueling procedures based in the fuel handling manual required by 17.2.3.
3. responses to potential contingencies.
4. tests of equipment intended for contingency response; and
5. reviews that assigned seafarers are trained to perform assigned duties during fueling, operation and contingency response.

Methanol related exercises are to be incorporated into periodical drills required by resolution IMO Resolution A.741(18).

(MSC.1/Circ.1621 2020)

7.10 Were the maintenance and repair procedures inclusive of the fuel containment system, adjacent spaces, and electrical equipment installed in explosion hazardous spaces and areas?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Maintenance and repair procedures should include considerations with respect to the fuel containment system and adjacent spaces. Special consideration should be given to the toxicity of fuel.

The procedures and information should include maintenance of electrical equipment that is installed in explosion hazardous spaces and areas. The inspection and maintenance of electrical installations in explosion hazardous spaces should be performed in accordance with recognised standards.

(MSC.1/Circ.1621 2020)

7.11 Were the officers and ratings familiar with the pre-bunkering verification requirements and the bunker operation safety checklist, and were there records available to demonstrate that the pre-bunkering and bunkering checks were conducted?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Prior to conducting bunkering operations, pre-bunkering verification including, but not limited to, the following should be carried out and documented in the bunker safety checklist:

1. all communications methods, including ship shore link (SSL), if fitted;
2. operation of fixed fire detection equipment;
3. operation of portable gas detection equipment;
4. readiness of fixed and portable fire-fighting systems and appliances;
5. operation of remote-controlled valves; and
6. inspection of hoses and couplings.

Documentation of successful verification should be indicated by the mutually agreed and executed bunkering safety checklist signed by both PICs.

(MSC.1/Circ.1621 2020)

RightShip recommends Ship Managers implement standardised bunker checklists. This initiative aims to minimise confusion that may result from following different rules and regulations at different ports and will likely increase personnel consciousness and familiarity with the standardized checklists contents. It is recommended that the ship be equipped with thermal cameras, lightning detectors, and CCTV at the bunkering station for leak and fire prevention. The crew must also be trained in the proper use of such equipment and familiarised during shipboard drills.

The International Association of Ports and Harbours (IAPH) has developed checklists for the use of Methanol as fuel in ship-to-ship and truck-to-ship operations. These checklists are readily available for download from their official website.

Section 7D: Fuel Management (Alternative Fuel- Ammonia)

This section is applicable when any equipment, machinery, or system aboard the ship operates using these alternative fuels.

7.1 Does the Ammonia Fuel Bunker Management Plan include procedures for the safe loading, transfer, and storage of ammonia; verification of equipment compatibility; and are the Master and Officers familiar with safe handling procedures, loading limits, equipment compatibility, emergency drills and communication protocols? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Bunker Management for Ammonia-Fuelled Ships

- > Bunker management plan for ammonia-fuelled vessels should address the unique hazards associated with ammonia, including its toxicity and flammability. The plan should incorporate
- > the following critical elements:
- > Safe Handling Procedures: Procedure for the safe loading, transfer, and storage of ammonia.
- > Loading Limits: Clearly defined loading limits in accordance with the IGF Code.
- > Communication Procedure: Effective communication procedures between ship and shore personnel.
- > Equipment Compatibility: Verification of compatibility between shipboard and shorebased systems.

6.8 Provisions on loading limit for fuel tanks

- > 6.8.1 Storage tanks for liquefied ammonia should not be filled to more than a volume equivalent to 98% full at the reference temperature as defined in 2.2.36 of the IGF Code.
- > 6.8.2 In cases where the tank insulation and tank location make the probability very small for the tank contents to be heated up due to an external fire, special considerations may be made to allow a higher loading limit than calculated using the reference temperature, but never above 95%.
- > (Class NK, 2024)
- > Bunkering procedures, and safety framework are to be developed with reference to the Ammonia Bunkering Management Plan (ABMP). Subsequent reviews and amendments in procedures are to be recorded in the ABMP.
- > (The Class NK 2025 Guidelines for the Safe Operation of Ammonia-Fuelled Vessels, Appendix 3)

Safety Management System Updates for Ammonia Fuel Use

- > The company's procedures should include fuel-specific operating procedures, maintenance requirements, and emergency preparedness measures, including regular drills. These procedures should specifically address residual risks that cannot be eliminated but must be mitigated through appropriate control measures. Formal risk assessments should be conducted to identify and address potential hazards associated with the use of ammonia as fuel. These assessments should consider risks to personnel onboard, the environment, and the structural integrity of the vessel. Particular emphasis should be placed on hazards related to the vessel's physical design, layout, operation, and maintenance, including any reasonably foreseeable failures. Given the toxicity of ammonia and the potential for its release, thorough evaluation is essential to safeguard crew and environmental safety.

Scope of Risk assessment

The assessment should cover all relevant aspects of the vessel, focusing on system interactions and available data. Key areas include:

- > Ammonia storage and vapor/pressure management systems
- > Venting and ventilation arrangements
- > Engine room and machinery spaces
- > Ammonia consumption equipment
- > Ammonia fuel supply and return systems
- > (ABS -Safety Insights For Ammonia as a Marine Fuel March 2025)

Risk assessment

- > 4.2.1 A holistic risk assessment should be conducted to ensure that risks arising from the use of ammonia as fuel affecting persons on board, the environment, the structural strength, or the integrity of the ship and its sub-systems are addressed. Consideration should be given to the hazards associated with physical layout, operation and maintenance, following any reasonably foreseeable failure.
- > 4.2.2 The risk assessment should specifically consider the ammonia system integrity with a focus on its ability to prevent and isolate leakages and also evaluate potential toxicity hazards, ignition mechanisms and consequences of ignition. Special consideration should be given, but not limited to, the following specific ammonia-related hazards and topics:
 - > 1- loss of function;
 - > 2- component damage;
 - > 3- fire;
 - > 4- explosion;
 - > 5- toxicity; and
 - > 6- electric shock
- > 4.2.3 Risks, which cannot be eliminated, should be mitigated as necessary. Details of risks, and the means by which they are mitigated, should be documented to the satisfaction of the Administration.
- > (MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.2 Is the vessel provided with a ship-specific bilge management system to prevent fuel from ammonia consumers mixing with bilge water, and are the Master and officers familiar with the associated procedures? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection**> Bilge Management for Fuel-Contaminated Areas**

- > Where there is a risk of fuel mixing with bilge water from ammonia fuel consumers, dedicated bilge storage tanks should be provided, separate from those used for general bilge collection.
- > This separation is essential to ensure safe bilge transfer operations, effective maintenance of bilge tanks, and proper onboard treatment of contaminated bilge water.
- > (Class NK 2024)
- > Procedures for bilge handling should be clearly documented and integrated into the vessel's SMS. The Master and officers should be trained in, and familiar with, these procedures. If alternative measures are implemented, they should provide an equivalent level of safety and operational effectiveness.

7.3 Were the Master, engineering officers, and all personnel involved in the Ammonia bunkering operation adequately trained according to the relevant IMO and international and local regulation applicable to the area of operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

19.2.2 The master, officers, ratings and other personnel on ships using ammonia fuel should have received training and be qualified in the use of gaseous fuel in accordance with the STCW Convention and the STCW Code, taking into account the specific hazards of ammonia.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

Considering the current regulatory gap in training requirements for seafarers handling ammonia as a marine fuel, RightShip recommends the adoption of an interim training framework. Vessel managers are encouraged to consider the following guidance when developing training programs for Masters and Officers involved in ammonia bunkering operations, until formal requirements are established by the IMO.

Safe Bunkering and Handling of Ammonia as Marine Fuel

- > Understand the properties and hazards of ammonia as a marine fuel.
- > Operate and maintain ammonia-related systems safely and effectively.
- > Respond appropriately to emergencies involving ammonia.
- > Comply with relevant safety, environmental, and operational procedure.

Training Content:

1. Introduction to Ammonia as a Marine Fuel

- > Role in maritime decarbonization
- > Benefits and challenges
- > Overview of current regulatory and gaps

2. Chemical and Physical Properties of Ammonia

- > Toxicity, flammability, explosivity
- > Corrosiveness and material compatibility
- > Behavior under varying temperature and pressure

3. Ammonia Fuel Systems and Equipment

- > Storage systems (pressurized, refrigerated, hybrid)
- > Transfer and bunkering systems
- > Instrumentation, control, and monitoring systems
- > Detection and alarm systems

4. Safety and Hazard Management

- > Hazard identification and risk assessment
- > Control measures and safety barriers
- > Occupational health and safety protocols
- > Personal Protective Equipment (PPE): selection, use, and maintenance

5. Operational Procedures

- > Pre-bunkering planning and checklists
- > Bunkering execution and monitoring
- > Fuel changeover procedures
- > Maintenance regimes for ammonia systems
- > Automation and remote operation considerations

6. Emergency Response

- > Leak, spill, fire, and explosion scenarios
- > Emergency shutdown systems
- > Use of water curtains and containment measures
- > Crew coordination and communication protocols

7. Environmental Protection

- > Spill prevention and containment
- > Pollution control technologies
- > Environmental compliance measures

8. Legal and Regulatory Awareness

- > Overview of applicable international and national regulations
- > Classification society guidelines
- > Documentation and reporting requirements

9. Practical Training and Simulation

- > Simulated bunkering and emergency scenarios
- > Use of PPE and emergency response tools

7.4 Does the vessel have an appropriate emergency response plan, are crew members familiar with their emergency duties, and are emergency drills conducted as required? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

20.2 Functional requirements

20.2.1 This chapter relates to functional requirements in 3.2.1, 3.2.12 and 3.2.16.

In particular the following apply:

1. for the protection of crew members who are engaged in operations, maintenance of ammonia fuel systems, and emergency response, the ship should have on board protective equipment suitable for ammonia exposure, taking the exposure risk of different operations into account;
2. for the protection and treatment of crew members affected by ammonia leakages, the ship should have on board suitable emergency equipment; and
3. suitable respiratory and eye protection for emergency escape purposes should be provided for every person on board.

7.5 Are the crew aware of the locations and operation of the decontamination showers and eyewash, and are the showers in good operational order in suitably marked locations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The showers and eyewashes should be operable in all ambient conditions. A heating system with temperature control is required if pipe routing of the water supply exposes the piping to freezing conditions. Water supply capacity should be sufficient for simultaneous use of at least two units. Thermal insulation is not considered as an alternative to a system with temperature control.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.6 Are officers and ratings aware of the locations of emergency stops and are these in good working order with recorded evidence of regular testing? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Compressors, pumps and fuel supply should be arranged for manual remote emergency stop from the following locations as applicable:

- > navigation bridge;
- > cargo control room;
- > onboard safety center;
- > engine control room;
- > fire-control station; and
- > adjacent to the exit of fuel preparation rooms.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.7

Is the vessel equipped with a Safe Haven that has sufficient capacity to accommodate all personnel, located and designed in accordance with safety requirements to prevent ammonia exposure, and are regular drills conducted to ensure crew readiness in the event of an ammonia release? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A Safe Haven should be provided on board to offer refuge in the event of an ammonia release. The following requirements apply:

- > The Safe Haven should consist of one or more enclosed spaces with a combined capacity sufficient to accommodate all persons on board.
- > It should be located at essential operational areas of the vessel and designed to minimize the risk of ammonia exposure. This can be achieved through:
 - > Dedicated ventilation systems, or
 - > Self-sustaining air supply systems.
- > Enclosed toxic areas should not be adjacent to accommodation spaces, service areas, electrical equipment rooms, or control stations. "Adjacent" includes facial, linear, or point contact.
- > Air intakes for the Safe Haven should be equipped with:
 - > Ammonia detectors, and
 - > Internal closing devices to seal the space when ammonia is detected.
- > If closing the air intakes poses a risk of suffocation, additional measures should be in place to:
 - > Prevent ammonia inflow, or
 - > Provide oxygen or breathable air to occupants.
- > Regular drills should be conducted to ensure crew members are familiar with Safe Haven procedures and can respond effectively in an emergency.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.8

Are the Master and officers, familiar with the procedure for safe entry into the fuel preparation room, including the maintenance and regular testing of the ammonia release mitigation system? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Safe Entry Procedure for Ammonia Fuel Preparation Room

Safe entry into an ammonia fuel preparation room requires a structured, multi-step procedure that prioritises ventilation, PPE, and emergency preparedness:

- > The room should be thoroughly ventilated prior to entry to eliminate any potential buildup of ammonia gas.
- > Personnel should wear appropriate PPE, including self-contained breathing apparatus (SCBA) and gas-tight protective suits.
- > A clearly defined emergency response plan should be in place, including designated escape routes and readily accessible safety equipment.

All personnel involved should be familiar with and trained in the emergency procedures.

Regular reviews and drills should be conducted to ensure ongoing compliance and readiness.

(China Classification Society -Key Points for Practical Application of Ammonia-fuelled Vessels 2025)

Provisions for fuel preparation rooms

5.7.1.2 When fuel preparation rooms cannot be located on open deck, or accessed from open deck, access should be provided through an airlock in compliance with 5.11.

5.7.1.3 Fuel preparation rooms should be designed to safely contain fuel leakages. The fuel preparation room boundaries should be gastight towards other spaces in the ship.

5.7.1.6 The fuel preparation room should be fitted with ventilation arrangements ensuring that the space can withstand any pressure build-up caused by vaporization of the liquefied fuel.

5.7.1.8 Fuel preparation room entrances should be arranged with water screens having constantly available water supply. The water screen should be possible to activate from a safe location outside the toxic fuel preparation room if an ammonia leak occurs. The water screens should be arranged on the outside of the fuel preparation room. The arrangement should include the means to safely manage any ammonia effluent produced in their operation.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.9 Are muster stations and access points for life jackets and immersion suits located outside all designated hazardous and toxic areas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

On ammonia-fuelled vessels, RightShip defines hazardous areas as locations where there is a credible risk of ammonia leaks or spills. These typically include fuel storage compartments, fuel preparation rooms, and areas surrounding fuel lines, valves, and associated equipment. Given the toxic and corrosive nature of ammonia, these zones must be clearly identified and equipped with appropriate safety controls to prevent leaks and minimise the risk of personnel exposure.

Toxic area means an area in which ammonia is or may be expected to be present.

Toxic space means an enclosed or semi-enclosed space in which ammonia is or may be expected to be present. A gas-safe machinery space is not considered to be a toxic space.

Mustering stations and life-saving equipment, and access to such stations and equipment, should not be located in toxic areas as specified in Gas Dispersion Plan.

Toxic areas and spaces are defined to allow for a safe arrangement preventing cross-contamination from ammonia releases, and to facilitate safe arrangement of life-saving appliances, emergency escapes, air intakes, outlets and other openings into the accommodation, service and machinery spaces, control stations and other non-toxic spaces.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.10 Are all access points to the machinery space and fuel preparation room adequately protected from exposure to hazardous and toxic areas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Engine room openings are not to open into other enclosed spaces. When applying this requirement, at least the following (1) and (2) are to be satisfied.

- (1) Elevator trunks connected to accommodation spaces are not to be installed.
- (2) Engine room escape trunks are provided with double doors.

(Guidelines for Ships using Ammonia as Fuel, 2024).

When fuel preparation rooms cannot be located on open deck, or accessed from open deck, access should be provided through an airlock.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

Airlock Room Requirements and Maintenance:

- > The airlock room must be a gas-tight space constructed of steel, equipped with two doors spaced a minimum of 1.5 meters and a maximum of 2.5 meters apart.
- > Airlock doors must be self-closing and fitted with mechanisms that prevent them from being held open.
- > The airlock space must be equipped with sensors to monitor the ingress of cargo vapours.
- > Mechanical ventilation is required, and the space must be maintained at a pressure higher than that of the surrounding areas.
- > Both audio and visual alarms must be installed to activate if both doors are opened simultaneously.
- > Testing and Inspection:
- > Regular testing of the airlock system is essential and should include:
- > Visual inspection of the doors and seals
- > Alarm functionality test
- > Ultrasonic leak detection test

7.11 Are drip trays installed in areas where ammonia leakage may occur to mitigate damage to the ship's structure and limit the impact of potential spills? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

5.9.1 Drip trays should be fitted where leakage may occur which can cause damage to the ship structure or where limitation of the area which is affected from a spill is necessary.
 5.9.2 Drip trays should be made of suitable material.
 5.9.3 The drip tray should be thermally insulated from the ship's structure so that the surrounding hull or deck structures are not exposed to unacceptable cooling, in case of leakage of liquid fuel.
 5.9.4 Each tray should be fitted with a drain valve to enable water to be drained over the ship's side where the tray is installed in a location where water may be retained.
 5.9.5 Each tray should have a sufficient capacity to ensure that the assumed maximum amount of spill according to the risk assessment can be handled.
 5.9.6 Drip trays should be provided with means to safely drain or transfer spills that contain ammonia to be contained or treated.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.12 Are the ventilation systems for air locks in good operational order, and are the audible/visual alarms function tested regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Airlocks should be mechanically ventilated at an overpressure relative to the adjacent hazardous/toxic area or space.
 (5.11.2, MSC.1-Circ.1687).

An audible and visual alarm system to give a warning on both sides of the airlock should be provided to indicate if more than one door is moved from the closed position. Audible and visual alarms should be given at a manned location to indicate both loss of pressure and opening of the airlock doors when pressure is lost.

(5.11.5, & 5.11.6, MSC.1-Circ.1687).

7.13 Are continuously manned spaces located outside the machinery space, and are remote monitoring systems for critical areas and equipment within the machinery space in working condition and maintained regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Manning and Monitoring Requirements for Engine Room Safety

Spaces that are required to be continuously manned by ship personnel must not be located within engine rooms. Specifically, engine control rooms should be situated outside of engine rooms where ammonia-consuming equipment is installed.

Remote monitoring systems, such as CCTV, must be provided to observe conditions within the engine room. These systems should be configured to monitor, at a minimum, the following equipment and areas:

- > Ammonia consumers
- > Purifier rooms
- > Boilers
- > Inert Gas Generators (IGG)
- > Incinerators
- > Other high fire-risk equipment, including associated piping routes

(Class NK, 2024)

7.14 Are the air intakes of air-conditioning systems serving accommodation spaces, service areas, and control stations equipped with internal closure mechanisms and gas detection systems, and are these systems regularly maintained and the gas detectors routinely calibrated? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Air intakes and exhaust vents in compartments where people are normally expected to be present, such as accommodation spaces, service spaces, electrical equipment rooms or control stations, shall be provided with devices to close the openings from inside the compartment.

Gas detectors are to be fitted at the air intakes of air-conditioning systems for accommodation spaces, service spaces, control stations, etc. These air intakes are to be automatically closed if ammonia in concentrations exceeding 25 ppm are detected. If all accommodation spaces are designed to be the safe havens specified in 5.11 (Protection from Toxic Areas), air conditioning should be designed to transition to air recirculation when all air intakes are closed.

(Class NK, 2024)

7.15 Are low-temperature fuel pipes thermally insulated, and is the insulation maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Low-temperature piping should be thermally isolated from the adjacent hull structure, where necessary, to prevent the temperature of the hull from falling below the design temperature of the hull material

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

Pipes which may contain low temperature fuel are to be thermally isolated from the adjacent hull structure to prevent the temperature of the hull from falling below the design temperature of the hull material.

Pipes, which may contain low temperature fuel, are to be thermally insulated to an extent which will minimize condensation of moisture or frosting.

7.16 Are the bunker manifold and its associated instrumentation, including the Emergency Shutdown (ESD) system and Emergency Release Couplers (ERC) /Emergency Release System (ERS), maintained in good condition? Are officers familiar with the operation of the ESD, ERC, and ERS systems, and are these systems regularly tested to ensure operational readiness, and are low-temperature fuel pipes thermally insulated, and is the insulation maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

8.4.1 The bunkering manifold should be designed to withstand the external loads during bunkering. The connections at the bunkering station should be arranged in order to achieve a dry-disconnect operation in one of the followings ways:

1. a dry-disconnect/connect coupling;
2. a manual connect coupler or hydraulic connect coupler, used to connect the bunker system to the receiving vessel bunkering manifold presentation flange; or
3. a bolted flange to flange assembly.

8.4.3 An emergency release coupler (ERC)/emergency release system (ERS) or equivalent means should be provided, unless installed on the bunkering supply side of the bunkering line. It should enable a quick physical disconnection "dry break-away" of the bunker system in an emergency event.

8.5.1 An arrangement for purging fuel bunkering lines with inert gas should be provided.

8.5.2 The bunkering system should be so arranged that no gas is discharged to the atmosphere during filling of storage tanks. Vapour return line, where fitted, should be sized adequately taking into consideration the expansion ratio of the fuel during bunkering operations.

8.5.3 A manually operated stop valve and a remote operated shutdown valve in series, or a combined manually operated and remote valve should be fitted in every bunkering line close to the connecting point. It should be possible to operate the remote valve in the control location for bunkering operations and/or from another safe location.

8.5.4 A bunkering-safety link (BSL), or an equivalent means for automatic and manual ESD communication to the bunkering source should be fitted.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.17 Is the Ammonia Release Mitigation System (ARMS) installed and maintained in good working condition, and are the Master and responsible officers familiar with its safe operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ammonia Release Mitigation System (ARMS)

The Ammonia Release Mitigation System (ARMS) is designed to safely manage ammonia by either consuming, collecting, or dispersing it through one of the following methods:

- > Thermal oxidation or decomposition
- > Catalytic oxidation or decomposition
- > Dissolution in water
- > Dilution with air

Under normal operating conditions, direct release of ammonia into the atmosphere is not allowed.

In the event that ammonia must be released and cannot be fully contained, ARMS must reduce the ammonia concentration to no more than 110 ppm to ensure safety.

ARMS is also an effective system for preventing ammonia leaks during purging operations prior to bunkering.

(Class NK 2024)

7.18 Is the fuel supply system designed with full redundancy and segregation from the fuel tanks to the fuel consumers and are the Master and Officers familiar with the actions to be taken in the event of a failure in any part of the system? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

9.3.1 For single fuel installations, the fuel supply system should be arranged with full redundancy and segregation all the way from the fuel tanks to the fuel consumer, so that a leakage in one system does not lead to an unacceptable loss of power.

9.3.2 For single fuel installations, the fuel storage should be divided between two or more tanks. The tanks should be located in separate compartments.

9.3.3 For type C tank only, one tank may be accepted if two completely separate tank connection spaces are installed for the one tank.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel)

7.19 Are the manually and automatically operated valves on the main fuel supply and return lines in good working condition, regularly maintained, and tested in accordance with the manufacturer's recommendations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Fuel Supply and Return Line Valves

Each main fuel supply and return line serving fuel consumers must be equipped with:

- > A manually operated stop valve, and
- > An automatically operated master fuel valve (or a combined manual/automatic valve).
- > These valves must be:
- > Installed in series on the fuel line,
- > Located outside the machinery space where the fuel consumers are housed,
- > Positioned as close as possible to the fuel heating installation, if one is fitted.
- > The master fuel valve must be connected to the safety system and should automatically shut off the fuel supply when triggered.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.20 Are fuel tanks equipped with a high liquid level alarm, an independent sensor that automatically activates a shutoff valve, and a gas detection system, all maintained in good condition, tested regularly, and supported by objective evidence of testing? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Each fuel tank should be fitted with a high liquid level alarm operating independently of other liquid level indicators and giving an audible and visual warning when activated. an additional sensor operating independently of the high liquid level alarm should automatically actuate a shutoff valve in a manner that will both avoid excessive liquid pressure in the bunkering line and prevent the fuel tank from becoming liquid full.

Where gas detection should cause shutdown in accordance with table 1, detector voting should be applied where two units should detect gas to activate shutdown. A failed detector should be considered as an active detection. The number of detectors in each space should be considered taking into account the size, layout and ventilation of the space.

If the fuel supply is shut off due to activation of an automatic valve, the fuel supply should not be opened until the reason for the disconnection is ascertained and the necessary precautions taken. A readily visible notice giving instruction to this effect should be placed at the operating station for the shutoff valves in the fuel supply lines.

A caution placard or signboard should be permanently fitted in the machinery space containing gasfuelled engines, stating that heavy lifting, implying danger of damage to the fuel pipes, should not be done unless the fuel supply lines are free from ammonia.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

7.21 Is the emergency and safety equipment provided suitable for use with ammonia fuel and maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

20.4.3 A stretcher that is suitable for hoisting an injured person from spaces, such as tank hold spaces, should be kept in a readily accessible location.

20.4.4 The ship should have onboard medical first aid equipment, including oxygen resuscitation equipment, based on the requirements of the Medical First Aid Guide (MFAG) for ammonia.

20.4.5 Suitable respiratory and eye protection for emergency escape purposes should be provided for every person on board, subject to the following:

1. filter-type respiratory protection is unacceptable;
2. self-contained breathing apparatus should have at least 15 minutes of service time; and
3. emergency escape respiratory protection should not be used for fire-fighting or cargo handling purposes and should be marked to that effect.

20.5.1 Sufficient, but not less than three complete sets of safety equipment, should be provided in addition to fire-fighter's outfits required by SOLAS regulation II-2/10.10. These additional sets should provide adequate personal protection to permit entry and work in a gas-filled space, and be equipped with two-way portable radiotelephone apparatus comprising of ear piece with microphone and push-talk units. This equipment should consider the nature of ammonia.

20.5.2 Each complete set of safety equipment should consist of:

1. one self-contained positive pressure air breathing apparatus incorporating full face mask not using stored oxygen and having a capacity of at least 1,200 litres of free air. Each set should be compatible with that required by SOLAS regulation II-2/10.10;
2. gastight protective clothing, boots and gloves to a recognized standard;
3. steel-cored rescue line with belt; and
4. explosion-proof lamp.

20.5.3 An adequate supply of compressed air should be provided and should consist of:

1. at least one fully charged spare air bottle for each breathing apparatus required by 20.5.1;
2. an air compressor of adequate capacity capable of continuous operation, suitable for the supply of high-pressure air of breathable quality; and
3. a charging manifold capable of dealing with sufficient spare breathing apparatus air bottles for the breathing apparatus required by 20.5.1.

20.5.4 The compressed air equipment should be inspected at least once a month by a responsible officer and the inspection should be logged in the ship's records. This equipment should also be inspected and tested by a competent person at least once a year.

(MSC.1-Circ.1687, Interim Guidelines For the safety of Ships Using Ammonia as Fuel 2025)

Section 8A: Cargo Operation- Solid Bulk Cargo other than Grain

8.1 This question has been removed from the current version of the document.

8.2 Has appropriate information about the cargo and its characteristics been provided to the Master prior to loading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The IMSBC Code categorises cargoes into three groups:

- > Group A –cargoes that may liquefy if shipped at a moisture content exceeding their transportable moisture limit (TML). TML is the maximum moisture content considered safe for carriage. Liquefaction means that the cargo becomes fluid or liquefies. On ships, this happens when the cargo is compacted by the ship's motion. Cargoes that are prone to liquefaction contain a certain quantity of moisture and small particles, although they may look relatively dry and granular when loaded. Liquefaction can lead to cargo shifting and even to the capsizing of the ship.
- > Group B—cargoes that involve a chemical hazard that could give rise to a dangerous situation on a ship
- > Group C—cargoes that are neither liable to liquefy (Group A), nor involve chemical hazards (Group B), but might still be hazardous.
- > Cargoes can be in Group A, B or C, or Group A and B.
- > Group B cargoes are those that meet either the IMDG Code's dangerous goods hazard criteria or the IMSBC Code's 'materials hazardous only in bulk' (MHB) criteria. MHB cargoes are materials that involve chemical hazards when transported in bulk, but that do not meet the criteria for inclusion in the IMDG classes above. However, they present significant risks to health and safety when carried in bulk and require special precautions.
- > The shipper must provide the Master with valid, up-to-date information about the cargo's physical and chemical properties. The exact information and documentation they must provide is listed in the IMSBC Code under 'Assessment of acceptability of consignments for safe shipment; Provision of Information', and includes the correct Bulk Cargo Shipping Name (cargo's official name used in the Code) and a declaration that the cargo information is correct.
- > To carry dangerous goods in solid form in bulk, the vessel must have a Document of Compliance for the Carriage of Dangerous Goods, supplied by the ship's flag or classification society. The Master must have a special list, manifest or stowage plan identifying the cargo's location, and there must be instructions on board for emergency response.

(IMSBC code, 2020)

8.3 Has the Master been provided with a signed certificate or declaration, indicating the moisture content, Transportable Moisture Limit (TML), angle of repose and density, and was the cargo free of evident damage? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the cargo was damaged for reasons other than the hatch cover's weathertight integrity.

Before loading is started, copies of all appropriate certificates for the cargo shall be provided, for example moisture content, transportable moisture limit (TML) and density.

The Master is to ensure the documents are in the correct format, can be clearly understood and gives all appropriate information for the safe loading and carriage of the intended cargo.

Certificates of transportable moisture limit must also be issued, with the interval between sample or testing and loading not exceeding six months. However, if it is suspected that the moisture content may have increased since the time of testing, or that the flow moisture properties of the cargo may have changed –possibly resulting from heavy rainfall or inefficient stockpiling – additional testing should be carried out to confirm the safety and suitability of the cargo to be loaded.

SOLAS requires that the shippers of bulk cargoes provide the Master in writing and sufficiently in advance of loading with information on any special properties of the cargo, including the likelihood of shifting, and, for concentrates* or other cargoes which may liquefy, additional information in the form of a certificate on the moisture content of the cargo and its Transportable Moisture Limit (TML). Cargoes which may liquefy shall only be accepted when the actual moisture content is less than the TML.

*Concentrates are materials obtained from a natural ore by a process of enrichment or beneficiation by physical or chemical separation and removal of unwanted constituents.

Angle of repose means the maximum slope angle of non-cohesive (i.e. free-flowing) granular material. It is measured as the angle between a horizontal plane and the cone slope of such material.

Cohesive material means materials other than non-cohesive materials.

Non-cohesive material means dry materials that readily shift due to sliding during transport, as listed in Appendix 3 of the Code.

Appendix 3 of the IMSBC Code lists a number of cargoes which are non-cohesive when dry. Each individual schedule of the cargoes listed in this section will state an angle of repose in the physical properties table and the trimming requirements in the loading section. Cargoes not listed in this section, but exhibit properties of non-cohesive material are subject to the same trimming requirements as non-cohesive cargoes.

To allow their safe carriage at sea, non-cohesive cargoes are required to be suitably trimmed in accordance with section 5 of the IMSBC Code.

Cargoes with an angle of repose less than or equal to 30° can free flow like a grain cargo. For this reason, the IMSBC Code requires these cargoes to be carried in accordance with the International Grain Code in addition to the requirements of the IMSBC Code.

The angle of repose stated on the shipper's declaration should be determined using a 'tilting box test'. The details of this testing procedure can be found in Appendix 2 of the IMSBC.

(Articles: Looking straight at the angle of repose, 2021)

8.4 Is information readily available on the ballasting and de-ballasting rate, the maximum allowable load per unit, the surface area of the tank-top plating, and the maximum allowable load per hold? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

This information should be prominently posted or readily available to the user.

8.5 This question has been removed from the current version of the document.

8.6 Is a Class-approved loading computer or programme in use and has the operational accuracy been regularly tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A loading instrument is an instrument, which is either analogue or digital, by means of which it can be easily and quickly ascertained that, at specified read-out points, the still water bending moments, shear forces, and the still water torsional moments and lateral loads, where applicable, in any load or ballast condition will not exceed the specified permissible values. A loading instrument comprises hardware and software.

SOLAS Requirement, regulation 11, Loading Instrument:

Bulk carriers of 150m in length and upwards shall be fitted with a loading instrument capable of providing information on hull girder shear forces and bending moments, taking into account the recommendation adopted by the Organization.

Bulk carriers of 150m in length and upwards, constructed before 1 July 1999, shall comply with the requirements not later than the date of the first intermediate or periodical survey of the ship to be carried out after 1 July 1999.

Bulk carriers of less than 150m in length, constructed on or after 1 July 2006, shall be fitted with a loading instrument capable of providing information on the ship's stability in the intact condition.

The computer software shall be approved for stability calculations by the Administration and shall be provided with standard conditions for testing purposes relating to the approved stability information.

(SOLAS 74, 2020)

It is the responsibility of the ship's Master to check the accuracy of the loading computer system at each annual survey by applying at least one approved test loading condition (other than light ship). If a surveyor is not present for the computer check, a copy of the test conditions results obtained by the computer check is to be retained on board as documentation of satisfactory testing for the surveyor's verification. At each renewal survey this checking for all approved test loading conditions is to be done in the presence of the Society surveyor.

(Computer Software for On-board Stability Calculations-IACS Unified Interpretations, 2017)

Regular on-board testing should also take place and records attesting to this should be maintained.

When a vessel has undergone any weight variations, such as the installation of a scrubber, a ballast water treatment plant, or major structural modifications, it is the responsibility of the vessel's manager to ensure that the loading instrument and stability booklet are updated as necessary.

8.7 Are the stresses, stability information and any limitations included in the cargo plan understood by the cargo watch officers, and are conditions being monitored and maintained within design limits throughout the cargo operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The officer in charge should closely monitor the ship's condition during cargo operations. If a significant deviation from the agreed loading/unloading plan is detected, all cargo and ballast operations must STOP.

The officer in charge should ensure that:

- > The cargo operation and intended ballast/de-ballast procedure are synchronised
- > Draught surveys are conducted at appropriate steps of the loading/discharge plan to verify the ship's condition
- > The draught readings, usually taken at amidships and the fore and aft perpendiculars, should be in good agreement with values calculated in the loading/discharging plan
- > Ballast tanks are sounded to verify their contents and rate of ballasting/de-ballasting
- > The cargo load is in agreement with the figures provided by the terminal
- > The SWSF, SWBM and, where appropriate, hold cargo weight versus draught calculations are performed at intermediate stages of the cargo operation. These results should be logged.
- > Any revised loading/unloading plan should be signed by a terminal representative and by the Master or Chief Officer.

Inspectors should make sure that regular monitoring of stress and stability have been taking place throughout cargo operation and verify if the vessel's conditions have been maintained within design limits.

There are three main problems associated with high loading rates which may result in over-stressing the ship's structure, namely:

- > The SWSF and SWBM may exceed the allowable limit
- > Overloading the local structure.
- > Synchronization of the ballasting operations may not be maintained.

High cargo loading rates may create problems with the ballasting operation as the pumping capacity of the ship may be relatively low compared to the cargo loading rate. In such cases the cargo operation must be stopped to ensure synchronisation with the ballasting operation is maintained.

When necessary, the loading rate must be adjusted to synchronise with the ship's pumping capacity.

(Bulk Cargo Loading and Discharging Guidance, 2012)

8.8 Are there procedures in place for loading, ballasting and de-ballasting of the designated ballast holds? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When cargo is to be carried in the ballast hold:

- > Proper steps should be taken to ensure that ballast water cannot be admitted to the hold by accident
- > Blanks or cover plates which were fitted to the bilge, the CO₂ smothering lines and to the hatch coaming drains must be removed so that these systems can operate whilst cargo is being carried.

Before ballasting of ballast hold:

- > It is vital to remove any blanks or cover plates which have been fitted to ballast suction within the hold
- > The bilge suction should be sealed to prevent ballast from leaking through the bilge system
- > The CO₂ injection and the coaming drains must be sealed
- > Cargo residue and rubbish could block the ballast suction, so they must be removed from the hold.

De-ballasting:

- > Hatch Cover Vents open.

Hold vents must be open when ballasting and/or de-ballasting the designated ballast hold.

8.9 Are there guidelines and procedures for hold cleaning after completion of unloading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record the list of hold cleaning equipment available on board in comments.

Bulk cargoes include a very wide range of commodities. The level of cleanliness required before loading and additional preparation will depend upon the type of cargo to be loaded. The vessel's manager shall provide comprehensive hold cleaning guidelines, procedures, a hold cleaning matrix for change of cargo and a hold cleaning inspection checklist.

Cargo hold cleaning plans shall consist of the following steps, where applicable:

1. Removal of dunnage, lashing material and / or cargo residues
2. Holds swept down
3. Holds swept down a second time (double swept)
4. Cargo residues that have set hard removed
5. Cleaning chemicals applied to hold surfaces and allowed to penetrate/react with stains prior to being washed off
6. Holds washed down with sea water
7. Holds washed down with detergents mixed in fresh water
8. Holds rinsed with fresh water to remove all traces of chlorides and detergents
9. Bilge wells and plates / strainers cleaned
10. Holds air dried
11. Loose paint flakes, loose rust scale and paint blisters removed
12. Paintwork touched-up
13. Barrier coat applied

Some vessels are equipped with fixed cargo hold washing machines, however, these vessels are in a minority and most vessels carrying solid bulk cargoes will need to manually wash the cargo holds. Where fixed washing machines are used, manual cleaning of shadow sectors within the holds may still be required.

(Cargo Hold Cleaning, 2017)

During the operation, the Master or Chief Officer should carry out inspections to ensure the cleaning is being carried out correctly and by use of the correct material and equipment. Inspections should be carried out at least once during each day by the Master or Chief Officer accompanied by the Bosun, to establish how the operation is progressing.

The cargo hold cleaning checklist should be incorporated in the vessel manager's hold cleaning procedure.

When end folding hatch cover panels are partially opened and not secured, they can place massive strains and back pressure on the hatch cover's hydraulic system, leading to failure of one or more hydraulic system components, such as hydraulic pipes, and accidental hatch cover closure. This unintentional closure may raise the risk of injury to those working near the hatch cover.

The hatch covers must be completely opened and secured in line with the manufacturer's recommendations during cargo hold cleaning.

8.10 Is the vessel free of any limitations or restrictions specified in the Loading Manual or Trim and Stability Booklet? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship shall be provided with an approved stability and loading booklet written in a language understood by the ship's officers. (The Code of Practice for the Safe Loading and Unloading of Bulk Carriers, 2011)
Important restrictions should be recorded in the inspector comments section.

Bulk carriers of 150 m in length and upwards of single-side skin construction, carrying cargoes having a density of 1,780 kg/m³ and above, if not meeting the requirements for withstanding flooding of any one cargo hold as specified in regulation 5.1 and the standards and criteria for side structures of bulk carriers of single-side skin construction, adopted by the Organization by resolution MSC.168(79), shall not sail with any hold loaded to less than 10% of the hold's maximum allowable cargo weight when in the full load condition, after reaching 10 years of age. The applicable full load condition for this regulation is a load equal to or greater than 90% of the ship's deadweight at the relevant assigned freeboard.

(SOLAS 74, 2020)

Strength of the inner bottom plating could be deteriorated due to corrosion wastage. The operational parameters and tank top strength(T/m2) of the effected vessels may be updated by the classification society.

8.11 Are officers familiar with the risk, hazard and carriage requirements of solid bulk cargo on board the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The IMSBC Code categorises cargoes into three groups:

- > Group A – cargoes which may liquefy if shipped at a moisture content exceeding their Transportable Moisture Limit (TML).
- > Group B – cargoes which possess a chemical hazard which could give rise to a dangerous situation on a ship.
- > Group C – cargoes which are neither liable to liquefy (Group A) nor possess chemical hazards (Group B). Cargoes in this group can still be hazardous.

Cargo-carrying can involve serious risk, which must be managed carefully to safeguard the crew and the ship. These risks include but are not limited to reduced ship stability and even capsizing due to cargo liquefaction, fire or explosion due to chemical hazards, and damage to ship structures due to poor loading procedures.

Officers shall be able to demonstrate a basic knowledge of the following:

- > Shipboard operations and cargo handling
- > MARPOL ANNEX V The discharge of wash water and any non-recoverable cargo residues
- > The IMSBC Code and BLU Codes
- > Cargo familiarity, i.e., hazard, stowage and segregation, hold cleanliness, weather precautions, ventilation, carriage, discharge and clean up requirements.
- > Sampling and testing of the moisture content for solid bulk cargo, where applicable and, as required:
- > Precautions for cargoes which may liquefy.
- > Precautions for cargoes with chemical hazards.
- > Limitations when loading high density cargoes.
- > Precautions when loading/unloading corrosive cargoes.
- > Hazards associated with solid cargo that give off toxic gas.
- > Handling high density cargoes
- > Can Test

A ship's Master may carry out a check test for approximately determining the possibility of flow on board the ship or at the dockside by the following auxiliary method:

Half fill a cylindrical can or similar container (0.5 to 1 litre capacity) with a sample of the material. Take the can in one hand and bring it down sharply to strike a hard surface, such as a solid table, from a height of about 0.2 m. Repeat the procedure 25 times at one- or two-second intervals. Examine the surface for free moisture or fluid conditions. If free moisture or a fluid condition appears, arrangements should be made to have additional laboratory tests conducted on the material before it is accepted for loading.

If samples remain dry following a can test, the moisture content of the material may still exceed the Transportable Moisture Limit (TML).

(IMSBC code, 2020)

Can test is a simple and useful check available to the ship's crew. When performed correctly, it can help determine if a cargo might be unsafe.

8.12 Have precautionary measures to minimise the risk of potential liquefaction and chemical reaction within the cargo during the voyage been incorporated in the procedures, and are these procedures being followed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

During the voyage, the Master and crew should continue to monitor the state of the cargo as per the vessel's manager's procedures and IMSBC Code, even if they are satisfied about the condition of the cargo they loaded.

Mechanical ventilation is to be provided for cargoes liable to emit flammable gases or vapours in an amount which can form an explosive atmosphere with air.

(IMSBC Code subsections 3.5.1 and 9.3.2.1.3).

For some cargoes like FERROSILICON 14082 or ALUMINIUM SILICON POWDER, UNCOATED 1398, the mechanical ventilation system must have a capacity of at least six air changes per hour based on an empty cargo space for removal of gases and vapours from cargo holds.

(SOLAS Regulation II-2/19.3.4.1 and the IMSBC Code Appendix

1. For the removal of gases and vapours, exhaust ventilation is recommended.
2. For other cargoes, a specific capacity is not clearly defined. In this case, the ventilation should be adequate to avoid the build-up of a flammable atmosphere.³
3. For cargoes with self-heating properties, mechanical ventilation should only be applied in special circumstances. In no case shall the ventilation be directed into the body of the cargo

(IMSBC Code subsection 3.5.6 and Appendix 1).

Continuous ventilation is required for cargoes that fall under IMDG Class 4.3 and are substances which, in contact with water, emit flammable gases, such as hydrogen gas, falling within the UN N.5 test as Dangerous Goods. In addition to Class 4.3 cargoes, there are cargoes assigned MHB (WF) such as 'FERROPHOSPHORUS (including briquettes)' and 'FERROSILICON with at least 25% but less than 30% silicon, or 90% or more silicon' that also require continuous ventilation. IMSBC references to continuous ventilation requirements can be found within IMSBC Code subsections 3.5.3 and 3.5.4.

(CARGO AND CARGO HOLD VENTILATION, 2020)

- 8.13** If the solid bulk cargo is not listed in the IMSBC Code, has the Master been provided with a certificate from the shipper, endorsed by the competent authority of the port, stating the characteristics of the cargo and the required conditions for carriage and handling? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If a solid cargo which is not listed in appendix 1 of the IMSBC Code is proposed for carriage in bulk, the shipper shall, prior to loading, provide the competent authority of the port of loading with the characteristics and properties of the cargo in accordance with section 4 of the IMSBC Code. Based on the information received, the competent authority shall assess the acceptability of the cargo for safe shipment.

When it is assessed that the solid bulk cargo proposed for carriage may present hazards, such as those defined by group A or B of the IMSBC Code as defined in 1.7 of the code, advice is to be sought from the competent authorities of the port of unloading and of the Flag State. The three competent authorities will set the preliminary suitable conditions for the carriage of this cargo.

When it is assessed that the solid bulk cargo proposed for carriage presents no specific hazards for transportation, the carriage of this cargo shall be authorised. The competent authorities of the port of unloading and of the Flag State shall be advised of that authorisation.

The competent authority of the port of loading shall provide to the Master a certificate stating the characteristics of the cargo and the required conditions for carriage and handling of this shipment. The competent authority of the port of loading shall also submit an application to the Organization, within one year from the issue of the certificate, to incorporate this solid bulk cargo into appendix 1 of the IMSBC Code.

Competent Authority means any national regulatory body or authority designated or otherwise recognized as such for any purpose in connection with the IMSBC Code. The competent authority shall operate independently from the shipper.

(IMSBC code, 2020)

- 8.14** Has as a cargo loading/unloading plan providing a detailed sequence of cargo and ballast transfer been prepared, understood, and signed off by the Master and deck officers? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

General requirements

A cargo loading/unloading plan should be laid out in such a way that for each step of the cargo operation there is a clear indication of:

- > The quantity of cargo and the corresponding hold number(s) to be loaded/unloaded.
- > The amount of water ballast and the corresponding tank/hold number(s) to be discharged/loaded.
- > The ship's draughts and trim at the completion of each step in the cargo operation.
- > The calculated value of the still-water shear forces and bending moments at the completion of each step in the cargo operation.
- > Estimated time for completion of each step in the cargo operation.
- > Assumed rate(s) of loading and unloading equipment.
- > Assumed ballasting rate(s)

The loading/unloading plan should indicate any allowances for cargo stoppage (which may be necessary to allow the ship to de-ballast when the loading rate is high), shifting ship, bunkering, draught checks, and cargo trimming.

(Bulk Cargo Loading and Discharging Guidance, 2012)

Loading plan consideration:

- > The arrangements at the port including the number of loaders and their range of movement, the least depth alongside and the air draft requirements
- > The loading sequence, including the number of pours per hold, where loading should begin and where the final trimming pours should be loaded
- > De-ballasting, including the timing of that operation, to coincide with the loading sequence, and the need for a substantial trim during stripping of the ballast tanks
- > The shear force, bending moments and stability of the ship at all stages of the operation, and
- > Trimming pours and the final draft requirements.

Unloading plan consideration:

- > The port arrangements, including the number of unloaders available and their range of movement, the maximum draft available and the minimum draft available
- > The weight of cargo to be unloaded at the port or ports and its distribution on board
- > Ballasting including the timing of that ballasting operation which should coincide with the unloading sequence and trim of the ship
- > The shear force, bending moments and stability of the ship at all stages of the operation, and
- > Final draft requirements and air draft requirements.

(Bulk Cargoes: A Guide to Good Practice, 2016)

The cargo loading/unloading plan should be completed by the responsible officer prior to arrival at port and the commencement of cargo operations and verified and approved by the Master.

8.15 Is an adequate record of all cargo operation activities maintained during loading and unloading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The followings should be recorded in the port logbook or deck log book:

- > Starting and stopping of work at each hold; times and dates
- > Tonnages loaded per pour into each hold, and a running total loaded; and in the case of unloading, tonnages offloaded per shift from each hold and a running total offloaded.
- > Weather conditions at intervals – for example, 6 hours
- > Use of ship's cranes, if appropriate
- > Movement of shore cranes, loaders or floating crane alongside
- > Movement of barges alongside, and of floating cranes or loaders if ship is at anchor
- > Opening and closing of hatches
- > Periods of precipitation
- > Draft readings
- > Any delays caused on board
- > Any surveyors attending or boarding with reason for attendance
- > Any stevedore's damage to ship's structure and/or fittings, and
- > Cargo temperature – in particular for seed cake and coal.

The values of SF and BM should be calculated at least at the end of each pour during cargo operation.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.16 Have details of cargo care during the voyage been adequately recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Where applicable, the following shall be recorded:

- > Records of ventilation on each day for each hold
- > All temperatures taken and dew points calculated
- > Whether or not ventilation has been carried out
- > Reason for not ventilating
- > Weather and sea conditions
- > Ventilation rule applied (e.g., three- degree rule or dew point rule)

Records relating to monitoring of cargo or hold atmosphere for each day (e.g., for coal or silicomanganese):

- > Result of measuring methane, oxygen, carbon monoxide and pH value; where applicable taken at each hold, with time reading taken
- > Any action necessary because of readings obtained, and the results of action taken
- > Findings during inspections of cargo in each hold, with time inspections carried out
- > Any action necessary because of findings, and the results of action taken

Records of bilge soundings and pumping operations:

- > Bilge sounding record
- > Time and dates of bilge pumping
- > Amount of water pumped out of each bilge well during each pumping
- > Sounding before and after each pumping

Regular hold bilge testing shall be systematically carried out during voyages carrying coal cargo. If the pH monitoring indicates that a corrosion risk exists, bilges shall be frequently pumped out during the voyage in order to avoid the possible accumulation of acids on tank tops and in the bilge system. Record of such monitoring should be available on board.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.17 Are the dangers associated with oxygen depletion of cargo understood by officers and crew, and have reasonable precautions been taken during routine inspections of the cargo, when entering the holds and adjacent spaces? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Some solid bulk cargoes are susceptible to oxidation, which may result in oxygen depletion, emission of toxic gases or fumes and self-heating. Some cargoes are not liable to oxidize but may emit toxic fumes, particularly when wet. There are also cargoes which, when wetted, are corrosive to skin, eyes and mucous membranes or to the ship's structure. When these cargoes are carried particular attention shall be paid to protection of personnel and the need for special precautions to be taken prior to loading and after unloading. Many solid bulk cargoes are liable to cause oxygen depletion in a cargo space or tank.

These include, but are not limited to, most vegetable products and forest products, ferrous metals, metal sulphide concentrates and coal cargoes. Emergency entry into a cargo space shall be undertaken only by trained personnel wearing self-contained breathing apparatus and protective clothing, and always under the supervision of a responsible officer.

(IMSBC code, 2020)

Below are examples of materials that can cause oxygen depletion:

- > Grain, grain products and residues from grain processing (such as bran, crushed grain, crushed malt, or meal), hops, malt husks and spent malt
- > Oilseeds as well as products and residues from oilseeds (such as seed expellers, seed cake, oil cake and meal)
- > Copra
- > Wood in such forms as packaged timber, round wood logs, pulpwood, props (pit props and other prop wood), woodchips, wood shavings, wood pulp pellets and sawdust
- > Jute, hemp, flax, sisal, kapok, cotton and other vegetable fibres, empty bags, cotton waste, animal fibres, animal and vegetable fabric, wool waste, and rags
- > Fishmeal and fish scrap
- > Guano
- > Sulphatic ores and ore concentrate.
- > Charcoal, coal, and coal products
- > Direct reduced iron (DRI)
- > Dry ice
- > Metal wastes and chips, iron swarf, steel and other turnings, borings, drillings, shavings, filings, and cuttings; and scrap metal.

Various forms of Direct Reduced Iron (DRI) are detailed in the International Maritime Solid Bulk Cargoes (IMSBC) Code. With the 2025 amendment to the IMSBC Code, DRI (D) has been newly introduced.

The principal forms of DRI are Type A (hot-moulded briquettes) and Type B (pellets). Fines generated during manufacturing and handling processes were conventionally shipped under DRI (C). However, the IMSBC Code stipulates that the moisture content of this cargo must not exceed 0.3%. Due to handling practices, such as outdoor storage and exposure to rain, the moisture content sometimes exceeds 0.3%. To transport such cargo, DRI (D) (by-product fines with a moisture content of at least 2%) was introduced as the fourth DRI schedule.

In the IMSBC Code, DRI (D) is classified both as Group A, which is liable to liquefy, and Group B, as Materials Hazardous only in Bulk (MHB). The primary hazard related to DRI (D) transport is the generation of flammable hydrogen gas in the upper space of the cargo hold. The code focuses on the measurement and control of hydrogen gas concentration through surface ventilation. For more detailed information, please click [here](#). (Japan P&I Club, 2025).

8.18

If coal is carried, is the ship equipped with adequate instruments for monitoring the temperature of the cargo, the atmosphere of the hold headspace, and the pH value of the cargo bilge, are the instruments maintained and in good working order? Are the atmosphere in the cargo hold, the temperature of the cargo, and the pH value of the cargo bilge sample measured, recorded, and monitored, with actions taken in case of self-heating or excessive gas concentration? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should examine the cargo logbook and verify that the necessary records have been kept. A finding should be recorded where incomplete records exist and if no action was taken where the methane (CH₄) level reached or exceeded 20% of the Lower Explosive Limit (LEL), or if the carbon monoxide (CO) level surpassed 50 PPM.

The ship shall be suitably fitted and carry on-board appropriate instruments for measuring the followings without requiring entry into the cargo space:

- > Concentration of methane in the atmosphere
- > Concentration of oxygen in the atmosphere
- > Concentration of carbon monoxide in the atmosphere
- > Temperature
- > pH value of cargo space bilge samples.

These instruments shall be regularly serviced and calibrated. Ship personnel shall be trained in the use of such instruments. It is recommended that means be provided for measuring the temperature of the cargo in the range 0°C to 100°C to enable the measurement of temperature of the cargo while being loaded and during voyage without requiring entry into the cargo space. (IMSBC code, 2020)

Glass thermometers, either mercury or alcohol filled, are too fragile and thus unsuitable for this purpose. "Pocket thermometers", in which the glass thermometer is held within a metal casing usually for mounting in tanks or pipes, are also unsuitable since the reaction time of the thermometer will be greatly increased due to the metal casing having to equilibrate with the temperature of the coal being measured. Infrared thermometers only measure the surface temperature. Probes can typically measure at depths up to 1m below the surface. (Monitoring of Self-Heating Coal Cargoes Prior to Loading, 2014)

Temperatures measured by lowering thermometers into sounding pipes may be useful in general terms but should not be relied upon to reflect any changes occurring in the bulk of the cargo, as temperature monitoring via sounding pipes will only detect heating coal in the immediate vicinity and will not provide information on the bulk of the cargo. (How to monitor coal cargoes from Indonesia, 2011)

Because coal self-heats as a result of carbon monoxide (CO) emissions, measuring gas concentrations is regarded to be a more reliable technique to check for self-heating than measuring temperature. Before measuring gas concentration, the hold ventilation should be turned off for around 4 hours. If CO levels are greater than 50 ppm (or have been rising steadily for three days), the cargo may be self-heating; in this case, ventilation should be shut off and ventilation openings sealed.

(Guidance on the carriage of coal 2021)

All temperature-measuring equipment should be regularly checked, serviced, and calibrated as recommended by the manufacturer, and the vessel should carry a sufficient quantity of spare parts.

If CH₄ is increasing and above 20% LEL, ventilation needs to be considered a priority to avoid explosion risks, which are more acute than self-heating. It is advisable to seek expert advice if the CH₄ level is increasing and close to or above 20% LEL.

If gas measurements indicate that CO is rising above 50 ppm in unventilated holds, this suggests that the coal has a tendency to self-heat. Expert advice is recommended if the CO level exceeds 50 ppm, as indicated in the IMSBC Code.

In such cases, gas measurements should be taken at least every 12 hours until the situation stabilises. (Coal cargoes: Avoiding explosion and self-heating 2023)

8.19

Is any special emergency equipment required by IMSBC on board (as applicable) and in a state of readiness during the cargo operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Section 8B: Cargo Operation - Bulk Grain

Note: This section can only be completed if the vessel is provided with a document of authorisation for the carriage of grain and a grain loading manual. The vessel must be carrying grain in bulk at the time of the inspection. However, a bulk carrier which, at the time of inspection is not actually carrying grain for a brief period, may be inspected as a bulk carrier (Bulk Grain), provided that an adequate assessment of the procedures on board for the carriage of grain can be made. In such cases, the report must clearly note the circumstances.

A ship without a document of authorisation for the carriage of grain and a grain loading manual shall not load grain until the Master demonstrates to the satisfaction of the Administration, or of the Contracting Government of the port of loading acting on behalf of the Administration, that, in its loaded condition for the intended voyage, the ship complies with the requirements of section A 8.3 and A 9 of The International Code for the Safe Carriage of Grain.

The International Code for the Safe Carriage of Grain applies to ships (regardless of size, including those of less than 500 tons gross tonnage) engaged in the carriage of grain in bulk, to which part C of chapter VI of the 1974 SOLAS Convention, as amended, applies.

The term grain covers wheat, maize (corn), oats, rye, barley, rice, pulses, seeds, and processed forms thereof, whose behaviour is similar to that of grain in its natural state.

Grain cargoes carried in bags are not considered as bulk cargo.

8.1 This question has been removed from the current version of the document.

8.2 Has appropriate information about the cargo and its characteristics been provided to the Master or Master's representative prior to loading? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The IMSBC does not cover carriage of grain in bulk; this is covered by the International Grain Code. Form of cargo information is required to be issued by shippers to the Master or Master's representative prior to loading. The cargo information should contain relevant information on inherent quality, safety risks and precautions. These forms should be the Master's initial point of reference in preparing to load the particular cargo. However, the Master should also refer to other accepted industry guidelines, such as Thomas' Stowage to verify the information stated on the form. (Carriage of Bulk Grain Cargoes, 2015)

General Rule: Grain cargoes shipped with average moisture content in the region of 12 to 14% have a high risk of going mouldy during the voyage. The risk increases significantly if average moisture content exceeds 14%.

Grain cargoes with an average moisture content of 10% or below has a low risk of going mouldy during the voyage.
(Carriage of Grain Cargoes, 2015)

8.3 This question has been removed from the current version of the document.

8.4 Is the approved document of authorisation and grain stability booklet (Grain Loading Manual) provided? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A document of authorisation shall be issued for every ship loaded in accordance with the regulations of the Grain Code either by the Administration or an organization recognized by it or by a Contracting Government on behalf of the Administration. It shall be accepted as evidence that the ship is capable of complying with the requirements of these regulations.

The document shall accompany or be incorporated into the grain loading manual provided to enable the Master to meet the requirements of A7 of the Code. The manual shall meet the requirements of A6.3. of the Code.

The intact stability characteristics of any ship carrying bulk grain shall be shown to meet, throughout the voyage, at least the criteria described in section 7 and Part B of the Grain code.

(International Code for the Safe Carriage of Grain in Bulk, 1991)

8.5 If the document of authorisation was not provided, can the Master demonstrate the compliance of the ship's stability with the Grain Code? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A ship without a document of authorisation shall not load grain until the Master satisfies the Administration, or the Contracting Government of the port of loading on behalf of the Administration, that the ship will comply with the requirements of the International Grain Code in its proposed loaded condition.

- > The total weight of the bulk grain shall not exceed one third of the deadweight of the ship.
- > All filled compartments, trimmed, shall be fitted with centreline divisions extending, for the full length of such compartments, downwards from the underside of the deck or hatch covers to a distance below the deck line of at least one eighth of the maximum breadth of the compartment or 2.4 m, whichever is the greater, except that saucers constructed in accordance with Grain code may be accepted in lieu of a centreline division in and beneath a hatchway except in the case of linseed and other seeds having similar properties;
- > All hatches to filled compartments, trimmed, shall be closed and covers secured in place.
- > All free grain surfaces in partly filled cargo space shall be trimmed level and secured in accordance with grain code.
- > Throughout the voyage the metacentric height after correction for the free surface effects of liquids in tanks shall be 0.3 m or that given by the following formula, whichever is the greater:

$$GMR = L B Vd (0.25 B - 0.645 \times \text{square root of } Vd B) / SF \times \Delta \times 0.0875$$

Where:

L = total combined length of all full compartments (metres)

B = moulded breadth of the vessel (metres)

SF = stowage factor (cubic metres per tonne)

Vd = calculated average void depth calculated in accordance with B 1 (metres-Note: not millimetres)

Δ = displacement (tonnes)

The Master must demonstrate to the satisfaction of the Administration or the Contracting Government of the port of loading on behalf of the Administration that the ship in its proposed loaded condition will comply with the requirements of this section.

(International Code for the Safe Carriage of Grain in Bulk, 1991)

8.6 Is a Class-approved loading computer or programme in use and has its operational accuracy been regularly tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Grain Code specifies how to determine the grain shift moment, and it sets the stability criteria for ships carrying grain in bulk. The loading computer or program shall be able to:

- > Calculate grain shift moments.
- > Calculate the allowable grain heeling moments.
- > Determine whether a loading condition complies with the grain stability criteria.
- > Create the table needed for the loading manual.

A loading instrument is an instrument, which is either analogue or digital, by means of which it can be easily and quickly ascertained that, at specified read-out points, the still water bending moments, shear forces, and the still water torsional moments and lateral loads, where applicable, in any load or ballast condition will not exceed the specified permissible values. A loading instrument comprises hardware and software.

(SOLAS Requirement, regulation 11, Loading Instrument)

Bulk carriers of 150 m in length and upwards shall be fitted with a loading instrument capable of providing information on hull girder shear forces and bending moments, taking into account the recommendations adopted by the Organization.

Bulk carriers of 150 m in length and upwards constructed before 1 July 1999 shall comply with the requirements not later than the date of the first intermediate or periodical survey of the ship to be carried out after 1 July 1999.

Bulk carriers of less than 150 m in length constructed on or after 1 July 2006 shall be fitted with a loading instrument capable of providing information on the ship's stability in the intact condition.

The computer software shall be approved for stability calculations by the Administration and shall be provided with standard conditions for testing purposes relating to the approved stability information.

(SOLAS 74,2020)

It is the responsibility of the ship's Master to check the accuracy of the loading computer system at each annual survey by applying at least one approved test loading condition (other than light ship). If a surveyor is not present for the computer check, a copy of the test conditions results obtained by the computer check is to be retained on board as documentation of satisfactory testing for the surveyor's verification. At each renewal survey this checking for all approved test loading conditions is to be done in the presence of the Society surveyor.

(Computer Software for On-board Stability Calculations-IACS Unified Interpretations, 2017)

Regular on-board testing should also take place and records attesting to this should be maintained.

8.7

Are the stresses, stability information and any limitations included in the cargo plan understood by the cargo watch officers and are conditions being monitored and maintained within design limits throughout cargo operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Inspectors should make sure that regular monitoring of stress and stability have been taking place throughout cargo operations and verify if the vessel's condition has been maintained within design limits.

The officer in charge should closely monitor the ship's condition during cargo operations. If a significant deviation from the agreed loading/unloading plan is detected, all cargo and ballast operations must STOP.

The officer in charge should ensure that:

- > The cargo operations and intended ballast/de-ballast procedure are synchronised.
- > Draught surveys are conducted at appropriate steps of the loading/discharge plan to verify the ship's condition.
- > The draught readings, usually taken at amidships and the fore and aft perpendiculars, should be in good agreement with values calculated in the loading/discharging plan.
- > Ballast tanks are sounded to verify their contents and rate of ballasting/de-ballasting.
- > The cargo load agrees with the figures provided by the terminal.
- > The SWSF, SWBM and, where appropriate, hold cargo weight versus draught calculations are performed at intermediate stages of the cargo operation. These results should be logged.
- > Any revised loading/unloading plan should be signed by a terminal representative and by the Master or Chief Officer.
- > The Master and Chief Officer should be aware of the worst-case damage condition for the existing cargo on board.
- > Cargo trimming is a mandatory requirement for grain cargoes. The loading and unloading plan should indicate any allowance for cargo trimming.

(Guidance and Information on Bulk Cargo Loading and Discharging to Reduce the Likelihood of Over-stressing the Hull Structure, 2018)

8.8

Are there procedures in place for loading, ballasting and de-ballasting of the ballast holds? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When cargo is to be carried in ballast hold:

- > Proper steps should be taken to ensure that ballast water cannot be admitted to the hold by accident.
- > Blanks or cover plates which were fitted to the bilge, to the CO₂ smothering lines and to the hatch coaming drains must be removed so that these systems can operate whilst cargo is being carried.

Before ballasting of ballast hold:

- > It is vital to remove any blanks or cover plates which have been fitted to ballast suction within the hold.
- > The bilge suction should be sealed to prevent ballast from leaking through the bilge system.
- > The CO₂ injection and the coaming drains must be sealed.
- > Cargo residue and rubbish must be removed from the hold, as they could block the ballast suction.

Deballasting:

- > Hatch cover ventilations must be open.

(Bulk Carrier Practice, Isbester, 2013)

Hold vents must be open when ballasting and/or de-ballasting the designated ballast hold.

8.9 Are there guidelines and procedures for hold cleaning in place (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record the list of hold cleaning equipment available on board in comments.

Bulk cargoes include a very wide range of commodities. The level of cleanliness required before loading and additional preparation will depend upon the type of cargo to be loaded. The vessel's manager shall provide comprehensive hold cleaning guidelines, procedures, hold cleaning matrix for change of cargo and hold cleaning inspection checklist.

Cargo hold cleaning plans shall consist of the following steps, where applicable:

1. Removal of dunnage, lashing material and/or cargo residues
2. Holds swept down
3. Holds swept down a second time (double swept)
4. Cargo residues that have set hard removed
5. Cleaning chemicals applied to hold surfaces and allowed to penetrate/react with stains prior to being washed off
6. Holds washed down with sea water
7. Holds washed down with detergents mixed in fresh water
8. Holds rinsed with fresh water to remove all traces of chlorides and detergents
9. Bilge wells and plates/strainers cleaned
10. Holds air dried
11. Loose paint flakes, loose rust scale and paint blisters removed
12. Paintwork touched-up
13. Barrier coat applied

Some vessels are equipped with fixed cargo hold washing machines, however, these vessels are in a minority and most vessels carrying solid bulk cargoes will need to manually wash the cargo holds. Where fixed washing machines are used, manual cleaning of shadow sectors within the holds may still be required. (Cargo Hold Cleaning, 2017)

During the operation, the Master or Chief Officer should undertake inspections to ensure the cleaning is being carried out correctly using of the correct material and equipment. Inspections should be conducted at least once during each day by the Master or Chief Officer accompanied by the Bosun, to establish how the operation is progressing.

The cargo hold cleaning checklist should be incorporated in the operator's hold cleaning procedure.

8.10 Have cargo holds been grain or hospital cleaned, where applicable, and has a ship's hold inspection certificate been issued by a 3rd party prior to loading grain? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Hospital clean is the most stringent cleaning standard, requiring the holds to have 100% intact paint coatings on all surfaces, including the tank top, all ladder rungs and undersides of hatches. The standard of hospital clean is a requirement for certain cargoes, for example kaolin/china clay, mineral sands including zircon, barites, rutile sand, ilmenite, fluorspar, chrome ore, soda ash, rice in bulk, and high grades of wood pulp. Generally, these high standards of cleanliness will only be met by vessels trading exclusively with such cargoes. It will rarely be required in the tramp trades.

Grain clean is the most common requirement. A ship will be required to be grain clean for the majority of bulk and break bulk cargoes, such as all grains, soya meal and soya products, alumina, sulphur, bulk cement, bauxite, concentrates, and bulk fertilisers. Some ports and shippers may allow a different standard of cleanliness.

The industry accepted definition of grain clean is provided by the National Cargo Bureau (NCB).

"Compartments are to be completely clean, dry, odour-free, and gas-free. All loose scale is to be removed." The definition is clear:

1. All past cargo residues and any lashing materials are to be removed from the hold
2. Any loose paint or rust scale must be removed
3. If it is necessary to wash the hold, as it generally will be, the holds must be dried after washing
4. The hold must be well ventilated to ensure that it is odour-free and gas-free

It is important to differentiate such scale from oxidation rust (i.e. light atmospheric rusting). Loose scale will break away when struck with a fist or when light pressure is applied with a knife blade or scraper under the edge of the scale. Oxidation rust will typically form on bare metal surfaces but will not flake off when struck or when light pressure from a knife is applied. Generally, the presence of hard-adhering scale within a hold is acceptable in a grain clean hold. The scale should not fall during the voyage or during normal cargo operations. (Bulk Cargoes Hold Preparation and Cleaning, 2011).

The 3rd party inspection company should be a member of the Grain and Feed Trade Association (GAFTA) or Federation of Oils, Seeds and Fat Associations (FOSFA) analyst and superintendent.

8.11 Is the vessel free of any limitations or restrictions specified in the loading manual or trim and stability booklet? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Important restrictions should be recorded in the inspector's comments section. The ship shall be provided with an approved stability and loading booklet written in a language understood by the ship's officers.
(The Code of Practice for the Safe Loading and Unloading of Bulk Carriers, 2011)

8.12 Are officers familiar with the risk, hazard and carriage requirements of grain cargo on board the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Shift of grain, contamination, ingress of water, moisture migration, and transfer of heat from engine room and / or bunker tank bulkhead and inadequate surface ventilation are the major risks and hazards associated with the handling grain cargo at sea.

Heat can be transmitted from engine room to the aft cargo hold and subsequently affect the cargo loaded against the aft bulkhead. The pattern of damage in cargoes situated close to fuel tanks which have overheated will be obvious as grain will discolour and clump where there has been heat transfer. If possible, stow grain cargoes in holds which will not be affected by heated fuel tanks. Fuel oil temperature should be closely regulated and recorded during the voyage.
(Carriage of Bulk Grain Cargoes, 2015)

As far as possible, the bunkers used during the voyage should be drawn from tanks situated well away from holds containing hygroscopic products. If impracticable, bunker tanks adjoining cargo spaces should be heated only when required, ensuring that the temperature does not rise above normal operational levels. (Cargo Ventilation and Precautions to Minimise Sweat, 2012)

8.13 Have hatch covers been ultrasonically tested for weather tightness before loading? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Before loading it will always be prudent to have the hatches tested for weather tightness by ultrasonic testing.
(Carriage of Grain Cargoes, 2015)

The technique is widely used throughout the industry to test and prove the weather tightness of hatch covers. The advantages of this method include:

1. The test identifies the exact location and extent of leakage
2. It indicates the compression status of the rubber seal; if compression is good, the rubber will be able to compensate for movements at sea and maintain a tight seal
3. The equipment is quick and easy to operate. One person operation is possible
4. The test may be carried out in loaded or empty holds
5. There are no weather/temperature limitations, and the test may be carried out during the day or night, and
6. There is no pollution risk.

The procedure comprises placing a transmitter in the cargo hold, switching it on, and properly closing and securing the hatch covers or access equipment to seaworthy requirements. The ultrasonic waves emitted by the transmitter within the enclosed space will leak through the smallest of apertures. Any leakage of sound may be detected by a receiver or detector between frequencies of 36.7 and 40.7 kHz and converted into aural frequencies or into digitally reproduced information. The location of leaks can be precisely detected from outside the hold by moving a hand-held detector along the periphery and cross seams of the covers. Evaluation of the extent of leakage can be established from reading a digital scale.
(UK P&I Club Carefully to Carry CONSOLIDATED EDITION 2018)

The use of ultrasonic equipment is a modern, viable means of testing for watertight integrity of hatch covers, access hatches, doors, ventilators, etc. It is preferable to use Class approved equipment operated by qualified personnel and to follow approved test procedures.

(Steamshipmutual.com, 2004)

8.14

Has the Master been provided with clear instructions regarding any fumigation, prior to arrival at the load port? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When a ship's cargo is fumigated, the following recommendations should be observed:

1- MSC.1/Circ.1264 – Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds, as amended by MSC.1/Circ.1396 (Amendment to the Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds).

2- MSC.1/Circ.1358 – Recommendations on the Safe Use of Pesticides in Ships.

3- MSC.1/Circ.1361/Rev.1 – Revised Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Transport Units.

Spaces connected to fumigated spaces should be treated as if fumigant gas could penetrate into them from the adjacent or connected cargo space. Apart from the fatalities due to mistaken, unauthorized or accidental entry into cargo spaces under fumigation, a significant proportion of the fatal accidents that have occurred during in-transit cargo fumigation have resulted from fumigant gas leaking into ships' accommodation areas, including cabins, as well as fore-castle head spaces, ballast tanks, other adjacent spaces and on deck. The mishandling of fumigant materials has also caused fires and explosions on ships. Continuous monitoring of the atmosphere of connected and adjacent spaces that are designed for continuous occupation, or are frequently visited working areas, is recommended.

(IMO, MSC.581(110), 2025)

Instructions regarding the fumigation, the type of fumigation, who has requested the fumigation and what company will carry out the operation, should be provided for the Master. The Master should check through the instruction and if everything is in order and it is safe for the operation to be carried out, the fumigation should be allowed to go ahead.

(Bulk Cargoes: A Guide to Good Practice, 2016)

The use of phosphine or any other fumigants is strictly forbidden by organic certification around the world. Currently, there are no organically approved fumigants that can be practically applied to organic bulk grain cargoes. As a result, all organic bulk cargoes must be transported without fumigation.

(Carriage of Organic Bulk Grain Cargoes, 2015)

Methyl bromide is an ozone-depleting chemical which was primarily used as a quarantine pesticide for soil, wood and grain. It is fast acting and fumigation exposure times can be as little as 24 hours, however in-transit fumigation with methyl bromide is prohibited due to safety concerns and the crew must also leave the vessel in the event that a methyl bromide fumigation is undertaken at berth. This can incur additional costs associated with accommodating the crew and lead to concerns for owners and charterers about the safety of their vessel during this time.

In 1992, the Montreal Protocol described the initial strategy to phase out the use of methyl bromide as a pesticide. The strategy was agreed by 160 countries. All developed countries agreed to a complete phase out of the chemical as a pesticide by 2005, while 2015 was the phase out date set for developing countries. Quarantine, pre-shipment, and critical uses of methyl bromide were totally prohibited in the USA, UK and EU by 2010. While the phase out of methyl bromide should be 100% effective in developing countries as of 1 January 2015, quarantine, pre-shipment, and critical uses of methyl bromide may be still permitted in certain circumstances.

(Carriage of Grain Cargoes, 2015)

The crew should remain ashore until fumigation has been completed and a gas freeing certificate has been issued by the fumigator-in charge of the operation or by another authorised person. Methyl bromide is only approved for fumigation in port and should never be used for fumigation continued in transit. Methyl bromide will be introduced into the cargo compartments as a gas and effective fumigation of the cargo is likely to be achieved within 24 hours to 48 hours. If it is proposed that methyl bromide is to be used for fumigation of cargo in transit, the Master should not allow the operation to be carried out.

(Bulk Cargoes: A Guide to Good Practice, 2016)

The task of ensuring cargo hold(s) integrity should not be taken lightly as this operation is key to ensuring crew safety. The company should establish procedures, plans and instructions, including appropriate checklists, for key shipboard operations concerning the safety of the personnel, ship and protection of the environment. The various tasks should be defined and assigned to qualified personnel, and the vessel should have procedures in place in their safety management system on how to handle fumigation jobs onboard.

It is strongly recommended that special attention is given to potential leakages from and/or through:

- > Cable locks
- > Ventilation systems
- > Ballast systems
- > Duct keels
- > Bilges
- > Wiring ducts
- > Dehumidifiers
- > Compartments of the engine room
- > Any other sort of piping arrangements connected to parts of the cargo hold

(Fumigants entering crew's spaces – a word of caution, 2020)

8.15 Has the vessel been provided with procedures and contingencies regarding fumigation of cargo holds and are the Master and Chief Officer familiar with the procedure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Master should familiarize themselves with the recommendations set out in the procedures given in the SMS manual and with the guidance set out in section 3 of MSC.1/Circ.1264. In addition, some individual countries, for example the USA and Canada, have produced their own requirements which should be followed when fumigation is being carried out on board a ship within their territorial waters; the Master should be familiar with these requirements if appropriate.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.16 Is crew familiar with major problems associated with fumigation of cargo in stowage on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

With respect to the use of phosphine, there are three major problems with the fumigation of cargo in stowage on board, these being its toxicity, its potential for fire or explosion and its effectiveness. The fumigation gas is toxic to insects but is also toxic to human and other animals. This being the case, safety of the crew, the operatives carrying out the fumigating operation, and other personnel on board must be ensured by following procedures strictly.

After the pellets or tablets have been distributed within the cargo or cargo compartment, they will react with the atmospheric moisture to produce the fumigant gas. The chemical reaction will also produce heat. If the fumigant is not distributed correctly, for example, is placed in piles rather than being spread around, the heat produced might cause heating of adjacent cargo which may produce combustion and fire. Alternatively, in extreme case, an explosion might be the result of spontaneous heating of the fumigation gas as phosphine gas is explosive at levels above 1.7% v/v in air. If the pellets or tablets become wet, for example by sea water ingress, rain or condensation, they can spontaneously ignite.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.17 Do on-board safety requirements for fumigation comply with sub-section 3.3.2.7 of the IMO recommendation on the safe use of pesticides? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship should carry:

- > Gas-detection equipment and adequate fresh supplies of service items for the fumigant(s) concerned as required by 3.3.2.12, together with instructions for its use and the occupational exposure limit values set by the Flag State regulations for safe working conditions.
- > Instructions on disposal of residual fumigant material.
- > At least four sets of adequate respiratory protective equipment; and
- > A copy of the latest version of the Medical First Aid Guide for Use in Accidents Involving Dangerous Goods (MFAG), including appropriate medicines and medical equipment.

(Recommendations on the safe use of pesticides in ships applicable to the fumigation of cargo holds, 2008)

It is reported that phosphine gas can be detected because it smells of garlic. This is not a reliable method, and the absence of a garlic smell does not mean the absence of phosphine gas. The only reliable method of detecting phosphine gas is by the use of the gas detection equipment which is on board for the purpose.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.18

Are the Master's appointed representatives for fumigation trained and is there evidence to show that they have been effectively performing duties associated with this task? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Before fumigation is conducted, one officer and one crew member, both of whom have been trained (i.e. shown the information on the fumigant Safety Data Sheet and the instructions for fumigant use), should be designated as the "trained representatives of the Master". These representatives must:

- > Brief the crew before a fumigation takes place and satisfy the fumigator-in-charge that this has been done.
- > Inspect and/or test empty cargo holds for leakage with instruments so that proper sealing can be done before or after loading. The fumigator-in-charge, accompanied by a trained representative of the Master or a competent person, should determine whether the cargo holds to be treated are or can be made sufficiently gastight to prevent leakage of the fumigant to the accommodation, engine-rooms and other working spaces in the ship. Special attention should be paid to potential problem areas such as bilge and cargo line systems.
- > Continue monitoring in the accommodation, engine room, etc. Though the initial check may not indicate any leaks, it is essential that monitoring is to be continued because concentrations may reach their highest levels after several days. Continue monitoring the gas levels in accommodation and working spaces after the fumigator has left the ship.

(Recommendations on the safe use of pesticides in ships applicable to the fumigation of cargo holds, 2008)

The Master representatives should be trained with regard to gas concentration checks which must be carried out before departure and during the voyage, where and when those checks must be done and what records must be kept to comply with other parts of the requirements.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.19

Have pre-fumigation and post fumigation statements been provided to the Master by the fumigator-in-charge? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The fumigator-in-charge, together with a trained representative, should carry out inspections and/or tests of cargo compartments to determine whether the holds to be treated can be made sufficiently gas-tight to prevent leakage of the fumigant from the holds into other compartments. Following such inspections, further discussion should be held between the Master and fumigator-in-charge, and the fumigator-in-charge should provide the Master with a signed document stating the following:

- > Details of inspections and tests conducted
- > Details of provisions and preparations for fumigation made
- > Confirmation that holds to be treated are or can be made satisfactorily gas tight for the fumigation.

If any holds cannot be made sufficiently gas-tight, a signed statement to this effect should also be supplied to the Master. The fumigator-in-charge should notify the Master, in writing, which cargo spaces are to be fumigated and which other spaces are considered to be unsafe.

The fumigator-in-charge, together with the trained representative of the Master should make an initial check for any gas leaks, using the gas detection equipment, around the hatches and if any leaks are found they should be sealed using appropriate material.

At an appropriate time after application of the fumigant, the fumigator-in-charge, accompanied by a representative of the Master, should check that accommodation, engine-rooms, and other working spaces remain free of harmful concentrations of gas.

(Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds, 2008)

When fumigators-in-charge are satisfied that the application of fumigation and the sealing of hatches has been completed, they should formally hand over to the Master in writing responsibility for maintaining safe conditions in all occupied spaces. The signed written statement should include following:

- > List of documents provided.
- > Confirmation that all spaces adjacent to treated spaces have been found gas free.
- > Confirmation that trained representative are fully conversant with the use of the gas detection equipment.
- > Confirmation that gas detection equipment and the respiratory equipment is in full working order.
- > Confirmation that adequate supplies of consumables for the equipment are available on board.

(Bulk Cargoes: A Guide to Good Practice, 2016)

The Master and fumigator-in-charge, or their representatives, should complete and sign the model checklist for in-transit fumigation.

(Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds, 2008)

8.20 Are visible means provided to prevent access to all entrances containing fumigant and other spaces that are considered unsafe to enter after fumigation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

On application of the fumigant, the fumigator-in-charge should post warning signs at all entrances to places notified to the Master as in 3.3.2.8 of MSC.1/Circ.1264. These warning signs should indicate the identity of the fumigant and the date and time of fumigation.

(Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds, 2008)

8.21 Has the Voyage Safety Plan (VSP), including the checklist for fumigation during the voyage, been discussed with and signed by the Master prior to sailing to the discharge port? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Voyage Safety Plan (VSP) including checklist confirming the following:

Before fumigant application

- > The inspection required before loading has been performed.
- > All the cargo spaces to be fumigated are satisfactory for fumigation.
- > Spaces, where found not be satisfactory, have been sealed.
- > The Master or his trained representatives have been made aware of the specific areas to be checked for gas concentrations throughout the fumigation period.
- > The Master or his trained representatives have been made familiar with the fumigant label, detection methods, safety procedures and emergency procedures.
- > The fumigator-in-charge has ensured that gas-detection and respiratory protection equipment carried on the ship is in good order, and that adequate fresh supplies of consumable items for this equipment are available to allow sampling.

The fumigator-in-charge has ensured that the necessary medicines and medical equipment, and the latest version of the Medical First Aid Guide for Use in Accident Involving Dangerous Goods (MFAG) are available on board the ship.

The Master has been notified in writing of:

- > The spaces containing cargo to be fumigated.
- > Any other spaces that are considered unsafe to enter during the fumigation.

After fumigant application

Presence of gas has been confirmed inside each hold under fumigation.

- > Each hold has been checked for leakage and sealed properly.
- > Spaces adjacent to the treated cargo spaces have been checked and found gas-free.
- > The responsible crew members have been shown how to take gas readings properly when gas is present and they are fully conversant with the use of gas-detection equipment provided;
- > Methods of application are described.
- > The Master or trained representatives have been briefed fully on the method of application and the spread of the gas throughout the hold.

The Master or trained representatives have been made:

- > Aware that even though the initial check may not indicate any leaks, it is essential that monitoring is to be continued in the accommodation, engine-room, etc. because concentrations may reach their highest levels after several days.
- > Aware of the possibility of the spreading of gas throughout the duct keel and/or ballast tanks.
- > Aware that the Master is responsible for all aspects of the safety of the fumigation once the "fumigator-in-charge" has formally handed over responsibility to them and left the vessel.
- > The fumigator-in-charge has supplied a signed statement to the Master conforming to the provisions of IMO Recommendations.

(Code of Practice on Safety and Efficacy for Marine Fumigation, 2010)

Degassing and checking results at the discharge port – should be done according to the guidance given by the fumigator in charge at the load port and included with the VSP. For example, there should be clear written instructions on how to handle and dispose of any fumigant containers or fumigation residues.

(Carriage of Grain Cargoes, 2015)

8.22 Have the air conditioning intakes for the accommodation, the engine room and other spaces been set to prevent the possibility of drawing in fumigant gas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ventilation procedures for accommodation, the engine room and other spaces should be reviewed to avoid the possibility of drawing fumigant gas into those spaces by incorrect ventilation. Further, it should be verified that ventilation flaps and closing devices are correctly set before the fumigation is carried out and they should be maintained in the correct arrangement throughout the fumigation period. A review of the ventilation regime should be completed before any ventilation of the cargo compartment is started and any necessary changes to those arrangements should be made.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.23 Are procedures in place for entering any cargo holds sealed for fumigation in transit? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Except in extreme emergency, cargo holds sealed for fumigation in transit should never be opened at sea or entered. If entry is imperative, at least two persons should enter, wearing adequate protection equipment and a safety harness and lifeline tended by a person outside the space, similarly equipped with protective, self-contained breathing apparatus.

(Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds, 2008)

8.24 Has the Master informed the appropriate authorities of the country of destination about the fumigation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Prior to the arrival of the ship, generally not less than 24 hours in advance, the Master should inform the appropriate authorities of the country of destination and ports of call that fumigation in transit has been carried out. The information should include the type of fumigant used, the date of fumigation, the cargo holds which have been fumigated, and whether ventilation has commenced.

Upon arrival at the port of discharge, the Master should also provide information about use of the fumigant, e.g., on the fumigant label or package itself, such as the recommendations of the fumigant manufacturer concerning methods of detection of the fumigant in air, its behaviour and hazardous properties, symptoms of poisoning, relevant first aid, special medical treatment, emergency procedures and instructions on disposal of residual fumigant material.

(Recommendations on the Safe Use of Pesticides in Ships Applicable to the Fumigation of Cargo Holds, 2008)

8.25 Did the vessel receive a fumigant gas-free certificate issued by the fumigator-in-charge at or prior to the discharge port or, in the absence of a gas-free certificate did the Master implement appropriate safety measures to manage potential residual fumigant risks before and during discharge operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Before discharge of grain cargoes can be approved, holds that have been subject to fumigation will have to be declared gas free. This is to ensure the holds are free from any gas that may make the holds unsafe.

After the holds have been declared gas free, the surfaces of the cargo may be subject to visual inspection. This can be carried out by the receiver, port officials and/or government inspectors as the hatches are open, prior to the approval of discharge.

(Carriage of Organic Bulk Grain Cargoes, 2015)

Fumigants are acutely toxic substances. Exposure to high concentrations can cause immediate and severe health effects.

Relying solely on periodic atmospheric monitoring may not detect the presence of fumigants in time to prevent fatal exposure.

(ICHCA International, 2025)

Master's Responsibilities

Where informed that a representative of the fumigation company is not available at the discharge port, and a gas-free certificate confirming the safe condition of the cargo holds and surrounding areas has not been issued, the Master should:

- > Immediately notify both the Port Authority and the Ship Manager clearly communicating the absence of gas-free certification and any associated safety concerns, following any instructions provided precisely.
- > Document all communications and actions taken.
- > Conduct a specific risk assessment prior to discharge operations, then based on the assessment determine and provide appropriate PPE which should protect against fumigant gases as well as other potential hazards, such as oxygen depletion.
- > Secure all entrances to unattended or hazardous areas to prevent unauthorized access.

Ensure that ongoing atmospheric testing is conducted at regular and frequent intervals using calibrated and certified equipment specific to the fumigant in use.

- > Empower crew members to stop any unsafe activity or behaviour by visitors or contractors that could lead to an incident.

Ship Manager Responsibilities

Vessel managers should be prepared for situations where a fumigation company's representative or fumigator-in-charge is unavailable at the discharge port. In such cases, they should:

- > Ensure The SMS should provide clear and explicit operational guidelines to ensure the safe conduct of cargo operations.
- > Support the Master in implementing appropriate safety measures

8.26 Are records maintained of fumigation operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Records for fumigation of cargo:

- > Communication relating to fumigation
- > Details of fumigation company, fumigator-in-charge, and operation
- > Fumigation plan
- > In the deck logbook record details of the operation
- > In a workbook record details of the operation including:
 - Details of the fumigator-in-charge and operatives.
 - Time of starting and finishing.
 - Holds involved.
 - Identity of the fumigant, application level and where and how applied.
 - All gas reading with location testing.
- > Plan and sketch for each hold fumigated, showing where the fumigant was applied and the amount of fumigant at each location.
- > Model checklist for in-transit fumigation signed by the Master and the fumigator-in-charge.
- > Certificate of fumigation
- > Formal written handover of responsibility.

Records relating to fumigation for each day:

- > Gas concentration readings obtained and location at which each reading was taken
- > Gas concentration safety checks at all appropriate locations, which should at least include:
 - Accommodation
 - Engine-rooms
 - Areas designated for use in the navigation of the ship
 - Frequently visited working areas and stores, such as the forecastle head spaces adjacent to cargo holds being subject to fumigation in transit should be continued
 - Throughout the voyage at least at eight-hour intervals or more frequently if so advised by the fumigator-in-charge. These readings should be recorded in the ship's logbook.
- > Time readings taken
- > Any action necessary because of high readings
- > Results of action taken
- > Details of ventilation holds after fumigation period

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.27 Are necessary instruments (with spare) to determine the dew point provided, maintained in good condition and are there records of the calibration of such instruments? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Dewpoint temperature may be measured by a variety of methods. Ships generally use a traditional wet and dry bulb arrangement consisting of two identical mercury thermometers, one of which has a damp muslin wick covering the bulb. These are normally housed in a protective marine screen on each bridge wing. The dewpoint temperature may then be determined by a "Dewpoint Table" to compare the wet and dry bulb temperatures. This figure is important when considering cargo ventilation requirements.

When using traditional wet and dry bulb thermometers, the accuracy of the dew point temperature will depend on the condition of the equipment. The muslin covering the wet bulb should be clean, the water in the reservoir should be distilled and the bulb itself should be wet. In order to ensure that the readings are correct, the device should always be positioned away from any exhaust vents, other draughts and all sources of heat. The readings should always be taken on the windward side of the vessel.

(Cargo Ventilation and Precautions to Minimise Sweat, 2012)

8.28 This question has been removed from the current version of the document.

8.29 Is ventilation of cargo holds, where required, being carried out and recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following should be recorded on each day for each hold:

- > All temperatures taken and dew points calculated
- > Whether or not ventilation carried out
- > Reason for not ventilating
- > Weather and sea condition

Ventilation rule applied (e.g., three –degree rule or dew point rule)

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.30 Is there evidence of a satisfactory grain stability calculation for the last voyage? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall verify that the following has been considered during a grain loading calculation:

- > Using the stowage factor provided (and any ship's experience factor is appropriate), and using the full hold volumes with trimmed ends, determine the weight of cargo to be stowed in each full hold, and then determine the weight of cargo to be stowed in each slack hold, if applicable.
- > Determine the ullage or sounding of each slack hold
- > Using the volumetric heeling moment data, determine the volumetric heeling moment for each hold, assuming each full hold has untrimmed ends and using the ullage or sounding for each slack hold
- > Apply the stowage factor to the volumetric heeling moment and obtain the total grain heeling moment
- > Complete stability calculations to determine the ship's fluid GM, the displacement and draft, and then determine from the data the maximum permissible grain heeling moment for the ship's loaded condition
- > Ensure the total grain heeling moment at each stage of the voyage is less than the corresponding maximum permissible grain heeling moment, sometimes referred to as the maximum allowable grain heeling moment

(Bulk Cargoes: A Guide to Good Practice, 2016)

Appropriate grain stability calculations, as required by the International Grain Code, should be carried out prior to loading grain.

8.31 Has a cargo loading/unloading plan providing detailed sequences of cargo and ballast transfer been prepared, understood, and signed off by the Master and deck officers? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

General requirements:

A cargo loading/unloading plan should be laid out in such a way that for each step of the cargo operation there is a clear indication of:

- > The quantity of cargo and the corresponding hold number(s) to be loaded/unloaded.
- > The amount of water ballast and the corresponding tank/hold number(s) to be discharged/loaded.
- > The ship's draughts and trim at the completion of each step in the cargo operation.
- > The calculated value of the still-water shear forces and bending moments at the completion of each step in the cargo operation.
- > Estimated time for completion of each step in the cargo operation.
- > Assumed rate(s) of loading and unloading equipment.
- > Assumed ballasting rate(s)

The loading/unloading plan should indicate any allowances for cargo stoppage (which may be necessary to allow the ship to de-ballast when the loading rate is high), shifting ship, bunkering, draught checks, and cargo trimming. (Bulk Cargo Loading and Discharging Guidance, 2012) (Bulk Cargo Loading and Discharging Guidance, 2012)

Loading plan consideration:

- > The arrangements at the port, including the number of loaders and their range of movement, the least depth alongside and the air draft requirements.
- > The loading sequence, including the number of pours per hold, where loading should begin and where the final trimming pours should be loaded.
- > De-ballasting, including the timing of that operation to coincide with the loading sequence and the need for a substantial trim during stripping of the ballast tanks.
- > The shear force and bending moments and stability of the ship at all stages of the operation.
- > Trimming pours and the final draft requirements.

Unloading plan consideration:

- > The port arrangements, including the number of unloaders available and their range of movement, the maximum draft available and the minimum draft available.
- > The weight of cargo to be unloaded at the port or ports and its distribution on board.
- > Ballasting, including the timing of that ballasting operation, which should coincide with the unloading sequence and trim of the ship.
- > The shear forces, bending moments and stability of the ship at all stages of the operation.
- > Final draft requirements and air draft requirements.

(Bulk Cargoes: A Guide to Good Practice, 2016)

The cargo loading/unloading plan should be completed by the responsible officer prior to arrival at port and commencement of cargo operations and verified and approved by the Master.

8.32 Are the hold bilges cleaned prior to loading and have all hold openings been made grain tight? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Hold bilges must be cleaned thoroughly to remove all residue, rust particles, and cargo stains etc. The filter (Rose Box) must be cleaned thoroughly. The bilge well sections shall be washed with continuous running of sea water to remove all odor and later rinsed with fresh water. The bilge well must be sponged dry to remove all trace of water and dried.

All tank-top and fuel tank sheathing must be grain tight. Where the condition of the sheathing renders this impracticable, the sheathing must be covered with hessian, polyethylene, paper, or other suitable material to prevent the ingress of grain.

Bilge spaces and bilge wells must be covered with hessian or similar porous material after inspection, in such a manner as to prevent the entry of grain into the bilge space or well, but to permit the entry of water.

Tween deck and other scuppers must be covered with hessian or similar porous material in such a manner as to prevent the entry of grain into the scupper opening but to permit the entry of drainage water.

(Bulk Cargoes Hold Preparation and Cleaning, 2011)

8.33 Do records on board verify that cargo lights in holds, where fitted, were properly isolated before cargo was loaded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Many bulk carrier/general cargo holds have fixed cargo lights. These can easily ignite combustible cargoes such as grain, animal feed, wood chips, pulp, and paper if they are too close to the light. Cargo lights in holds need to be properly isolated before cargo is loaded. This is best done by removing fuses or other physical links in the electrical circuits so that the lights cannot be switched on by mistake.

(Fire! A Guide to the causes and prevention of cargo fire, 2017)

There are certain types of cargo hold lights that can be considered safe for continuous use without posing a risk to the cargo. Examples include:

LED Lights

- > Low heat emission
- > Energy-efficient and long-lasting

Fluorescent Lights

- > Generally safe for use around combustible cargo
- > Commonly used in car carriers
- > Low risk of ignition due to lower operating temperatures.

If such lights are installed inside the cargo hold, and there is objective evidence of regular maintenance in accordance with the manufacturer's recommendations, along with procedures in the SMS that address the requirement for isolating these lights based on the type of cargo, and these procedures have been followed, then the inspector should select "Yes" as the answer to this question.

The inspector should also record in the comments section the measures in place to ensure the safety of the cargo within the hold.

8.34 Is an adequate record of all cargo operation activities maintained during loading and unloading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following should be recorded in the port logbook or deck logbook:

- > Starting and stopping of work at each hold, times, and dates
- > Tonnages loaded per pour into each hold, and a running total loaded; and in the case of unloading, tonnages offloaded per shift from each hold and a running total offloaded.
- > Weather conditions at intervals – for example 6 hours.
- > Use of ship's cranes, if appropriate
- > Movement of shore cranes, loaders, or floating crane alongside
- > Movement of barges alongside and of floating cranes or loaders if ship is at anchor
- > Opening and closing of hatches
- > Period of precipitation
- > Draft readings
- > Any delays caused on board
- > Any delays caused ashore
- > Any surveyors attending or boarding with reason for attendance
- > Any stevedore's damage to ship's structure and/or fittings
- > Cargo temperature in particular for grain, seed cake and coal

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.35 Are the dangers associated with oxygen depletion of grain cargo understood by officers and crew, and have reasonable precautions been taken during routine inspections of the cargo, when entering the holds and adjacent spaces? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Most grain cargoes may deplete the oxygen levels in cargo holds and possibly in adjacent spaces. All cargo holds and adjacent spaces must be treated as enclosed spaces and subject to controlled entry. (Carriage of Bulk Grain Cargoes, 2015)

Section 8C: Cargo operation - general cargo

Note: This section should only be completed if the vessel is a general cargo ship, a roll on roll off (Ro-Ro) ship, a timber carrier or a non-cellular ship fitted for the carriage of containers.

Poor planning, improper supervision of stowage and securing of cargo are the common causes of incidents on board multi-purpose ships. The safe stowage and securing of cargoes depend on proper planning, execution and supervision. All cargoes should be stowed and secured in such a way that the ship and persons on board are not put at risk.

Dangerous goods carried as cargo, which are listed or classified in the latest edition of the International Maritime Dangerous Goods (IMDG) Code as amended, are ascribed with the primary hazard characteristics of a class in the IMDG Code or as solid substances in Appendix B of the Code of Safe Practice for Solid Bulk Cargoes (BC) Code (also published in the Supplement to the IMDG Code), which would also be subject to the provisions of the IMDG Code when such goods are carried in packaged form.

Ships of the following descriptions:

1. All other ships of 500 tons or over constructed on or after 1 September 1984; and
2. All other ships of under 500 tons constructed on or after 1 February 1992.

which are intended for, or which have cargo spaces which are intended for, the carriage of dangerous goods on international voyages must carry a document of compliance. The document of compliance will certify that the ship complies with regulation 54 of Chapter II-2 of the International Convention for the Safety of Life at Sea 1974 and be limited to 5 years from the date of issue, in accordance with IMO MSC/Circ.1027.

In the case of ships carrying a document of compliance without an expiry date, vessels' managers are advised to seek renewal no later than the expiry date of the Cargo Ship Safety Construction Certificate, where carried. In other cases, owners are advised to seek a replacement certificate at a convenient survey, e.g. renewal of the passenger ship safety certificate or within five years. (MGN.36 (M), Document of Compliance for Ships Carrying Dangerous Goods in Packaged or Dry Bulk Form, 1997)

This section must not be used for cellular container ships.

8.1 This question has been removed from the current version of the document.

8.2 Has appropriate cargo information been provided to the vessel prior to loading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The shipper shall provide the Master or his representative with appropriate information on the cargo sufficiently in advance of loading, to enable the precautions which may be necessary for proper stowage and safe carriage of the cargo to be put into effect. Such information shall be confirmed in writing and by appropriate shipping documents prior to loading the cargo on the ship.

In the case of general cargo, and of cargo carried in cargo units, a general description of the cargo, the gross mass of the cargo or of the cargo units, and any relevant special properties of the cargo units. For the purpose of this regulation, the cargo information required in sub-chapter 1.9 of the Code of Safe Practice for Cargo Stowage and Securing, adopted by the Organisation by resolution A.714 (17), as may be amended, shall be provided.

Prior to loading cargo units on board ships, the shipper shall ensure that the gross mass of such units is in accordance with the gross mass declared on the shipping documents. (SOLAS 74/2020)

Prior to shipment the shipper should provide all necessary information about the cargo to enable the shipowner or ship operator to ensure that:

- > The different commodities to be carried are compatible with each other or suitably separated
- > The cargo is suitable for the ship
- > The ship is suitable for the cargo, and
- > The cargo can be safely stowed and secured on board the ship and transported under all expected conditions during the intended voyage.

The Master should be provided with adequate information regarding the cargo to be carried so that its stowage may be properly planned for handling and transport.

(CSS code, 2011)

If the vessel is loading heavy lift cargo, the shipper should provide the following information to the Master:

- > A general description of the cargo
- > The gross mass of the item or of each item if there are more than one
- > The principle dimensions of the item or items and, if possible, scale drawings
- > The location of the centre of gravity of each item
- > Particulars of the bedding area of the cargo units and details of any precautions with regard to the bedding of the item(s)
- > Details of lifting points or slinging positions and, if possible, information on how best to lift each item
- > Details of securing points, including their strength and radius of strength.

(Bliault and North of England P & I Association, 2007)

- > The following information as applicable for each parcel of timber cargo should be provided by the shipper and collected by the Master or his representative:
- > Total amount of cargo intended as deck cargo
- > Typical dimensions of the cargo
- > Number of bundles
- > Density of the cargo
- > Stowage factor of the cargo
- > Racking strength for packaged cargo
- > Type of cover of packages and whether non-slip type, and
- > Relevant coefficients of friction, including covers of sawn wooden packages if applicable.

(Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2012)

8.3 If dangerous goods are carried in packaged form, have appropriate documents been provided to the vessels? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The dangerous goods transport document shall include a certification or declaration that the consignment is acceptable for transport and that the goods are properly packaged, marked, and labelled, and in proper condition for transport in accordance with the applicable regulations.

(IMDG Code, 2020)

When dangerous goods are packed or loaded into any container or vehicle, those responsible for packing the container or vehicle shall provide a "container/vehicle packing certificate" specifying the container/vehicle identification number(s) and certifying that the operation has been carried out in accordance with the following conditions:

- > The container/vehicle was clean, dry and apparently fit to receive the goods.
- > Packages which need to be segregated in accordance with applicable segregation requirements have not been packed together onto or in the container/vehicle (unless approved by the competent authority concerned, in accordance with 7.3.4.1);
- > All packages have been externally inspected for damage, and only sound packages have been loaded.
- > Drums have been stowed in an upright position, unless otherwise authorised by the competent authority, and all goods have been properly loaded and, where necessary, adequately braced with securing material to suit the mode(s) of transport for the intended journey.
- > Goods loaded in bulk have been evenly distributed within the container/vehicle.
- > For consignments including goods of class 1 other than division 1.4, the container/vehicle is structurally serviceable in accordance with 7.1.2.
- > The container/vehicle and packages are properly marked, labelled, and placarded, as appropriate; .8 When substances presenting a risk of asphyxiation are used for cooling or conditioning purposes (such as dry ice (UN 1845) or nitrogen, refrigerated liquid (UN 1977) or argon, refrigerated liquid (UN 1951)), the container/vehicle is externally marked in accordance with 5.5.3.6; and
- > A dangerous goods transport document, as indicated in 5.4.1, has been received for each dangerous goods consignment loaded in the container/vehicle. Note: The container/vehicle packing certificate is not required for portable tanks.

(IMDG Code, 2020)

Each ship carrying dangerous goods in packaged form shall have a special list or manifest setting forth, in accordance with the classification set out in the IMDG Code, the dangerous goods on board and the location.

A detailed stowage plan, which identifies by class and sets out the location of all dangerous goods on board, may be used in place of such a special list or manifest. A copy of one of these documents shall be made available before departure to the person or organisation designated by the port state authority.

(SOLAS 74, 2020)

The AMSA Information Sheet reminds ship managers about the requirements for the safe carriage of Ammonium Nitrate on board ships, with particular emphasis on:

Stowage under deck in a clean cargo space capable of being opened up in an emergency.

Protection from sources of heat.

Stowage out of direct contact with metal engine room bulkheads.

Carriage of Ammonium Nitrate in Flexible Intermediate Bulk Containers (FIBC).

The full document can be accessed by clicking [here](#).

The atmosphere in any space containing dangerous goods may pose serious risks to the health or life of anyone entering. Hazards can include flammable, toxic, corrosive, or asphyxiant gases or vapors, residues on packages, and spilled materials. These same dangers may also exist in spaces adjacent to or connected with the cargo spaces. Information on the hazards of specific substances can be found in the IMDG Code, the Safety Data Sheet (SDS), and the Shipper's Declaration. If there is evidence or suspicion of leakage of dangerous substances, the precautions specified in MSC.581(110) (2025) should be followed.

Personnel tasked with handling spillages or removing defective or damaged packages should be properly trained and equipped with suitable breathing apparatus and protective clothing appropriate for the task.

8.4 Are procedures for safe lashing and securing operations incorporated in the ship's SMS? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Procedures for safe lashing and securing operations should be included in the ship's Safety Management System as part of the ISM Code documentation.

(CSS code, 2011)

8.5 Is an approved ship-specific Cargo Securing Manual available and are officers thoroughly familiar with the contents of the manual? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

In accordance with the SOLAS chapters VI, VII and the Code of Safe Practice for Cargo Stowage and Securing (CSS Code), cargo units, including containers shall be stowed and secured throughout the voyage in accordance with a Cargo Securing Manual (CSM), approved by the Administration.

The CSM should be developed, taking into account the recommendations given in these Guidelines, and should be written in the working language or languages of the ship. If the language or languages used is not English, French or Spanish, a translation into one of these languages should be included.

- > The guidance given in the CSM should by no means rule out the principles of good seamanship, neither can it replace experience in stowage and securing practice.
- > The information and requirements set forth in the manual are consistent with the requirements of the vessel's trim and stability booklet, International Load Line Certificate (1966), the hull strength loading manual (if provided) and with the requirements of the International Maritime Dangerous Goods (IMDG) Code (if applicable).
- > The CSM specifies arrangements and cargo-securing devices provided on board the ship for the correct application to and the securing of cargo units, containers, vehicles and other entities, based on transverse, longitudinal and vertical forces which may arise during adverse weather and sea conditions.
- > It is imperative to the safety of the ship and the protection of the cargo and personnel that the securing of the cargo is carried out properly and that only appropriate securing points or fittings should be used for securing cargo.
- > The cargo-securing devices mentioned in this manual should be applied so as to be suitable and adapted to the quantity, type of packaging, and physical properties of the cargo to be carried. When new or alternative types of cargo-securing devices are introduced, the manual should be revised accordingly. Alternative cargo-securing devices introduced should not have less strength than the devices being replaced.
- > There should be a sufficient quantity of reserve cargo-securing devices on board the ship.
- > Information on the strength and instructions for the use and maintenance of each specific type of cargo-securing device, where applicable, is provided in this manual. The cargo-securing devices should be maintained in a satisfactory condition. Items worn or damaged to such an extent that their quality is impaired should be replaced.
- > The Cargo Safe Access Plan (CSAP) is intended to provide detailed information for persons engaged in work connected with cargo stowage and securing. Safe access should be provided and maintained in accordance with this plan.

(MSC.1/Circ.1353/Rev.1, Revised Guidelines for the Preparation of the Cargo Securing Manual, 2014)

Note: The Cargo Safe Access Plan (CSAP) is only required for containerships with keel-laying date on or after 1 January 2015. The Cargo Securing Manual should be updated as appropriate. Whenever the ship's outfit of portable cargo securing devices changes, those changes should be recorded in the appropriate section of the CSM. An appropriate record should be completed whenever routine visual examinations or periodic detailed examinations and re-testing of the devices are carried out.

(Bliault and North of England, P & I Association, 2007)

Lashing plans contained within the approved Cargo Securing Manual should be compatible with the current design of the ship and the intended container securing method is both safe and physically possible. The CSM, lashing plans and the CSAP are kept up to date.

Lashing plans and the CSAP are compatible with the design of the vessel and the equipment available.

(CSS code, 2011)

8.6 Are records maintained of the regular inspection and maintenance of the cargo-securing devices on board the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Equipment considered to be in poor condition is to be marked and not used. All relevant actions for replacement of such equipment are to be taken. The equipment in use should be in good condition and inspected as appropriate, in accordance with Class and/or maker's guidance.

It is important that each device used for securing cargo should be marked clearly with its SWL and a batch mark or number, where that mark can be verified by a test certificate. RightShip recommends that these test certificates be clearly labelled and kept in an easily accessible file. A method of correct identification and matching of individual certificates with the cargo-securing devices should be established on board. Cargo-securing devices without certificates must not be used on board.

There should be a procedure in place for removing damaged lashing devices from service. The inspection and maintenance schemes of the cargo-securing devices on board the ship shall be carried out as specified in the Cargo Securing Manual.

Regular inspections and maintenance should be carried out under the responsibility of the Master.

Inspection of cargo-securing devices should include as a minimum:

- > routine visual examinations of components being utilised; and
- > periodic examinations/re-testing as required by the Administration. When required, the cargo securing devices concerned should be subjected to inspections by the Administration.

The inspection and maintenance the ship's cargo-securing devices should be documented. Entries should be made in a record book, which should be kept with the Cargo Securing Manual. This record book should contain the following information:

- > procedures for accepting, maintaining and repairing or rejecting cargo-securing devices; and
- > record of inspections.

The record should contain information for the Master regarding inspections and adjustment of securing arrangements during the voyage.

(MSC.1/Circ.1353/Rev.1, Revised Guidelines for the Preparation of the Cargo Securing Manual, 2014)

As with lashings, dunnage material that is part of the ship's outfit, and is not discarded at the end of a voyage, should be stored in a suitably clean and dry storage space, away from any chemicals or other items that might cause damage. At appropriate intervals, the dunnage materials should be visually examined to determine whether or not any damage has been sustained. Damaged pieces should be discarded. Whenever dunnage materials are brought into use, and when new dunnage is brought on board, the items should be thoroughly inspected for defects and for their suitability for the intended purpose.

Timber carriers

All equipment, lashings, hog wires, uprights, deck fittings etc should be in good condition when taken into use. Routine inspections of all loose equipment should be carried out. All moving parts should be lubricated as appropriate, in accordance with the requirements set out in the Cargo Securing Manual.

Uprights and their base-foundations, lashing points and all other fixed equipment should be routinely examined. Any defects, such as worn or damaged lashing points or wasted or deformed uprights, should be repaired to the satisfaction of classification society as appropriate. Appropriate inspection and maintenance record sheets should be completed and retain on board.

(Bliault and North of England, P & I Association, 2007)

The cell guides, loose lashing and securing equipment including twist locks should be in good condition and free of excessive wear and corrosion. The twist locks, lashing and securing equipment of the same type and number, as specified in the approved Cargo Securing Manual, should be available on board.

Twist locks can be rated for different tensile loads up to 20 or 25 tonnes. It is important not to use a mix of twist locks that have different strength ratings.

(A Master's Guide to: Container Securing, 2012)

8.7 This question has been removed from the current version of the document.

8.8 Is there evidence to show that samples of the timber cargoes are being weighed during loading and what is the actual weight compared to the weight stated by the shipper? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If deemed necessary, samples of the timber cargo should be weighed during loading and their actual weight should be compared to the weight stated by the shipper, in order to correctly assess the ship's stability.
(Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2012)

8.9 If the vessel is a timber carrier, have up to date lashing plans for each stowage and securing arrangement been incorporated in the Cargo Securing Manual? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Lashing plan means a sketch or drawing showing the required number and strength of securing items for the timber deck cargo to obtain safe stowage and securing of timber deck cargoes.

In the Cargo Securing Manual, each stowage and securing arrangement should additionally be documented by a lashing plan showing at least the following:

- > Maximum cargo weight for which the arrangement is designed.
- > Maximum stowage height.
- > Required number and strength of blocking devices and lashings as applicable.
- > Required pretension in lashings.
- > Other cargo properties of importance for the securing arrangement such as friction, rigidity of timber packages, etc.
- > Illustrations of all securing items that might be used; and
- > Any restrictions regarding maximum accelerations, weather criteria, e.g., for non-winter conditions only, restricted sea areas, etc.

(Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2012)

8.10 If the vessel is a timber carrier, has a lashing plan according to the ship's Cargo Securing Manual been prepared? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A lashing plan according to the ship's Cargo Securing Manual should be prepared and the following calculated:

- > Weight and height of stows per hatch.
- > Number of sections in longitudinal direction per hatch.
- > Required number of pieces of lashing equipment; and
- > Required number of uprights, if applicable.

(Resolution A.1048 (27), Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2011)

8.11 Is a Class-approved loading computer or programme in use and has its operational accuracy been regularly tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A stability instrument installed on board should cover all stability requirements applicable to the ship. The software is subject to approval by the Administration. An operation manual should be provided for the stability instrument. The language in which the stability calculation results are displayed and printed out as well as the operation manual is written should be the same as used in the ship's approved stability booklet. A translation into a language considered appropriate may be required. In case of modifications of the ship which cause alterations in the stability booklet, the specific approval of any original stability calculation software is no longer valid. The software should be modified accordingly and re-approved. Any change in software version related to the stability calculation should be reported to and be approved by the Administration.

It is the responsibility of the ship's Master to check the accuracy of the stability instrument at each annual survey by applying at least one approved test condition. If an Administration's representative is not present for the stability instrument check, a copy of the test condition results obtained by this check should be retained on board as documentation of satisfactory testing for the Administration's representative's verification. At each renewal survey this checking for all approved test loading conditions should be done in the presence of the Administration's representative.

(Resolution MSC.267 (85), Adoption of the International Code on Intact Stability, 2008)

8.12 Are officers aware of the strength limits of tank tops, tween decks, hatch covers and weather decks with regards to safe cargo stowage and is this information posted in the ship's office/ ballast control room? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The maximum permissible loadings, given in tonnes per square metre (t/m²) for each deck and the tank-top, and the hatch covers should be posted in the Ship's office. The maximum permissible load figures can be found in the capacity plan, the midship section plan and on-deck plans. This information is also quoted in the ship's trim and stability booklet and in the Cargo Securing Manual.

If the maximum permissible loading is exceeded by cargo units, the deck, 'tween deck or tank-top plating, and the under-deck stiffening members will sustain damage.

(Blaugt and North of England P & I Association, 2007) (UK P&I CLUB, Carefully to Carry Consolidated Edition 2018, 2018)

Strength of the inner bottom plating could be deteriorated due to corrosion wastage. The operational parameters and tank top strength (T/m²) of the effected vessels may be updated by the classification society.

8.13 Have pre-stowage and stowage plans been prepared and completed effectively? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Consideration must be given to the ship's stability in advance of loading when planning a voyage. The stability condition should include the departure as well as arrival condition at discharge ports. The inspector should check that effort has been made to identify the point in the voyage when stability was lowest. The stability analysis should not just amount to a determination of the vessel's GM but must also consider the curve of righting levers (GZ). Both are to be checked for compliance with the criteria.

The ship's condition should be updated regularly by careful consideration of the weight and centres of all deadweight on board. Tanks' soundings should be checked regularly. The contents of tanks should be determined from soundings and the calculated stability condition is only valid if the tank status remains unchanged.

The pre-stowage plan should be prepared prior to vessel's arrival to load port.

The pre-stowage plan shall contain the vessel's name, voyage number, date, name of loading ports, name of discharging ports, dimension of holds, capacity of cranes, total cargo on board, total cargo for each port and the signatures of Chief Officer and Master.

The following should be considered when preparing a pre stowage plan:

- > The complete cargo specifications, including description, weight and overall dimensions, special handling instructions, hazardous classifications, and deck option.
- > Weight of cargo in relation to permissible load of each decks including tank top; each of the decks has weight limitations.
- > proper weight distribution and load spreading measures to prevent the decks from being overloaded.
- > Securing arrangements for the particular cargo.
- > The intended load and discharge port rotation to avoid shifting of cargo and prevent possible damage as a result of shifting cargo.
- > Details of all heavy lifts and out of gauge units, including lifting instructions and the centre of gravity.
- > SWL of ship's cranes and availability of shore crane at the load port/discharge port if needed.
- > Weight distribution, trim, stability and stress levels throughout the voyage and measures preventing from becoming excessively stiff or tender.
- > Cargo compatibility - incompatible cargoes are not stowed next to one another.

The objectives of pre-stowage plan are to minimise broken stowage and to prevent overload of tank top/tween decks and hatch covers.

The final stowage plan should include details of the final cargo distribution, the total weight and cube in each compartment and the total weight and cube for each discharge port. The plan will show the location of all heavy lifts and hazardous cargo.

- > Before loading a timber deck cargo: A pre-loading plan according to the ship's Trim and Stability Book should be done and the following should be calculated and checked:
 - > Stowage height.
 - > Weight per m².
 - > Required amount of water ballast; and
 - > Displacement, draught, trim and stability at departure and arrival.

When undertaking stability calculations, variation in displacement, centre of gravity and free surface moments due to the following factors should be considered:

- > Absorption of water in timber carried as timber deck cargo according to special instruction, see annex C.
- > Ice accretion, if applicable.
- > Variations in consumables; and
- > Ballast water exchange operations, in accordance with approved procedures.

(Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2012)

8.14

Is there evidence to show that evaluation of forces acting on the cargo unit have been calculated, and correct cargo-securing devices are being used to secure the cargo to the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Evaluation of forces acting on cargo units shall be incorporated in the Cargo Securing Manual and contain the following information:

- > Tables or diagrams giving a broad outline of the accelerations which can be expected in various positions on board the ship in adverse sea conditions and with a range of applicable metacentric height (GM) values.
- > Examples of the forces acting on typical cargo units when subjected to the accelerations referred to in paragraph 3.2.1 of MSC/Circ.745 and angles of roll and metacentric height (GM) values above which the forces acting on the cargo units exceed the permissible limit for the specified securing arrangements as far as practicable.
- > Examples of how to calculate number and strength of portable securing devices required to counteract the forces referred to in 3.2.2 of MSC/Circ.745 as well as safety factors to be used for different types of portable cargo securing devices. Calculations may be carried out according to Annex 13 to the CSS Code or methods accepted by the Administration.
- > It is recommended that the designer of a Cargo Securing Manual converts the calculation method used into a form suiting the particular ship, its securing devices and the cargo carried. This form may consist of applicable diagrams, tables, or calculated examples; and
- > Other operational arrangements such as electronic data processing (EDP) or use of a loading computer may be accepted as alternatives to the requirements of the above paragraphs 3.2.1 to 3.2.4 of MSC/Circ.745, providing that this system contains the same information.

It is important that securing devices meet acceptable functional and strength criteria applicable to the ship and its cargo. It is also important that the officers on board are aware of the magnitude and direction of the forces involved and the correct application and limitations of the cargo-securing devices. The crew and other persons employed for the securing of cargoes should be instructed in the correct application and use of the cargo securing devices on board the ship. "Maximum Securing Load (MSL)" is a term used to define the allowable load capacity for a device used to secure cargo.

To a ship, "Safe Working Load (SWL)" may be substituted for MSL for securing purposes, provided this is equal to or exceeds the strength defined by MSL.

(MSC.1/Circ.1353/Rev.1, Revised Guidelines for the Preparation of the Cargo Securing manual, 2014)

8.15

Have personnel engaged in cargo securing operations been provided with relevant training and familiarisation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends that personnel engaged in cargo securing operations attend a formal training program.

Personnel engaged in cargo-securing operations should be trained in the lashing and unlashings of containers as necessary to carry out their duties in a safe manner. This should include the different types of lashing equipment that are expected to be used.

Personnel engaged in cargo-securing operations should be trained in the identification and handling of bad order or defective securing gear in accordance with each ship's procedures, to ensure damaged gear is segregated for repair and maintenance or disposal.

Personnel engaged in cargo-securing operations should be trained to develop the knowledge and mental and physical manual handling skills that they require to do their job safely and efficiently, and to develop general safety awareness to recognise and avoid potential dangers.

Personnel should be trained in safe systems of work. Where personnel are involved in working at heights, they should be trained in the use of relevant equipment. Where practical, the use of fall protection equipment should take precedence over fall arrest systems.

Personnel who are required to handle thermal cables and/or connect and disconnect temperature control units should be given training in recognising defective cables, receptacles and plugs.

Personnel engaged in containership cargo operations should be familiarise with the ship's unique characteristics and potential hazards arising from such operations necessary to carry out their duties.

(CSS code, 2011)

8.16

If the vessel is carrying timber deck cargo, are relevant regulations of the applicable Load-Line Convention for stowage and securing of timber as prescribed in the ship's Cargo Securing Manual being followed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A complete stowage of tightly stowed timber will increase the ship's reserve of buoyancy. A ship with such a stowage of timber may be safely loaded to a deeper draught than would normally be allowed. If the ship does not have a timber load-line, then it cannot load deeper than the appropriate load-line or the load port and the voyage.

The timber deck cargo must be compactly stowed, lashed and secured. The timber deck cargo must not affect the safe navigation, day to day operation on board the ship and stability of the ship at any stage of the voyage. Due regard must be given to additions of weight, such as those due to absorption of water; and to losses of weight, such as those due to consumption of fuel and stores.

The height of the timber deck cargo above the weather deck on a ship within a seasonal winter zone in winter should not exceed one third of the extreme breadth of the ship. Otherwise, the height of the timber deck cargo should be restricted so that:

- > Adequate visibility is assured.
- > A safe margin of stability is maintained at all stages of the voyage.
- > Any forward-facing profile does not present overhanging shoulders to a head sea, and
- > The weight of the timber deck cargo does not exceed the designed maximum permissible load on the weather deck and hatches.

During the course of the voyage, if there is no convenient passage for the crew on or below the deck of the ship giving safe means of access from the accommodation to all parts used in the necessary working of the ship, guard lines or rails, not more than 330 mm apart vertically, should be provided on each side of the deck cargo to a height of at least 1 m above the cargo. In addition, a lifeline, preferably wire rope, set up taut with a tightening device should be provided as near as practicable to the centreline of the ship. The stanchion supports to all guardrails or lifelines should be spaced so as to prevent undue sagging. Where the cargo is uneven, a safe walking surface of not less than 600 mm in width should be fitted over the cargo and effectively secured beneath, or adjacent to, the lifeline.

(Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2012)

8.17

If the vessel is carrying timber, are instructions for ballast water exchange operations for the intended voyage available in the Ballast Water Management Plan? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

All ballast tanks required for the voyage and included in the stability calculations should be filled before the commencement of loading on deck and it should be ensured that free surfaces are eliminated in all tanks intended to be completely full or empty. Proper instructions for ballast water exchange operations, if applicable for the intended voyage, should be available in the Ballast Water Management Plan.

(Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2012)

8.18

Can timber deck cargo be jettisoned into the sea in a controlled manner in an emergency? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Crew going out on deck in the conditions that would possibly necessitate the need to jettison part or all of a cargo of timber put themselves in danger. Remotely operated jettisoning systems should be considered.

8.19 If the vessel is a non-cellular ship, have the containers been stowed correctly on deck? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

At no time should deck-loaded containers overstress the hatch covers or the hatchway structure. In cases of doubt, details of stress limitations should be obtained from the Classification Society.

(Merchant Shipping Notice No. M.1167 Carriage of Containers and Flats in Ships Not Designed or Modified for the Purpose, 1985)

- > Containers carried on deck or on hatches of such ships should preferably be stowed in the fore-and-aft direction.
- > Containers should not extend over the ship's sides. Adequate supports should be provided when containers overhang hatches or deck structures.
- > Containers should be stowed and secured so as to permit safe access for personnel in the necessary operation of the ship.
- > Containers should at no time overstress the deck or hatches on which they are stowed.
- > Bottom-tier containers, when not resting on stacking devices, should be stowed on timber of sufficient thickness, arranged in such a way as to transfer the stack load evenly on to the structure of the stowage area.
- > When stacking containers, use should be made of locking devices, cones, or similar stacking aids, as appropriate, between them.
- > When stowing containers on deck or hatches, the position and strength of the securing points should be taken into consideration.

(CSS code, 2011)

8.20 If refrigerated containers are carried, is there a procedure in place for monitoring their temperature, are records of temperature readings maintained, and are sufficient spare parts available on board? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

Regular monitoring and recording of the temperatures of the reefer containers should be carried out and properly documented. In the event of a claim these can be compared against the reefer unit download data and shipper's mobile temperature devices.

(Refrigerated Containers, 2013)

In the event of reefer container breakdowns, ships should have adequate spares onboard and the relevant skills to carry out emergency repairs to the reefer onboard.

The ship should also give prompt notification of reefer problems or malfunctions that cannot be repaired on board.

(Refrigerated Containers, 2013)

The inspector shall record in comments if the reefer containers had been equipped with IoT Device. For additional information, reference should be made to the DCSA IoT data standard for remote Reefer container monitoring on board a vessel. [Click here.](#)

8.21 This question has been removed from the current version of the document.

8.22 If refrigerated containers are carried, is the electric power supply permanently installed from the engine room and are the ship's electrical distribution system and electric container sockets in good condition and undamaged? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

The ship's electrical distribution system and container supply sockets should be in good working order and undamaged.

(Refrigerated Containers, 2013)

8.23 Are pre-loading/acceptance procedures for the carriage of vehicles on board a ro-ro cargo ship incorporated in the SMS? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

A well-documented pre-loading/acceptance procedure should cover basic checks for the carriage of new and used vehicles.

This may include, but not be limited to:

- > Ignition switched off and the key removed to an agreed location. Consideration should be given to keeping the keys inside the vehicle in a visible place to avoid the potential of delays resulting from the loss of keys
- > Disconnection of all battery cables; isolation of battery terminals
- > Inspection of battery for visible signs of damage
- > Prohibiting the carriage of spare/excess fuels or flammable liquids
- > Checking the integrity of seals and pipelines in order to ensure there are no visible leaks - are there visible signs of leaking oils or fuels? Is the engine bay lagging oil-soaked? Is the engine bay relatively clean?
- > Checking interior to ensure that flammable material such as oily rags spare fuel, undeclared chemicals etc. aren't stored inside a vehicle.

(Ro-Ro Fires, 2017)

8.24 Is the ro-ro cargo ship equipped with CCTV remote monitoring to monitor the vehicle decks? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

It is vitally important that alarms are treated seriously, and the appropriate action taken to establish the current condition of the space where the alarms have been activated. This may involve the use of CCTV equipment or sending someone to go to the area to investigate further. Whilst the use of a lookout offers a valuable first-hand onsite appraisal, it has a number of limitations that should be understood, amongst which are:

- > The time taken to get onsite, especially on large vessels, may add considerable time to a first response for fighting any potential fire
- > You may be placing the lookout in a potentially dangerous situation where they may be overcome by smoke or heat
- > It is essential that if using a lookout then they are briefed about the situation and equipped with functioning two-way communications with the OOW.

(Ro-Ro Fires, 2017)

8.25 Is the ro-ro cargo ship drainage system in good order, tested regularly and are effective measures in place to prevent blocking of drains? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Resolution MSC.256 (84) introduced changes to SOLAS Chapter II-2, Regulation 20 concerning the drainage of fire-fighting water from fixed pressure water-spraying systems within vehicle, special category and ro-ro spaces. For cargo and passenger ships constructed on or after January 1, 2010, which have vehicle, ro-ro or special category spaces fitted with a fixed pressure water-spraying system, there are requirements for the drainage of the spaces.

For closed vehicle and ro-ro spaces and special category spaces, where fixed water-spraying systems are fitted, effective measures should be in place to ensure floating debris does not block drains in spaces.

- > An easily removable grating, screen or other means should be installed over each drain opening in the protected spaces to prevent debris from blocking the drain. The total open area ratio of the grating to the attached drainpipe should be at least 6 to 1. The grating should be raised above the deck or installed at an angle to prevent large objects from blocking the drain. No dimension of the individual openings in the grating should be more than 25 mm.
- > No grating or screen is required when a fixed mechanical system is provided to unblock the drainage system, or when other than a gravity drain system is provided with its own filter.
- > A clearly visible sign or marking should be provided not less than 1,500 mm above each drain opening stating, "Drain opening – do not cover or obstruct". The marking should be in letters at least 50 mm in height.

In-service examination requirements:

- > Drainage systems should be visually examined periodically for blockage or other damage. If obstructions are noted, then they should be flushed with hoses to confirm that the system is functional.

(MSC.1/Circ.1320, Guidelines for the drainage of firefighting water from closed vehicle and ro-ro spaces and special category spaces of passenger and cargo ships, 2009)

8.26 Is the ro-ro cargo ship provided with an approved Operating and Maintenance Manual (OMM)? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The requirement for an Operating and Maintenance Manual (OMM) is applied to shell doors such as bow doors with the associated inner doors, side shell doors and stern doors with respect to the IACS Unified Requirements S8, S9, S15 and S16 as well as SOLAS 74 as amended. In addition to common operating instructions, the OMM shall provide full information to the ship's staff for maintaining and monitoring the sound condition of all elements of the shell door systems, relevant for the safety of the vessel. Detailed information on the procedures of maintenance and function tests are to be incorporated in the OMM.

The OMM is applicable to:

- > Ro-Ro passenger ships
- > Ro-Ro cargo ships with bow door, and keel laid on or after 01 July 1996, if the bow door/inner door gives access to an enclosed superstructure
- > Ro-Ro cargo ships with side or stern door, and keel laid on or after 01 July 1997, if the side or stern door gives access to enclosed spaces.

The OMM is subject to Class approval. The OMM must be prepared in a language understood by the users. If this language is not English, a translation into English is to be included.

Operational instructions for the Master: Special safety precautions shall emphasize the importance of closed openings for granting seaworthiness of the vessel. It shall be pointed out that special care must be taken when opening of shell doors at sea becomes necessary e.g. for embarkation of pilots or in case of emergency.

The operating panels for the operation of doors are to be inaccessible to unauthorised persons.

(SOLAS 74, 2020) (IMO A. 793(19) Strength and securing and locking arrangements of shell doors on ro-ro passenger ships, 1995)

8.27

Are procedures in place to carry out function and tightness testing of bow, inner, side shell, stern doors and main to lower deck cargo elevators of ro-ro cargo ships and is there evidence of regular testing? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

In comments, record the date and frequency of function and tightness testing of the doors, and function testing of the indicator system and tightness test. Also, record the opening and closing time of the doors.

It is recommended that recorded inspections of the door supporting and securing devices be carried out by the ship's staff at monthly intervals or following incidents that could result in damage, including heavy weather or contact in the region of side shell and stern doors. Any damage recorded during such inspections is to be reported to the Classification Society.

Maintenance, function, and tightness tests of the doors shall be incorporated in the PMS system as per manufacturer's recommendation.

The following shall be incorporated in the function test procedures.

Function test of the doors:

- > Proper working of the hinging arms and hinges
- > Proper engagement of the thrust bearings
- > Device for locking the door in the open position
- > Securing, supporting and locking devices
- > Proper sequence of the interlock system for the opening/closing system and the securing and locking devices
- > Mechanical lock of the securing devices
- > Proper locking of hydraulic securing devices in the event of a loss of the hydraulic fluid, according to the procedure provided by the OMM
- > Correct indication of open/closed position of doors and securing/locking devices at the navigation bridge and other control stations
- > Isolation of the hydraulic securing/locking devices from other hydraulic systems
- > Confirmation that the operating panels are inaccessible to unauthorised persons
- > Verification that a notice plate giving instructions to the effect that all securing devices are to be closed and locked before leaving harbour is placed at each operating panel and supplemented by warning indicator lights
- > Examination of electrical equipment for opening, closing and securing the doors.

Function test of the indicator system – indicator systems where fitted should be incorporated in the procedure and tested regularly:

- > Proper visible indication and audible alarm on the navigation bridge panel, according to the selected function "harbour/sea voyage" and on the operating panel
- > Lamp test function on both panels
- > Verification that it is not possible to turn off the indicator light on both panels
- > Verification of failsafe performance, according to the procedure provided by the OMM
- > Confirmation that power supply for indicator system is supplied by the emergency source or other secure power supply and independent of the power supply for operating the doors
- > Proper condition of sensors and protection from water, ice formation and mechanical damage.

Where fitted, the water leakage detection system is to be tested, including proper audible alarm on the navigation bridge panel and on the engine control room panel, according to the procedure provided by the OMM.

Tightness test: A hose test or equivalent is to be carried out at regular interval.

8.28

Is an operation manual for the ventilation system in a ro-ro cargo space provided and do records on board verify that the air quality is tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Maintenance of the ventilation system shall be incorporated in the PMS system as per manufacturer's recommendation.

An operation manual should be supplied and should include a plan of the ventilation system, showing fans, air supply and exhaust air openings and doors, ramps, hatches, etc.

The location of the control panel for the ro-ro cargo space ventilation system should also be marked. The plan should show the various options for operation of the ventilation system. It should include details of the air flow design and of the estimated number of different types of vehicles in the different ro-ro cargo spaces under various loading and unloading conditions.

The plan should be periodically revised and/or supplemented on the basis of the experience gained from the normal vehicle loading and unloading conditions. A number of blank drawings should therefore be kept on board. On the basis of such experience, it should also be possible to draw up guidelines for the maximum number of vehicles that should be allowed to operate simultaneously. Whenever possible, places which are sheltered from the air flow should be indicated on the plans.

The operation manual should include guidance for the service and maintenance of the systems.

Shipowners and operators should consider testing the air quality in conjunction with tests of the ventilation system to ensure proper maintenance and functioning of the ventilation system. Situations which indicate the necessity to conduct air-quality monitoring include worker complaints (e.g. headache, dizziness, stinging of the eyes or respiratory system), indications that the ventilation system itself has deteriorated, and changes in vessel operation which are substantially different from that for which the original ventilation system was verified.

(MSC.1/Circ.1515, Revised design guidelines and operational recommendations for ventilation systems in ro-ro cargo spaces, 2015)

8.29

Is the ro-ro ship fitted with an automatic system to control air quality in the cargo holds and are records of inspection, testing, calibration, and maintenance of the system being maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ro-ro ships to be fitted with an automated system to control air quality in ro-ro cargo holds by analysing the hold atmosphere and varying the ventilation rate accordingly. The air-quality control system should comply with the revised design guidelines and operational recommendations for ventilation systems in ro-ro cargo spaces contained in MSC.1/Circ.1515.

Inspection, maintenance, and repairs should be carried out in a professional manner. Owners should ensure that this is done and that the necessary skills, equipment and spares are available.

Annual testing of the vehicle space ventilation system should be conducted by the ship's safety delegate. Third-party testing of the vehicle space ventilation system should be undertaken before entry into service of a new ship and at periodical intervals of five years thereafter.

(MSC.1/Circ.1515, Revised design guidelines and operational recommendations for ventilation systems in ro-ro cargo spaces, 2015)

8.30 Are vehicles on the ro-ro car decks safely stowed and secured? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

All vehicles should be secured to the ship as per the approved ship specific cargo-securing manual and the condition of lashings should be monitored regularly.

Before being accepted for shipment, vehicles should be inspected externally by a competent and responsible person or persons to check that they are in satisfactory condition for shipment.

Second-hand vehicles may be transported on top of or within other second-hand vehicles. These vehicles, carried as cargo, should be subject to the same rigorous checks as other vehicles being shipped.

Labels, placards and marks that indicates the carriage of dangerous goods should be properly displayed.

Details of hazardous units should be reflected on the stowage plan and the crew should be aware of the location of and be vigilant against the carriage of dangerous goods.

All vehicles should be secured to prevent movement. Contact between vehicles during a voyage may damage and rupture the fuel tanks or damage the electrical systems.

Vehicles should, so far as possible, be aligned in a fore and aft direction. They should not be parked on permanent walkways or in such a way as to obstruct fire-fighting equipment or scuppers.

If water spray fire curtains are installed, then vehicles should not be parked across them.

(Ro-Ro Fires, 2017)

Battery-powered vehicles :**1. Risks associated with battery-powered vehicles:**

- > High voltage shocks
- > Direct jet flames
- > Fires develop in intensity quickly and rapidly reach their maximum intensity (typically within 2-3 minutes)
- > Toxic gases
- > Gas explosion (if the released gas accumulates for a while before being ignited)
- > Long lasting re-ignition risk (can ignite or re-ignite weeks, or maybe months after the provoking incident)
- > Once established fires are difficult to stop/extinguish
- > Thermal runaway

2. Thermal runaway: It is a phenomenon in which the lithium-ion cell enters an uncontrollable, self-heating state.

3. Stability checks and loading limitations:

Electric Vehicles (EVs) are expected to be on average 25% heavier than conventional vehicles. Appropriate considerations should be made in relation to loading limitations and stability calculations.

4. Identification of vehicles:

The information on the type of fuel or energy supply should be provided during booking and confirmed at the check-in if possible. The ship's Operator should update in the most appropriate manner its website/booking system.

The crew should be able to quickly identify the type of Alternative Fuel Vehicles (AFV) based on the information provided.

Battery Electric Vehicles (BEVs) should be clearly marked and should ideally be located in a designated area known to the crew. The location should, where possible, be on weather decks, away from dangerous goods, have sufficient drainage, adequate separation between vehicles for crew members wearing breathing apparatus to access from multiple access paths and be well clear of all emergency muster points.

5. Conditions for carriage:

AFVs should only be allowed onboard if they comply with the provisions of the IMDG Code as also described in 1.4.3. Particular attention should be paid to the following:

- > If there is suspicion that the battery of EVs is damaged or their battery is defective, they should only be allowed if their battery is removed
- > are free from any leakages of fuel/gases

6. Charging onboard:

Charging onboard ro-ro passenger ships should not be allowed unless the ship operator conducts a comprehensive risk assessment and approves and implements the relevant risk control measures. The vessel's manager may refer to the ALBERO project and its relevant deliverable on the requirements of charging stations onboard.

7. Video monitoring:

The installation of closed-circuit television (CCTV) systems with flame recognition capabilities/ thermal detection properties where vehicles, including BEVs, are located can allow early detection of fires. Typically, thermal runaway can be detected at 60-70 degrees centigrade. The position of CCTV systems should be considered. Typically, these are placed fore and aft on a vehicle deck. Operators should consider fitting increased numbers of CCTV cameras with thermal detection capabilities in the athwartship directions to enable rapid detection of thermal runaway. This also allows for the precise location of the fire to be identified.

8. Fire patrol routines:

Crew members who conduct safety patrols of the vehicle decks should be made familiar with the early signs of thermal runaway and if possible equipped with thermal handheld detectors. Early indications of thermal runaway are: Off-gassing. This is a release of various gases from the battery, including carbon dioxide, carbon monoxide, hydrogen, and volatile organic compounds. During the early phase of gas generation, the off-gases can be heavier than air and accumulate at deck-level or be lighter than air and dissipate or accumulate at deck-head level if the space is not appropriately ventilated. Owing to the various battery chemistries, it is not possible to predict which will dominate. A damaged battery can create rapid heating of the battery cells. If you notice hissing, whistling, or popping, a possible sweet chemical smell, black "smoke" (nanoparticles of heavy metals, not smoke) or white vapour coming from the high-voltage battery or the vehicle generally, assume that it is in thermal runaway and take appropriate firefighting measures. Early intervention will minimise the spread of any fire to adjacent vehicles. Fire patrols should pay special attention to look for evidence of battery coolant leakage, smoke or heat from the areas of vehicles where a battery is normally located, for example the underside. They should also listen for "popping sounds" which may be indicative of a potential thermal-runaway event.

Crew should conduct frequent emergency drills and training in the identification and initial response to a BEV battery fire. Early detection and prompt action can minimise the spread of a fire.

(DCV Safety Alert 02/2023 – risks associated with the carriage of Battery Electric Vehicles 2023)
(Guidance on the carriage of afvs in Ro-Ro Spaces 2022)

8.31 Has the vessel been provided with procedures and contingencies with regards to fumigation of cargo holds and are the Master and Chief Officer familiar with the procedure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Master should familiarise themselves with the recommendations set out in the procedures given in the SMS manual and with the guidance set out in section 3 of MSC.1/Circ.1264. In addition, some individual countries, for example the USA and Canada, have produced their own requirements which should be followed when fumigation is being carried out on board a ship within their territorial waters; the Master should be familiar with these requirements if appropriate.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.32 Have the air conditioning intakes for the accommodation, the engine room and other spaces been set to prevent the possibility of drawing in fumigant gas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ventilation procedures for accommodation, the engine room and other spaces should be reviewed to avoid the possibility of drawing fumigant gas into those spaces by incorrect ventilation. Further, it should be verified that ventilation flaps and closing devices are correctly set before the fumigation is carried out and they should be maintained in the correct arrangement throughout the fumigation period. A review of the ventilation regime should be completed before any ventilation of the cargo compartment is started and any necessary changes to those arrangements should be made.

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.33 Are necessary instruments (with spares) to determine the dew points provided, maintained in good condition and are there records of calibration of such instrument? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Dewpoint temperature may be measured by a variety of methods. Ships generally use a traditional wet and dry bulb arrangement consisting of two identical mercury thermometers, one of which has a damp muslin wick covering the bulb. These are normally housed in a protective marine screen on each bridge wing. The dewpoint temperature may then be determined by a "Dewpoint Table" to compare the wet and dry bulb temperatures. This figure is important when considering cargo ventilation requirements.

When using traditional wet and dry bulb thermometers, the accuracy of the dew point temperature will depend on the condition of the equipment. The muslin covering the wet bulb should be clean, the water in the reservoir should be distilled and the bulb itself should be wet. In order to ensure that the readings are correct, the device should always be positioned away from any exhaust vents, other draughts and all sources of heat. The readings should always be taken on the windward side of the vessel.

(Cargo Ventilation and Precautions to Minimise Sweat, 2012)

8.34 This question has been removed from the current version of the document.

8.35 Is ventilation of cargo holds being carried out and recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following should be recorded on each day for each hold:

- > All temperatures taken and dew points calculated.
- > Whether or not ventilation carried out.
- > Reason for not ventilating.
- > Weather and sea condition.
- > Ventilation rule applied (e.g., three –degree rule or dew point rule)

(Bulk Cargoes: A Guide to Good Practice, 2016)

8.36 Are the hold bilges cleaned prior to loading and are cleaning and checks being recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.37 Do records on board verify that cargo lights in holds were properly isolated before cargo was loaded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Many bulk carrier/general cargo holds have fixed cargo lights. These can easily ignite combustible cargoes such as grain, animal feed, wood chips, pulp and paper if they are too close to the light. Cargo lights in holds need to be properly isolated before cargo is loaded. This is best done by removing fuses or other physical links in the electrical circuits so that the lights cannot be switched on by mistake.

(Fire! A Guide to the causes and prevention of cargo fire, 2017)

There are certain types of cargo hold lights that can be considered safe for continuous use without posing a risk to the cargo. Examples include:

LED Lights

- > Low heat emission
- > Energy-efficient and long-lasting

Fluorescent Lights

- > Generally safe for use around combustible cargo
- > Commonly used in car carriers
- > Low risk of ignition due to lower operating temperatures.

If such lights are installed inside the cargo hold, and there is objective evidence of regular maintenance in accordance with the manufacturer's recommendations, along with procedures in the SMS that address the requirement for isolating these lights based on the type of cargo, and these procedures have been followed, then the inspector should select "Yes" as the answer to this question.

The inspector should also record in the comments section the measures in place to ensure the safety of the cargo within the hold.

8.38 Is an adequate record of all cargo operation activities maintained during loading and unloading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following should be recorded in the port logbook or deck logbook:

- > Starting and stopping of work at each hold, times, and dates
- > Tonnages loaded per pour into each hold and a running total loaded and in the case of unloading, tonnages offloaded per shift from each hold and a running total offloaded
- > Weather conditions at intervals, for example 6 hours
- > Use of ship's cranes, if appropriate
- > Movement of shore cranes, loaders, or floating crane alongside
- > Movement of barges alongside and of floating cranes or loaders if ship is at anchor
- > Opening and closing of hatches
- > Period of precipitation
- > Draft readings
- > Any delays caused on board
- > Any delays caused ashore
- > Any surveyors attending or boarding with reason for attendance
- > Any stevedore's damage to ship's structure and/or fittings
- > Cargo temperature in particular for grain, seed cake and coal

(Bulk Cargoes: A Guide to Good Practice, 2016)

Section 8D: Cargo Operation - Cellular Container Ships

Note: There have been incidents in recent years in which the stowage of containers did not comply with the approved arrangements. Such practices compromised the effectiveness of cargo stowage and securing arrangements and increased the risk of cargo being lost overboard while at sea. The impact of such events on safety and the environment was often significant.

The nature and practices of the container-ship trade i.e. pre-planning of the stowage positions of containers by terminal, tight operating schedules, the short turnaround time of ships, constantly changing information, containers arriving up to the last minute, all in all shall not release the Master from the duty of care for the vessel's safety.

This chapter can only be completed if the vessel is a cellular container ship and must not be used for non-cellular ships that carry containers.

8.1 This question has been removed from the current version of the document.

8.2 Has appropriate cargo information been provided to the vessel prior to loading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The shipper shall provide the Master or the Master's representative with appropriate information on the cargo sufficiently in advance of loading to enable the precautions which may be necessary for proper stowage and safe carriage of the cargo to be put into effect. Such information shall be confirmed in writing and by appropriate shipping documents prior to loading the cargo on the ship.

In case of general cargo, and of cargo carried in cargo units, this information shall include a general description of the cargo, the gross mass of the cargo or of the cargo units, and any relevant special properties of the cargo units. For the purpose of this regulation, the cargo information required in sub-chapter 1.9 of the Code of Safe Practice for Cargo Stowage and Securing, adopted by the Organisation by resolution A.714 (17), as may be amended, shall be provided.

Prior to loading cargo units on board ships, the shipper shall ensure that the gross mass of such units is in accordance with the gross mass declared on the shipping documents.

(SOLAS 74,2020)

The AMSA Information Sheet reminds ship managers about the requirements for the safe carriage of Ammonium Nitrate on board ships, with particular emphasis on:

Stowage under deck in a clean cargo space capable of being opened up in an emergency.

Protection from sources of heat.

Stowage out of direct contact with metal engine room bulkheads.

Carriage of Ammonium Nitrate in Flexible Intermediate Bulk Containers (FIBC).

The full document can be accessed by clicking [here](#).

8.3 If dangerous goods are carried in a container, have appropriate documents been provided to the vessel, and is safety in relation to stowage of declared dangerous goods and higher risk cargoes being considered? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The dangerous goods transport document shall include a certification or declaration that the consignment is acceptable for transport and that the goods are properly packaged, marked, and labelled, and in proper condition for transport in accordance with the applicable regulations.

(IMDG Code, 2018)

Each ship carrying dangerous goods in packaged form shall have a special list or manifest setting forth, in accordance with the classification set out in the IMDG Code, the dangerous goods on board and the location thereof. A detailed stowage plan, which identifies by class and sets out the location of all dangerous goods on board, may be used in place of such a special list or manifest. A copy of one of these documents shall be made available before departure to the person or organisation designated by the port State authority.

(SOLAS 74,2020)

These safety considerations include the following measures in relation to stowage of declared dangerous goods and higher risk cargoes:

- > Dangerous goods which cannot be extinguished by CO2 should be stowed on deck.
- > Dangerous goods which cannot be extinguished by either water or CO2 should be stowed on deck.
- > Dangerous goods prone to fire or explosion should be segregated from known ignition sources.
- > Explosives should be stowed furthest from the accommodation and primary life-saving appliances.

The Cargo Incident Notification System (CINS) document "Safety Considerations for Ship Operators Related to Risk-Based Stowage of Dangerous Goods on Containerships" provides information related to safe stowage of dangerous good on container ships.

(Safety Considerations for Ship Operators Related to Risk-Based Stowage of Dangerous Goods on Containerships, 2019)

8.4 Is there a procedure for checking the container's seals and is there documented evidence of compliance? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Master should be satisfied that the Company has in place a procedure for checking the container seals in compliance with the SSP. Any irregularities should be notified immediately to the stevedores or terminal operators responsible for the loading, as well as the vessel's agent and the Company. Seals should likewise be checked at discharge to evidence that they have remained intact whilst on the vessel.

(Guidance to Masters, 2006)

8.5 Is cargo on flat racks, where applicable properly secured? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If flat racks are loaded, the Master should ensure that the cargo on these units is properly lashed, secured, and protected against external elements. This includes locating a suitable stowage position to avoid damage by the impact of waves. If the Master is in any doubt as to whether the cargo on the flat racks is sufficiently lashed, the Master should call the Company to arrange for a surveyor to attend and check the securing of the cargo on the flat racks. Tarpaulins, if in use, should be tight and not torn and need to be checked and adjusted at regular intervals during the voyage. These checks should be recorded.

(Guidance to Masters, 2006)

8.6 Is the verified gross mass communicated in shipping documents sufficiently in advance to the Master? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Master or Master's representative and the terminal representative should enter into arrangements to ensure the prompt sharing of verified container gross mass information provided by shippers. Existing communication systems may be used for the transmission and sharing of such verified container gross mass information.

A container packed with packages and cargo items should not be loaded onto a ship to which the SOLAS regulations apply unless the Master or Master's representative and the terminal representative have obtained, in advance of vessel loading, the verified actual gross mass of the container.

(GUIDELINES REGARDING THE VERIFIED GROSS MASS OF A CONTAINER CARRYING CARGO, 2013)

8.7 Have containers carried on deck or on hatch covers been stowed in the fore-and-aft direction? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if any containers were loaded in an athwartships direction.

Containers carried on deck or on hatches of such ships should preferably be stowed in the fore-and-aft direction. This stowage method is sensible regarding the interplay of stresses in rough seas and the loading capacity of containers. Stresses in rough seas are greater athwartships than fore and aft and the loading capacity of container side walls is designed to be higher than that of the end walls.

However, on many ships the containers are stowed in athwartships bays or are transported athwartships for other reasons. This must be taken into consideration when packing containers and securing cargo.

8.8 Are containers stowed in block stowage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if containers were stowed in isolated stacks, especially in outboard locations.

Generally, container stacks do not depend on each other for support. However, they do provide protection to each other from wind and waves, so stowage in isolated stacks, especially in outboard locations, should be avoided.

Making block stowage may be difficult for coastal container ships when limited containers are available for loading.

The question shall be answered YES, where isolated stowage is loaded on board a coastal container ship, provided that the calculated lashing stress is within allowable range. The inspector shall record in comments if the isolated stowage is loaded on board a coastal container ship.

8.9 Are procedures for safe lashing and securing operations of containers incorporated in the ship's SMS? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Procedures for safe lashing and securing operations should be included in the ship's Safety Management System as part of the ISM Code documentation.

(CSS Code, 2011)

8.10 Is an approved ship specific Cargo Securing Manual available, and are officers thoroughly familiar with the contents of the manual and is the lashing plan compatible with the design of the vessel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the vessel has deviated from the approved lashing arrangements shown in the Cargo Securing Manual, except to add additional lashings.

In accordance with the SOLAS chapters VI, VII and the Code of Safe Practice for Cargo Stowage and Securing (CSS Code), cargo units, including containers shall be stowed and secured throughout the voyage in accordance with a Cargo Securing Manual (CSM), approved by the Administration.

The CSM should be developed, considering the recommendations given in these Guidelines and should be written in the working language or languages of the ship. If one of the working languages is not English, French, or Spanish, a translation into one of these languages should be included.

- > The guidance given in the CSM should by no means rule out the principles of good seamanship, neither can it replace experience in stowage and securing practice.
- > The information and requirements set forth in the manual should be consistent with the requirements of the vessel's trim and stability booklet, International Load Line Certificate (1966), the hull strength loading manual (if provided) and with the requirements of the International Maritime Dangerous Goods (IMDG) Code (if applicable).
- > The CSM should specify arrangements and cargo-securing devices provided on board the ship for the correct application to and the securing of cargo units, containers, vehicles, and other entities, based on transverse, longitudinal and vertical forces which may arise during adverse weather and sea conditions.
- > It is imperative to the safety of the ship and the protection of the cargo and personnel that the securing of the cargo is carried out properly and that only appropriate securing points or fittings should be used for securing cargo.
- > The cargo-securing devices mentioned in the manual should be suitable and adapted to the quantity, type of packaging, and physical properties of the cargo to be carried. When new or alternative types of cargo-securing devices are introduced, the manual should be revised accordingly. Alternative cargo-securing devices introduced should not have less strength than the devices being replaced.
- > There should be a sufficient quantity of reserve cargo-securing devices on board the ship.
- > Information on the strength and instructions for the use and maintenance of each specific type of cargo-securing device, where applicable, should be provided in the manual. The cargo-securing devices should be maintained in a satisfactory condition. Items worn or damaged to such an extent that their quality is impaired should be replaced.
- > The Cargo Safe Access Plan (CSAP) is intended to provide detailed information for persons engaged in work connected with cargo stowage and securing. Safe access should be provided and maintained in accordance with this plan.

(MSC.1/Circ.1353/Rev.1, Revised Guidelines for the Preparation of the Cargo Securing Manual, 2014)

Note: The Cargo Safe Access Plan (CSAP) is only required for containerships with keel-laying date on or after 1 January 2015.

- > The Cargo Securing Manual should be updated as appropriate. Whenever the ship's outfit of portable cargo securing devices changes, those changes should be recorded in the appropriate section of the CSM. An appropriate record should be completed whenever routine visual examinations or periodic detailed examinations and re-testing of the devices are carried out.

(Blaught and North of England, P & I Association, 2007)

- > Lashing plans contained within the approved Cargo Securing Manual should be compatible with the current design of the ship and the intended container securing method must be both safe and physically possible. The CSM, lashing plans and the CSAP must be kept up to date.
- > Lashing plans and the CSAP are compatible with the design of the vessel and the equipment available.

(CSS code, 2011)

8.11 Are records maintained of the regular inspection and maintenance of the cargo-securing devices on board the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Equipment considered to be in poor condition is to be marked and not used. All relevant actions for replacement of such equipment are to be taken. The equipment in use should be in good condition, inspected as appropriate in accordance with Class and/or maker's guidance.

It is important that each device used for securing cargo should be marked clearly with its SWL and a batch mark or number, where that mark can be verified by a test certificate. RightShip recommends that these test certificates be clearly labelled and kept in an easily accessible file. A method of correct identification and matching of individual certificates with the cargo-securing devices should be established on board. Cargo-securing devices without certificates must not be used on board.

Portable fittings should be certified by some form of type-approved system, usually coming from manufacturer (when approved), a Classification Society or other accepted testing body.

The inspection and maintenance schemes of the cargo-securing devices on board the ship shall be carried out as specified in the Cargo Securing Manual. Regular inspections and maintenance should be carried out under the responsibility of the Master.

Inspection of cargo-securing devices should include as a minimum:

- > Routine visual examinations of components being utilised; and
- > Periodic examinations/re-testing as required by the Administration. When required, the cargo-securing devices concerned should be subjected to inspections by the Administration.
- > Inspection and maintenance of the ship's cargo-securing devices should be documented. Entries should be made in a record book, which should be kept with the Cargo Securing Manual. This record book should contain the following information:
 - > Procedures for accepting, maintaining, and repairing or rejecting cargo-securing devices; and
 - > Record of inspections.

The record should contain information for the Master regarding inspections and adjustment of securing arrangements during the voyage.

(MSC.1/Circ.1353/Rev.1, Revised Guidelines for the Preparation of the Cargo Securing Manual, 2014)

The cell guides, loose lashing and securing equipment including twist locks should be in good condition and free of excessive wear and corrosion. Twist locks, lashing and securing equipment of the same type and number as specified in the approved Cargo Securing Manual should be available on board.

Twist locks can be rated for different tensile loads up to 20 or 25 tonnes. It is important not to use a mix of twist locks that have different strength ratings.

(A Master's Guide to: Container Securing, 2012)

8.12 Is there a sufficient quantity of reserve cargo- securing devices on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

There should be a sufficient quantity of reserve cargo-securing devices on board to deal with unexpected circumstances as per the Cargo Securing Manual.

8.13 Is the Cargo Safe Access Plan (CSAP) prepared and was it followed on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Applicable to ships with keel laid on or after 01 January 2015.

8.14 Are appropriate securing points being used for cargo securing and is there recorded evidence of regular inspection and maintenance of them? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Deck securing points must provide effective leads in terms of the axes of the forces being resisted and be so arranged to avoid chafing. The securing points must not be overloaded by holding more lashings than they can safely take, and, if necessary additional points are to be welded. Securing points on deck to be marked as appropriate. Safe Working load to be measured as per Class instructions. Maintenance to be conducted as required. Regular inspection of fixed deck fittings is essential to establish whether progressive wear has undermined their integrity.

(Container carriage- A selection of articles previously published by Gard AS, 2014)

8.15 Are there procedures for reporting and removal of damaged lashing devices from service and are there records to demonstrate that damaged lashing devices have been removed from service? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A container ship Master must be prepared to use all available tools in the ISM system in order to report defective stowage to the vessel operators and designated person ashore. It is a fundamental requirement of ISM that defects of this type are reported.

(Container lashing and stowage, 2004)

8.16 Is a Class-approved loading computer or programme in use and has its operational accuracy been regularly tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the class approved container stowage calculations for lashing, stack weight and visibility were not incorporated in the shipboard loading computer or program.

A stability instrument installed on board should cover all stability requirements applicable to the ship. The software is subject to approval by the Administration. An operation manual should be provided for the stability instrument. The language in which the stability calculation results are displayed and printed out as well as the operation manual is written should be the same as used in the ship's approved stability booklet. A translation into a language considered appropriate may be required. In case of modifications of the ship which cause alterations in the stability booklet, the specific approval of any original stability calculation software is no longer valid. The software should be modified accordingly and re-approved. Any change in software version related to the stability calculation should be reported to and be approved by the Administration.

It is the responsibility of the ship's Master to check the accuracy of the stability instrument at each annual survey by applying at least one approved test condition. If an Administration's representative is not present for the stability instrument check, a copy of the test condition results obtained by this check should be retained on board as documentation of satisfactory testing for verification by the Administration's representative. At each renewal survey this checking for all approved test loading conditions should be done in the presence of the Administration's representative.

(Resolution MSC.267 (85), Adoption of the International Code on Intact Stability, 2008)

8.17 Was the pre-loading plan forwarded to the ship prior to loading and is there evidence to show that the ship's loading plan was updated and reviewed by the Chief Officer prior to loading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The pre-loading plan received from the terminal planner should be entered into the shipboard loading computer. The draft ballast plan should also be entered, and an initial evaluation of the ship's condition should be performed and consequential corrective action identified.

Stack weight limitations and lashing limitations should be identified. Any corrective actions or changes that need to be taken to ensure compliance should be discussed with the terminal planner and stowage co-ordinator as required. Bridge visibility rules in accordance with SOLAS must be observed and checked.

(Safe Transport of Containers by Sea-Guidelines on Best Practices, 2008)

Ships' staff should not allow loading operations to commence until they have received a copy of the proposed stowage plan. It may be the case that a full stowage plan has not been completed, but a loading terminal should be able to give the plan for the bays about to be worked. A relatively quick inspection should show whether heavy containers have been planned over light ones; and whether the stack and tier weights are within the permissible limits.

A reason for this is that the system for container loading is entirely driven from ashore by the planner, who creates a stowage plan and has the ability to vary and modify it right up to the moment a particular unit is picked up by a crane. It is frequently the case that the final bay plan, received after work has been completed, bears only passing resemblance to the pre-load plan which was received just as work was commencing.

(UK P&I CLUB, Carefully to Carry Consolidated Edition, 2018)

In November 2014, the IMO Maritime Safety Committee adopted an amendment to SOLAS to require that shippers obtain the 'verified gross mass' (VGM) of packed containers and communicate it to the ocean carrier sufficiently in advance of the ship's stowage planning. Ocean carriers are obliged to use the VGM in the stowage plan and, together with the terminal operator, ensure that any container that does not have a VGM is not loaded on a ship. The revised regulation entered into effect on 1 July 2016.

8.18 Is there recorded evidence to show that the pre-loading plan has been checked for 'heavy' container stacks? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Container stacks are containers which are stacked vertically and secured horizontally by stackers, lashing etc. Prior to loading cargo, stacking weights of containers must be checked against the allowable stack weights on board the vessel, both on deck and under deck. Neglecting this procedure may cause serious damage to the ship's structure, hull and eventually overall stability of the ship may be affected. Maximum allowable stack weights of tank tops, hatch covers, and decks shall not be exceeded at any time.

Ships' staff should always check the pre-loading plan for 'heavy' container stacks. These should be identified and, if possible, the container numbers in these stacks checked during loading. If a different container appears in the upper tier then it may be a heavy unit stowed by mistake and of sufficient weight to overload the stack and the lashing system.

(UK P&I CLUB, Carefully to Carry Consolidated Edition, 2018)

8.19 Has the vessel avoided loading of heavy containers above light containers or at the top of a stack in a deck stow? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Heavy on light can only be accepted when specifically permitted in the Cargo Securing Manual.

It is essential to avoid loading heavy containers over light, or at the top of a stack in a deck stow, unless specifically permitted in the Cargo Securing Manual. This is because the securing system would normally have been designed on the assumption that light containers are stowed on top. Stowage may allow for 'heavy-heavy-light'; however, loading 'heavy-medium-medium' may result in the same stack weight but would produce different strain on the securing system, especially if the GM is high.

(A Master Guide to: Container Securing, 2012)

Loading heavy containers top stow may cause stability problem and excessive lashing strains during ship's motions at sea.

8.20 Is there recorded evidence to show that the lashing pattern from the Cargo Securing Manual has been provided to the terminal and is the information for the lashing pattern posted at the ship's access? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.21 Had the Ballast Water Management Plan for the present port stay been executed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Based on pre-loading plan information, the ship shall execute a ballast water management plan for the coming port stay. This includes optimisation of ballast water distribution to allow for minimal discharges in port.

(Safe Transport of Containers by Sea - Guidelines on Best Practices, 2008)

8.22 Is there documented evidence of carrying out spot checks of actual loading against loading plan? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

During the loading operation, spot checking of the actual loading compared to the loading plan should be undertaken throughout the cargo area, with particular attention paid to OOG, DG and reefer containers. Discrepancies should be resolved with the terminal planner and stowage coordinator, paying due regard to the health and safety implications of any solution.

(Safe Transport of Containers by Sea - Guidelines on Best Practices, 2008)

8.23 Is there documented evidence of signing off completed lashing per bay and is all lashing gear used compatible with the lashing plan? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The lashing arrangement of each bay should be inspected and adjusted if necessary, by the crew following completion of work by the terminal personnel. (Safe Transport of Containers by Sea-Guidelines on Best Practices, 2008)

All gear should be uniformly compatible with the lashing plan, i.e., all twist locks of the same type, semi-automatic twist locks used with suitable lashing bars.

8.24 Is the bridge visibility condition for the next port confirmed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Bridge visibility rules accordance with SOLAS must be observed and checked. In order to confirm sufficient visibility conditions, a check of the arrival conditions for the next port should also be made.

(Safe Transport of Containers by Sea - Guidelines on Best Practices, 2008)

8.25 Is there evidence to show that evaluation of forces acting on the containers have been calculated and the correct cargo-securing devices are being used to secure the containers to the ship? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

An evaluation of forces acting on cargo units shall be incorporated in the Cargo Securing Manual and contain the following information:

- > Tables or diagrams giving a broad outline of the accelerations which can be expected in various positions on board the ship in adverse sea conditions and with a range of applicable metacentric height (GM) values.
- > Examples of the forces acting on typical cargo units when subjected to the accelerations referred to in paragraph 3.2.1 of MSC/Circ.745 and angles of roll and metacentric height (GM) values above which the forces acting on the cargo units exceed the permissible limit for the specified securing arrangements as far as practicable.
- > Examples of how to calculate number and strength of portable securing devices required to counteract the forces referred to in 3.2.2 of MSC/Circ.745 as well as safety factors to be used for different types of portable cargo securing devices. Calculations may be carried out according to Annex 13 to the CSS Code or methods accepted by the Administration.
- > It is recommended that the designer of a Cargo Securing Manual converts the calculation method used into a form suiting the particular ship, its securing devices and the cargo carried. This form may consist of applicable diagrams, tables, or calculated examples; and
- > Other operational arrangements such as electronic data processing (EDP) or use of a loading computer may be accepted as alternatives to the requirements of paragraphs 3.2.1 to 3.2.4 of MSC/Circ.745, providing that this system contains the same information.

It is important that securing devices meet acceptable functional and strength criteria applicable to the ship and its cargo. It is also important that the officers on board are aware of the magnitude and direction of the forces involved and the correct application and limitations of the cargo-securing devices. The crew and other persons employed for the securing of cargoes should be instructed in the correct application and use of the cargo securing devices on board the ship. 'Maximum Securing Load (MSL)' is a term used to define the allowable load capacity for a device used to secure cargo to a ship. 'Safe Working Load (SWL)' may be substituted for MSL for securing purposes, provided this is equal to or exceeds the strength defined by MSL.

(MSC.1/Circ.1353/Rev.1, Revised Guidelines for the Preparation of the Cargo Securing Manual, 2014)

8.26 Is there documented evidence that lashings were tightened after departure once the lashings and containers settled in? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

It is good practice to verify the tightening of the lashing after departure once the lashing and containers have settled in. This is especially the case before ocean crossings, after receipt of bad weather outlooks, and after bad weather has been encountered (Safe Transport of Containers by Sea - Guidelines on Best Practices, 2008)

Lashings should be checked and tightened within 24 hours after leaving port and regularly thereafter. This is especially true before the onset of bad weather.

8.27 Have personnel engaged in cargo securing operations been provided with relevant training and familiarisation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends that personnel engaged in cargo securing operations attend a formal training course.

Personnel engaged in cargo-securing operations should be trained in the lashing and unlashings of containers as necessary to carry out their duties in a safe manner. This should include the different types of lashing equipment that are expected to be used.

- > Personnel engaged in cargo-securing operations should be trained in the identification and handling of bad order or defective securing gear in accordance with each ship's procedures, to ensure damaged gear is segregated for repair and maintenance or disposal.
- > Personnel engaged in cargo-securing operations should be trained to develop the knowledge and mental and physical manual handling skills that they require to do their job safely and efficiently, and to develop general safety awareness to recognise and avoid potential dangers.
- > Personnel should be trained in safe systems of work. Where personnel are involved in working at heights, they should be trained in the use of relevant equipment. Where practical, the use of fall-protection equipment should take precedence over fall-arrest systems.
- > Personnel who are required to handle thermal cables and/or connect and disconnect temperature control units should be given training in recognising defective cables, receptacles, and plugs.
- > Personnel engaged in containership cargo operations should be familiar with the ship's unique characteristics and potential hazards arising from such operations necessary to carry out their duties.

(CSS code, 2011)

8.28 Is the vessel equipped with sufficient portable radio equipment for use and has a direct radio communication capability been established between the terminal (planner, foreman, and watchman) and the ship duty's officer? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.29 Has an IMDG spotting plan been prepared, updated, and made available for emergency preparedness? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The IMDG spotting plan should be updated for emergency preparedness. Segregation requirements should be confirmed in accordance with the IMDG Code.

(Safe Transport of Containers by Sea - Guidelines on Best Practices, 2008)

8.30 Is there a procedure for monitoring the temperature of refrigerated containers and are records maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Regular monitoring and recording of the temperatures of the reefer containers should be carried out and properly documented. In the event of a claim these can be compared against the reefer unit download data and shipper's mobile temperature devices. (Refrigerated Containers, 2013)

The inspector shall record in comments if the reefer containers had been equipped with IoT Device. For additional information, reference should be made to the DCSA IoT data standard for remote Reefer container monitoring on board a vessel. [Click here.](#)

8.31 If refrigerated containers are carried, are sufficient spare parts available on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

In the event of reefer container breakdowns, ships should have adequate spares on-board and the relevant skills to carry out emergency repairs to the reefer on-board.

The ship should also give prompt notification of reefer problems or malfunctions that cannot be repaired on board.

8.32 If refrigerated containers are carried, is the electric power supply permanently installed from the engine room and are electric container sockets in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship's electrical distribution system and container supply sockets should be in good working order and undamaged.

(Refrigerated Containers, 2013)

The vessel's manager shall use a Power Pack Unit (PPU) if the loading quantity of reefer cargo exceeds the ship's power capacity as indicated in the drawing and charter party. The PPU has an independent power supply and will not require the electric power supply permanently installed from the vessel's engine room.

The generator set shall be maintained in good working condition and free of oil leaks.

8.33 Are officers familiar with the exothermic chain reaction and the stowage and segregation requirements for calcium hypochlorite in containers? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

An exothermic reaction is a chemical reaction that releases energy through light or heat. An example of an exothermic reaction is a reaction between water and calcium chloride.

The incorrect stowage of exothermic dangerous goods such as calcium hypochlorite caused many fires in recent years.

Calcium hypochlorite or calcium hypochlorite mixtures shall be transported in compliance with the stowage and segregation requirements set out in the IMDG Code.

IMDG Code Special Provision 314, under Part 3, Chapter 3.3, of the Code applies to calcium Hypochlorite.

The International Group of P&I Clubs and the Cargo Incident Notification System (CINS) document 'Guidelines for the Carriage of Calcium Hypochlorite in Containers' provides information for the carriage of calcium hypochlorite in containers.

(Guidelines for the Carriage of Calcium Hypochlorite in Containers, 2018)

8.34 Is an adequate record of all cargo operation activities maintained during loading and unloading? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Section 8E: Cargo Operation - Self-Unloading Transshipment

Note: This section is only applicable to gravity-based self-unloading and/or hybrid self-unloading vessels discharging bulk cargo on shore or offshore.

This section shall be completed along with 'section 8:-cargo operation - solid bulk cargo other than grain' and/or section 8:-cargo operations grain', as appropriate, for gravity-based self-unloading and/or hybrid self-unloading vessels.

For the purpose of uniformity in this section, the generic term "self-unloading system" has been used instead of 'gravity-based self-unloading' and/or 'hybrid self-unloading' system.

8.1 This question has been removed from the current version of the document.

8.2 Is the vessel provided with self-unloading isolation procedures? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The objectives of the ISM Code are to ensure safety at sea, prevention of human injury or loss of life, and avoidance of damage to the environment, in particular to the marine environment and to property.

(ISM Code 2018)

Rightship recommends that the isolation procedures should be about the need to isolate, de-energise, lockout and tag-out the system, with the aim of reducing the risk of death or injury during system inspections, repairs, maintenance, assessments, adjustments, or cleaning.

The isolation procedures shall incorporate the followings:

- > The isolation procedure, check list and risk assessment documented and accessible to the relevant crew on board.
- > Providing information, instruction and training to crew involved with the system.
- > Appointing a person as a supervisor to make sure the crew strictly follows isolation procedures.

Rightship recommends the vessel's manager provide ship-specific procedures, detailing steps for isolating cargo equipment prior to maintenance.

8.3 Have personnel engaged in the operation of the self-unloading system been provided with relevant training and familiarisation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Rightship recommends officers and engineers attend a combined on-board/shore-based training course consisting of theoretical and practical training. Theoretical training shall include the mechanical system, hydraulic system, electrical system, maintenance, spare parts, function, trouble-shooting and normal and emergency operation. Practical training shall include hands-on training sessions and trouble-shooting different components.

8.4 Are the ship's crew aware of the safe operating requirements of tunnel conveyor watertight bulkhead doors and/or watertight doors in tunnel conveyor room, where applicable, and is a record of inspections and maintenance available on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Doors provided to ensure the watertight integrity of internal openings which are used while at sea are to be sliding watertight doors capable of being remotely closed from the bridge and are also to be operable locally from each side of the bulkhead. Indicators are to be provided at the control position showing whether the doors are open or closed, and an audible alarm is to be provided at the door closure. The power, control and indicators are to be operable in the event of mains power failure. Particular attention is to be paid to minimising the effect of control system failure. Each power-operated sliding watertight door shall be provided with an individual hand-operated mechanism. It shall be possible to open and close the door by hand at the door itself from both sides.

(SOLAS 74,2020)

Access doors and access hatch covers normally closed at sea, intended to ensure the watertight integrity of internal openings, shall be provided with means of indication locally and on the bridge showing whether these doors or hatch covers are open or closed. A notice is to be affixed to each such door or hatch cover to the effect that it is not to be left open. (SOLAS 74,2020)

Woodchip carriers are installed shuttle conveyor and shuttle conveyor room. The shuttle conveyor swings out from the ship's side hull through opened watertight doors. There should be evidence of watertightness testing of the watertight doors at regular intervals.

8.5 Are the officers and ratings aware of the location of the emergency stop devices; are the emergency stop devices in good order and is there recorded evidence of regular testing? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The emergency stop devices at each operator-controlled stations should be tested regularly to ensure that they are functioning correctly, and the result of these tests should be recorded.

8.6 Are light fittings in the tunnel conveyor Ex'd' rated and in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ex 'd' ex-proof type of protection, which responds to the European standard EN 60079-1, is based on the concept of containment. In this mode of protection, the electrical equipment is enclosed in special boxes, designed to contain a possible internal explosion and to prevent its transmission to the outside atmosphere.

The manufacturers or Administration's certificate approving the fittings for use in gas-hazardous areas will be invalidated if the correct bolts for securing the cover, or the correct light bulb size, are not used.

Particular attention should be paid to the following:

- > Cracks in metal, cracked or broken glass or failure of cement around cemented glass in flameproof or explosion-proof enclosures
- > Covers of flameproof enclosures to ensure that they are tight, that no bolts are missing, and that no gaskets are present between mating metal surfaces
- > Each connection to ensure that it is properly connected
- > Possible slackness of joints in conduit runs and fittings

8.7 Are lights and warning devices of the self-unloading system regularly tested and the result recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall witness the testing of lights and warning devices where possible.

Tunnel rate lights, tunnel horn, deck horn, tunnel warning beacon, deck warning beacon and boom lighting and the end light (red flashing) of boom conveyor where applicable should be, tested regularly.

The electrical equipment inside the tunnel shall be intrinsically safe/explosion proof.

8.8 Is there a procedure for suppression of dust during cargo operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the method of dust suppression in the conveyor tunnel is not effective.

Record the type of suppression system that is employed on-board the ship.

There is always an element of dust produced by the product during cargo operation. Procedures, personnel protection and effective methods to suppress dust in the conveyor tunnel and on the boom conveyor shall be provided by the vessel's manager.

Example of methods used to suppress dust are:

- > Dry fog system
- > Water mist-spray system
- > Dust extraction system
- > Foam dust suppression system
- > Belt conveyors and boom conveyors were protected by cover – steel plates across the conveyors or portable ones for example tarpaulin made.

Maintenance of dust suppression systems shall be incorporated in the PMS as per manufacturer recommendation.

- 8.9 Is the belt cleaning system/scrapper in good order and maintained as per manufacturer recommendation and is there evidence to show that the belt cleaners have been inspected regularly for proper function? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The necessary maintenance and repairs as per manufacturer recommendation shall be incorporated in the PMS.

Conveyor Skirt Boards are designed to secure material on conveyor belting and mitigate excessive spillage during cargo operations. The skirt board arrangement could consist of wooden boards, chute lining, worn belts, or skateboard rubber, and should be maintained in good condition as per the manufacturer's recommendations.

- 8.10 Are the belt speed- sensors, belt scales and slip detecting device, where applicable, in good order and maintained as per manufacturer recommendations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A belt scale is used to measure the flow rate of the cargo.

The vessel's manager shall provide guidelines for selecting various belt speed and discharge rates and precautions to avoid overloading the self-unloading system.

- 8.11 Are the cargo hold discharge gates and watertight doors clearly marked to indicate the hold and spaces they serve? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Rightship recommends that the ship shall be provided with the hold discharging gates and watertight doors identification numbers used in the loading or unloading plan. The location, size and colour of these numbers should be chosen so that they are clearly visible to the cargo-gate operators.

- 8.12 Has the hydraulic oil of the self-unloading system been tested regularly for contamination and deterioration as per manufacturer recommendation? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Monitoring hydraulics with oil analysis is the only way to establish whether the hydraulic fluid is impacting the anticipated performance. Contamination is the main cause of failure, typically water contamination or dirt ingress. In addition to the Elemental and Contaminant tests, the hydraulic oil of the self-unloading system shall also be subjected to a Particle Count test.

- 8.13 Are the tunnel-conveyor emergency stop, speed switch, and belt misalignment switch in good order and inspected regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

- 8.14 Is the tunnel-conveyor hydraulic system arrangement in good condition and free from oil leakage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The pipelines, valves and shafts should be in good condition. If there is evidence of oil leakage this must be recorded as a Finding.

8.15 Are boom dust cover, spill tray and telescopic chute arrangements, where applicable, adequate? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The belt and boom conveyors shall be covered by fixed and/or portable means, i.e. fixed steel plate across the conveyors or portable type, for example made of tarpaulin.

8.16 Has the sequence of the cargo plan been followed by the cargo-gate operators, and is the movement of the cargo-gate operators reported to the duty officer and is there evidence to confirm that movement of cargo-gate operators has been monitored to ensure compliance with the cargo plan? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.17 Is the vibrator system in good order and is there recorded evidence of regular inspection as per manufacturer recommendations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.18 Is the lift conveyor in good order, free of corrosion, damage and leakage and is the spillage chute of the lift conveyor, where applicable, collecting the cargo spillage effectively? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Any cargo spillage from the cleats shall be collected in the spillage chute, located near the base of the vertical belt, and directed back onto the belt.

8.19 Are walk-rounds of the entire system carried out regularly, recorded and are the staff conducting the rounds aware of their duties? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A walk around checklist that covers the most important aspect of the check should be incorporated in the SMS. There should be a field inspection patrol schedule developed to detect any conditions or components that need attention.

8.20 Is the condition of belt tension checked during cargo operations and is there evidence to show that the belt tension and alignment is inspected regularly for proper function? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Material spillage from the loop belt may be as a result of incorrect belt alignment and low belt tension. The belt shall not move in a zigzag direction.

The operational hours of the conveyor belt should be logged, and essential maintenance tasks, including alignment and replacement of the conveyor belt, should be carried out according to the manufacturer's recommendations.

8.21 Are the tunnel trash pumping and disposal arrangements in good order and is the overboard valve provided with a notice warning against discharge of oil or oily mixture? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The trash pumps are provided at the tunnel bilge wells and are also capable of pumping out trash.

The accumulated water inside the tunnel may be contaminated. The water may consist of oil from hydraulic oil leaks that may occur at each unloading or contain cargo residue.

Adequate arrangements should be in place for the disposal of water from the tunnel conveyor. The bilge strum box shall be clean and free of cargo residue.

Care must be taken to ensure that the water is free of contamination before disposal.

8.22 Is a water-level alarm system provided for the tunnel bilge well, and is the alarm system regularly tested and the result recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should test the bilge-well high-water-level audible and visual alarms when possible.

8.23 Is the boom conveyor in good order, free of corrosion, damage, and oil leakage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The maintenance and inspection of slewing and luffing components shall be incorporated in the PMS as per manufacturer recommendation. Baffle plates should be intact, in good condition and free of damage, deformation and holed. The boom platform and railing should be maintained in good order.

8.24 Are the boom conveyor and /or shuttle conveyor, where applicable, walkway lights and floodlights in good working condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.25 Is all electric equipment including cable box, cable run and cable tray and shore interlock connection system on the boom conveyor in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The shore connection system links to shore conveyor and it makes the ship's conveyor to stop if shore or ship's conveyor stop.

8.26 Are the officers aware of the slewing/swing out limits of the boom conveyor/shuttle conveyor and is there recorded evidence of verification of function of slew/luff limit switch? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Inspector shall review the evidence of regular testing and maintenance of the limit switches.

8.27 Are the idlers and return rolls of the self-unloading system free from build-up material and are they rotating freely? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Idlers and return rolls should be checked for material build-up and freeness of rotation. Rollers can be overheated due to either a bearing failure or to being jammed with refuse and overheating causing and sustaining a fire for a period of time.

The vessel should carry a sufficient quantity of spare parts.

The rollers grease fittings should be in place and in good order. The recommendation of manufacturer for selection, application and replenishment of correct lubricant shall be followed.

8.28 Is the ventilation system of the tunnel area in order, and is there documented evidence to confirm that the ventilators are working to their operational capacity? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The wearing of paper face masks by operators inside the tunnel proves poor air quality. Record a Finding if the quality of air inside the tunnel was poor. Record in comments whether the vessel was equipped with dust- removal or air-filtration system.

RightShip recommends that capacity of ventilation system checked regularly, and fans were operating to capacity.

8.29 Is the tunnel area clean free from refuse and is the overall standard of housekeeping satisfactory? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Particular attention should be paid to any loose items which may fall on the belt and damage the belt or hoppers between the belts.

8.30 Is the vessel is provided with guidelines for 'the prevention, early detection and suppression of fire in the cargo conveyor tunnel' and are the crew familiar with the guidelines? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Rightship recommends shipowners consider fitting a remote fire-fighting system in the tunnel and associated areas. Record in comments whether the cargo conveyor tunnel was protected with a fixed fire detection and fire extinguisher system.

Risk of fire shall be controlled by regular monitoring to ensure effectiveness of control measures taken with respect to potential ignition sources. If the vessel was provided with a fixed fire-detection and firefighting system, the inspector shall review the documented evidence of regular checks.

Except for ro-ro and vehicle spaces, cargo spaces on cargo ships of 2000 gross tonnage and upwards shall be protected by a fixed carbon dioxide or inert gas fire-extinguisher system complying with the provisions of the Fire Safety System Code, or by a fire-extinguishing system which gives equivalent protection.

The administration may exempt from the requirements of paragraphs 7.1.3 and 7.2 cargo spaces of any cargo ship if constructed, and solely intended, for the carriage of ore, coal, grain, unseasoned timber, non-combustible cargoes or cargoes which, in the opinion of the Administration, constitute a low fire risk. Such exemptions may be granted only if the ship is fitted with steel hatch covers and effective means of closing all ventilators and other openings leading to cargo spaces. When such exemptions are granted, the administration shall issue an Exemption Certificate, irrespective of the date of construction of the ship concerned, in accordance with regulation 1/12(a)(vi) and shall ensure that the list of cargoes the ship is permitted to carry is attached to the Exemption Certificate.

(SOLAS 74,2020)

8.31 Does the vessel stability information provide guidance for shipboard personnel on the effect of flooding the tunnel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The introduction of water into the tunnel in order to submerge the conveyor system in case of fire creates a free surface area which, under the sheltered conditions of the harbour where there is no motion of the ship, presents little danger of significant stability loss. Under different circumstances, where motion could be imparted by external forces, such as wind and seas or by a cargo shift, and particularly if at sea when subject to rolling and pitching, the dynamic surging of the water will not only cause massive adverse effects on stability but can also cause considerable structural damage.

8.32 If the electric voltage power of vessel is greater than 1000 volts (high voltage), is staff suitably trained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comment which officers have undergone Shipboard High Voltage Training and the nature of such training.

A vessel equipped with high voltage systems above 1,000 volts should have on board certificated engineer officers as per STCW Table A-III/2 of chapter. Holders of a Certificate of Competency of Electro-technical officer according to regulations III/6 of the annex to the STCW-Convention fulfil this requirement.

8.33 Is the vessel equipped with sufficient intrinsically safe portable radios for use inside the tunnel and, are the radio 'dead zones' within tunnel identified and safeguards put in place? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Section 8F: Cargo Operation - Heavy lift vessel

Note: This section pertains to specialised ships that can transport non-standardized heavy cargoes. These specialised ships can be divided into four main categories: project cargo ships, open deck cargo ships, dock ships, and semi-submersible ships. This section does not apply to General cargo ships when transporting heavy lift (heavy lift Carrier).

8.1 Has the final lifting and rigging plan been developed and documented as the procedure for the safe lifting of heavy cargo? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

For any heavy lift, it is crucial to develop and document a lifting and rigging plan. This plan should outline the procedures for the lift and include precise information regarding the centre of gravity of the unit, the intended lifting spread to be used, and the calculations for rigging stability, often referred to as the lifting triangle.

The plan should include a defined method for assessing vessel stability. The following recommended approach is provided for consideration:

- > For the initial stability check: virtual centre of gravity or Kaps method,
- > For two-chain suspensions: Nikitib method,
- > For detailed and critical lifts: numerical simulation.

It is crucial to assess all the disturbing factors, including wind, rigging tolerances, crane movement, vessel movement, shifts in the centre of gravity, steering line forces, and friction at lift points.

For every heavy lift, the wind is an important factor for the safety of the operation. The maximum acceptable limits of wind speed are typically established through experience or by following established guidelines. To analyse the impact on the stability of lifts, various analytical formulas may be utilised to calculate the forces acting on the lifted structure and the resultant tilt angles of the suspended rigging arrangement.

The calculations for the wind profile, which include the logarithmic profile with a reference height of 10 metres, as well as the force equations, should be executed with accuracy, ensuring that conservative limits are set.

The plan should clearly define the operational wind and weather limits by detailing the maximum Beaufort scale or specific wind speeds, and this information should be thoroughly documented in each lifting and rigging plan.

The sliding and tilt angles should be determined for each shape—flat, cylindrical, conical, and box—utilising conservative friction coefficients.

The stability assessment should consider vessel motions and heel effects by examining vessel pitch and roll, as well as changes in trim and heel, and their influence on GM and GZ curves.

The plan should specify the rigging tolerances, SWL, MBL, and friction values. Furthermore, load assumptions, such as SWL, MBL, friction factors, and tolerance ranges, should be documented clearly.

For further information, reference should be made to the Guidance on Stability of Lifts (click [here](#)), which provides an overview of methods for assessing the stability of lifting arrangements.

8.2 Was a specific risk assessment provided for the heavy lift vessel, and are all lifting operations formally risk assessed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should record a Finding if any task was initiated when the residual risk rating, after implementing all control measures, was above medium risk.

The risk assessment document should evaluate the risks associated with a heavy lift operation at each stage of the process and should include a contingency plan to address unforeseen internal and external circumstances. Each hazard should include a description of the potential risks, the existing controls in place to mitigate those risks, an initial risk rating calculation, and an identification of any additional controls required to further reduce the risk.

8.3 Does the heavy lift vessel have a competence and training matrix that addresses the planning and execution of lifting operations, as well as the maintenance and operation of lifting equipment and ballast control operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should verify that the crew is adequately trained and certified to conduct inspections of lifting equipment. The inspector should check that refresher training is conducted for specialised roles, such as offshore crane operators and ballast control operators.

All persons involved in planning/performing lifting operations and maintaining lifting equipment shall be trained and competent for their role. Refresher training and periodic assessment is necessary to assure competence. The Safety Management System should establish clear competency standards for all critical roles and set out a defined process to ensure that personnel involved in planning and executing lifting operations are suitably qualified and capable. The inspector should check for the availability of a training matrix and confirm that the crew engaged in lifting operations, including crane and ballast operations, has received the requisite training. Additionally, it is essential to ensure that records of training and certifications are maintained on board.

8.4 Are there documented procedures in place by the vessel's manager to ensure that the Master receives the necessary pre-voyage guidance? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The vessel manager's procedure should include guidelines on various issues such as weather restrictions, voyage routing, motion limitations, project-specific lift analysis, and the transport manual.

8.5 Was the heavy lifting vessel equipped with an effective lifting equipment management system? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

An effective lifting equipment management system should include the following key elements:

- > All equipment must be clearly marked to correspond with its certification.
- > Valid certificates for each item should be readily available on board.
- > Defined inspection and maintenance standards should apply to lifting equipment, including loose gear and associated components.
- > Clear guidelines should be established for the safe retirement of wire ropes.
- > Robust controls should be in place to prevent unauthorised alterations to lifting equipment.

8.6 Does the heavy lift vessel have procedures in place that cover the general safety arrangements for activities conducted on the exposed working decks? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The procedure should include requirements for PPE, lifejackets, lifebuoys, safety signage, and the availability of fall arrestor equipment for multiple users in exposed areas.

8.7 Do the emergency procedures address additional scenarios related to the vessel heavy lift operations, including loss of stability or watertight integrity, loss of moorings or station keeping, emergency disconnect, helicopter accidents, severe weather, tidal waves, tsunamis, and solitons? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Evidence should exist showing that personnel have been familiarised with scenarios related to emergencies associated with heavy lifting.

8.8 Is there a qualified individual responsible for ballast control and stability calculations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the vessel is classified as a Mobile Offshore Unit (MoU), the ballast control operator must comply with the competency requirements outlined in Table 6.4 of IMO Resolution 891. For vessels that do not fall under the MoU category, RightShip recommends that the vessel's manager arrange specific training for the individual responsible for ballast control operations to ensure safe and effective practices.

8.9 Do the training records of the Ballast Control Operators (BCOs) show their understanding of the vessel ballast system, the management of free surface effects, and the implications of unintentional ballast shifts? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector needs to examine the records for induction and vessel-specific training. If the heavy lift vessel is semi-submersible, the BCO should be capable of taking the unit to survival draft if necessary.

8.10 Does the latest cargo operation plan include stress and stability information, have the relevant calculations been carried out for the current operation, and are the BCOs fully aware of any operational limitations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Inspectors should verify that stress and stability calculations have been completed prior to cargo transfer, encompassing the initial, intermediate, and final stages. Continuous monitoring throughout the operation is essential to ensure conditions remain within the vessel's design limits.

8.11 Is there a calibrated inclinometer positioned near the ballast control panel? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

8.12 Is the draft remote gauging system in good condition, and does it accurately reflect the vessel's actual draft? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The vessel's manager should establish the procedure for calibrating the remote draft gauging system and for regular maintenance according to the manufacturer's guidelines. The inspector should cross-reference the visual readings with the remote draft gauging system.

8.13

Are the tank remote gauging systems, including the level alarms, regularly maintained and calibrated according to the manufacturer's recommendations, is there objective evidence confirming the regular verification, calibration of tank gauging systems, and testing of level alarms? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Tanks should be manually checked at least once a week, and the readings should be compared to those from remote gauges. Discrepancies should be documented and accessible to the BCO.

The sounding tubes must be clear, and the sounding pipes should be labelled to indicate the tank they serve, as well as being equipped with a cap.

Section 9A: Hatch Cover and Lifting Appliances

9.1 Are the cargo holds, including the underside of hatch covers, free of loose rust scale and paint flakes? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Underside and internal structure of hatch panel should be free of loose rust scale, paint flakes or blistering of paint coatings.

Hatch covers with a double skin, in the form of a closed box, are filled with inert gas. After structural repair, the inner spaces must be re-inerted. This is done by inserting special tablets (available from the hatch cover manufacturer) into the space and welding shut. Never allow water to penetrate the box construction.

(A master's guide to hatch cover maintenance, 2002)

In general, the holds should be cleaned so that there are no residues of previous cargoes, no loose rust scale, paint flakes or blistering of paint coatings.

9.2 Is there a procedure in place for the safe operation and inspection of hatch covers and, has the Master/Chief Officer received appropriate training in hatch cover inspection and maintenance?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends the Master and Chief Officer complete a hatch cover inspection and maintenance training course pertinent to the type installed on the vessel being inspected. The focus of the course should be to familiarise them with the operational principles, limitations, maintenance requirements, and occupational safety considerations related to the shipboard installations.

Where not bespoke, the formal training course can be of a generic nature but this must be supplemented with type-specific material for the hatch covers fitted. The vessel's manager should strongly consider using audiovisual materials, as they have been proven to enhance knowledge retention and the learning experience.

The term "Audiovisual" is used to describe the combination of audio, video, and digital technologies that deliver information in an engaging and creative manner, assisting learners in better understanding concepts.

Participants should receive a training certificate(s) that clearly indicates the duration of the course(s), method of delivery, provider, type of hatch cover (where applicable) and date of training.

Record which officers have undergone Hatch Cover Inspection and Maintenance Training and the nature of such training.

Rubber Packing: The design compression of rubber packing in a hatch cover, as specified by the manufacturer, can be found in the manufacturer's hatch cover manual along with other design criteria for the specific rubber packing.

Bearing Pads: Bearing pads will wear down based on their type, age, and the loads acting upon them. As bearing pads wear down, the compression of the packing rubber will increase; therefore, the wear process should be regularly monitored by checking the skirt clearance. Information regarding the skirt clearance and maximum allowed wear on the bearing pads can be found in the manual.

Locators: The locators (sliding positioners) ensure that the panels are properly guided and positioned when being closed. This means that the packing rubber at panel intersections is properly compressed up to the manufacturer's design compression and that panels are properly aligned and centred. The locators are important for achieving tightness; therefore, regular inspection and measurement of clearance are required.

Stoppers: Once the hatch cover is properly closed in port (static condition), the panels should remain in their correct sealing position throughout the voyage while the ship is in a dynamic condition at sea. Relative movements will tend to distort the panel arrangement, alignment, and geometry, but to prevent forces acting on the hull girder from being transferred directly into the panels, some limited movement should be allowed. The clearance of these stoppers, sometimes called rolling or pitching stoppers depending on their working condition, should be measured regularly.

The vessel manager should establish a procedure and ensure that the clearance measurements of hatch cover components are carried out regularly every six months.

The Master and/or Chief Officer shall be able to produce appropriate documentation that Hatch Cover Inspection and Maintenance Training has been undertaken.

The training should provide the ship's staff with some useful information on the role of key elements, such as what to look for, how to undertake a proper inspection prior to sea passage and make corresponding entries in the logbook as ultimate proof of due diligence, which would greatly contribute to ship and cargo safety.

On vessels with hatch covers, the SMS should include the following:

- > A risk assessment for opening and closing hatch covers and working in cargo holds;
- > Instructions and procedures to ensure the safe operation of hatch covers and associated equipment; and
- > Defined levels of authority and lines of communication amongst shipboard personnel during such activities.

The procedure should emphasize the following points:

- > Assuring hatch covers are secured at all times, whether open or closed, unless they are being operated;
- > Keeping employees away from moving or pressured machinery; and
- > Practicing safe working at height techniques when working near an open hatch cover.
- > Communication between seafarers is critical for the safe operation of hatch covers. Coordination between the crews of vessels and establishing an environment where crew members speak out when they sense something risky can help prevent these types of serious incidents.
- > Attention is drawn to the dangers of proceeding to sea without fully secured hatch covers. Securing of all covers shall always be completed before the commencement of a sea passage.
- > During voyages, especially on loaded passages, cover securing devices and tightness of cleating and securing arrangements shall be checked, especially in anticipation of, and following periods of, severe weather.
- > Voyage checks shall consist of an external examination of the closed hatch covers and securing arrangements in anticipation of, and after, heavy weather but in any event at least once a week, weather permitting. Particular attention shall be paid to the condition of hatch covers in the forward 25% of the ship's length, where sea loads are normally greatest.

(Annex 2 resolution MSC.169(79) 2004)

Subject to meeting any special provisions detailed in the manufacturer's operation manual, hatch covers should only be opened on passage, when necessary, during favorable sea and weather conditions.

Each vessel should carry and maintain an up-to-date list of the minimum recommended spare parts for the safe operation and to support both planned and corrective maintenance of type of hatch cover, as specified by the manufacturer.

9.3 Are adequate procedures in place for carrying out hose and / or ultrasonic testing of hatch covers and is documented evidence of such testing available? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends that the weather tightness of hatch covers should be checked at least once every 3 months. Record the date and type of last test in comment.

In case of steel cargo shipment, weather tightness of hatch covers should be carried out by means of ultrasonic testing, before loading and in the presence of the loading surveyor.

The "means for securing weather-tightness" of hatch covers is defined in regulation 16.4 of the load line convention which states: "The means for securing and maintaining weather-tightness shall be to the satisfaction of the Administration. The arrangements shall ensure that the tightness can be maintained in any sea conditions, and for this purpose tests for tightness shall be required at the initial survey and may be required at periodical surveys and at annual inspections or at more frequent intervals."

(Load Lines, 2005)

The following parameters can be used for a hose test: Water pressure 2 bar, Nozzle size 15 - 18 mm, Spraying distance 1 - 1.5 m.

(Wet Damage on Bulk Carriers, 2018)

When carrying out an ultrasonic test, the instructions of the manufacturer should be followed. The transmitter is first placed in the cargo compartment, an open hatch value (OHV) is then obtained, after which the detector is passed along each seal in turn to find any leakage of ultrasonic sound. When the test is completed, any area giving reading in excess of 10% OHV, indicates a point where water ingress is possible. The use of ultrasonic testing equipment operated by a certified person is widely recommended when weather tightness testing of hatch covers is required. (Bulk Cargoes: A Guide to Good Practice, 2016)

The tests are only part of the loss-prevention process. Well-maintained seals, cleats, supports, drains and other hatch cover components are the key to achieving weather tightness.

9.4 Are the compression bars and the coaming tops' water channels clean, free of corrosion and maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Effective sealing is only possible with a straight, undamaged, and non-corroded compression bar. Compression bars which are not in this condition should be repaired or replaced, taking care to align the bars properly.

Hatch coaming tops and the double drainage channels should be clean, free of corrosion and obstructions.

(Hatch Cover Maintenance, 2015)

9.5 Are the drain holes of coamings clean, the inboard coamings faces free from any vertical rust staining and are the non-return valves in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Damaged, missing, or defective non-return drain valves should be repaired or renewed. Where applicable, the fire cap of the drain valve should be firmly connected by a lanyard to the valve and maintained in good condition.

Check that the drainpipes and drain valves are not clogged. Maintenance of non-return valve should be incorporated in the plan maintenance system.

Any rust stains on the inboard coaming could be a sign of water leakage, especially at the cross-joints or split-joints.

(Hatch Cover Maintenance, 2015)

9.6 Are quick-acting cleats and crutches all in place and in good working condition with the rubber washers flexible and free from paint and cracks? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Quick acting cleats should be fitted at their original positions free of any corrosion or bending. The rubber washers that are fitted to the cleats should have proper elasticity. Other component parts to which cleats are welded or acting on such as coaming table, crutches, snugs, panel side plating should be in good condition.

(Hatch Cover Maintenance, 2015)

9.7 If the hatch panel side and end plates are in steel-to-steel contact with the coaming tops when closed, are the coaming tops free from grooving or wear? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

9.8 If the hatch covers are supported by bearing pads, are they free from wear or damage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Bearing pads are fine pieces of engineering and replacing them with non-original spares or non-compatible steel could result in serious problems. Often bearing pads are replaced or repaired by the ship's crew with only one thing in mind, i.e., restoring the height of the bearing pads, whereby the correct size and use of appropriate material for the mating surfaces is overlooked. For bearing pad adjustment it is strongly recommended that manufacturers or specialists are called in for advice.

(Hatch Cover Maintenance, 2015) (Vervloesem, 2017)

9.9 Are the side and cross-joint rubber seals in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Packing rubbers are designed to be compressed to a certain depth, which is generally referred to as the design compression (rule of thumb for estimating the design compression of ordinary box-type packing rubbers is as follows: design compression = 25% of the nominal thickness of the packing rubber). Depending on the type of rubber packing, design compression will generally be in the range of 4-20mm, and this is either specified in the maker's manual or indicated in the drawing.

The minimum length of replaced rubber packing should be one metre.

Rubber packing and adhesive has a limited shelf life, so check the date stamp and discard if beyond the use-by date. Manufacturers approved spare parts should be used.

The corner pads, joints and end pieces of rubber sealing should be intact, properly glued and in the correct position.

Whenever packing rubbers need replacing, it is extremely important to ensure that not only are the dimensions compatible, but also that the alternative product will meet with the required performance criteria.

(Hatch Cover Maintenance, 2015) (Vervloesem, 2017)

9.10 Are hatch cover panels free of misalignment? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

The permanent deflection of rubber seals should be in the centre of the rubber and not to one side. Non-central permanent set may indicate misalignment of panel.

(Hatch Cover Maintenance, 2015)

9.11 Are seal retaining channels in good condition and free of corrosion? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

Where the rubber seal retaining channel (socket) is corroded, rubber gaskets cannot be stuck in the socket, and consequently will lose their function.

(Marine Order 32 (Cargo handling equipment) 2016, 2017)

The sides of the panels in the way of the rubber seal retaining channel and edges should be in good condition and free from distortion.

9.12 Are the cross-joint seal retaining channels and the cross-joint compression bar straight, free of corrosion and damage, and are the channel supports and brackets in good condition? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

9.13 Are the following parts of the hatch covers, where applicable, all in good order and do they appear to be well maintained? (V)

- > Wheels/bearings or trackway
- > Hydraulic system including hoses
- > Chains
- > Link pin and safety pin
- > Cargo holds' ventilator on the side and end of hatch panel

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

When agreeable by Master, inspectors are required to seek that the hydraulic hose protector's wrapping be removed at random in order to conduct a visual and random check of these pipes."

Hydraulic hoses shall be inspected and replaced in accordance with manufacturer recommendations; in the absence of manufacturer recommendations, ISO/TS 171652:2018 table 1 and clause 8 shall be followed. The requirements for hydraulic hose maintenance must be included into the PMS. RightShip recommends that all flexible hydraulic hoses exposed on deck be replaced at least every 5 years.

9.14 Are cross-joint wedges and their wedge bridge (where fitted) in place, operational and effective? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

9.15 If hatch covers are hydraulically operated, has the hydraulic oil been tested regularly for contamination and deterioration? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The oil tank of the hydraulic system should be kept filled to the operating level and with hydraulic oil recommended by the manufacturer. The cleanliness and viscosity of hydraulic oil must be checked. Samples of the oil should be sent to a chemist for testing as per manufacturer recommendation. The hydraulic system is provided with bleed points.

Hydraulic oil should be replaced every five years or sooner if significant repairs have been carried out, such as piping or cylinder replacement. Hydraulic oil filters should be changed every twelve months or in accordance with the manufacturer's recommendations.

If the vessel's manager has a procedure in place for regular hydraulic oil analysis and the results confirm the oil remains within acceptable limits, the renewal interval may be extended, provided this does not exceed the maximum lifespan specified by the oil manufacturer. Objective evidence should include documented analysis results and confirmation from the oil manufacturer supporting continued use within the approved limits. All extensions should be recorded and retained for verification.

In addition to the Elemental and Contaminant tests, the hydraulic oil of the hatch cover system shall also be subjected to a Particle Count test.

It is essential that samples be taken from the proper location inside the system, and not from any bleed point. As bleed points exist just for captive loop or dead end air removal.

9.16 Are officers familiar with emergency hatch cover operation arrangements and is there evidence of effective training of personnel available on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The emergency hatch cover operation procedure should be incorporated in the ship's manual.

Emergency or auxiliary operation of hatch covers can be executed either by traditional hand pumps and wire pullers or by portable electric pump units, connected either to the on-board electric power system or to shore.

The portable electric pump unit makes the operation easy, swift and safe, as the unit is connected directly to the control stand with quick couplings including all safety enhancing valves active.

Rightship recommends vessels to carry an emergency portable pump unit (portable electric and/or hydraulic type) for situations when the standard pump unit cannot be used. The manufacturers' instructions for the maintenance and test of the emergency portable pump unit shall be incorporated in the PMS.

9.17 Has a thorough examination and load test of lifting appliances been carried out and is the record of the test and examination being maintained properly? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

All lifting appliances and every item of loose gear shall be thoroughly examined by a competent person at least once in every twelve months and five yearly load testing shall be carried out when the safe working load (SWL) of the lifting appliance is more than one tonne. Flag State may impose the quadrennial load test for the lifting appliances.

The term "competent person" means a person possessing the knowledge and experience required for the performance of thorough examinations and tests of lifting appliances and loose gear and who is acceptable to the competent authority.

(Register of Lifting Appliances and Items of Loose Gear, 1985)

The lifting appliance maintenance records should be updated and available on board.

9.18 Are the hoist and luffing wires of cranes, where fitted, reported in good order and is there recorded evidence of regular inspection and maintenance? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Poor maintenance of wire leads to weakening of the wire over time. Inappropriate operation of the crane and incorrect use of the wire leads to damage being sustained by the wire. Wire ropes should be examined for defects, and if those defects are excessive, advice should be obtained and/or the wire should be discarded. Example of defects that may be found on the wires are:

1. Broken wires and fractured strands
2. External and internal wear/corrosion
3. Decrease in elasticity
4. Kinks and other mechanical damages

There should be an appropriate test certificate for all wire ropes on board; that is for all of those in use and for all spare wire ropes. The certificates should give the date of manufacture, the material strength, the construction of the wire and the breaking load test of a sample. There should be an inventory of all wire ropes on board and records of the dates of renewal of the wires in use on all cranes.

(Cranes, Their Operation and Reasons for Failures, 2015)

Such inspection should be included under the PMS system. Crane wires and sheaves should be in apparent satisfactory condition.

9.19 Are the main structures, foundation structures and mountings of the cargo cranes free of apparent defect or damage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a finding if the rocking test was not conducted at least once every six months.

The main structures, foundation structures and mountings should be inspected to determine whether any defects or damage are present and, if any such defects/damages are present, appropriate repairs should be carried out.

(Cranes, Their Operation and Reasons for Failures, 2015)

The holding down bolts and slewing ring should be free of significant corrosion and crane access ladders and platforms in apparent satisfactory condition.

Such routine inspections should be included under the PMS system.

Rocking Test

To ensure that slewing ring bearings are not becoming worn, the vessel's planned maintenance system should include a rocking test, and each crane tested every six months. A record of these tests must be entered into the Registry of Lifting Appliances. The rocking test comprises the measurement of the deflection between the crane pedestal and the rotating crane housing in the way of the slewing ring using a dial test indicator to determine the bearing clearances.

Measurements are taken at the two locations on the slewing ring susceptible to the most significant bearing wear, under the jib's centre and diametrically opposite. A set of eight readings are taken with the crane pointing dead ahead, dead astern, to port and then to starboard, initially with the jib luffed to its minimum radius with no weight on the hook and then with the jib luffed to its maximum radius with the difference between the two readings giving the bearing clearance. The readings are compared against the manufacturer's maximum permissible clearance, which will depend on the type and size of the slewing ring and bearing size. The original bearing clearances should be available to determine the degree of wear. Consider shortening the testing intervals when the noted wear has increased appreciably between consecutive rocking tests. Some manufacturers have produced integrated devices to measure unacceptable wear of the bearing in their slewing rings to provide early warning of excessive bearing wear.

(Crane Slewing Gear Maintenance and Rocking Tests 2018)

9.20 Is the cargo crane's machinery and operator's cab, including the operator's cab controls, in good working order and inspected, tested, and maintained in accordance with manufacturer's recommendations?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspection of the crane cabinet's structural integrity should be included in the PMS.

The machinery of a crane includes all electrical control equipment and systems, all motors, hydraulic oil pumps, filters and coolers, and winches, together with winch brakes and control gear, all limit switches, cut-out switches and other pieces of equipment. Routine maintenance of these various pieces of machinery is essential for their continuing correct operation. Such routine inspection, testing and maintenance should be included under the PMS system as per manufacturer recommendation.

(Cranes, Their Operation and Reasons for Failures, 2015)

Operating instructions for use of a crane shall be posted inside the operator's cab. The crane / derrick safety devices should be in apparent operation and regularly tested. The window wiper, window's glass and driver's chair should be maintained in good condition.

9.21 Are the ship's grabs being maintained as per manufacturer recommendations and have the maintenance requirements been incorporated in the PMS? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship's grabs should be part of the ship's planned maintenance system and routine inspections and maintenance should be carried out. That routine maintenance and inspection regime should include the thorough inspection of all structures of the grabs and their mechanical parts and of any associated equipment, including any umbilical cables and control systems to ensure:

1. All parts are without defect or damage
2. Hydraulic oil reservoirs are filled to the appropriate level
3. All machinery and control systems are functioning correctly, and
4. All moving parts are free and well-greased.

Before each ship's grab is taken into use for cargo operations, it should be rigged to the cargo hoist wire of the ship's crane and should be carefully tested to demonstrate its full functioning capability. A record of that testing should be kept.

The technical specification for a clamshell grab will include its capacity in cubic metres, its weight in tonnes, its dimensions in metres and details of its operation. The capacity might be a single figure, or might be two or more figures, if spill plates or moveable panels are fitted to the grab, which can be removed or put in place to alter the capacity of the grab when closed. The capacity of typical grabs used for the loading and discharging of bulk cargoes using ship's cranes ranges from about 4m³ to about 16m³. The weight of the grab might be about 2 tonnes or as much as 12 tonnes. This should be shown on the name plate attached to the grab.

The weight of cargo which can be lifted by a grab depends upon the capacity of the grab and the density, or the stowage factor, of the commodity. When calculating the weight of cargo lifted by a grab, it should be remembered that it is likely that the surface of the cargo in the grab will be slightly peaked or crowned; such that a greater weight than the volume of the grab might indicate will be lifted. An allowance for this of 25% should be included in any calculation.

If the crane is designed for both hook operation and for grab operation, it is likely to be given two different SWL ratings by the manufacturer – one for hook operations and one for grab operations. It is usual for the grab operation rating to be 20% less than the hook operation rating. The two SWL ratings should be stated on the plate on the crane jib.

(Cranes, Their Operation and Reasons for Failures, 2015)

9.22 Are the loose gears of lifting appliances clearly marked and are the certificates of the loose gears available and traceable on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

It is important that the loose gears used for lifting appliances (including deck cranes and the engine room crane) are marked clearly with its SWL and a batch mark or number, where that mark can be verified by a test certificate. RightShip recommends that these test certificates are clearly labelled and kept in an easily accessible file. A method of correct identification and matching of individual certificates with the loose gears should be established on board. Loose gears without certificates must not be used on board.

Section 9B: Gantry Cranes

Note: This section only applies to a vessel equipped with gantry cranes that can travel along the main deck and be used for handling of general cargo like containers, paper bales and bulk cargo and/or opening of the hatch cover.

This section shall be completed along with "section 9- Hatch cover and Lifting appliance".

9.1 Has the vessel been provided with procedures for the safe operation of the gantry crane? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should randomly check the implementation of the procedures.

Every company should develop, implement, and maintain a Safety Management System (SMS) which includes instructions and procedures to ensure safe operation of ships and protection of the environment in compliance with relevant international and Flag State legislation.

The company should identify equipment and technical systems the sudden operational failure of which may result in hazardous situations. The safety management system should provide for specific measures aimed at promoting the reliability of such equipment or systems. These measures should include the regular testing of standby arrangements and equipment or technical systems that are not in continuous use.

(ISM code 2018)

Details of the following should be included in the SMS:

- > On-board training in the operation (driving) of the gantry cranes.
- > On-board training of companion persons to guide crane drivers when attaching and landing each load.
- > A familiarisation process and associated checklist related to safe operation of the gantry cranes.
- > Safety features that can be used to stop a gantry crane in an emergency.
- > Familiarisation of new crew members and contractors with the gantry cranes.
- > Permit to work system related to the gantry cranes.
- > Precautions required when working on deck while the gantry cranes are in operation.
- > Crane-handling operation (preparation, during operation and work after crane operation)/or
- > Hatch cover lifting and travelling operation.
- > Emergency travelling operation of gantry cranes.
- > Appropriate checklists.

9.2 If gantry cranes are used for lifting hatch covers, are the hydraulic cylinders used for lifting the hatch covers free from leaks and are the hatch covers being marked correctly for precise positioning of the lifting beams and hooks where applicable? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if a person rides on a hatch cover when it is being lifted by crane.

The hatch cover lifting devices consists of lifting frames and hydraulic cylinders. The maintenance of hydraulic cylinders shall be incorporated in the PMS in accordance with the manufacturer's recommendations.

9.3 Are check lists in place and being used to cover the check before use, starting the crane and releasing the crane from sea stowage? Is a procedure in place for safe travelling of the gantry crane on deck and are crew familiar with the procedure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Trained companion persons on the ship and yard should be used to guide the crane driver when attaching and landing each load.

Poor communication is one of the factors that contribute to the root cause of incidents related to gantry cranes.

There must be enough persons on each side of the deck with adequate means of communication when intending to travel the gantry crane. The crane operator must not start moving the crane until receiving confirmation that the crane rail is clear.

Installation of a track sweeper can minimise the risk of injury in event of collision.

9.4 Is the vessel provided with a specific isolation procedure for the gantry cranes? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The objectives of the ISM Code are to ensure safety at sea, prevent human injury or loss of life, and avoid damage to the environment, in particular to the marine environment, and to property. (ISM Code 2018)

RightShip recommends that the isolation procedure should cover the need to isolate, de-energise, lockout and tag out systems, with the aim of reducing the risk of death or injury during operation, inspection, repair, maintenance, and assessment of gantry cranes.

The isolation procedures shall incorporate the following:

- > Isolation procedure, check list and risk assessment documented and accessible to the relevant crew on board.
- > Information, instruction, and training provided to crew involved with the system.
- > Supervisor appointed to make sure the crew strictly follows isolation procedures.

9.5 Are the visual and audible warning signals provided for gantry cranes in the deck area in good order and tested regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Locating the alarm bell near the wheels reduces the risk of collision, as it makes it easier to hear the alarm at the point of danger. All necessary warning signs shall be located on the crane legs.

Gantry cranes must be fitted with an acoustic warning device, such as horn that can be sounded by the crane operator, that:

- (i) emits an audible sound before travelling motion is commenced; and
- (ii) continues to sound until travelling motion has ceased.

(Marine Order 32 (Cargo handling equipment) 2016, 2017)

Gantry cranes and similar cranes shall be provided with a horn or other audible warning device operated by the crane operator to warn or attract the attention of any personnel within the operational area.

In case of travelling cranes moving at ground level, a continuous audible warning shall automatically be given when the crane is to move/is moving along the track/rails. The warning signal shall be distinctly different from other audio signals on the installation.

(Code for Lifting Appliances in a Marine Environment, 2009)

To alert personnel of gantry movement, the crane shall be equipped with four flashlights and warning bells mounted at each corner of the gantry. A signal horn operated from the driver's cab shall be provided. Red indicator lights shall be fitted on the extremes of the jibs and be automatically actuated when the jibs are in operation position.

9.6 Are the safety devices of gantry cranes in good order and is there recorded evidences of regular testing? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A list of safety devices shall be incorporated in the PMS and maintenance intervals as per manufacturer's recommendation shall be followed. The inspector should verify whether these tests have been included within the PMS.

Crane emergency stop pushbuttons/trip wires: The emergency stops should be located at all control positions and local panels around the cranes, clearly marked to indicate the crane they serve, and within easy reach.

Cow catchers: The cow catchers are mechanical switches mounted on each corner of the gantry operated by a collision bar mechanism.

End stop limit switches: Each gantry has hardwired slow down and end stop limits at both sides of its movement range. Hoist, trolley and shift trolley movements have software limit switches called smart slow down. Hardwired end stop limits are activated only in fault situations.

The over hoisting/lowering end stop limit switches will activate if the main hoist would rise/lower over its normal stop position. Main trolley traversing and shift trolley shifting end stop limit are rotating limit switches mounted on the machinery. Gantry travelling has a rotating limit switch, after crane backward direction has also proximity limit switch as end position.

Overspeed protection: The main hoist, main trolley traversing and gantry travelling motions are protected against overspeed during operation.

A gantry crane shall be fitted with a clearly labelled emergency stop switch that can easily be operated by persons at deck level.
(Marine Order 32 (Cargo handling equipment) 2016, 2017)

9.7 Are the main hoist overload protection and unbalance detection systems in good order and tested as per manufacturer's recommendation? Are officers aware of the safe operation parameters of gantry cranes and is this information posted in the ship's office/ ballast control room? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship's condition during load handling, hatch cover lifting, jib turning and crane sea stowing as well as the wind speed during crane operation shall be posted in the ship's office/ballast control room.

The overload protection is designed to prevent lifting loads that exceeds the lifting capacity of the crane. The Chief Officer shall be aware of the maximum load that the main hoist can load at all possible places and the heaviest load that must be loaded at the centre position. This information shall be posted in the cargo control room.

The load unbalance detection system shall prevent movement of eccentric loads.

During storm wind events, the gantry cranes and their associated structures are exposed to loading which may not have been considered in the design and construction or during the operation of the crane. Crew should be familiar with and understand the design/operating limitations of the gantry cranes with regards to wind speed criteria.

9.8 Is each gantry cranes provided with a storm locking arrangement and, does the arrangement appeared to be well maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The gantry cranes shall be fitted with devices that lock the crane in position when exposed to wind pressure.

(Marine Order 32 (Cargo handling equipment) 2016, 2017)

If a hydraulically operated locking system has been provided, the components of the system, i.e., hydraulic cylinder actuation and hydraulic hoses, shall be in good condition and the system shall be free of oil leaks.

9.9 Are the gantry cranes electrically bonded and earthed to the vessel's structure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A bridge or gantry crane used in loading or unloading must be electrically bonded and earthed to the vessel's structure.

(Marine Order 32 (Cargo handling equipment) 2016, 2017)

ANSI/NFPA 70 National Electric Code within Article 610, Cranes and Hoists, section 610.61: "The trolley frame and bridge frame shall not be considered electrically grounded through the bridge and trolley wheels and its respective tracks. A separate bonding conductor shall be provided."

9.10 Are the mechanical brakes for the gantry crane motions in good order, tested regularly, and recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Mechanical brakes for the crane movements are primarily designed as holding brakes. Deceleration forces are normally provided by the electrical braking of the drive motors. During an emergency stop, the mechanical brakes are engaged immediately, but without the deceleration force from the motor. The condition of the brake linings should be checked regularly.

Procedures for checking the air gap and the thickness of the brakes' lining materials should be incorporated in the PMS as per manufacturer's recommendation.

9.11 Are the anti-collision and deceleration devices of the gantry cranes in good order and tested regularly, and are track wheels at the deck level fitted with foot guards? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If there is more than one crane on the same trackway – each must be fitted with a device to prevent collision. Any track wheels at deck level shall be fitted with foot guards.

(Marine Order 32 (Cargo handling equipment) 2016, 2017)

The Anti-collision system will stop the cranes if the fore crane and aft crane would come too close each other. Deceleration devices, such as limit switches and two-stage decelerators, slow the gantry crane down regardless of what control is being pressed when a travelling crane is approaching the end of the runway, to prevent end stop collision or overrun.

9.12 If the operator's cabin of a gantry crane travels with the horizontal movement of a load, is the crane operator able to leave the cabin safely in case of power failure or another emergency? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If the operator's cabin of a track-mounted crane travels with the horizontal movement of a load, the structure must enable the operator to leave the cabin safely if there is a power failure or other emergency.

Example of an appropriate structure: A mobile or portable access or means of returning the cabin from the track to the point of access.

(Marine Order 32 (Cargo handling equipment) 2016, 2017)

The emergency escape routes shall be clearly marked.

9.13 Are portable fire extinguishers located around the gantry cranes and in good condition, and are they maintained regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

9.14 Are the driver's cabs and local operating stations of the gantry cranes maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The cabin panel shall be labelled, and the function lights shall be in working condition. The window wiper, window's glass and driver's chair should be maintained in good condition.

The local operating stations are used for the purpose of the gantry drive, hatch cover and JIB operation.

The switches, instruments such as controllers, telephone, microphone, speaker, heater, air-conditioner, electric window wiper, fire extinguisher, rotating chair and cabin's lights should be in good condition. The maintenance of driver's cab should be incorporated into the PMS.

9.15 Are the retractable jib roofs, sides' curtains and their securing arrangement being maintained in good condition, are checklists for the operation of the jib roof and curtains available and is the evidence of their consistent use? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The jib roof end limit switches and emergency stop pushbutton shall be function tested regularly and evidence of such tests shall be recorded.

The gantry cranes are fitted with retractable roofs, which cover the whole working area including the outreach area which enables operation in unfavourable weather. Some vessels are also equipped with curtains on the sides that enable work in the event of most adverse weather conditions. The side curtains are roller mounted horizontally on the outside of the main girders. At the bottom, the curtains are equipped with counterweights and supported by the girder.

The jib roofs' locking arrangements shall be maintained in good condition.

9.16 Is the telephone system provided for communication between driver's cab, electrical control room, hatch cover control station and deck level in good order and is there evidence of regular testing and maintenance? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following considerations should be taken into account if radios are to be used as a means of communication:

- > The radio should be tested prior to use to ensure clear and reliable transmission.
- > The crane should use a separate working channel, if possible, and if frequencies are available.
- > The crane cab operator's radio should be equipped with a hands-free system.
- > Radios should not be used for personal communication or discussions unrelated to the crane operation at hand.
- > Ensure that the crane operator and crew communicating with the crane operator are trained in the proper use of the radio equipment.

9.17 If a platform has been provided on the top of the crane roof for helicopter winching operations, does the area comply with the requirements of ICS guidelines? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The positioning and marking of the winching operations shall comply with the requirement of section 4.4.1 and 4.4.3 of the Guide to Helicopter/Ship Operations.

9.18 If the gantry crane is equipped with a pilot ladder hoist, is the system being by-passed and not in use? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

In December 2010 IMO MSC 88 adopted MSC.308(88), which contains amendments to SOLAS regulation V/23 and IMO resolution A.1045(27). This amendment provides additional recommendations for pilot ladder arrangements and pilot transfer arrangements. Amendments and changes to SOLAS chapter V/23 and IMO Resolution A.1045(27) have banned the use of mechanical pilot hoists.

9.19 Have a Crane Monitoring System and a port/crane performance logger been provided, are they in good condition and regularly monitored? Are faults recorded in the fault log, verified and fault-finding rectification procedures recorded appropriately? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Crane Monitoring System gives more detailed information about faults. All defects must be repaired before the crane is taken back into operation.

9.20 Is the emergency pump for the operation of the crane maintained in good condition, are crew familiar with their duties in the using the emergency pump and are emergency drills being carried out? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The crane is equipped with an emergency pump that can be used instead of the main hydraulic pumps. In case of electric fault, driving of hydraulic cylinders with a manually controlled valve is possible. However, the person operating the manual controlled valve will not be able to see the crane's operation clearly enough. To prevent accidents there must be an assistant(s) observing the motion of the hydraulic cylinders and all working parts. The assistant(s) must have phone communication with the person operating the emergency pump.

9.21 Is the steel structure of girders and trolleys free of deformation? Are the access and service platforms of the gantry cranes, including vertical ladders, ladders cages, rungs, stations and platforms being maintained, free of corrosion and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Random checks of following should be made by inspector to ensure that the gantry crane is free of apparent structural defects.

- > Cracks in the welds,
- > Deformed structural members,
- > Any negative camber in the bridge beam,
- > Operational damage due to collision with another crane,
- > Excessive structural corrosion

9.22 Are the cranes' transverse rail, rack, tooth-rack, travelling pinion and travelling wheel free of cracks, misalignment, and abnormal wear; and is there evidence of regular inspection and maintenance? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Parts of a crane are subject to wear by operation and deformation due to vibration. If wearing or deformation of any part have reached the prescribed limit, the part must be renewed. This limit is called the wearing limit. A procedure shall be in place for measuring the wear and comparing the measurements with the manufacturer's recommendation. The inspector should randomly check whether the wearing of parts is within the acceptable limit, as recommended by the manufacturer.

The travelling rail and rack shall be free of 1- obstructions on the rail, 2-cracks, 3-damage of the rail brackets, 4-deformation of the rail and 5-have adequate clearance at rail joint, as per manufacturer's recommendation.

The travelling wheel shall run smoothly without any abnormality in bearing revolution, free from 1-abnormal wearing on the wheel surface, 2-cracks, and 3- the fitting bolts should be tightened.

The travelling pinion shall be free of 1-cracks, 2-abnormal wearing or pitching and 3-abnormal engaging with travelling rack.

The gear boxes' oil level shall be maintained at normal level.

9.23

Is there evidence to show that regular maintenance has been conducted on the AC motors, electromechanical brake, gear reducer and the blower motors? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall check if the maintenance of main hoist machinery, turntable slewing machinery, shift-trolley shifting machinery, main trolley traversing machinery, gantry travelling machinery and hydraulic unit are incorporated in the PMS and being carried out as per manufacturer's recommendation.

9.24

Has the hydraulic oil of the gantry crane system been tested regularly for contamination and deterioration as per manufacturer recommendation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Monitoring hydraulics with oil analysis is the only way to establish whether the hydraulic fluid is impacting the anticipated performance. Contamination is the main cause of failure, typically water contamination or dirt ingress.

Section 10: Mooring Operations

This section is compiled with reference to SOLAS Regulation II-1/3, MSC.1/Circ.1619, and 1620, the IACS Recommendation No. 10 Rev. 5, and Section 5 (Mooring lines), Section 6 (Mooring winch), and Appendix B (Guideline for the purchasing and testing of mooring lines and tails) of The Mooring Equipment Guidelines Edition 4 (MEG4).

10.1 Has the company established guidelines and procedures for the inspection, maintenance and wear zone management of the mooring lines and are they being implemented? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall conduct random checks on the ship's mooring line management plan against Section 2 of INTERTANKO's Guidance on Line Management Plans. [Click here](#) to download the INTERTANKO guideline.

Record a Finding if the ship's Mooring Line Management Plan does not meet the requirements of section 2 of the INTERTANKO guideline.

Record a Finding, if there is no evidence that the Line Management Plan is being implemented on board the ship.

Line Management Plan (LMP) is used to manage the operation and retirement of mooring lines and tails. The LMP also documents the requirements, assumptions and evaluation methods used in determining the line retirement criteria. The LMP is specific to a vessel's manager, ship type, and trade route; however, Oil Companies International Maritime Forum (OCIMF) has given general guidance on establishing a LMP in the Mooring Equipment Guideline Edition 4(MEG4).

The vessel's manager is responsible for the development and implementation of the ships Line Management Plan (LMP). The LMP will contain the vessel's manager requirement for the management of mooring line maintenance, inspection, and retirement during the operational phase of the mooring line lifecycle.

The LMP can be a standalone tool, or it may be integrated into existing safety or maintenance management systems. It can be available as hard or electronic copy, or both. Whatever the format, the LMP should be capable of being updated. It should be accessible for internal and external compliance verification, ship personnel training and communication with manufacturers. LMP information should be stored in a location that is easy for all users to access, e.g., on a computer system that can be accessed from both the ship and shore or compiled in a single physical location. It should be easy for the system users to access the LMP information from a single physical or virtual location.

Table 5.2 of the Mooring Equipment Guideline (MEG4) gives an overview of the type of information that could be included in the LMP for maintenance, inspection, and retirement, as well as general considerations that apply to the safe use and maintenance of mooring lines. Operators can use the table as a starting point for the development of their LMP but should recognise that this list it is not considered exhaustive.

All types of mooring lines experience localised fatigue and damage caused by common line routeing and deployment processes. The location and extent of localised damage on the mooring line can vary due to various factors, i.e., trading patterns, berth layout and design, mooring pendent length and material, ship's movements while at the port, environmental condition, and laden and ballast ship.

The wear zone management should be incorporated in the LMP. Section 5.4.4 of MEG4 has provided further guidance about the wear zone management.

(Mooring Equipment Guidelines (MEG4), 2018)

Conventional fibre lines: At routine intervals, the entire length of line should be inspected by a competent person*. Attention should be paid to those sections of line that are proven by experience to be the main areas of deterioration, such as spliced eyes and interface area with winches, capstans, bollards, fairleads, and rollers. In the absence of sufficiently detailed inspection processes from the line manufacturer, the vessel's manager should refer to the Cordage Institute guideline, "Fiber Rope Inspection and Retirement Criteria – The Guideline that can Provide Enhanced Fiber Rope Durability and Important Information for the Safer Use of Fiber Rope".

(Mooring Equipment Guidelines (MEG4), 2018)

*Personnel assigned the responsibility for rope inspections should be properly trained to recognise rope damage and to understand the rope inspection procedures and retirement criteria.

Every vessel shall have a minimum of two spare mooring lines available and ready for immediate use.
(33 CFR 401.12)

10.2 This question has been removed from the current version of the document.

10.3 Are the certificates of mooring lines and mooring tails available on board? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A mill certificate or certificate of conformity issued by the rope manufacturer does not necessarily guarantee the quality of the mooring line or the proper execution of quality control. RightShip recommends that mooring lines should be accompanied by a type approval certificate issued by one of the IACS members. This type approval certificate not only assesses the product's performance but also includes requirements for quality control and production processes, thereby providing the most reliable evidence of consistent performance and quality over time. Mooring lines that do not possess type approval certificates could still comply with Q10.3 by providing a verification statement. This statement should confirm that a surveyor from a classification society, or a third party approved by the classification society, has witnessed the product test in accordance with specific standards listed in the statement. This verification statement should pertain to a specific batch, and the rope in use should originate from that batch.

It is important that all ropes and wires used for mooring have a certificate. It is considered good practice for these certificates to be clearly labelled and kept in an easily accessible file. A method of correct identification and matching of individual certificates with the mooring ropes and wires should be established on board. Mooring ropes, mooring pendants and wires without certificates must not be used on board.

An example of a mooring line base design certificate, mooring tail base design certificate, mooring line certificate and mooring tail certificate as developed by OCIMF and the template documents can be found on the [website here](#).

The mooring line and mooring tail data sheet shall be provided by the line manufacturer. The recommended contents of data sheet can be found in B4 of appendix B of MEG4.

10.4 Do mooring lines and mooring tails comply with industry guidelines and are they in good order? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Any increase in the LDBF of the mooring lines above the specified limits, i.e., 100 to 105 percent of the Ship Design MBL, will require the classification society to conduct a review of the mooring equipment and fittings' operational characteristics and load limitations, as well as their hull supporting structures.

Record a comment if one or more mooring lines carried onboard, either in service or as spares, have a Line Design Break Force (LDBF) greater than 105% of the ship's design MBL. Specify the number of lines affected and the actual LDBF expressed as a percentage of the ship's design MBL.

Record a comment if one or more mooring tails carried onboard, either in service or as spares, have a Tail Design Break Force (TDBF) greater than 130% of the ship's design MBL. Specify the number of tails affected and the actual TDBF expressed as a percentage of the ship's design MBL.

Record a Finding if any mooring lines onboard, mounted on a winch, loose, or carried as spares, have an LDBF lower than 100% of the ship's design MBL.

Record a Finding if any mooring tails onboard, either in service or carried as spares, have a TDBF lower than 125% of the ship's design MBL.

Record a Finding if the vessel was equipped with HMPE mooring lines but without mooring tails fitted.

The Line Design Break Force (LDBF) of mooring lines fitted on board shall be 100-105% of the ship design MBL. The ship designed minimum breaking strength of each mooring line is available in the mooring arrangement plan.

Nylon (polyamide) mooring lines and tails should be specified as break tested wet, because nylon lines and tails change strength characteristics once exposed to water and generally do not fully dry to their original construction state.

Mooring tails experience more wear in services than lines, and for this reason the Tail Design Break Force (TDBF) should be higher than the LDBF. The TDBF of tails should be 125% - 130% of the ship's design MBL. The increase in TDBF will not necessarily increase tail fatigue life and may undermine the integrity of the mooring system by reducing system compliance.

The standard recommended overall length for mooring tails is 11 metres, however for exposed mooring a 22-metre tail provides additional elongation in the mooring arrangement. Mooring tails should be inspected before every use and according to the requirements of the LMP.

The vessel's manager should arrange for the retirement of the mooring lines and tails when their residual strength reaches 75% of the ship's design MBL.

Tails can be attached directly to HMSF mooring lines using a cow hitch. A cow hitch is a method of joining two fibre ropes without using connection devices. Worked cow hitch connection can become very tight and the inclusion of a pigtail is recommended to help separate the tail and mooring line for replacement or inspection.

(Mooring Equipment Guidelines (MEG4), 2018)

The mooring ropes should be stowed clear of the deck, preferably on a pallet. The area should also be free of any obstructions which could hinder the view of the mooring deck. The mooring stations so far as possible should be painted with a non-slip treatment.

(Safe Mooring Practice, 2009)

10.5 Is there a procedure for testing the winch brake rendering setting and is it being tested regularly? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Rightship recommends the vessel's manager to set the brake at 60% of the Ship Design MBL as per OCIMF guidelines in section 6.4.6 of the MEG 4. RightShip recommends that tests are conducted at least annually.

Each winch manufacturer will have their own test equipment and procedure, which should be followed by the operator. Details of the equipment and procedures should be in the instruction manual for the mooring winch.

The winch test procedure should include:

- > Preparing the winch for testing
- > Setting up the test gear and applying the test load
- > A curve or table relating hydraulic jack test pressure to line pull
- > Hydraulic Jack pressure at which the brake is designed to render
- > Setting of the winch brakes including the values for torque wrench or pressure gauge fitted for setting up the brakes
- > Recording the test results and ensuring markings on the drum are correct and visible

For an undivided winch drum, OCIMF recommended asking the manufacturer for guidance on maintaining the OCIMF recommendations for brake rendering. This may require ship operational experience to identify the normal layer in use for most mooring operations.

The winch will need to be marked with a marking device after the test is done and the brake setting calibrated to ensure that the brake is applied to the same torque during every mooring operation.

The marking device should be built so that further tightening in emergency conditions is not hampered, and it should be of suitable construction to remain in place and in good working condition between routine tests. The RightShip Lessons Learned and Best Practice circular "What is Mooring Winch Brake Render Testing" includes examples of both unsuitable and acceptable marker designs.

The main purpose of brake testing is to verify that the brake will render at a lower load than the Ship Design MBL. For conventional screw brakes, a tag should be attached stating the torque value. For spring applied brakes the spring compression distance should be recorded, and the mechanisms secured with a seal. A stopper arrangement, i.e. locking nut on the threaded end, should not be used on the tightening screw. Stopper arrangement can impede the brake setting and reduce the brake holding load. (Mooring Equipment Guidelines (MEG4), 2018)

The stall load or stall heaving capacity of a winch is the maximum short term or instantaneous pull of the winch, and the stall setting is used to set the maximum heave power of the winch hydraulic motor. The stalling hydraulic pressure is controlled by a proportional valve, however over a period of years this can drift and change the winch setting. The winch stalling load should never exceed 50% of ship design MBL.

The Rightship Lessons Learned and Best Practice circular "What is Mooring Winch Brake Render Testing" must be taken in to account when performing mooring winch brake rendering test.

Please [CLICK HERE](#) to download the document.

10.6 Are mooring lines correctly deployed and tended? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

To minimise the occurrence of line failures and risks of injury or loss of life from a resulting snapback, lines of the same lead should be made of the same material/rope type. As an example, if a vessel runs four headlines, all four lines must be constructed with the same material/rope type.

Lines should be led, so far as possible, without sharp changes of direction. Wires and synthetic fibre ropes need to be kept separate and not allowed to cross or be led through the same lead.

- > Breast lines should be run, so far as practicable, from as far forward and aft and at right angles to the fore and aft line of the vessel
- > Spring lines should be run, so far as practicable, parallel to the fore and aft line of the vessel
- > Where synthetic fibre ropes and wires are available, the same type and size of lines should be used for the same service
- > The mooring operations must be properly planned. For large ships on tidal berths, mooring plans are developed and agreed prior to vessel arrival, and the roles and responsibilities of each crew and officers are explained.
- > The vertical angle of the mooring lines should be kept to a minimum. The flatter the mooring line angle, the more effective the line will be in resisting horizontally- applied loads on the ship.
- > Mooring lines of the same size and material should be used for all leads. If this is not possible, all lines in the same service, i.e., breast lines, spring lines, etc. should be the same size and type.
- > The mooring tails should be the same material and size.

10.7 This question has been removed from the current version of the document.

10.8 Are all mooring lines on winches correctly reeled on drums, and if made fast ashore, are winch brakes secured and winches out of gear? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The effectiveness of the brake is dependent upon the rope being correctly reeled onto the drum. The direction of reeling the mooring line on the drum in accordance with manufacturer's instructions is important to ensure that the brake will hold or render at the correct load.

The winch brake should be applied, and the winch motor should be out of gear after the mooring operation.
(Mooring Winch Brake Holding Capacity, 2015)

10.9 On split drum winches, have the split drums been set up correctly after the completion of mooring operation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Split drum winches are designed so that the line under tension is on the first wrap on the drum, providing maximum holding power. Excessive turns should not be left on the working side of a split mooring winch (tension drum).

(Safe Mooring Practice, 2009)

Guidance on the minimum number of turns on the tension drum should be obtained from the line manufacturer and documented in the Line Management Plan.

Split-drum winches should not have more than one layer of mooring line on the tension section of the drum because it can reduce the brake holding capacity of the mooring winch.

(Mooring Equipment Guidelines (MEG4), 2018)

10.10 If mooring tails are used, have they properly connected to the main mooring lines in accordance with industry guidance? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

It is critical that the mooring tails are to be connected to the primary mooring line by cow hitch or by shackle or link as per manufacturer's instruction. The SWL of the joining shackle should always be equal to or greater than the Working Load Limit (WLL) of the lines in the mooring system, so that the SWL will never be exceeded within the working load range of the lines to which they are attached. Although WLL values for wires and synthetic lines are slightly different (55% and 50% of Ship Design MBL respectively) it is not intended that joining shackle manufacturers or ship operators attempt to match the SWL of the shackle to the WLL.

The eyes of mooring pendants(tails), and in the absence of mooring pendants(tails), the mooring lines should be covered with chafe protection.

10.11 This question has been removed from the current version of the document.

10.12 Have the heaving lines been constructed with either a monkey's fist or a small, high-visibility soft pouch at one end, with a total weight not exceeding 0.5 kg? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The "monkey's fist" should be made with rope only and must not contain added weighting material. Safe alternatives include a small high-visibility soft pouch, filled with fast-draining pea shingle or similar, with a weight of not more than 0.5 kg. Under no circumstances is a line to be weighted by items such as shackles, bolts or nuts, or twist locks.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

There may be alternative, for example throwing rings of soft material, however under no circumstances shall the weight exceed 0.5 KG.

10.13 Are the mooring stations maintained in a safe working condition, and is the entire mooring deck area clearly marked with visible signage and appropriately identified as a hazardous zone? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The entire area of the mooring deck should be considered a potential snap-back zone. All crew working on a mooring deck should be made aware of this with clearly visible signage.

The painting of snap-back zones on mooring decks should be avoided because they may give a false sense of security. In effect, the whole mooring deck may be considered a danger zone.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

The mooring station should be well lit, clean, and free from oil leaks and the deck suitably prepared to prevent slips or trips. Mooring areas naturally contain many trip hazards, and all surfaces are painted the same colour, hiding trip hazards such as save-alls, windlass platforms, forecastle access hatch and bitts.

Physical hazards such as bulkhead frames, mooring bits, pedestal fairleads and cleats, platforms, and hawse pipe covers are to be highlighted.

(Safe Mooring Practice, 2009)

10.14 Are appropriate stoppers in use and are the mooring ropes turned up to bitts correctly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Given the variety of synthetic fibre lines available, it is crucial to use the appropriate type of stopper. Polyamide, polypropylene, and polyethylene mix lines lack the necessary stiffness to function effectively as stoppers. HMSF ropes are generally too slippery to be used as stoppers.

Ideally, a stopper should be:

- > Made of polyester
- > Used in a double configuration
- > Flexible yet made of a stiff material
- > Approximately two metres in effective length from the securing point
- > Thinner than the mooring line
- > About half as strong as the mooring line

(Effective Mooring 2019)

When laying up the line onto the mooring bitts, the first two turns should be taken directly around the first post of the bitts before the rope is laid up in a figure eight around the bitts. Once a rope is laid up on the bitts the stopper should be released from the rope.

Drum ends are not designed to have mooring ropes secured to them for long periods of time. Ropes should never be left on drum ends when not tensioned; they must always be laid up on the bitts.

10.15 Are the controls, linkages, operating levers, brake drums, brake linings, and pins of the winches, as well as the working access arrangement to the winches, in good working order?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The brake drum should be free of corrosion, pitting and rust scale.

The grease nipples should be free of rust, salt, paint and grit.

Check the brake lining for significant wear. Brakes should be closely examined to ensure all linkages are working correctly, brake band material thickness is adequate, and the condition of the brake lining is satisfactory. Equipment manufacturer's manuals will provide details of the permitted minimum brake band thickness.

Clutches should operate smoothly and pins for securing the clutches should be attached to the clutch control levers ready for use.

Winch control levers must be marked with the direction of operation for both paying out and heaving in.

Drum ends should be kept free from damage and rust.

The bed plates of winches should be regularly inspected for deterioration or damage.

10.16 Are the pedestal fairleads, roller fairleads and other rollers free of grooving, well-greased and free to turn? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Regular visual inspections of all types of fairleads should be incorporated in the Plan Maintenance System. Excessive clearance between roller and pedestal table is an indication that the roller pin is worn, meaning it will have lost part of its original strength and may fly back when under tension.

Grooving over part of the surface of the roller indicates that the roller is frozen and that the line/wire is always chafing against the roller in the same area. Grooving or corrosion and scale accelerates damage to the mooring lines/wires. If the fairlead is subsequently used for lines with a different diameter, the sharp edges of the groove will damage the line through abrasion and chafing.

10.17 Are the fairleads, rollers, bitts, chocks and other items of mooring equipment clearly marked with the relevant SWL? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The SWL of each shipboard fitting should be marked (by weld bead or equivalent) on the deck fittings used for mooring. The SWL should be expressed in tonnes (letter 't') or in kilo newtons (letter 'KN').

10.18 Are the windlasses, anchors, locking bars, and cables, as well as the working access arrangement, in good working order and are they maintained as part of the plan maintenance system? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following should be incorporated in the plan maintenance system:

- > Inspection of the anchor, anchor shackle, enlarged links, swivel, joining shackles (Kenter / Baldt / Lugless), anchor D shackles, shackle pins, crown pin, joining shackles, flukes, and shanks for damage, wear down and missing items.
- > Inspection of the cable for damage, wear down and missing studs. IACS class societies require anchor cable replacement when wear down exceeds 12% diameter.
- > Inspection of cable markings, both permanent and painted.
- > Brake band thickness and condition of mating surface.
- > Maintenance of hoist motors in accordance with manufacturer's recommendations.
- > Gypsy for wear down.
- > Ranging of cables for inspection at routine dry dockings in line with Class requirements.
- > Check wear down of guillotine bar, hinge and securing pin.
- > Inspection of devil's claw – if the devil's claw is damaged it should be replaced. Repair of devil's claw is not an acceptable practice.
- > Renewal of wire lashings periodically when damaged / corroded.

10.19 Have the anchors been tightly secured in the hawse pipe? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Whilst moored alongside and anchors are not in use, they should be properly secured by brake and guillotine. The anchors should be housed in hawse pipes properly.

Tight securing of the anchor in the hawse pipe during voyage will avoid excessive vibrations and prevent possible detachment of D-shackle.

(Anchor loss-technical and operational challenges and recommendations, 2016)

10.20

Are the chain locker doors firmly battened down and are the bitter end securing arrangements located outside the chain locker and accessible? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The bitter end should be inspected regularly; the tools for quick release should be available.

The fastening should be provided with a means suitable to permit, in case of emergency, an easy slipping of the chain cables to sea, operable from an accessible position outside the chain locker. A specially marked (red painted) sledge hammer should always also be installed in an accessible position to allow the release of the cable in any emergency.

(Anchoring, Mooring and Towing Equipment, 2016)

10.21

Is the Master aware of the limitations of anchoring equipment? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Anchoring equipment is only to be used for the temporary mooring of a vessel, within a harbour or a sheltered area, when awaiting berth, tide, etc. It is particularly emphasised that the equipment is not designed to hold a ship off a fully exposed coast in bad weather or to stop a vessel from drifting.

The rules for anchoring equipment, including the grade, length and size of chain, the number and weight of the anchors, the strength of the chain stoppers and the power of the anchor windlasses and the brakes are established by the class societies.

They can be found in the rules of the individual societies, or in the unified rules of IACS, the International Association of Class Societies. It is important to be aware that these are minimum requirements, and to know the assumptions made in the calculations.

For each vessel, the class society will calculate an equipment number by using a formula, which includes the displacement of the vessel, the breadth of the ship and the height from the summer load waterline to the top of the uppermost house, as well as the profile view area of the hull, superstructures, and houses above the summer load waterline.

Thus, the forces on the ship by current and wind from both the front and the sides are taken into account. The formula is based on an assumption that the speed of the current may reach 2.5 m/sec, and wind speed of 25 m/sec (which represents quite high forces), but it is also assumed that the vessel can use a scope between 6 and 10 – the scope being the ratio between lengths of chain paid out and water depth.

However, large ships at deep anchorages do not have sufficient chain on board to reach scopes of such magnitude. If a ship is at anchor in ballast condition, the Master should also bear in mind that wind forces acting on the ship may be much larger than the calculations have accounted for, as larger ship side areas are now exposed while the measurements entered in the formula were taken from the summer load water line.

(Limitations of a vessels' anchoring equipment, 2010)

10.22

Has the vessel been provided with a ship-specific Emergency Towing Booklet? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Applicable to cargo ships constructed on or after 1st January 2010; and cargo ships constructed before 1st January 2010, with effect from 1st January 2012.

- > The Emergency Towing Booklet (ETB) should be ship-specific and presented in a clear, concise, and ready-to-use format (booklet, plan, poster etc.).
- > A minimum of three copies should be kept on board and located in:
 1. The Bridge
 2. A forecastle space
 3. The ship's office or cargo control room

The emergency towing procedure shall include:

1. Drawings of fore and aft deck showing possible emergency towing arrangements
2. An inventory of equipment on board that can be used for emergency towing
3. Means and methods of communication
4. Sample procedures to facilitate the preparation for and conduct of emergency towing operations.

(SOLAS 74,2020)

10.23

This question has been removed from the current version of the document.

Section 11: Radio and Communication

11.1 This question has been removed from the current version of the document.

11.2 Is communication equipment, listed in the Record of Equipment attached to the Safety Radio Certificate or Safety Certificate (Form R or Form C), in good condition and has the GMDSS Logbook (the Radio Log) been maintained correctly and are daily, weekly and monthly tests being carried out? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The master must nominate one or more crew members, normally the person/s qualified for distress and safety radio communications, to maintain the radio log and to carry out the tests and checks of the equipment.

Daily test:

- > The proper functioning of the Digital Selective Calling (DSC) facilities without radiation of signals, by the use of the equipment's internal test facility.
- > Battery voltage checks. Mainly the battery ON LOAD and OFF LOAD voltages should be checked by a voltmeter connected to the charger.
- > Check that all printers are in a working condition.

Weekly test:

- > Proper operation of the DSC facilities by means of a test call when within the communication range of a coast station fitted with DSC equipment.
- > If batteries are not the reserve source of energy for the GMDSS equipment, the reserve source shall be tested.

Monthly test:

- > The EPIRB should be examined by carrying out a self-test function without using the satellite system.
- > The Search and Rescue Transponder (SART) is equipped with a self-test mechanism to test the operational function of the beacon. The SART to be tested using the ship's X band radar.
- > Each survival craft should carry two-way VHF equipment to ensure proper operation in case of a distress situation. It should be tested on a frequency other than vhf channel 16 (156.8 MHz). The expiry date of the battery needs to be checked and changed when required.
- > Battery: The battery connections and compartment should also be checked. The level of the electrolyte and the specific gravity of each cell should be checked and recorded.
- > All antennas for security of mounting and visible damage to the cables and insulators.

11.3 Has the Satellite EPIRB been correctly installed, tested and maintained? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Satellite EPIRBs shall be tested at intervals not exceeding 12 months for all aspects of operational efficiency, with particular emphasis on frequency stability, signal strength and coding.

Satellite EPIRBs are subject to shore-based maintenance at intervals not exceeding five years. (SOLAS 74,2020)

406 MHz EPIRBs are to be physically examined and the self-test function checked at least once per month.

Check that the EPIRB ID and other information (include call sign and MMSI of the ship) is clearly marked on the body of the equipment.

Check for the presence of beacon operating instructions.

11.4 Is the most current edition and up to date List of Radio Signals available on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The record of Equipment for Cargo Ship Safety (Form E) attached to the Cargo Ship Safety Equipment Certificate should be endorsed, if electronic nautical publications are provided.

11.5 Is the vessel equipped with sufficient hand-held radios, for use in general on-board operations? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The primary means of communication between the personnel involved in shipboard operations and between the terminal and ship are VHF or UHF radio.

The main difference between UHF and VHF radio is their range. UHF radio waves are smaller than VHF, which means that UHF frequencies have smaller waves that produce a wider reception, while VHF has longer wavelengths. UHF signals are more likely to pass barriers like steel, bulkheads, and building walls more easily.

VHF radio is generally a better signal for long-distance communication. UHF is particularly effective when using radios for indoor use, such as shipboard operations within the ship. An advantage of using UHF is that the crew is less likely to experience interference from other two-way radios.

Sufficient portable radios with chargers and spare batteries should be available to allow communications between the Chief Officer, deck officer in charge of cargo operations, the deck watch, and the Master.

If a vessel uses VHF radio for shipboard operations, the emergency channels and designated port operational channels must not be used.

The GMDSS portable survival craft VHF units are designed for emergency use only. These radios are not for use in general on-board operations.

11.6 Are Search and Rescue Radar Transponders (SARTs) in good order and tested regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

On vessels equipped with a minimum of two (2) search and rescue locating devices and free-fall lifeboats, one of the search and rescue locating devices should be stowed in a free-fall lifeboat. The other device should be situated in the immediate vicinity of the navigation bridge, ensuring it can be utilized on board and is readily available for transfer to any other survival craft.

The Search and Rescue Radar Transponder as a part of GMDSS is subject to annual testing.

(IMO Resolution A.802 (19).

One SART is required for ships of between 300 and 500 gross tons. Two SARTs are required for ships 500 gross tons or greater. Each SART should have self-test capability

(Resolution A.802 (19) Performance Standards for Survival Craft Radar Transponders for use in Search and Rescue Operations, 1995)

Check that the battery expiry label shows sufficient battery life to cover the next routine voyage.

11.7 Are survival craft portable VHF radios in good order and charged? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Equipment for which the source of energy is intended to be user-replaceable should be provided with a dedicated primary battery for use in the event of a distress situation. This battery should be equipped with a non-replaceable seal to indicate that it has not been used.

(Resolution A.809 (19) Performance Standards for Survival Craft Two-Way Radiotelephone Apparatus, 1995)

11.8

Is the AIS static, dynamic and voyage data up to date and has an AIS annual test been performed and the record available on board? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Automatic Identification System (AIS) shall be subjected to an annual test by an approved surveyor or an approved testing or servicing facility. A copy of the test report shall be retained on board and should be in accordance with a model form set out in the annex to MSC.1/Circ.1252.

(SOLAS 74,2020) (MSC.1/Circ.1252, Guidelines on Annual Testing of the Automatic Identification System (AIS), 2007)

Static data that is set up during equipment installation and includes information such as MMSI, IMO Number, International call sign, beam, and ship type.

Dynamic data that is current navigation information including position, course, speed, and navigational status (at anchor, moored, underway or special condition); and Voyage data relates to the specific voyage and include information on draft, destination, ETA and hazardous cargo.

It is important that the AIS is operated correctly and that watch keepers are familiar with the equipment, including how to check that all information being transmitted by AIS is both accurate and update.

Bridge Procedures Guide, 2022)

According to IMO guidelines provided by Resolution A.917(22), AIS should always be in operation when ships are underway or at anchor. Only if the Master believes that the continual operation of AIS might compromise the safety or security of the ship, the AIS may be switched off.

The Master should report the switch-off and the reason for doing so to the competent authority. Actions of this nature should always be recorded in the ship's logbook together with the reason for doing so.

Rightship recommends that the date and time of switching on (and off as per above) should be recorded in the deck logbook. Deliberately turning off the transmitter signal without legitimate reason represents a breach of SOLAS and puts the ship in breach of flag state regulations.

11.9

Is there a Shore-Based Maintenance Agreement in place to fulfil the maintenance requirements? (M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A shore-based maintenance agreement / certificate is needed on board to fulfil the maintenance requirements as mentioned in the "SOLAS GMDSS" regulations (CHAPTER IV Reg. 15) and the Radio Maintenance Guidelines (RES. A702-17), for GMDSS equipment sailing in Sea Area A2-A3-A4.

Section 12: Security

The inspector should not sight the sensitive security materials.

12.1 Is access to the ship being controlled by an adequate deck watch? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Even if it is not applicable under local regulations for some vessels to comply with the ISPS Code, it must be borne in mind that it is good practice to have a member of the vessel's crew permanently stationed at the gangway for safety purposes. They will be able to assist persons transiting the gangway as required and to monitor any dangerous practices. Gangway watch personnel must keep in mind that they are the first point of contact on the vessel for anyone boarding. If a vessel is alongside a berth affected by tidal conditions, constant reassessment of the situation should be carried out. In addition, the watchman must have access to the times of high and low waters and be aware of any cargo operations which may affect the vessel's trim. If a watchman is not present at the gangway and an incident occurs, the vessel's crew may carry on with their duties unaware of the situation.

(Gangways, 2014)

The deck watch has a responsibility to make all visitors aware of any specific hazards of the cargo or operations onboard the vessel and point out instructions to visitors what to do in the event of an emergency.

12.2 Has a Ship Security Officer (SSO) been appointed and trained adequately to perform the duties of SSO and have all crew received security-related training and instructions? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The duties and responsibilities of the SSO shall include, but are not limited to:

- > Undertaking regular security inspections of the ship to ensure that appropriate security measures are maintained
- > Maintaining and supervising the implementation of the SSP, including any amendments to the plan
- > Coordinating the security aspects of the handling of cargo and ship's stores with other shipboard personnel and with the relevant PFSOs
- > Proposing modifications to the SSP
- > Reporting to the company's security officer (CSO) any deficiencies and non-conformities identified during internal audits, periodic reviews, security inspections and verifications of compliance and implementing any corrective actions
- > Enhancing security awareness and vigilance on board
- > Ensuring that adequate training has been provided to shipboard personnel, as appropriate
- > Reporting all security incidents
- > Coordinating implementation of the SSP with the CSO and the relevant Port Facility Security Officer (PFSO)
- > Ensuring that security equipment is properly operated, tested, and calibrated, and ensuring the occurrence of ship security drills and exercises.
- > Ensuring the proper maintenance of all records pertaining to the ship's security
- > Notifying the CSO of ship security incidents and any breaches of this regulation. In the absence of a CSO, notify law enforcement agencies and other law enforcement respondents of ship security incidents and any breaches of this regulation, and
- > Ensuring that all security measures set forth in this regulation are implemented and enforced.

(ISPS Code, 2003)

In accordance to the revised STCW 2010 Code as of 1st January 2014 all seafarers must receive approved security awareness training. (STCW code Reg A-VI/6-1)

12.3 Are deck officers familiar with the function and use of the Ship Security Alert System and is the Ship Security Alert System being tested regularly? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall not ask for the details and location of the ship's Security Alert System.

All ships constructed after 1st July 2004 shall be fitted with a ship security alert system.

The ship security alert system shall, when activated, initiate, and transmit a ship-to-shore security alert to a competent authority, which in these circumstances may include the Company, identifying the ship, its location and indicating that the security of the ship is under threat or it has been compromised.

It shall not send the security alert to other ships or raise the alarm on board, and it shall continue until deactivated or reset.

The ship security alert system shall be capable of being activated from the navigation bridge, and in at least one other location.

(SOLAS 74, 2014)

12.4 This question has been removed from the current version of the document.

12.5 If the vessel has transited or is expected to transit a high-risk piracy area, have updated security charts and relevant publications been provided, and has a voyage risk assessment been completed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

ADMIRALTY Maritime Security Charts contain safety-critical information to assist bridge crews in the planning of safe passages through high-risk areas. All information has been gathered by the UKHO through work with NATO and other government organisations, ensuring each chart has the most accurate, up-to-date, and verified information available.

Each Maritime Security Chart includes:

- > Information about dangers to the security of navigation including piracy, terrorism, embargoes, mine warfare, exclusion zones, blockades, and illegal fishing. This information, when used alongside official navigational charts, can help to ensure the safety of ships, crew, and cargo.
- > General security advice, self-protective measures, security procedures and regional contacts, as well as routeing and reporting requirements implemented by military or security forces.
- > Weekly updates and new editions to help maintain high levels of accuracy and safety. Guides also include ADMIRALTY Quick Response (QR) codes for quick access to a list of all Notices to Mariners (NMs) that affect the specific chart or publication.
- > Maritime Security Charts should be kept up to date with the latest security-critical navigational information. The Security Related Information to Mariners (SRIM) service provides all the data needed to maintain your charts from official government sources.

(Admiralty.co.uk, 2018)

The company's security officer (CSO) and the vessel's Master have the combined responsibility to produce a voyage risk assessment. The procedure for this should be outlined in the vessels SMS. The risk assessment should include:

- > Highlighting areas of increased threat to the vessel. Identify the high-risk areas for that region
- > Identifying methods often used by pirates in these areas, and vulnerable areas where pirates could board
- > The ships own characteristics including handling, freeboard, speed, and general arrangement
- > Military or official organisation cooperation and reporting requirements
- > Existing guidelines and information sources
- > Ship and company procedures, communication, and chain of command.

The vessel's manager should implement appropriate measures to meet the threat of piracy by adopting IMO and other industry-recommended practices suitable for the circumstances of the voyage and ship type.

(Maritime Security – General Recommendations, 2017)

Maritime security threats vary across regions and within them, both in terms of the nature of the threats and their severity. For additional information, reference should be made to the Introduction to Best Management Practices Maritime Security, the BMP Marine security publication. Please click [here](#) to access the publication.

12.6 Have preventive measures been taken by the Master and crew during the stay in port and prior to departure to prevent stowaways? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The issue of stowaways is one which has existed ever since vessels began to trade. Procedures for the prevention of stowaways should be incorporated in the Safety Management System and should be effectively implemented by the Master and the crew on board the ship.

12.7

Are cyber security policies and procedures integrated into the safety management system, and has the cyber security management system been evaluated and certified? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record Finding if cyber security management has not been incorporated into the vessel's SMS by the company's first annual verification of the DOC after January 1, 2021.

The cyber security management shall:

- > Identify the roles and responsibilities of users, key personnel, and management both ashore and on board
- > Identify the systems, assets, data and capabilities, which if disrupted, could pose risks to the ship's operations and safety
- > Implement technical measures to protect against a cyber-incident and ensure continuity of operations. This may include configuration of networks, access control to networks and systems, communication and boundary defence and the use of protection and detection software
- > Implement activities and plans (procedural protection measures) to provide resilience against cyber incidents. This may include training and awareness, software maintenance, remote and local access, access privileges, use of removable media and equipment disposal
- > Implement activities to prepare for and respond to cyber incidents.

(The Guidelines on Cyber Security On board Ships, 2017)

The IMO have urged the maritime industry to refer to the requirements of Member Governments and Flag Administrations, as well as applicable international and industry standards and best practices, for detailed guidelines on cyber risk management. Additional guidance and standards may include, but are not limited to:

1. The Guidelines on Cyber Security Onboard Ships produced and supported by ICS, IUMI, BIMCO, OCIMF, INTERTANKO, INTERCARGO, InterManager, WSC and SYBAss.
2. Consolidated IACS Recommendation on cyber resilience (Rec 166).
3. ISO/IEC 27001 standard on Information technology – Security techniques – Information security management systems – Requirements. Published jointly by the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC).
4. United States National Institute of Standards and Technology's Framework for Improving Critical Infrastructure Cybersecurity (the NIST Framework).

Reference should be made to the most current version of any guidance or standards utilized.

(The additional guidance and standards are listed as a non-exhaustive reference to further detailed information for users of these Guidelines. The referenced guidance and standards have not been issued by the Organization and their use remains at the discretion of individual users of these Guidelines.)

(IMO Guidelines on Maritime Cyber Risk Management 2021)

As computer technology advances, the nature of digital attacks will continue to evolve. To secure the safety of the digital infrastructure, shipping companies are strongly encouraged to go above and beyond regulatory compliance and implement a more proactive cyber-risk management approach.

RightShip urges vessel managers to create a robust cyber security management system to avoid and reduce cyber threats to their ships by engaging cyber security expert firms. The system should undergo an operational, technical, and physical evaluation in accordance with industry standards, and be certified by an expert cyber security firm. Example of cyber security expert firms include classification societies and other firms specializing in this domain.

The term "Cyber Security Expert Firm" denotes a professional entity with proven proficiency in certifying management systems under the ISM Code and/or cyber security management systems in accordance with ISO/IEC 27001 standards. These firms possess the requisite skills, technological acumen, and training to safeguard an enterprise's sensitive data from both internal and external threats.

12.8

Are measures in place for controlling the use of removable media such as USB memory sticks, CDs, DVDs, and diskettes on shipboard computers? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Removable media is a collective term for all methods of storing and transferring data between computers. This includes laptops, USB memory sticks, CDs, DVDs, and diskettes.

Transferring data from uncontrolled systems to controlled systems represents a major risk of introducing malware. Removable media can be used to bypass layers of defences and can be used to attack systems that are otherwise not connected to the internet.

A clear policy for the use of such media devices is essential; it must ensure that media devices are not normally used to transfer information between un-controlled and controlled systems.

To avoid unauthorised access, removable media blockers should be used on all physically accessible computers and network ports. (The Guidelines on Cyber Security on board Ships, 2017)

ECDIS should be protected from malware and virus attack. Access to USB and RJ-45 ports shall be controlled – i.e., disable or lock the ports.

Section 13: Machinery Space

13.1 Are adequate engineering procedures, instructions and guidelines included in the SMS? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Engine room procedures shall provide specific guidance on how to operate and maintain engine rooms and all associated equipment, in a safe and responsible manner.

At a minimum, the SMS and safety management manual should include the following:

- > A system for allocating engineering watch keeping duties and responsibilities for operational procedures;
- > Guidelines for ensuring that crew members are trained and competent to undertake their duties on board;
- > Procedures for engine room operations including checklists
- > Procedures for critical operations, including bunkering, port arrival and departure;
- > Procedures for preventing and controlling pollution, including activities such as oil spill response and disposal of waste;
- > Emergency response procedures and instructions
- > A defect reporting procedure and system for rectifying defects;
- > Procedures for change management
- > Procedures for control, validity and changes for documentation
- > Reporting procedures for accident and near misses
- > Maintenance procedures, including control of work and permit to work systems;
- > Identification of critical machinery/equipment and procedures to ensure availability and for isolation/maintenance of critical equipment;
- > Procedure for management of minimum critical and essential spares
- > A Planned Maintenance System and a method for recording maintenance activities
- > Procedures for crew familiarization and handover
- > A recognized system for identifying training needs.

(Engine Room Procedures Guide 2024)

Overridable engine power limitation (EPL) or overridable shaft power limitation (SHaPoLi) are commonly used to improve a ship's attained Energy Efficiency Existing Ship Index (EEXI) performance.

The company should provide guidelines for the appropriate use of overridable power limitation.

Overridable Shaft Power Limitation (SHaPoLi) system means a verified and approved system for the limitation of the maximum shaft power by technical means that can only be overridden by the ship's master or the officer in charge of navigational watch (OICNW) for the purpose of securing the safety of a ship or saving life at sea.

Overridable Engine Power Limitation (EPL) system means a verified and approved system for the limitation of the maximum engine power by technical means that can only be overridden by the ship's master or OICNW for the purpose of securing the safety of a ship or saving life at sea.

2.2.1 The SHaPoLi / EPL system should be non-permanent but should require the deliberate action of the ship's master or OICNW to enable the use of unlimited shaft / engine power (power reserve) of the ship. For systems that use a Password/PIN to control access to the power reserve override, attention should be paid to ensure that the necessary Password/PIN is always available when override is required.

2.2.2 For SHaPoLi / EPL system for the electronically controlled engine, the control unit should inform the ship's master or OICNW clearly and conspicuously when the ship's shaft / engine power exceeds the limited shaft / engine power as stated in the Onboard Management Manual (OMM) for SHaPoLi / EPL or in any case of system malfunction.

2.2.3 For EPL for the mechanically controlled engine, the sealing device should either:

- > visibly indicate removal of the sealing when the ship's engine power exceeds the limited engine power as stated in the OMM for EPL or in any case of system malfunction; or
- > be equipped with other systems such as an alert-monitoring system which can indicate when the ship's engine power exceeds the limited engine power as stated in the OMM for EPL or in any case of system malfunction and recording the use of unlimited mode, verified by the Administration or the RO

3.2 Any use of a power reserve should be recorded in the record page of the OMM for SHaPoLi / EPL, signed by the master and should be kept on board. The record should include:

- > ship type;
- > IMO number;
- > ship size in DWT and/or GT, as applicable;
- > ship's limited shaft / engine power and ship's maximum unlimited shaft / engine power;
- > position of the ship and timestamp when the power reserve was used;
- > reason for using the power reserve;
- > Beaufort number and wave height or ice condition in case of using the power reserve under adverse weather condition;
- > supporting evidence (e.g. expected weather condition) in case of using the power reserve for avoidance action;
- > records from the SHaPoLi / EPL system for the electronically controlled engine during the power reserve was used; and
- > position of the ship and timestamp when the power limit was reactivated or replaced.

3.3 Where an EPL/ShaPoLi override is activated but the power reserve is not subsequently used, this event should be recorded in the bridge and engine-room logbooks. The engine-room logbook should record power used during the period when the override was activated. The EPL/ShaPoLi should be reset as soon as possible, and details of the reset should also be recorded in the bridge and engine-room logbooks.

3.4 In case of having used a power reserve, the ship should without delay notify its Administration or RO responsible for issuing the relevant certificate and the competent authority of the relevant port of destination with the information recorded in accordance with paragraph 3.2.

Onboard Management Manual (OMM) for SHaPoLi / EPL

4.1 The SHaPoLi / EPL system should be accompanied by the OMM for SHaPoLi / EPL that should be permanently on board the ship for inspection.

4.2 The OMM for SHaPoLi / EPL should be verified by the Administration or the RO after a survey verifying the ship's attained EEXI, as required by regulation 5.4 of MARPOL Annex VI.

(MEPC.335(76) 2021)

The Resolution MEPC.335(76) provides further guidelines on the shaft/engine power limitation system to comply with the EEXI requirements and use of a power reserve. [Please download the document here.](#)

It is recommended that OPL override trainings are conducted every 3 months. If there has been a crew change of key members such as navigation or engine room officers, the training should be carried out immediately after sailing.

Such a training should include:

- > An explanation of the purpose, effects and limitation of the OPL.
- > Examples of circumstances when the OPL may need to be overridden.
- > An on-site demonstration to the extent possible so that the relevant officer can sight the location of the OPL and become familiar with its parts and tools required to override it.
- > Awareness of the time required to override the power limitation considering the type and location of the OPL.
- > Officers should be able to demonstrate that they are able to reset the limitation to the original conditions.
- >

The senior maintenance engineer should hold toolbox talks with the maintenance team at the start of each day and before starting any major work. A toolbox talk is a short meeting to review the work, identify the risks, and discuss how those risks will be managed.

Maintenance and day work ratings should maintain a high standard of safety while carrying out their assigned duties. During toolbox talks, the officer in charge of maintenance should encourage the crew to raise any doubts about the procedures, PPE and safety measures.

(Engine room Procedures Guide 2024)

13.2 Are the responsibilities of watch standing engineers and engine ratings well-defined and clearly posted in the Engine Control Room? Is there a manning matrix for engineers that takes into account both planned and unplanned changes? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Chief Engineer or designated representative should increase manning levels when required, whether planned or unplanned. As far as possible, the work/rest hour requirements should still be met.

Planned Changes:

The Chief Engineer should identify planned changes in manning in consultation with the Master. The planned changes should be identified for every passage of the ship. Examples of events/operations requiring planned manning changes are:

- > Arrival/departure;
- > Cargo operations
- > Bunkering
- > Fuel change overs;
- > Planned machinery overhauls; and
- > Docking for surveys and trial runs
- > Unplanned Changes:
- > In an emergency, the EOOW on duty should set a manning level appropriate to the situation. The EOOW should be encouraged to take these measures as early as necessary. Examples of events that may require an unplanned manning change include:
- > Navigation hazards;
- > Machinery faults
- > Adverse weather;
- > Electrical blackout;
- > Fire; and
- > Flooding.

(Engine Room Procedures Guide 2024)

13.3 Has the Chief Engineer prepared specific standing orders, night and day orders, and have these orders been read, understood, and signed by the watch standing engineers, engine crew, and electrical engineer, where applicable? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Chief Engineer should issue written standing orders for the engineering team. These should reflect the Chief Engineer's own requirements, and take into account the Master's standing orders, the circumstances of the ship and trade, and the experience of the engineering team on board.

Standing orders and instructions should not conflict with the SMS. However, they provide a good opportunity to give specific guidance about the occasions when the Chief Engineer should be consulted or called to the engine room.

On joining the ship, all relevant engineer officers should read, sign and date the standing orders. A reference copy of the order should be readily available in the ECR.

The Chief Engineer should issue night orders and day orders in the engine department order book, to provide specific instructions to address circumstances and requirements outside the normal routines. All EOOWs should fully understand and acknowledge these orders when going on or off watch.

(Engine Room Procedures Guide 2024)

13.4 If the vessel has been certified for periodically unattended machinery spaces operation (UMS), is the machinery space being operated in that mode? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the machinery space is not being operated in UMS mode, as result of defective equipment/machinery or unreliability of the UMS system.

The finding should not be raised if the vessel's machinery space is manned for operational reasons, such as transiting high-risk piracy areas, hold cleaning, post dry-dock (for a period of up to one month), or manoeuvring, provided there are sufficient engineers and crew on board to safely man the machinery space.

Before changing over to unattended operation, the EOOW should complete a round of the machinery spaces, following a checklist all the parameters to be tested and verified. A reference to checklist B2: Preparation for UMS of Engine Room Procedures Guide shall be made (Second Edition 2024)

(Engine Room Procedures Guide , 2024)

13.5 If the engine room is not being operated in UMS mode, are there sufficient engineers and crew on board for safe operation of the machinery space? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

In case the Engine Room (ER) is not suitable for unmanned operation, the composition of the crew should be adapted for manned ER services.

13.6 Have the entry requirements to the engine room when operating in UMS mode been documented, posted at the entrance to the engine room and understood by all crew? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Seafarers should never enter or remain in an unmanned machinery space alone unless they have received permission from or been instructed by the engineer officer in charge at the time. They may only be sent to carry out a specific task that they may be expected to complete in a comparatively short time.

Before entering the space, at regular intervals whilst in the space and upon leaving the space, they must report by telephone, or other means provided, to the duty deck officer. Before they enter the space, the method of reporting should be clearly explained. If it is the engineer officer in charge who enters the machinery space alone, they too should report to the deck officer before entry, at regular intervals whilst in the space and upon leaving the space.

Notice of safety precautions to be observed by seafarers working in unmanned machinery spaces should be clearly displayed at all entrances to the space. Warning should be given that in unmanned machinery spaces there is a likelihood of machinery suddenly starting up.

If there is a personnel alarm system in place, reporting at regular intervals may be omitted. A personnel alarm is a system that will indicate a person's presence and their well-being in unmanned machinery spaces. Vessels without a personnel alarm system should have additional guidance recorded in the safety management system.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

13.7 If an engine room dead man alarm (personnel alarm) is provided, is it correctly set and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The EOW should report to the bridge whenever entering or leaving an unattended machinery space. On entering the space for any reason, they should use the deadman alarm system if fitted.
On ships without a deadman alarm, the bridge should be contacted at least once every 15 minutes.

(Engine Room Procedures Guide 2024)

13.8 Is an engineer's calling alarm system fitted and is it tested regularly, in good order and the results recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When it is safe and agreeable by the Master, the inspector shall test the engineer call alarm.
Engineers' alarm.

In addition to manual operation from the machinery space, the engineers' alarm on vessels with periodically unattended machinery spaces should operate when the machinery alarm is not accepted in the machinery spaces or control room in a specified period of time, e.g., two minutes.

(Code on alerts and indicators 2009, 2010)

13.9 Is the engine room logbook, as well as other required records being properly maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Engine room alarm histories are necessary for machinery space inspections and audits to determine management's level of control over engine room activities and to ensure the machinery space has been operated safely and responsibly. Record a Finding, if there is evidence that manual logging of alarms is not being undertaken in the absence of an engine room alarm logging printer and if historical records for the last 12 months cannot be provided.

The followings should be recorded correctly in the engine logbook:

- > Readings of main propulsion engine
- > Readings of auxiliary engines
- > Readings of other running machineries
- > Main engine RPM and load on the engine
- > Daily entry for all the lube oil ROB
- > Daily entry for all grade of fuel oil ROB
- > ROB value of sludge and bilge
- > Running hour counter for important machinery
- > Record of any major breakdown and reason for the same
- > Record of all bunkering operation (time, place, and quantity)
- > Record of soot blow for the boiler tube (Soot blowing should be performed when the vessel is at sea and/or outside the port limit)
- > Record the start and stop times of UMS mode

The engineer watch keeper should sign the logbook after completion of watch and the Chief Engineer should sign the logbook on a daily basis.

Errors made in the log should be struck through with a single line and initialled and dated. Correction fluid must not be used for correction of error made in the logbook.

Examples of required records include the following:

- > Fuel change over log;
- > Cooling water and boiler water logs;
- > Daily fuel and bilge tank sounding log;
- > Fresh water log;
- > Stern tube bearing temperature records;
- > Bio Fouling record book;
- > NOx technical files records, record book of engine parameters;
- > Ozone Depleting Substance(ODS) records;
- > Inventory of hazardous material records;
- > Grey and black water discharge log;
- > Machinery defect log
- > The EOW should ensure that the alarm printers and automation system are set to the correct time and date at the start of each watch. This is critical during inspections and incident investigations to demonstrate compliance.

13.10 Are procedures to recover essential equipment documented and posted in the engine room? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector shall check if the engineers are familiar with the equipment which is shed on the operation of the preferential trip.

A ship's specific procedure should be readily available and posted in the engine control room and at the local position near equipment which, where applicable, covers the following:

- > Restoring power from the emergency to the main switchboard
- > Charging the air receivers and starting the main diesel generators in order to provide electrical power to all auxiliaries (fuel and lubricating oil pumps and the boiler supply)
- > Restarting all auxiliaries
- > Restarting the main engine

Use of schematic diagram or photograph along with instructions is an example of effective procedure/instructions.

The preferential trip is a part of the ship's generator protection system. It is the electrical arrangement on ships which is designed to disconnect the non-essential circuits (i.e. supplying non-essential load) from the main bus bar in case of partial failure or overload of the main supply. The non-essential circuits or loads on ships are air conditioning, exhaust and ventilation fans, and galley equipment which can be disconnected momentarily and can be connected again after fault finding. The main advantage of preferential trip is that it helps in preventing the operation of main circuit breaker trip and loss of power on essential services and thus prevents blackout and overloading of the generator. The preferential trip operates at timed intervals and the load is removed accordingly. If the overload persists, then an audible and visual alarm is sounded. The preferential trip is an important electrical circuit which helps remove excessive load from the main bus bar, thus preventing a blackout.

The crew should be familiar with the equipment which is shed on the operation of the preferential trip. This is often a multi-stage process with first and second stage tripping arranged to shed load. The items are usually indicated on the switchboard to show what is shed for each level of trip.

(Reducing the Risk of Propulsion Loss, 2017)

13.11 Is an effective and up to date planned maintenance system available and being followed on board the vessel? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends a computer-based planned maintenance system on board the vessel.

Record Finding if:

- > The vessel is not equipped with a computer-based planned maintenance system
- > The officers and engineers are not familiar with the use of the software and have not received any familiarisation training.
- > The automatic data feed in almost real time cannot be take place between the vessel and the vessel's manager.

The planned maintenance system is mandatory as per the International Safety Management Code (ISM) Section 10.1. The planned maintenance system (PMS) shall be a paper or software-based system which allows ship owners or the vessel's manager to carry out maintenance in intervals according to manufacturers and classification society requirements.

An effective PMS streamlines the planning, documentation and implementation of maintenance work and surveys on board ship. The followings minimum requirement shall be incorporated in an effective PMS:

- > The description and documentation of the planned maintenance system are to be in the English language and/or working language of the crew
- > Planned maintenance program must include equipment manufacturers' requirements
- > Inventory content, i.e., items/systems
- > Maintenance time intervals, i.e., time intervals at which the maintenance jobs are to take place
- > Maintenance instructions, i.e., maintenance procedures to be followed
- > Maintenance documentation and history, i.e., documents specifying maintenance jobs carried out and their results
- > Reference documentation, i.e., performance results and measurements taken at certain intervals for trend investigations from delivery stage
- > Document flow chart, i.e., chart showing flow and filling of maintenance documents such as planning cards, job cards etc
- > Signing instructions, i.e., who signs documents for verification of maintenance work carried out

In addition to the above, the computerised planned maintenance systems shall provide:

- > A unique login ID and password for each person performing the maintenance/inspection
- > Adequate backup – either backup copy on board or a regular exchange of data between ship and office
- > Automatic Data Transfer – synchronisation of data between the fleet of vessels and ship's management office using the import/export functionality automatically or manually when required, enabling the vessel's manager to monitor the status of maintenance on board the ship.

The planned maintenance system must be approved when the vessel entered the planned maintenance scheme of a classification society. A type approval certificate for the software of the planned maintenance system is required.

If the vessel is accepted by the classification society for an approved planned maintenance scheme for machinery (PMS), as an alternative to the continuous machinery survey (CMS), it considers surveys to be carried out on the basis of intervals between overhauls recommended by manufacturers, documented operator's experience and a condition monitoring system, where fitted. Access to computerized systems for updating the maintenance documentation and maintenance program shall only be permitted by the Chief Engineer or other authorised person. A computerised and approved planned maintenance system shall be provided. Computerised systems shall include back-up devices, such as CDs which are to be updated at regular intervals.

Condition Monitoring System:

Machinery or technical installations, which are subject to a condition monitoring system, shall be surveyed in line with the requirements described in the "Guidelines for Machinery Condition Monitoring" of the classification society. Prerequisite for this special survey arrangement CM is the existence of a computerized planned maintenance system (PMS). The elements of the PMS considering the machinery components or part of them covered by condition monitoring shall be approved by the classification society. When a vessel entered the condition monitoring scheme of the classification society, the vessel manager shall consider following:

- > The maintenance strategy adopted must be clearly documented in the PMS / SMS – the full scope must be documented and fully understood by all engineers and shore-based technical
- > CBM/CBO inspection intervals must be adhered to and proper records must be maintained (measurements, observations, clearances, oil/fuel analysis, performance reports and photos)
- > There must be evidence that CBM/CBO reports are factored in to determining predictions of revised next full overhaul hours, and these should be clear in the PMS
- > There must be evidence that the maker's instructions for CBM/CBO strategy are complied with
- > The maker's associated Service Letters must be readily at hand to support the CBM/CBO strategy as associated reference guides
- > There must be objective evidence to show that shore technical are actively involved in the ongoing review of inspection records/results
- > CMB/CBO does not absolve the ship from maintaining a sufficient number of spare parts on board if overhaul is unexpectedly required

When PMS notation was assigned to the vessel, the latest version of the PMS shall be installed on board, and the Type Approval certificate for the specific PMS version should be available on board.

Critical Equipment and Associated Spare Parts:

It is recommended that a proactive risk-based approach to the carriage of safety critical spare parts is taken for the management of hazardous situations. This approach may need to be above and beyond minimum regulatory requirements. Companies should apply this approach to both new-builds and to existing vessels.

(Safety critical equipment and spare parts guidance 2018)

To minimize equipment failures, original spares should be used whenever possible. An important part of engine room procedure is the storage and upkeep of technical spare parts. A list of critical spares needs to be carried on board and a minimum quantity of these spares kept at all times.

All members of the engineering team should be trained on how to use spares and consumables properly. Companies can often source spare parts directly from Original Equipment Manufacturer(OEMs) rather than the machinery supplier. However, while this can reduce costs, there is a risk that if the specification of the spares changes, the OEM might not be able to supply the correct spares. Companies should be aware of the risk of sourcing spare parts directly from Original Equipment Manufacturer(OEMs) rather than the machinery supplier.

(Engine Room Procedures Guide , 2024)

The OCIMF information paper "Safety Criteria Equipment and Spare Parts Guidance" provides further information. Please download the document via [HERE](#).

13.12 This question has been removed from the current version of the document.

13.13 Is the main engine maintained as per manufacturer's recommendations and records of periodic maintenance kept? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Engineers should be familiar with the utilisation of the override function of engine power limiters. They should also understand that the activation of this override may be necessary while navigating in pilotage waters.

Random checks should be made by inspectors to ensure that the periodic maintenance of fuel valve, fuel pump, exhaust valve, cylinder cover, pistons, liner, cross head, bottom end, main bearing, turbo charger and governor of the main engine units had been done as per manufacturer recommendation. The main engine shall appear well maintained and free of any leaks.

The watchkeeping team should conduct frequent rounds to check engine operating parameters. Any deviations or abnormalities should be noted, and adjustments made. This includes the temperatures and pressures for fuel oil, lube oil, jacket cooling water and scavenge air.

The most frequent indicator of engine problems is the exhaust temperature, with high deviation either above or below average, along with abnormal color of the exhaust smoke. Any excessive deviations in the exhaust temperature should be investigated. This is done by using indicator cards or electronic devices to check peak and compression pressures.

The engine should be operated within the parameters specified by the manufacturer. Specific guidelines for low load operations should be followed for ships that continuously run on reduced speed.

(Engine Room Procedures Guide , 2024)

13.14 Are the auxiliary engines maintained as per manufacturer's recommendations and records of periodic maintenance kept?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Periodic maintenance intervals recommended by the manufacturer shall be followed.

Random checks should be made by inspectors to ensure the periodic maintenance of fuel valve, fuel pump, cylinder cover, piston, liner, bottom end, main bearing, and turbo blower of the auxiliary engines had been done as per manufacturer recommendation.

Inspector shall check the engine logbook entries to see that any idle generators had been run recently. The auxiliary engines shall appear well maintained and free of any leaks.

Check that the automatic switch over arrangements and protection devices such as reverse power relays are in good order.

13.15 Are all areas of the machinery space well illuminated, emergency escape routes clearly marked, unobstructed and are ship's crew familiar with the escape routes? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Emergency escape routes should be regularly checked to ensure they are properly marked and clear of obstructions. When an escape trunk is fitted, check doors for ease of operation. Door seals should be effective and lighting within the trunk should be operational.

Doors should not normally be locked in port. However, if there are security concerns, the measures to prevent unauthorised access should also ensure that personnel have an escape route.

Locking arrangements and accessibility to embarkation decks

1. The escape routes are routes for escape and also for access. Accordingly, the locking arrangement should be such that it does not obstruct these two objectives (escape and access). Doors along any designated escape routes which require keys to unlock them when moving in the direction of escape should not be permitted.

2. The embarkation deck should be accessible from the open decks to which escapes routes.

13.16 This question has been removed from the current version of the document.

13.17 Is the emergency equipment tested, in good condition and the result recorded? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Testing of the emergency generator should be carried out under load at least once every quarter. During the inspection, the inspector should ask the accompanying engineer to explain the method used for automatic sequential testing. Please note that this testing is not to be conducted during a RightShip inspection.

The emergency fire pump, main fire and foam pumps, emergency air compressor, emergency generator, emergency generator switchboard, emergency steering, emergency stops, engineers' alarms and bilge pumping system, where applicable, shall be tested.

The emergency air compressor, if fitted, should be regularly tested to the starting pressure of the diesel generator. The emergency air reservoir should be permanently maintained at the required pressure.

Special attention may be paid to the correct operation of the priming device attached to the emergency and main fire pumps. They should also have visible and legible operating instructions.

(Engine Room Procedures Guide, 2024)

13.18 Are engine room emergency stops for ventilation fans and the closing mechanism of ventilation supply and exhaust ducts clearly marked, in working condition, and do records indicate that they have been regularly tested? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

13.19 Are engine exhausts and other hot surfaces effectively shielded against oil spray and are flanges and connections of flammable liquid pipelines adequately protected with guards and spray tape? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Spray from engine room equipment can be at relatively high pressures and can spray many metres from the source of the leak. Almost invariably there is a hot exhaust or some other hot surface nearby. Typically, these can be at a temperature greater than the auto ignition temperature of the sprayed liquid, resulting in a fire. (Swedishclub.com, 2018)

Surfaces with temperatures above 220°C which may be impinged as a result of a fuel system failure shall be properly insulated. Precautions shall be taken to prevent any oil that may escape under pressure from any pump, filter, or heater from coming into contact with heated surfaces.

(SOLAS 74,2020)

A perfect insulation of all exhaust pipes and other hot surfaces will make an engine room more fire safe.

When installing the spray tape on flanges its width should cover the entire surface of the flange and a minimum of 100 mm on both sides of the pipe joint. This is required to ensure enough adhesive surface strength in case when high pressure expands installed material.

13.20 Is the lagging and insulation in good condition and free of oil impregnation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Steam and other hot pipes should be lagged properly with the appropriate insulating material, and the lagging should be kept clean and free of oil. To avoid energy loss, it is important to keep lagging and insulation in good condition. Additionally, steam traps should be maintained in good condition.

(Engine Room Procedures Guide , 2024)

13.21 Are the main engine bearing temperature monitors or crankcase oil mist detector(s) in good condition and tested on a regular basis as specified by the manufacturer, and are engineers familiar with the procedure to follow in the event of oil mist in the crankcase? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Internal combustion engines of 2,250 kW and above or having cylinders of more than 300 mm bore shall be provided with crankcase oil mist detectors or engine bearing temperature monitors or equivalent devices.

(SOLAS 74,2020)

Oil mist or vapors coming into contact with heated surfaces, can ignite inside engine crankcases and cause explosions. Engineers should be familiar with the procedures to follow if oil mist is detected in crankcase. In this case, the Engine Room Procedure Guide's (First Edition 2024) checklist C6 should be used.

(Engine Room Procedures Guide , 2024)

Regardless of if the vessel has a UMS notation assigned, if engine bearing temperature monitors or an Oil Mist Detector is fitted, they should be regularly tested and maintained as per manufacturer instructions and the inspector should satisfy that they are in good working order.

13.22 Were the electrical switchboards surrounded by suitable deck insulation, and was the insulation in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Insulating matting is required only at the front and rear of switchboards. Individual machinery starter boxes located throughout the machinery space are not classified as switchboards and, as such, do not necessitate the use of insulating matting either in front of or behind them.

In certain instances, decks may be constructed from composite insulating materials. Where such materials are employed, additional insulation may not be necessary. In these cases, and where insulating matting is not provided, ship documentation should be available to verify the extent of the composite deck covering and its corresponding safe working voltage.

The insulating matting used should be appropriate for the voltage of the switchboard in question. As a minimum standard, it should comply with IEC 61111:2009 Class 0 or an equivalent specification, which provides a safe working voltage of up to 1,000 volts. For switchboards operating at voltages exceeding 1,000 volts, matting of a higher classification—such as Class 1, rated for up to 7,500 volts—should be used.

Acceptable evidence of the safe working voltage rating of the deck insulation includes manufacturer certification, appropriate markings, or other relevant documentation.

13.23 Are the spring-loaded drain valves and gauge glass self-closing valves/ cocks being maintained and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Self-closing valves are fitted between the lower end of an oil tank and its gauge glass. The purpose of these valves is to isolate the tank gauge glass from the tank. In normal operation they should be shut and only opened to check the tank contents, after which they should shut automatically under spring pressure or counterbalance gravity.

Chocks of wood, pieces of wire and purpose-made clamps shall not be used to keep these valves open. Self-closing valves are essential safety devices. They should be properly maintained and should never be tampered with.

(Quick Closing and Self Closing Valves, 2011)

13.24 Are the sounding pipes and self-closing sounding devices in good order and closed?

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The 'deadman' weight of a self-closing sounding device must not be removed, reversed, or lashed open. If spring-loaded types self-closing sounding device are in use, the spring must be fitted.

Sounding rods/tapes or funnels are frequently found to be left inside the open sounding pipe for ease of operation, or for dumping of residues back into the tanks via the sounding pipes.

(Engine Room Sounding Pipes, 2009)

The inspector shall record Finding if the above practices are noticed in the engine room.

13.25 Where moving machinery presents a hazard, is it guarded effectively? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Correct safety guards should be securely fixed to appliances requiring them and should be checked for security before starting any operation. Such guards should only be removed when the equipment is not operating.

No machine should be used when a guard or safety device is missing, incorrectly adjusted or defective, or when it is itself in any way faulty. If any defect is identified, the machine should be isolated from its power source until it has been repaired.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

13.26 Is the workshop clean and tidy, and are the engine room workshop tools' protective guards, shields, and emergency stops in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Workshop and bench machines should only be operated by competent personnel. The operator should check a machine every time before use and ensure that all safety guards and devices are in position and operative; that all tool pieces (drill bits, cutting blades, etc.) are in good condition, and that the work area is adequately lit and free from clutter.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

The absence of protective guards and shields on workshop machinery can cause serious eye or bodily injury not only to operators but also other crew members present in the workshop.

The guards fitted to the lathe, drill and grinder should be well maintained, transparent and made from impact-resistant material.

Regular checks on the condition of workshop machinery guards should form part of the shipboard planned maintenance system.

13.27 Is the engine room crane, other lifting equipment, and hydraulic tools inspected, tested, and maintained on a regular basis? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The engine room crane is subject to annual thorough examination every 12-month period and load test every 5 years, unless otherwise stated by the ship's flag administration.

Portable lifting gear including chain blocks, strops and slings shall be inspected and maintained on a regular basis and record of such inspection and test should be available on board.

- > The engine room transverse crane beam should be clearly marked in several locations – not only on the travelling beam itself and, on the hook, but also on the internal fore and aft 'I' beam – so that no matter where you are standing on the engine room top plates the SWL of the equipment is clearly visible.
- > Shackles are by their nature a 'link' between two components and therefore play an essential role in terms of safety. They should be marked and stamped with the safe working load (SWL).

(Lifting equipment – shackles and other loose gear, 2013)

The spring-loaded retaining 'tongue' of hooks should be in good order.

A lifting appliance log should be maintained on board.

Hydraulic tools, such as hydraulic jacks and bending machines, are often used in the engine room. They should be thoroughly inspected before use and the correct grade of oil should be topped up to the required level. Hydraulic hoses should be maintained in good condition and they should not be twisted or entangled during operation. The pressure gauge installed on hydraulic tools should be calibrated regularly to ensure safe working pressures. The manufacturers' instruction should be followed.

(Engine Room Procedures Guide, 2024)

13.28 Are all spare parts and loose gear in the machinery spaces, stores and steering compartment properly secured? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The improper handling of Steel plates, kept as spares for the fabrication and maintenance of a variety of ship components, could lead to serious injury or even death.

It is essential that the vessel's manager identifies potentially hazardous manual handling operations, including the development and implementation of procedures for the storing, securing and manual handling of spare parts and steel plates. The manual handling of these items must only be performed after a formal risk assessment has been conducted by a trained and competent individual using the company's approved form, ensuring that records are kept in compliance with procedure.

Prior to undertaking a manual handling task, RightShip recommends that the vessel manager assess the following four factors: Task, Individual, Load, and Environment (TILE) to ensure a thorough evaluation of the task and its associated risk.

Refer to Section 10 and Annex 10.1 of the latest edition of the UK MCA Code of Safe Working Practices (COSWP), Skull P&I, for additional information, or [click here](#) to learn more about TILE.

13.29 Is the standard of housekeeping in the machinery space and steering gear room satisfactory, are these areas clean, free from obvious leaks, and is the engine room bilge clean and free of oil and sediment? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Housekeeping in the engine room is critical. Keeping debris from building up in the corners of the main space or machinery flats eliminates potential fuel. Workshops, spare part storeroom, chemical stores, electrician's store/workshop should be maintained clean. Hydraulic or other types of oil may accumulate in the bilge wells of the steering compartment. Appropriate arrangements should be made for its safe disposal. If overboard discharge valves are installed, they must be securely closed and locked, and pollution prevention notices should be clearly posted nearby.

Engine room Bilge:

The presence of oil accumulated in bilges or drip trays act as additional fuel to sustain burning and increase the likelihood of the fire reaching further areas in the engine room.

The bilges, especially in the engine room, should be kept clean and free of oil at all times.

(Hazards associated with dirty engine room bilges - The Shipowners' Club, 2018)

13.30 This question has been removed from the current version of the document.

13.31 Is the bilge high level alarm system in good order, regularly tested and are records of test maintained? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

High-level alarms for engine room bilges should be tested at least once daily when the engine room is manned, and prior to leaving the engine room in Unmanned Machinery Space (UMS) mode, as part of the pre-UMS checks.

13.32 Are the sea chests, seawater pumps, and associated seawater lines and valves in good working order, with no leaks, hard rust, or temporary repairs?

☐ Yes ☐ No ☐ N/A ☐ N/V

13.33

Is the following machinery/equipment, where applicable, in good order and well maintained? (V)

- > Shaft generator and emergency generator
- > Boilers, including waste heat and domestic boilers (Boilers should be operated in automatic mode where the automated boilers are installed)
- > Boiler safety system and instrumentation
- > Boiler water safety system
- > Main and emergency air compressors
- > Purifiers and fuel oil handling equipment
- > Stern tube sealing arrangements
- > Incinerator
- > Sewage system
- > Air condition and heating system
- > Refrigeration plant
- > Accommodation service systems (i.e., Calorifiers, Portable water equipment, heating etc)
- > Any other items of machinery, including stand-by machinery.
- > Burners, tubes, uptakes, exhaust manifolds and spark arrestors.
- > Engine control console including the control and monitoring system
- > Steering gear system
- > Battery maintenance and renewal.

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The inspector should record a finding if boiler or cooling feedwater management, including the maintenance of regular testing records, is not being conducted.

Feedwater for Boiler and Cooling Systems

Water contamination remains a constant threat to boiler and cooling systems. To safeguard system integrity and performance, distilled water should be the standard quality used as feedwater.

Using non-distilled or untreated fresh water—containing elevated levels of minerals (e.g. calcium, magnesium, carbonates, chlorides) and other contaminants (e.g. dissolved organics and suspended solids)—can lead to serious water-related issues if not addressed promptly through an integrated chemical and operational strategy.

Risks of Using Non-Distilled or Untreated Water

Boiler Systems

- > Corrosion: Caused by poor feedwater quality, lack of heating/de-aeration, and inadequate chemical treatment.
- > Scaling: Mineral buildup reduces steam generation efficiency and can result in tube rupture.
- > Carryover: Inadequate water level control or treatment allows contaminants into the steam, causing downstream damage.

Cooling Systems

- > Corrosion: Often due to seawater ingress or absence of proper water treatment.
- > Scaling: Especially detrimental to plate heat exchangers, reducing engine cooling capacity.
- > Biofouling: Microbial growth in closed-loop systems disrupts heat exchange and compromises hygiene.

Use High-Quality Feedwater: Always use distilled or properly treated water to limit corrosion, scaling, and contamination risks. Avoid untreated freshwater or seawater, which may carry harmful minerals and biological agents.

Apply the Correct Chemical Treatment: Use appropriate chemical dosing to manage pH levels, prevent scaling, and inhibit corrosion. Ensure dosing systems are calibrated and tailored to your vessel's specific needs.

Conduct Regular Monitoring and Testing: Implement daily onboard testing, supported by periodic laboratory analysis. This enables early detection of issues such as scale, corrosion, or biofouling, and ensures water chemistry stays within safe parameters.

Boiler Blowdown Procedures

Boiler blowdown is the process of removing water from an operating boiler to eliminate sediment, chemical concentrations, and dissolved solids. This practice is essential for preventing carryover, corrosion, and scaling, thereby ensuring the production of high-quality steam and maintaining the reliability and longevity of the boiler system.

Types of Blowdown:

1. Bottom Blowdown (Blowoff) : The primary objective of bottom blowdown is to remove sludge and sediment that accumulate at the bottom of the boiler. The recommended frequency and duration should be determined in accordance with the guidelines provided by the chemical treatment provider.

2. Continuous Blowdown (Surface Blowdown / Skimmer) :Continuous blowdown targets the removal of impurities from the water surface. In smaller boilers, this may involve a simple dip tube positioned just below the water level. In larger systems, a skimmer tube runs the length of the boiler just beneath the surface, with evenly spaced holes to extract water slowly. This method helps eliminate scum and control the concentration of Total Dissolved Solids (TDS), which can negatively impact boiler performance.

Recordkeeping and Monitoring:

To ensure optimal water treatment and system performance, log sheets for both boiler water and cooling water systems should be maintained either as separate records or within the Engineering Logbook.

Procedure for Loss of Economizer Water

The company should establish a documented procedure outlining the actions to be taken in the event of economizer water loss. Inspectors should verify that ship-specific procedures for monitoring economizer water levels are clearly documented within the vessel manager's SMS. The procedure should include clearly defined corrective actions to be implemented in the event of water loss.

Waste heat boiler (Economizer): A particularly high-risk event on a ship is an economizer soot fire. The soot fire cannot happen in an economizer free of soot deposits.

The best way to avoid soot deposits is to soot-blow frequently. This is especially true in ships that are often slow steaming, which raises the likelihood of soot deposits. Except in emergencies before soot-blowing economizers or boilers the bridge permission should be sought.

Exhaust gas inlet and outlet temperature and pressure differential between gas inlet and outlet of the economizer should be checked at least once every watch. The economizer should be isolated and manually cleaned with fresh water if the differential pressure is observed to be significantly high.

Economizers may have bypass arrangements for when the ship is maneuvering or to control steam output. To avoid them seizing, the bypass should be tested on a regular basis.

Refrigeration and air conditioning: MARPOL Annex VI Regulation 21 requires refrigeration systems that use refrigerants that are classified as ozone depleting substances (ODS) to maintain an ODS record book. Any intentional charging or discharge of these refrigerants should take place only from/into approved containers. This includes venting the system to remove any trapped air. All of these details, including any maintenance, should be recorded in the ODS record book.

Incinerator:

- > Waste oil or sludge should not be incinerated when in ports, harbours or estuaries;
- > Sludge from Exhaust Gas Cleaning System(EGCS) should not be incinerated.
- > Plastics and PVC should only be incinerated in IMO-approved incinerators.
- > A copy of the IMO type approval certificate can be found in the incinerator's manufacturer manual.

(Engine Room Procedures Guide , 2024)

Draining water from the ship's main air receiver can result in catastrophic injury to humans if the observation window for the air receiver drainage pot explodes as a result of the following design issues:

1. If the nominal bore of the inlet is greater than the nominal bore of the outlet, or
2. If the output line of the drain pot becomes clogged,

The inspector shall record a Finding if this arrangement is fitted.

RightShip strongly urges the vessel's manager to remove the sighting glass totally and replace it with a discreet steel baffle to dampen any emulsion blow back

Battery Maintenance:

Thermal runaway occurs when a battery generates more heat than it can dissipate, often due to overcharging or high temperatures. This can lead to catastrophic failure. Valve Regulated Lead Acid (VRLA) and Absorbed Glass Mat (AGM) batteries are particularly prone to this issue, especially with mismatched chargers and poor ventilation. Prolonged float charging also increases the risk.

It is important to select the right type of lead acid battery for the intended purpose:

- > Flooded or wet cell batteries contain an electrolyte of sulphuric acid/distilled water and require regular maintenance to retain the correct electrolyte level in the cells. These batteries must be stored upright to prevent leakage. This type of battery is most suitable for installations that utilise float charging in the battery circuit to maintain the charge. The service lifespan is 5 years to 7 years.
- > Low maintenance wet cell VRLA batteries have sealed cells with a valve arrangement to release the gases created by charging the battery. The electrolyte level cannot be maintained by topping up. These batteries are most suitable for intermittent use where the battery circuit is not maintained by a float charge. The service lifespan is 5 years to 7 years, although batteries installed in a system using a continuous float charge should be replaced after 2 years of use due to the increased risk of explosion under high discharge loads.
- > Maintenance free gel type VRLA batteries have similar properties to wet cell VRLA batteries and are most suitable for intermittent use. Batteries installed in a system using a continuous float charge should be replaced after 2 years of use.
- > Absorbent glass mat (AGM) batteries are a type of VRLA battery. The electrolyte is absorbed and suspended between AGM plates to provide extended life and durability.

These batteries are designed for intermittent charging use and should be replaced after 2 years if installed in a system using a continuous float charge. Maintain.

Batteries are a piece of machinery and require a documented service and maintenance log. The record should individually list the type of battery; where it is located on board; when it entered service; dates and details of any inspections or maintenance and, if applicable, what corrective action was taken; and when it is due for replacement.

Hazard : Safe stowage of battery installations is crucial to contain and minimise impact in the event of a fire or explosion. To prevent a build-up of gases from the charging process batteries should be kept in a purpose-built, well-ventilated locker in a cool, dry location away from the elements. Light fittings in a battery compartment should be corrosion-resistant and flame/explosion-proof.

(MAIB 2025)

To access the ATSB accident report regarding the explosion of the air receiver pot observation window glass—caused by exposure to air receiver pressure, [click](#) here.

13.34 Is the pipe work in the machinery space, including but not limited to steam, fuel, lubricating oil, sewage, drain and air lines well maintained, in good condition and free of temporary repair and leakage? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

13.35 Are engineers familiar with operation of the main engine from the local manoeuvring control position? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record the date of last drill in comments.

All members of the engineering team should be trained and proficient in the local and emergency procedures for starting and maneuvering the main engine. Periodic drills will help to maintain this proficiency. Clear instructions on this procedures should be posted next to the manual/emergency starting and maneuvering stations.

(Engine Room Procedures Guide , 2024)

13.36 Are crew familiar with the starting procedure for the emergency generator and how to put power on the emergency switch board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

When agreeable by the Chief Engineer and safe to do so, the inspector shall witness the running test (but not on load) of the emergency generator.

SOLAS II-1/Reg.43.7 requires that provisions for the testing of the emergency source of electrical power, including its automatic starting arrangement, are to be made. Such testing can be conducted using a test switch provided in the Emergency Switch Board (ESB) that enables automatic starting and connecting of the emergency generator to the ESB during simulated blackout conditions, in general.

It is recommended that tests to ensure automatic starting as well as connecting of the emergency generator to the ESB shall be carried out at appropriate intervals using the test switch in the ESB.

(Operation test for automatic starting arrangement of emergency generator (Blackout simulation test), 2018)

Inspector shall review of the suitable evidence of such test onboard and question the engineer to explain the process of sequential test start and loading of the ESB.

All crew members must be familiar with starting procedure of the emergency generator.

Each emergency generating set arranged to be automatically started shall be equipped with starting devices approved by the Administration with a stored energy capability of at least three consecutive starts. A second source of energy shall be provided for an additional three starts within 30 minutes unless manual starting can be demonstrated to be effective.

(SOLAS 74,2020)

The brief instruction should be simple, clear, and understandable by all crew. The instruction shall incorporate how to put power on the emergency switch board, if the system is not automatic.

13.37 If the starting source of the emergency generator relies on a single starter motor, has a spare starter motor been provided? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

RightShip recommends that a spare starter motor be provided if the starting source relies on that one starter motor. The spare starter motor should be tested quarterly to ensure operational readiness.

13.38 If an emergency generator is not fitted, are engine room emergency batteries in good order, fully charged and capable of supplying the designed power load up to a minimum 18 hours? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

13.39 Is the main and emergency switchboard earth fault monitoring equipment operational with no earthing faults indicated? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Minimum insulation resistance as per classification society requirements is 1 megohm. It is good practice to maintain the insulation resistance more than 5 megohms for 440 system and 2 megohms for a 220 Volte system. Alarm settings should be 0.2 MOhm for 220V systems and 0.5 MOhm for 440V systems. This meets the minimum insulation resistance requirement for 1000 Ohm per Volt.

13.40 Is an emergency steering gear drill being carried out every three months? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Emergency steering drills shall take place at least once every three months in order to practice emergency steering procedures. These drills shall include direct control within the steering gear compartment, the communications procedure with the navigation bridge and, where applicable, the operation of alternative power supplies.

Simple operating instructions with a block diagram showing the change-over procedures for remote control systems and steering gear power units shall be permanently displayed on the navigation bridge and in the steering gear compartment.

(SOLAS 74,2020)

13.41 Is the emergency reserve tank of the steering gear system fully charged and is the manual transfer pump operational?? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A fixed storage tank shall be provided having sufficient capacity to recharge at least one power actuating system including the reservoir.

(SOLAS 74,2020)

13.42 Is a heading indicator and communication system provided in the steering gear room and are they in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ships with emergency steering positions shall at least be provided with a telephone or other means of communication for relaying heading information to such positions.

(SOLAS 74,2020)

In addition, ships of 500 GT and upwards constructed after 1st February 1992 shall be provided with arrangements for supplying visual compass readings to the emergency steering position.

(SOLAS 74,2020)

13.43 Is the rudder angle indicator at the emergency steering position in good order, and is the display clearly legible? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

13.44 This question has been removed from the current version of the document.

13.45 Are suitable handrails, gratings or other non-slip surfaces provided for the steering gear compartment? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The steering gear compartment shall be provided with suitable arrangements to ensure working access to steering gear machinery and controls. These arrangements shall include handrails and gratings or other nonslip surfaces to ensure suitable working conditions in the event of hydraulic fluid leakage.

(SOLAS 74,2020)

Section 14: General Appearance - Hull and Superstructure

14.1 Is the ship's hull clean, free of significant corrosion, extensive coating breakdown and marine growth? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Implementing practices to control and manage biofouling can greatly assist in reducing the risk of the transfer of invasive aquatic species.

Such management practices can also improve a ship's hydrodynamic performance and can be effective tools in enhancing energy efficiency and reducing air emissions from ships. This concept has been identified by the IMO in the "Guidance for the development of a ship energy efficiency management plan (SEEMP).

Hull resistance can be optimized by new technology-coating systems, possibly in combination with cleaning intervals. Regular in-water inspection of the condition of the hull is recommended.

(GUIDANCE FOR THE DEVELOPMENT OF A SHIP ENERGY EFFICIENCY MANAGEMENT PLAN (SEEMP), 2009)

The vessel should be provided with effective, environmentally safe, and practical biofouling management procedures that are based on industry recommendations for in-water cleaning of the ship's hull to reduce the spread of invasive aquatic species.

The vessel is required to maintain a Biofouling Record Book in which all inspections and biofouling management measures are recorded.

14.2 Are the following permanent markings on the ship's hull, where applicable, plainly visible and painted in a contrasting colour? (V)

- > The vessel's name
- > Port of registry
- > Load lines
- > Draft marks
- > Thruster warnings
- > Tug push points
- > IMO number
- > Bulbous bow mark

☐ Yes ☐ No ☐ N/A ☐ N/V

14.3 Are the weather decks free of loose rust scale and maintained in a satisfactory condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

14.4 Are the pipes on deck free of significant corrosion, pitting, soft patches, leakage or temporary repair and maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The following deck pipes should be checked for external indications of corrosion, pitting and temporary repair:

- > Hydraulic and pneumatic pipework
- > Fire mains and associated fittings
- > Pneumatic lines
- > Electrical conduit lines
- > Ballast lines
- > Fresh water line
- > Steam pipe including heating system

Pipe securing arrangements should be maintained in good condition and allow free movement of the pipes, as necessary.

14.5 Are all watertight doors (where fitted), fire doors, weathertight doors, portholes and wheelhouse windows maintained in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Fire-resistant divisions constructed in accordance with SOLAS II-2 are utilized to contain the fire and reduce the risk of fire spread. These divisions' openings, such as engine room access doors, are equipped with self-closing devices. Fire doors should not be fastened or wedged open in any way.

Doors positioned on a weather deck, particularly the main deck, are also critical to the vessel's safety. As a result, they should never be fastened or tied open when underway.

14.6 Are the vents and air pipes on weather decks maintained in good order and are they clearly marked to indicate the compartment they serve? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Vent head should be maintained in good condition. The flame screen, if fitted, should be clean and in good condition. The closing device which prevents the ingress of water into the space through the vent head should be in good condition and operating correctly.

14.7 This question has been removed from the current version of the document.

14.8 Are the hatch numbers clearly indicated and correctly placed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship shall be provided with the hatch identification numbers used in the loading manual and loading or unloading plan. The location, size and colour of these numbers should be chosen so that they are clearly visible to the operator of the loading or unloading equipment.

(BLU Code, 2011)

Cargo spaces to be included in the computation of net tonnage are enclosed spaces appropriated for the transport of cargo, which is to be discharged from the ship, provided that such spaces have been included in the computation of gross tonnage. Such cargo spaces shall be certified by permanent marking with the letters CC (Cargo Compartment) to be so positioned that they are readily visible and not to be less than 4 inches in height.

(International Convention on Load Lines (1966). Protocols and Organization, 2005)

14.9 Are the mast heads and their fittings, including but not limited to wire stays, as well as the flood lights, deck lights, emergency lights, and hold lights (if installed), in good working order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Wire stays should be inspected, maintained, tightened, and replaced as needed. Some manufacturers sheath wire stays in plastic. While the sheathing repels water when new, its effectiveness decreases over time. Deterioration of the plastic coating can allow seawater penetration resulting in corrosion undetectable to external observation. The vessel's manager should establish a procedure for the regular inspection and maintenance of wire stays sheathed in plastic.

Hold lighting system shall be fully operational and properly maintained. The inspector shall test the lights to make sure the lighting system is operative and there is no significant earth on the switch boards.

14.10 Are portable and fixed cargo lights used for illumination of cargo holds inspected regularly and maintained in good condition? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a finding if the lamp holder, drip shield, and shade of portable cargo lights are not non-conductive, do not isolate the crew from electrical shock hazards by the installation of an isolation transformer, or have a voltage greater than 50V AC (1-1000Hz) or 120V DC.

The human hazard of electric current depends on the intensity and duration of current flow in a specific current path through the body. The technical specification IEC TS 60479-1 comprises the permissible touch currents and the required data to calculate the permissible touch voltages under several conditions (e.g., body resistance, current path, skin moisture (see Parameters for effects of electric current) for alternating current and direct current. A touch voltage of 50 V AC (1-1000 Hz) or 120 V DC for long shock duration (> 3 s) should not be exceeded in healthy adults otherwise a life-threatening condition may occur.

"Many bulk carrier / general cargo holds have fixed cargo lights. These can easily ignite combustible cargoes such as grain, animal feed, wood chips, pulp, and paper if they are too close to the light. Self-decomposition of fertiliser has been initiated in this manner. Cargo lights in holds need to be properly isolated before cargo is loaded".

"This is best done by removing fuses or other physical links in the electrical circuits so that the lights cannot be switched on by mistake. In container ships the lights need to be properly placed so that they do not overheat cargo or other combustibles and thus cause damage or fire. Lights in car carriers and ferries are usually fluorescent, which are unlikely to cause ignition. Nonetheless it makes sense to leave lights switched off when they are not needed, particularly in cargo areas where combustibles are present"

(A guide to the causes and prevention of cargo fires, 2017)

14.11 Is the condition of electrical equipment including lights, switches, sockets, junction boxes, plugs, conduits, cabling and wiring on weather decks and around the accommodation satisfactory? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

14.12 Are the chemicals and gases stored correctly, and are the paint locker, battery room, oxygen and acetylene rooms, and other flammable lockers and storage spaces equipped with ventilation systems, explosion-proof lights, and other fittings in good working order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Applicable to ships constructed on or after 01 Jan 2007:

No electrical equipment shall be installed in any space where flammable mixtures are liable to collect, for example in compartments assigned principally to accumulator batteries, in paint lockers, acetylene stores or similar spaces, unless the administration is satisfied that such equipment is:

1. Essential for operational purposes
2. Of a type which will not ignite the mixture concerned
3. Appropriate to the space concerned, and
4. Appropriately certified for safe usage with the dusts, vapours, or gases likely to be encountered.

(SOLAS 74,2020)

It is essential before use of any hazardous substance that the manufacturer's safety data sheet (SDS) is referred to, to select appropriate personal protective equipment (PPE) and working methods.

(Code of Safe Working Practices for Merchant Seafarer's, 2024)

All stores on board where hazardous or toxic substances are kept, such as paint and chemical stores, shall have readily accessible MSDS.

2.3 The importance of the proper storage of Dangerous Goods, Chemicals and Materials on board ship should not be underestimated. If the dangerous goods are subject to the provisions of the IMDG Code, the stowage provisions of chapter 7.1 of the IMDG Code apply. However, a wide variety of chemicals (materials), not subject to the provisions of the IMDG Code, which are commonly used in the marine industry, can react violently together should the packaging become damaged or involved in a fire. Their safe storage should be subject to a risk assessment.

The following points should be considered:

- 2.3.1 Storage areas** – are designated and controlled areas for portable machinery and equipment containing chemicals, materials, waste, flammable substances e.g. foam plastics, flammable liquids and gases such as propane and hazardous substances e.g. pesticides and timber treatment chemicals; such areas should be arranged so that in the event of a spillage or leakage the substance concerned is contained locally and does not react violently with any nearby substances or materials.
- 2.3.2 Accommodation** – storage areas should not be located in or close to accommodation areas;
- 2.3.3 Access routes** – corridors and other walkways should not be used as storage areas. Do not store materials where they obstruct access routes or where they could interfere with emergency escape routes;
- 2.3.4 Segregation** – store incompatible materials in separate areas; flammable materials will usually need to be stored away from other materials and protected from accidental ignition;
- 2.3.5 Safe stowage/Storage at height** – all stores should be securely stowed, if materials are stored at height (e.g. on shelving) make sure necessary guard rails are in place to stop items falling and, if the storage area is fitted with a fixed sprinkler system, ensure that the maximum stowage height limit is adhered to at all times.
- 2.3.6 Tidiness** – keep all storage areas tidy, whether in designated stores areas or at a workstation on board the ship; and
- 2.3.7 Stock control** – plan deliveries to keep the amount of hazardous materials on board to a minimum, taking into account the operating pattern of the vessel.

(Maritime & Coastguard Agency 2013)

When battery room ventilators are equipped with a closing device, these devices should be left open and a clear warning notice installed to prevent accidental closing. The closing device should be used only in an emergency.

The battery locker should contain personal protection equipment (PPE) for testing and handling the batteries.

The PPE includes a face shield or eye-glasses, chemical handling gloves, chemical resistant shoes or boots, a suitable apron and a valid bottle of eye wash.

The PPE must be stowed clear of the batteries to avoid possible contamination from battery acid.

(Battery rooms ventilation and proper upkeep, 2013)

14.13 Are the stores located inside the accommodation and on the weather decks clean and tidy? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

14.14 This question has been removed from the current version of the document.

14.15 Are galley appliances, audio-visual equipment, and other electrical equipment inside the accommodation in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The electrical cooking appliances inside the galley must all be in working condition. The deep fat fryer is equipped with a safety thermostat. The thermostat should be in working condition.
Deep-fat cooking equipment installed onboard ships constructed on or after 01 Jul 2002 in enclosed spaces or on open decks shall be fitted with the following:

1. an automatic or manual fire-extinguishing system tested to an international standard acceptable to the Organization;
2. a primary and backup thermostat with an alarm to alert the operator in the event of failure of either thermostat;
3. arrangements for automatically shutting off the electrical power upon activation of the fireextinguishing system;
4. an alarm for indicating operation of the fire-extinguishing system in the galley where the equipment is installed; and
5. controls for manual operation of the fire-extinguishing system which are clearly labelled for ready use by the crew.

(SOLAS II-2 Reg 10 6.4)

14.16 Are the door seals, catches and alarm system of the refrigerated space in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Separate refrigerators should be used for cooked and raw food. Refrigeration chambers must be kept at recommended temperatures, which should be regularly checked and to ensure good air circulation. Door seals and catches should also be checked regularly.

Safe temperatures for cold stores are generally considered to be 5°C or colder and minus 18°C or colder for chill and freezer cabinets respectively but a slight tolerance of one or two degrees is unlikely to create any significant risk to food safety. If cabinets do not have a means of checking temperatures, a suitable thermometer should be provided. Thermometers should be calibrated periodically but a simple check monthly, using boiling water (99°C to 101°C) or melting ice (-1°C to +1°C) will verify the accuracy of the thermometer.

In freezer units, the combination of high humidity and fluctuating temperatures (warmer than minus 10°C) accelerate mould and other spoilage bacterial growth. Fluctuating temperatures may also cause an accumulation of ice deposits. Food should never be stored in front of cooling units as this restricts the circulation of air. Suitable packaging is essential to avoid the loss of moisture from the surface of food which can produce a freezer burn effect on exposed meat cuts or joints.

If defrosting is not an automatic process, equipment should be defrosted regularly to maintain its efficiency. Although fridges and freezer cabinets should be maintained according to the ship's planned maintenance system, cooks and others working in the galley should regularly check the condition of door seals and closing devices as well as routinely monitoring temperatures.

14.17 Is the elevator, where fitted, inspected, tested and in good order? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Each lift shall be tested and thoroughly inspected before being brought into use and after repair work and important modifications. The inspection should preferably be carried out at 12-month intervals, but at intervals not exceeding 18 months unless rules from classification societies and other rules, as referred to under clause 0, require otherwise.

The maintenance operations shall be carried out by authorised lift maintenance personnel.

The basic characteristics of the lift shall be recorded in a register or file; drawn up, at the latest, at the time the installation is brought into service. This register or file shall be kept up-to-date and shall comprise of inspection, test, and maintenance information.

(ISO 8383:1985 / Lifts on ships -- Specific requirements, 2016)

A procedure clearly defines elevator maintenance, responsibilities and safety barriers shall be incorporated in the SMS. Evidence of permit to work and risk assessment related to maintenance shall be available.

14.18 If provided, is the ship's hospital properly equipped, clean, hygienic and for medical use only? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The MLC, 2006 requires regular inspection of the vessel's medicine chest by the competent authority. RightShip recommends that the annual inspection of medical chest conducted by a vessel's supplying pharmacist or a doctor.

The ship's hospital shall not use the hospital as a cabin or storage space. Vessels are required to carry a medicine chest and medical equipment that complies with the requirements in the current edition of the WHO 'International Medical Guide for Ships' and / or flag State.

Ships carrying dangerous goods

Ships carrying dangerous goods have additional medicines, specific antidotes, and special equipment on board, as prescribed in the International Maritime Organization's Medical First Aid Guide for Use in Accidents Involving Dangerous Goods (MFAG). These special items, which are not listed in this guide, should be stored, and registered together with the regular medicines and medical supplies carried on board.

(International MEDICAL Guide for Ships, 2007)

For additional information, reference should be made to the Medical First Aid Guide for use in Accidents Involving Dangerous Goods. Rightship recommends all ships shall carry the latest edition of the Ship Captain's Medical Guide.

14.19

Are the ship's guard rails, walkways, and access ladders, as well as the steps and railings, maintained and in good working order, and are the securing arrangements for deck cargoes on the deck also maintained and in satisfactory condition?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if the chains strung between guard rail stanchions, in lieu of a fixed railing, are sagging and fail to provide a minimum clearance of one meter from the deck.

According to paragraph 2 of Regulation 25 "Protection of the Crew" in Annex I of the Load Line Convention, guard rails must be installed around all exposed decks and must be at least one meter in height from the deck. Chains installed between two fixed stanchions and/or bulwarks are allowed in lieu of guard rails where necessary for the ship's normal operation.

Section 15: Health and Welfare of Seafarers

15.1 Do the Seafarer Employment Agreements (SEA) comply with the requirements of MLC 2006, do crew salaries meet or exceed the current ILO Minimum Wage Scale, and are individual monthly statements provided to all seafarers on board, detailing their wages and any authorised deductions, such as allotments? (V & M)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if unauthorised deductions, such as payments for travel to or from the ship was recorded on the monthly statement.

Record in the comment the duration of the seafarers' employment agreement for the key personnel (Master, Chief Officers, Chief Engineer, and Second Engineers), other Officers, and crew on board.

Collective agreements established by the ITF can prescribe the salary and working conditions for all crew of Flag of Convenience (FOC) vessels, regardless of their nationality. All vessels covered by an ITF-approved agreement receive a certificate denoting the agreed-upon salaries and working conditions. If the vessel is covered by any form of ITF agreement (Green Card, Blue Card, or Collective Bargaining Agreement), the inspector is not required to assess the crew contract for conformity with ILO pay rates.

When the vessel is not covered by any form of ITF agreement, inspectors shall randomly check to verify if the seafarer's pay is in accordance with the ILO's minimum recommended wage scale.

The ILO minimum wage scale is published annually. [Click Here](#) for the ILO rates applicable from January 1, 2025. Original copies of the SEA shall be provided to all mariners.

- > If the 'employer' is a 3rd party manning agent, then the shipowner must guarantee to meet the employer's obligations if the employer fails to do so
- > Must be paid at least monthly in full
- > Late payments incur 20% p.a. interest
- > Schedule of duties, with hours of work/rest prominently posted
- > Payment in lieu cannot replace leave entitlement
- > Shore leave must be granted where consistent with operational requirements
- > Duty to repatriate at no cost to seafarer
- > Insurance in place to cover liabilities relating to repatriation

(Crew Health and Welfare 3, 2016)

All seafarers are entitled to repatriation:

- > After a maximum 12 month period
- > As stated in the SEA
- > In case of termination for justified reasons (by the shipowner or seafarer)
- > When they are not able to carry out their duties on board due to illness, injury, etc.

(ILO MLC Pocket Checklist, 2012)

15.2 Are the accommodation spaces safe, provided to a respectable level of health and hygiene and regularly inspected, including checks of ventilation, noise, heating, lighting, and sanitation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record a Finding if records of the weekly Master's inspections of the vessel's accommodation are not available. The inspector shall conduct a random check of cabins to ensure they are clean and fully functional.

The Master or Master's representative shall conduct a weekly accommodation/cabin inspection with due diligence to ensure a respectable level of health and hygiene.

- > Accommodation spaces shall be kept clean and free of dirt and dust
- > All cabin portholes shall be checked for water tightness
- > Hot and cold water in the washrooms of cabins must be in working condition
- > The bed must be checked for clean sheets, washed linen and overall tidiness
- > The laundry equipment should be in working order. Separate washing machines for civil clothes and boiler suits shall be provided. Sufficient detergent shall be provided. The build-up of lint inside and under the dryer can cause fire. Dryer vents, vent hoses and filters should be cleaned regularly.
- > The heating and ventilation ducts inside the cabins and common accommodation spaces should be in working condition
- > Adequate natural and artificial light shall be available
- > Private / common toilets and shower rooms shall be in good order. Soap, detergents, and other cleaning material to keep the space clean should be supplied to the ship's staff regularly.
- > Food store handling areas, refrigerated areas, galley, and pantries should be well illuminated, clean, tidy, hygienic, and free of obstructions. To minimise the proliferation of Legionella bacteria weekly flushing of all freshwater outlets should be undertaken, particularly of 'dead legs' (a length of pipe leading to an outlet which has been removed or is rarely used or unused entirely), infrequently used outlets e.g. showers in medical rooms, and any areas of long runs within the water distribution system

- > The condition of portable electrical equipment located within the cabins, whether ship-owned or personal items, should be inspected.
- > The condition of pipe and cable trunks inside the accommodation space.
- > The elimination of the risks of food cross-contamination involves checking the cleanliness of the cooking equipment each time different foods are used, especially raw and cooked foods. Separate work surfaces and chopping boards should be used, and consider using colour-coded boards for different food types. Discard boards that become excessively worn or develop deep grooves, as these can harbour bacteria.

15.3 Are the ship's staff provided with adequate recreational facilities on board, and has the Master been allocated a monthly welfare budget? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record the most recent group social activities that were carried out on board. Record in comments the monthly welfare budget available to the Master.

RightShip strongly encourages the provision of free internet access to the crew providing them with the opportunity to communicate with their families and friends while away from home.

It is however crucial to implement mitigation measures and controls to ensure that such access does not compromise the safety of navigation or the operational efficiency of the ship.

The measures and controls should consider issues such as access restricted rest periods, workplace distractions, offensive posts, internet piracy and cyber security.

The following recreational facilities shall be provided on board:

- > Separate smoking room and bars
- > TV, radio, video, CD, DVD and PC equipment
- > Sports facilities
- > Table and deck games
- > Library, and
- > Communication facilities including email and internet access.

(Crew Health and Welfare 3, 2016)

15.4 This question has been removed from the current version of the document.

15.5 Are seafarers being provided with sufficient food and water free of charge and does the cook hold appropriate qualifications? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments the food budget of the vessel per person/day. Record a Finding if the only water offered free of charge for human consumption on board the ship was non-potable.

Where a potable water management plan is not in place and there is no evidence that water quality testing is being conducted at least twice a year a finding should be recorded.

Potable water is fresh water that is intended for human consumption, drinking, washing, teeth brushing, bathing or showering; for use in fresh-water recreational water environments; for use in the ship's hospital; for handling, preparing or cooking food; and for cleaning food storage and preparation areas, utensils and equipment. Potable water, as defined by the WHO Guidelines for drinking-water quality (2008) does not represent any significant risk to health over a lifetime of consumption, including different sensitivities that may occur between life stages.

Ships may be equipped with two or three different water systems: potable water, non-potable water used for other operational procedures and water for firefighting. Whenever practicable, only one water system should be installed to supply potable water for drinking, personal hygiene, culinary purposes, dishwashing, and hospital and laundry purposes. Non-potable water, if used on the ship, needs to be loaded and distributed through a completely different piping system, which should be colour coded according to existing international standards.

(Handbook for the inspection of Ships and issuance of ship sanitation certificates 2005)

Food and catering preparation standards should include, but are not limited to:

- > Sufficient quantities of good quality food including fresh fruit, vegetables and drinking water should be supplied free of charge
- > Food is to be nutritious, varied and prepared and served in hygienic conditions
- > Religious and cultural considerations should be considered
- > The cook is over 18 years of age and holds appropriate qualifications, in accordance with the flag state's laws and regulations
- > All other catering staff are adequately trained (a training programme, posters, etc. may be available)
- > For ships with less than 10 crew, no cook is required, but the crew handling food are to be trained in food hygiene.

(ILO MLC Pocket Checklist, 2012)

The Merchant Shipping Notice, MSN 1845(M), "Maritime Labour Convention, 2006: Food and Catering: Provision of Food and Fresh Water" provides further guidance. EU Directive 98/83/EC of 3 November 1998 on the quality of drinking water defines drinking water as all water, whether in its original state or after treatment, that is intended for drinking, cooking, food preparation, or other domestic purposes, regardless of its source or whether it is supplied via a distribution network, a tanker, or in bottles or containers. Notably, this term includes water used for other domestic uses, such as personal hygiene – tooth brushing, showering, etc.

To safeguard the health of the crew, potable water on ships should be protected by multiple layers of safety measures. This "multiple-barrier system" spans from the water source on shore, through the ship's distribution system, treatment, and storage, and finally to each water supply outlet. These barriers prevent contamination throughout the entire process.

The vessel's manager should develop a Potable Water Management Plan that considers the delivery, sampling, and analysis of the water, storage and cleaning of storage tanks at regular intervals, and the distribution of drinking water on board. To ensure water quality stability, RightShip urges the vessel's manager to carry out the analysis of potable water samples at least twice a year.

Sampling should be performed from locations considered highly hazardous. Specific sampling points should include:

- > Potable water supply lines
- > Freshwater tank systems
- > Toilets and showers
- > Galley
- > Crew cabins
- > Any other connection or tank for potable water loaded on board

15.6 Are ship's staff provided with appropriate medical care and access to health promotion programs, and is there evidence that visits to a qualified medical doctor or dentist are arranged without delay in ports of call, when required? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments the method of health promotion and related education programs on board the ship.

- > The medical care should be provided free of charge and be comparable to workers ashore
- > Include health promotion and education programmes
- > An up-to-date list of radio contacts where medical advice can be obtained should be readily available

(ILO MLC Pocket Checklist, 2012)

Health promotion might include:

- > Health Awareness Material displayed in crew rest rooms/ mess rooms
- > Training films shown to crew
- > Working in hot and sunny environments -Heat Stroke/Sunburn.
- > Dangers of dehydration
- > Healthy lifestyle - balanced diet, adequate sleep and regular exercise
- > Protection from Mosquito/insect bites

Health protection and medical care, including essential dental care should be available and free of charge to all seafarers. The medical log and visit reports are kept up to date. A standard medical report form is used for both onshore and on-board medical personnel and the completed forms are kept confidential.

(ILO MLC pocket checklist, 2012)

15.7 This question has been removed from the current version of the document.

15.8 This question has been removed from the current version of the document.

15.9 Has the vessel's manager developed a policy and procedure addressing violence and harassment, including sexual harassment, bullying, and sexual assault, are the requirements of this policy being disseminated to the crew in their native language, and, are the crew familiar with the complaint procedure and how to report a complaint? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Violence and Harassment, Including Sexual Harassment, Bullying, and Sexual Assault

Potential harassment incidents, including sexual harassment, on ships are a critical concern that demands a well-organised and effective strategy. Harassment is harmful behavior towards a person that creates a risk to health and safety and is based on personal characteristics. The vessel manager should develop a policy and procedure, adopting a zero-tolerance approach to dealing with bullying and harassment at sea. The following should be considered when developing such a policy and procedure:

- > The policy and procedure should clearly define the types and examples of violence and harassment, and outline how to deal with such incidents onboard.
- > Disseminating the company's policies regarding harassment to everyone onboard, in the native language of crew members.
- > Assigning responsibilities to senior management and administrations for addressing reported cases, as well as providing adequate resources to respond, including medical care and mental health support for victims.
- > Ensuring training and familiarisation of seafarers and designated shoreside personnel on the company's policy, procedure, and their implementation.
- > Establishing channels for reporting and actions to be taken when a complaint is filed.
- > Ensuring privacy and confidentiality to encourage disclosure.

Effective training is essential to combat harassment. Training should not only cover harassment policies but also include modules on gender sensitivity, communication skills, and conflict resolution. It is crucial that training programs are mandatory for all crew members, regardless of rank or position. Participants should learn to recognise early warning signs of harassment, bullying, and bias, including behavioral changes, isolation, and diminished morale or performance.

Each seafarer should be given a copy of this procedure. The complaints should be handled in a timely, fair and effective manner. The contact details of the flag state and the competent authority in the seafarer's country of residence for complaints should be available on board and posted in the seafarer's recreation rooms.

A complaints log shall be maintained on board.

15.10 Has the vessel been provided with adequate policies addressing mental health and mental disorders, and have key on-board personnel received training to help them recognise signs of mental health issues? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The aim of mental health and mental disorders policy shall be:

- > To promote the health, safety, and welfare of seafarers
 - > To foster a company culture that is conducive to improving the mental health of seafarers
 - > To ensure awareness of the importance of good mental health among company managers
 - > To provide support for staff who are identified as having mental health problems, ensuring that they are treated with sympathy and respect and in confidence
 - > To increase awareness among all staff of the potential signs of mental health problems
 - > To provide training to staff in having conversations with others about their mental health.
- (Guidelines to shipping companies on mental health awareness, 2018)

Key personnel including the Master, Chief Engineer, Chief Officer and Second Engineer.

The company should provide, or arrange training for management-level personnel on-board ships and ashore in recognising signs of mental health problems, facilitating discussions in staff meetings about mental health and having sensitive and supportive conversations with sufferers of mental health problems.

(Guidelines to shipping companies on mental health awareness, 2018)

15.11 Are seafarers provided with free access to external sources of support, whom they can contact in confidence while on board? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The company should consider providing free access to external sources of support for seafarers, whom they can contact in confidence. These may include maritime trade unions, seafarer welfare organisations or organisations specialising in the provision of support to those with mental health problems.

(Guidelines to shipping companies on mental health awareness, 2018)

15.12 Has company provided training for on-board key personnel in recognising signs of mental health problems? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Key personnel including the Master, Chief Engineer, Chief Officer and Second Engineer.

The company should provide, or arrange training for management-level personnel on-board ships and ashore in recognising signs of mental health problems, facilitating discussions in staff meetings about mental health and having sensitive and supportive conversations with sufferers of mental health problems.

(Guidelines to shipping companies on mental health awareness, 2018)

15.13 This question has been removed from the current version of the document.

15.14 Does the Air Handling Unit (AHU) maintain a comfortable temperature and is there recorded evidence of regular maintenance and cleaning of AHU available?

Guide to Inspection

Certain forms of bacteria flourish in a ship's air conditioning system. These organisms or bacteria grow or multiply in stagnant water or moist slime or sludge formations. If these germs are not eliminated, they impact negatively on the ship's living conditions, making them dangerous for the crew.

The primary hazard areas are the air input systems, filter, cooler unit (dehumidifier), humidifier, and plenum insulation. The system should be inspected and cleaned on a regular basis not exceeding three months.

For additional information, please refer to MGN 38 (M+F) Legionella Bacteria Contamination of Ships' Air Conditioning Systems by [Clicking Here](#)

Section 16: Ice or polar water operations

This section shall be completed only if the vessel meets one or more of the following conditions:

- 1 An Ice class notation was assigned to the vessel, or
- 2 The vessel is or intends to navigate in an icy area, or
- 3 The vessel is in possession of a polar water certificate.

16.1 Is the vessel provided with an approved ship-specific Polar Water Operation manual or an Ice Operation manual? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The Polar Water Operations Manual shall be approved by the vessel's Flag State.

This is a ship-specific manual carried on board which outlines the ship's capabilities and limitations.

The manual must also cover procedures for the use of ice breaker assistance vessels whilst trading in the region.

Procedures to be followed in the event of an incident or emergency occurring within the Polar Regions should also be included.

- > Risk-based procedures that are contained in the manual should include:
- > Voyage planning instructions and guidance for operating in such regions, including any vessel limitations
- > The potential lack of reliable chart information that is possible in some polar areas
- > How to gather weather reports in higher latitudes
- > Any additional equipment to be carried
- > Any procedures required for machinery and equipment to ensure its continued safe operation in Polar Regions
- > Emergency contact details for any areas the vessel will operate in

Voyage planning is covered in the Code, and as such bridge teams should familiarise themselves with its contents.

The goal of the PWOM is to provide the owner, operator, Master, and crew with sufficient information regarding the ship's operational capabilities and limitations in order to support their decision-making process.

(The Polar Code, 2017)

If the vessel is operating outside polar waters, but in ice water the ice operation manual shall include the following:

- > Ship handling and navigation in ice
- > Ice and snow accretion prediction and calculation
- > Masters standing instruction
- > Managing ballast water
- > Engine room systems
- > Safety and lifesaving equipment
- > Firefighting equipment
- > General precautions
- > Task Risk Assessment

Arctic Council has developed a guideline contains best practice methods and data sources for conducting regional and area-wide risk assessments concerned with ship traffic and operations in Arctic. For additional information, reference should be made to the Guideline for Arctic Marine Risk Assessment [via link](#)

16.2 Is the Master aware of the operational limitations specified in the Polar Ship Certificate? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comment the following operation limitation of vessel in polar waters:

1. Category of ship
2. Ice condition
3. Temperature, and
4. High latitude

The Polar Ship certificate shows a vessel's Polar Category, operational limitations and capabilities, and any required additional safety, communications and navigation equipment needed for operating in Polar Regions.

(The Polar Code, 2017)

16.3 Is the vessel appropriately manned by adequately qualified, trained, and experienced personnel? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

If the vessel is not operating in the polar region at the time of inspection, this question should be answered with N/A. The inspector should only comment if the officers are in possession of the relevant certificates.

There are two levels of training and certification:

- > Certificate in Basic Training for ships operating in polar waters as per STCW Code, A-V/4, paragraph 1
- > Certificate in Advanced Training for ships operating in polar waters as per STCW Code, A-V/4, paragraph 2

A Certificate of Proficiency (CoP) will be issued to persons qualified in accordance with the requirements.

CoPs may be issued by training providers and there is no requirement that they be issued by Administrations. CoPs issued under Regulation V/4 are not required to be issued with endorsements attesting to the recognition of the certificate (i.e., Flag State endorsement).

Basic Training:

When required by the Polar Code, Masters, Chief Mates and Officers in charge of a navigational watch on ships operating in polar waters are to hold the CoP in Basic Training for ships operating in polar waters. Every candidate for the CoP in Basic Training for ships operating in polar waters shall have completed an approved basic training course. There are no special seagoing service or experience requirements for this level of training.

Advanced Training:

When required by the Polar Code, Masters and Chief Mates on ships operating in polar waters are to hold the CoP in Advanced Training for ships operating in polar waters.

(Information and guidance training requirements for personnel on ships operating in polar waters, 2017)

16.4 Is polar water operation incorporated in the approved SOPEP manual? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

On-board documentation concerning pollution prevention needs to be updated to take operation in polar waters into account, including requirements from MARPOL Chapters I, II, IV and V.

Documents such as Oil Record Book and SOPEP on board the existing ships are to be revised, taking into account operation in polar waters and the Occasional Survey of existing ships to confirm the documents for compliance with Part II is to be carried out prior to entering polar waters on or after 1 January 2017.

Although the item relating to the Polar Code was added to the form of IOPP Supplement (Form A, Form B) on 1 January 2017 in accordance with the Resolution MEPC.265 (68), the IOPP Certificate is not necessary to be rewritten at the Occasional Survey of Existing Vessels for compliance with Part II to comply with Polar Code (in other words, the current IOPP Certificate is valid on board until expired) based on MEPC.1 / Circ.856, unless so instructed by the Flag Administration. Regardless of whether the vessel enters polar waters or not, the IOPP Supplement amended by the Resolution MEPC.265 (68) will be issued at the next IOPP renewal survey.

(Technical Information - Polar Code, 2016)

16.5 Is the vessel provided with a means of detecting floating ice? (V)

☐ Yes
 ☐ No
 ☐ N/A
 ☐ N/V

Guide to Inspection

Record in comment the means available on board the ship.

Radar, searchlights and lookouts are examples of means for detecting ice.

All ships intended to operate in periods of prolonged darkness should be equipped with at least two suitable searchlights, which should be controllable from conning positions. The searchlights should be installed to provide, as far as is practicable, all-round illumination suitable for docking, astern manoeuvres, or emergency towing.

(Guidelines for ships operating in arctic ice-covered waters, 2002)

A standard marine radar does not provide a good picture of ice conditions. As visibility is frequently limited by darkness, snow or fog, other navigational aids should be considered. Cross-polarised radar systems can provide a much better resolution of ice features, including bergy bits and dangerous free-floating ice. These are now becoming available from specialised radar suppliers.

(Vessels operating in low temperature environments, 2006)

16.6 Is the vessel able to receive up-to-date information including ice reports for safe navigation? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments the means available on board the ship.

All ships should be provided with equipment capable of receiving ice and weather information charts.
(Guidelines for ships operating in arctic ice-covered waters, 2002)

Vessels shall be equipped with a weather telefax receivers or equivalent capable of receiving high resolution ice weather charts.
(Vessels operating in low temperature environments, 2006)

16.7 Are main engine sea chests provided with steam heating systems and is a record of check available? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Steam heating systems for sea chests are to be checked in good working condition and kept in operation when the vessel is in ice waters.

16.8 Are personnel provided with appropriate protective equipment suitable for sub- freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Cold weather can endanger lives and destroy the well-being of seafarers whose jobs put them in the midst of frigid temperatures and extreme weather conditions.

Frostbite most often affects fingers, toes, the nose, ears, chin, and cheeks. The condition is a bodily injury that is caused by freezing and it can damage the body permanently.

Hypothermia results when body temperature is below 35°C and often occurs from prolonged exposure to cold temperatures. Low body temperature has an adverse effect on the brain, compromising the victim's ability to think clearly or to move well.

16.9 Are the accommodation spaces provided with adequate heating systems? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

All sleeping rooms, mess rooms, day rooms, recreation rooms, rooms for watching films and television, hobbies and games rooms, offices, studies, sanitary accommodation, and hospitals shall be installed with a main heating system capable of ensuring that:

1. The ventilation system provided for the room or crew accommodation is working as to supply at least 25 cubic metres of fresh air per hour for each person which the room or crew accommodation is designed to accommodate at any one time
2. When the temperature of the ambient air is -1° C the temperature in that room or crew accommodation can be maintained at 21°C
3. The main heating system shall be operated by steam, hot water or electricity, or shall be a system supplying warm air
4. The means for turning on or off or varying the heat emitted by a radiator or other heating device without using a tool or key shall, wherever reasonably practicable, be provided in the space in which that radiator or other device is fitted. All heating equipment shall be so constructed that its operation is not affected by the use or non-use of propelling machinery, steering gear, deck machinery, calorifiers or cooking appliances.

Heating equipment shall be constructed and installed, and if necessary shielded, so as to avoid the risk of fire, danger or discomfort to the crew.

(Maritime Labour Convention 2006, 2006)

16.10 Is the vessel equipped with suitable material and / or equipment for cleaning the ice and snow from critical areas? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Example of critical area are handrails, steps, ladders, and walkways.

16.11 Is effective vision enhancement equipment provided on the Navigation Bridge? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

All Polar Class ships should be fitted with a suitable means to de-ice sufficient conning position windows, to provide unimpaired forward and astern vision from conning positions.

The windows should be fitted with an efficient means of clearing melted ice, freezing rain, snow, mist, and spray from outside and accumulated condensation from inside. A mechanical means of clearing moisture from the outside face of a window should have operating mechanisms protected from freezing, or the accumulation of ice that would impair effective operation.

All persons engaged in navigating the ship should be provided with adequate protection from direct and reflected glare from the sun.

(Guidelines for ships operating in arctic ice-covered waters, 2002)

16.12 Is exterior electronic equipment, such as communication transmitters / receivers exposed to rotating radar scanners and fog horns protected from sub-freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments how the equipment was protected.

Special consideration may be warranted for exterior electronics equipment. Communication transmitters and receivers may require anti-icing features to provide continual functionality, although whip type antennas can usually be de-iced with a strike of a wooden mallet or shaken to remove ice build-up. Other communications, including antennas with horizontal surfaces or dish shaped configurations, may require built-in heat elements. Exposed rotating radar scanners normally require no special measures, even at extremely cold temperatures, due to internal heating elements. However, the smaller enclosed type arrays can become encrusted with ice and can be difficult to de-ice due to their inherent fragility of construction.

(Vessels operating in low temperature environments, 2006)

16.13 Are procedures in place to safeguard the operation of critical equipment in sub-freezing temperatures? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Particular attention should be given to power generation/distribution, rescue boat and lifeboats.

Insulating, heating, and/or adding antifreeze to any lines exposed to freezing temperatures may be required for any engine, and particularly for those using freshwater cooling systems. Where batteries are used to provide power for emergency equipment, they should be suited and sized for low temperature operation.

(Vessels operating in low temperature environments, 2006)

16.14 Are procedures in place to safeguard the readiness of lifesaving appliances and survival arrangements in sub-freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Ice build-up in the way of all lifeboats, life rafts, cradles, davits, and other launching gear should be regularly removed so that launching arrangements are not hindered. This may include ensuring a wooden mallet is available at each station or in the vicinity of the lifesaving appliances.

Additional rations stored in the deckhouse / accommodation are recommended so that water is readily available to the crew as the water stored onboard the lifeboats will be frozen.

Note that most EPIRBs work only down to -20°C according to manufacturer's instructions. Care must be taken to verify that the selected EPIRBs are suitable for the design service temperature. The manufacturer should be consulted for guidance.

Air-cooled engines provide additional heating and can reduce problems associated with frozen valves, piping, and water intakes.

The lifeboat's propeller is susceptible to damage from ice, particularly when operating astern.

Another issue will be condensation, as humidity from survivor breathing touches the cold hull and canopy.

This can render survivors even more uncomfortable and can fog the windows at the coxswain station (and elsewhere). Consideration should be given to installing supplementary ventilation or air circulation features, and to heaters for the craft interior.

Lifeboat engine distillate fuel should have a cloud point well below the design service temperature. Lifeboat engine lubricating oil should have the correct viscosity at the design service temperature without the use of a heater.

The IMO Life Saving Appliances Code requires inflatable life rafts to be capable of inflation within 3 minutes at a temperature of -30°C (-22°F). Lower design service temperatures may result in an inability to inflate properly at low temperatures, and so operators should verify that adequate air or other proven cold temperature gas is used for the inflation of life rafts. Manual inflation pumps are to be suitable for operation at the design service temperature.

(Vessels operating in low temperature environments, 2006)

16.15 Are procedures in place to safeguard the readiness of firefighting equipment in sub-freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Door gaskets should be treated with de-ice treatments at least each month or when required.

All snow and ice accumulation on equipment should be removed using steam, compressed air or equal.

Fire water hoses that have been used should be drained and dried immediately after use or stored at a frost-free location.

Fire mains should be drained until needed when the temperature is 0°C (32°F) or below.

When the temperature drops below 0°C (32°F), all external fire equipment should be checked daily, or more often when required.

All the fire dampers directly exposed to the weather are to be checked and their function tested every day when the temperature is 0°C (32°F) or below.

Fire pumps, including emergency fire pumps, are to be located in heated compartments. The pumps and their auxiliaries in the compartment are to be adequately protected from freezing at the design service temperature.

Isolating valves are to be in accessible locations. Isolating valves located in exposed locations are to be protected from freezing spray. The fire main is to be arranged so that external sections can be isolated and means of draining are to be provided.

Hydrants are to be positioned or designed to remain operable at the design service temperature. Ice accumulation and freezing are to be considered.

All hydrants are to be installed with a two-handed valve lever or hand wheels and provided with quick connects for hoses. Valves and hydrants exposed to design service temperatures less than or equal to -30°C (-22°F) are not to be of cast iron.

(Vessels operating in low temperature environments, 2006)

16.16 Are procedures in place to safeguard the ballast lines, hydraulic lines, fire lines and bunker lines in sub-freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

If fitted, piping connecting the upper wing tank and lower wing tanks is to be protected from freezing. The supports for ballast piping systems on deck are to be arranged so that free expansion and contraction of the pipes during ballast operations cannot be blocked by accumulated ice or snow.

For hydraulic equipment, the hydraulic oil is to be suitable for the minimum anticipated temperature.

If the hatches are hydraulically operated, the hydraulic oil should be suitable for the minimum anticipated temperature. A heater or other suitable means for heating is to be provided for the hydraulic oil sump, where necessary.

Fire mains should be drained until needed when the temperature is 0°C (32°F) or below.

Piping systems and equipment prone to freezing are to be able to be drained and are to be provided with drain cocks to facilitate drainage.

(Vessels operating in low temperature environments, 2006)

16.17 Are means and procedures in place to safeguard the readiness of the ballast systems in sub-freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Record in comments how the system was protected.

Ballast water tanks for design service temperatures equal to or above -30°C (-22°F) but lower than -10°C (-4°F) are to be provided with arrangements to prevent freezing. These arrangements may be heating systems or turbulence-inducing systems, such as bubbler systems.

(Vessels operating in low temperature environments, 2006)

16.18 Are means and procedures in place to prevent the blockage of vent pipes in sub-freezing temperature? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The blockage of vent pipes by ice accumulation at the deck or by the freezing of plugs inside the pipe can result in safety hazards due to over-pressurization. (Vessels operating in low temperature environments, 2006)

16.19 Are the emergency drill procedures amended prior to entering sub-freezing / polar areas and are the crew being regularly trained with such a procedure? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Crew members are to be provided with proper on-board instructions and be regularly trained in the operation of the vessel's evacuation, survival at sea and on ice / ashore, fire and damage control equipment and systems with appropriate cross-training of crew members with an emphasis on changes to standard procedures made necessary by operations in low temperature environments. (Vessels operating in low temperature environments, 2006)

Prior to entering a polar area, emergency drill procedures should be amended to cover additional topics such as:

- > Donning immersion suits and thermal protective clothing
- > Prevention of cold-related injuries and hypothermia
- > Cold climate survival
- > Lifesaving craft launching

16.20 If the vessel intends to trade in Polar Regions, have the hull underwriters and P&I Club been informed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The ship-owner has to inform their Hull underwriter and P&I club before trading in the Polar Regions

Section 17: Ship To Ship Operation

This section has been compiled, utilising relevant publications from OCIMF and incorporating insights garnered through consultations with experts in the field.

This section is applicable when the bulk carrier has been involved in STS operations in the last 12 months, is involved at the time of inspection, or intends to be involved in Ship-to-Ship operations. These operations include lightering/topping-off to and from barges, as well as the transshipment of cargoes at sea. This involves specialized vessels with conveyors/cranes or the use of offshore floating cranes.

17.1 Does the vessel have an STS operations plan, and are the Master, Officer, and deck rating familiar with this plan? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

STS operations, which involve the transfer of cargo between two seafaring vessels, carry inherent risks that must be effectively managed. The vessel manager should develop an 'STS Plan'. This plan should provide clear policies, procedures, and guidelines to ensure that STS operations are conducted safely and efficiently.

The contents of the STS Plan should cover the following aspects:

Safety: Safety is paramount in STS operations due to the potential risks involved, such as collision, pollution, and personal injury. The plan should outline safety measures, including risk assessments, emergency procedures, and personnel training requirements, to ensure the protection of personnel, ships, and the environment.

Regulatory Compliance: The plan should align with international regulations and industry standards that govern STS operations, such as those established by the IMO or other recognized industry organizations. Compliance ensures that operations meet legal requirements and adhere to best practices.

Operational Procedures: The plan should outline step-by-step procedures for conducting STS operations safely and efficiently. This should include pre-transfer checks, equipment inspection and maintenance, communication protocols between ships and shore authorities, and cargo transfer techniques.

Risk Assessment: The plan should include risk assessment procedures to identify potential hazards associated with STS operations and mitigate risks through appropriate measures. Risk assessment factors may include weather conditions, sea state, proximity to other vessels or structures, and cargo characteristics.

Emergency Response: The procedure should encompass handling emergencies, such as spills, fires, or equipment failures, which are vital components of the STS Plan. These procedures should include protocols for notifying authorities, initiating emergency shutdowns, deploying firefighting equipment, and conducting personnel evacuations if necessary.

Training and Competence: The plan should specify training requirements for personnel involved in STS operations, including the ship's crew, cargo handlers, and supervisors. These training requirements should ensure that personnel possess the necessary skills, knowledge, and competence to perform their roles effectively and safely.

Documentation and Record-Keeping: Proper documentation of STS operations is crucial for regulatory compliance and risk management. The plan should include requirements for maintaining records of safety inspections, risk assessments, operational procedures, incident reports, and personnel training.

17.2

Is the vessel provided with a location assessment, either in the STS plan or the company's SMS? Is there objective evidence to confirm that the location assessment for the last STS operations was conducted, and were the hazards identified in the location assessment addressed through a risk assessment?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

A location assessment prior to a ship-to-ship transfer operation for bulk carriers, such as lightering to or topping from barges or transloaders, is crucial for ensuring the safety, efficiency, and success of the operation. The Master or vessel's manager should present a location assessment either in a generic form or in a form of Risk Assessment adapted at the SMS of the company.

The following key components indicate the importance of the location assessment:

- > Safety
- > Environmental Concerns
- > Operational Efficiency
- > Legal and Regulatory Compliance

Components of a Location Assessment are indicatively shown below:

- > Physical Environment
- > Geographical Considerations
- > Infrastructure and local Resources
- > Environmental Sensitivity
- > Regulatory Compliance
- > Potential location hazards
- > Emergency Response Preparedness

By conducting a comprehensive location assessment that addresses these components, stakeholders involved in ship-to-ship transfer operations for bulk carriers can mitigate risks, ensure compliance, and optimize the efficiency of the operation while prioritizing safety and environmental protection.

17.3

Does the SMS define the duties of the STS superintendent, including the criteria that a Master should meet to be appointed as an STS superintendent, and is the Master aware of the responsibilities of the STS superintendent and the master's overriding authority to make decisions about the vessel's safety?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The STS Organiser appoints the STS Superintendent, who will have overall advisory control of the STS Operation. The role of the STS Superintendent can be delegated to any of the Masters involved in the transfer. The responsibilities of the STS Superintendent include the following:

- > Advising both vessel Masters on safe mooring.
- > Reviewing the mooring apparatus and configuration of both vessels.
- > Assessing vessel compatibility.
- > Reviewing the STS-specific Risk Assessment.
- > Ensuring the correct positioning of fenders.

The STS Superintendent should have, at a minimum, prior experience of the following:

- > Mooring operations with similar types of vessels.
- > Similar loading and discharging operations.
- > Assessing fendering requirements.
- > Location assessment.
- > Drafting a JPO, Risk Assessment, and vessel compatibility analysis.
- > Handling emergency situations, including contingencies.

The qualifications of the STS Superintendent should be assessed in advance by both Masters. In this regard, it is recommended that STS Superintendents maintain detailed records of past STS Operations.

- 17.4 Does the vessel have a compatibility assessment procedure in place, and was the outcome of this assessment communicated with the STS organiser and the participating vessel, and is there objective evidence that any aspect of the compatibility assessment requiring particular management was addressed? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The compatibility assessment should evaluate whether the two vessels involved in the transfer are compatible in terms of size, structure, equipment, and operational capabilities. Prior to Ship-to-Ship (STS) operations, several key aspects should be assessed at a minimum to mitigate risks and ensure successful transfers:

- > Size and Stability
- > Draft and Freeboard
- > Cargo Compatibility
- > Mooring and Fendering Systems
- > Maneuverability and Thruster Capability
- > Communication and Navigation Equipment
- > Environmental Conditions
- > Emergency Response Capabilities

The Vessel Compatibility assessment should be presented by means of a report, which might be specific for each STS type or location.

- 17.5 Was a Joint Plan of Operation (JPO) readily available onboard, and did the master and deck officers demonstrate familiarity with its contents, and were the operations executed in accordance with the plan's requirements?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The JPO should also include as a minimum, the following information:

- > Location assessment
- > Risk Assessment
- > Weather forecast
- > A Mooring plan
- > Fendering Plan
- > Certificates of STS Equipment
- > Last test of cranes
- > Details of involved tugs
- > Sequence of cargo discharging
- > Contingency Plan

The STS Superintendent or any of the participating Masters should ensure that the Joint Plan has been thoroughly communicated and agreed, over the toolbox meeting prior to the STS Operation.

- 17.6 Was the vessel equipped with STS specific contingency plans to address all potential emergencies during the STS operation and, were the Master and deck officers familiar with the contents of these plans and the various contingency scenarios?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The contingency plan should outline the control measures, responsibilities, and actions required on both ships. The plan should be relevant to the location of the operation, taking into account the resources available both at the transfer area and in terms of nearby backup support. It should also cover the requirement for drills in various emergency scenarios.

Emergency scenarios should cover the full scope of the operation. Examples of such scenarios are as follows:

- > Vessel collision during mooring/unmooring maneuvers,
- > Cargo pollution at sea,
- > Fire/explosion,
- > Multiple mooring line failure,
- > Emergency unmooring,
- > Emergency on own vessel or other vessel involved in operation,

17.7

Has the company identified additional roles and responsibilities that necessitate training and familiarisation for the Officers to conduct the STS operation safely and effectively? Have the Master and Officers undergone suitable training for these roles?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The crew may need to take on additional or different tasks, roles, and responsibilities compared to those required during routine port and cargo operations. It's crucial that any additional roles and responsibilities are identified, and that appropriate training is provided before the operation. The training requirements for each ship will vary based on the recent experience of the individuals on board. Factors such as location, services provided, and equipment may necessitate additional training for experienced personnel. If there is little or no experience with Ship-to-Ship (STS) operations, consideration should be given to bringing in additional experienced STS personnel prior to the operation. This can assist with the training of personnel and the execution of the STS operation.

Training may include items such as:

- > Roles and responsibilities of involved parties.
- > Bridge watchkeeping procedures.
- > Deck watchkeeping procedures.
- > Machinery operation.
- > Mooring and unmooring, which includes procedures for passing lines between vessels, properties of mooring lines, fender management, measures to minimise chafing of lines, and awareness of snap-back zones.
- > Operation of cranes /conveyors.
- > Personnel transfer.
- > Transfer equipment.
- > Emergency operations, such as aborting mooring operations, collision, cargo spill, and emergency departure.

The level of knowledge and training required will depend on a person's role and past experience.

17.8

Has the vessel's manager developed procedures that require post-operation feedback and assessment by the Master and, where applicable, was there objective evidence demonstrating that such feedback was provided after the last STS operation?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

The vessel's manager should develop a procedure that details the content of the post-operation feedback and assessment by the master. This may include, but is not limited to, the following:

- > Environmental Conditions (e.g., weather, sea state),
- > Vessel Handling and Maneuvering,
- > Performance/ suitability of the STS Location,
- > Performance of participating vessel,
- > Performance of the STS Organizer/ STS Superintendent,
- > Communication Effectiveness,
- > Equipment Performance and Reliability,
- > Safety Procedures and Emergency Response,
- > Crew Competency and Training,
- > Compliance with Regulations and Guidelines,
- > Incident Reporting and Lessons Learned.

17.9

Was the vessel provided with STS operational and safety checklists, and was there documented evidence that demonstrated the effective use of the checklist?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Checklists are crucial risk management tools designed to ensure that operations are conducted safely. They serve as essential reminders of the principal safety factors to be considered, but they should be supplemented by continuous vigilance throughout the entire operation. Checklists should be developed to specifically address factors that are relevant to the STS operation.

17.10

Was the STS equipment, including fenders, cargo transfer systems, mooring equipment, personnel transfers, and ancillary equipment, in good condition, and was there documented evidence indicating that the STS equipment was maintained and checked prior to the STS operation?(V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Floating Pneumatic Rubber Fenders:

Bulk carrier(s) should be equipped with primary fenders. These fenders must be capable of withstanding anticipated berthing energies and evenly distributing the forces over the appropriate areas of both bulk carriers' hulls. It is recommended to use fenders that are constructed and maintained according to ISO 17357. In line with industry best practices, the safety valve on pneumatic fenders should be inspected at intervals not exceeding two years, and a certificate should be provided as evidence of this inspection. The fenders, and their pennants should be maintained in good condition. Objective evidence of a visual inspection prior to the STS operation should be available on board.

Use of gravity-based self-unloading and/or hybrid self-unloading system:

The components of the gravity-based self-unloading system or hybrid self-unloading system (including conveyor belts and cranes) used for the STS transfer of bulk cargos should be maintained and kept in good working condition, as per manufacturer or industry recommendations.

Mooring Equipment:

It is important that ships involved in STS operations are equipped with good quality mooring lines, efficient winches, well-placed and sufficiently strong fairleads, bollards, and other associated mooring equipment. All mooring equipment should be in good order and free of defects.

Personnel transfer-at Sea operation:

A plan for personnel transfer should be prepared in advance and incorporated into the SMS. This plan should encompass a detailed risk assessment of the entire operation and receive approval from the masters of both ships involved. It should be communicated to all personnel involved. The plan should identify all potential risks and include pre-testing of equipment, establish clear communication lines, and devise a contingency plan for emergencies.

The frequency of personnel transfers between ships should be minimised as much as possible. The safest method of transfer should be determined through a risk assessment, taking into account the prevailing conditions and circumstances at the time and location of the proposed transfer. The risks associated with such transfers should be compared with the risks associated with other available transfer methods.

Potential means of transfer could include the accommodation ladder, a workboat, or a personnel transfer basket. If a cargo or provision crane is used, it must be upgraded prior to transferring personnel and certified for personnel transfer by the classification society.

The Personal Transfer Basket (PTB) should be approved and certified specifically for the transportation of personnel. The PTB should not be repurposed for other uses on board, such as serving as work platforms.

Ancillary Equipment:

Ancillary equipment refers to the additional apparatus used in conjunction with the main equipment to form a complete system. For the STS operation, the ancillary equipment, which includes wires, messengers, stoppers, stops, and shackles, should be inspected for their condition before initiating the STS operation.

17.11

Has the STS Unit Questionnaire been completed by either the STS Organiser or the STS Unit Manager, and is there available objective evidence from the managers, demonstrating that any identified gaps in the questionnaire have been assessed and addressed through appropriate risk mitigation measures to ensure the safety of the Ocean-Going Vessel during operations with the unit(s)? (V)

☐ Yes ☐ No ☐ N/A ☐ N/V

Guide to Inspection

Definitions and Purpose of the STS Unit Questionnaire

The term "unit" refers to any floating structure involved in transshipment operations, including barges, floating cranes, and offshore support vessels. While these may not be classified as traditional vessels, they are essential to the transshipment process.

STS Organiser: A shore-based company responsible for coordinating and managing STS transfer operations. The STS Organiser may also serve as an STS Service Provider.

STS Unit Manager: A company or individual responsible for the technical and operational management of a transshipment unit such as a transloader, floating crane, or barge engaged in STS transfer operations.

This questionnaire is intended to be completed prior to transshipment operations involving the transfer of dry cargo between the unit and an oceangoing vessel. Its primary objective is to assess the condition and operational standards of each unit participating in the operation.

Responsibilities and Requirements

The questionnaire should be submitted by the vessel's manager through established commercial channels typically via the Charterer or Cargo Owner, requesting its completion for each transshipment unit involved in the STS operation.

Given that there may be no direct contractual relationship between the vessel's manager and the STS Unit Manager, all communications should be routed through the party holding the commercial agreement with the STS Unit. This practice is consistent with standard procedures across other sectors of the maritime industry and helps ensure clarity, accountability, and proper coordination.

Once completed, the questionnaire remains valid for a period of 12 months. If multiple operations are conducted with the same STS Unit within that timeframe, it is not necessary to complete a new questionnaire for each operation.

The information gathered is used to:

- > Assess the level of compliance of all parties involved
- > Evaluate the operational standards of participating units
- > Identify the necessary risk mitigation measures, taking into account all parameters that may impact the safety and efficiency of the STS operation.
- > Improve onboard STS practices on the mother vessel.

Documentation and Inspection Protocol

For any STS operation involving non-ocean-going vessels, inspectors must upload the completed questionnaire to Q 17.11. A copy of the completed form should be readily available onboard and included in the final inspection documentation.

Exceptional Circumstances

In cases where, despite the vessel manager's due diligence, the questionnaire cannot be completed in time, the vessel manager should:

- > Conduct and document a risk assessment addressing the absence of the STS Unit Questionnaire
- > Notify relevant stakeholders as required

Click [here](#) to access a sample of the questionnaire for reference and review.

Annex A

This table outlines the major changes introduced in RISQ Revision 3.2. All references should be made directly to the RISQ 3.2 document.

Please note that this table is intended solely to raise awareness of key updates and should not be used during inspections or vessel preparation. For any questions regarding the table below or any aspect of RISQ, please contact risq@rightship.com

Section	Question No.	Revision Description
Table of content	N/A	A new Section 8F, designed for the inspection of Heavy Lift Vessels, has been added.
Abbreviations	N/A	Multiple numbers of abbreviations have been added or updated.
Introduction	N/A	The Type, Scope, and Timing of Inspection has been updated to address the expected time of inspection for each type of inspection. Standard Inspection A full onboard inspection by a single inspector, should last 10–12 hours. Hybrid Inspection Requires submission of all documentation at least 72 hours before boarding. If documentation is complete and the vessel is prepared per RISQ standards, the inspection can be completed in 8–10 hours. Dual Inspection Two inspectors share responsibilities, aiming to complete the inspection within 6–7 hours. Note: Durations refer to net inspection time and exclude interruptions such as meal breaks, PSC, or Flag State inspections.
Introduction	N/A	A new topic "RightShip Inspection Enhanced Training Requirements" added into the introduction section that define the acceptable training format including CBT, classroom and online learning model as well as the requirement of following the IMO model course when such model course available and use the subject matter expert in the absence of IMO model.
Section One	1.5	Question 1.5 has been deleted from the RISQ.
	1.17	The guidelines of Q 1.17 has been updated and now define the Carbone Intensity Indicator (CII) and its application and the place where the attained EEDI can be find.
	1.18	Question 1.18 has been deleted from the RISQ.
	N/A	Two new questions have been added to the inspection process: • Is the vessel manager a subscriber to the Dry Bulk Centre of Excellence? • Has a DryBMS self-assessment been completed? The responses to these questions are not recorded in the list of findings and do not impact the inspection outcome.
	1.21	A new guideline has been added and requires the inspector to record the finding under Q4.3, if the vessel has not been inspected through a combination of marine and technical superintendent visits at regular intervals, as required by the inspection guide.
	1.22	Q1.22 that captures the name of inspector has been updated to capture the name of Co- inspector in case of the dual inspection.
	1.24	The question was redesigned to capture the type and time of inspection. For hybrid inspections, it records the time spent reviewing documents prior to boarding the vessel, the time spent reviewing documents onboard, and the reason for that time allocation. For all inspection types, it also captures the dates of commencement and completion of the opening and closing meetings.
Section Two	2.1	The inspection guide has been updated. Inspectors are now required to record any findings if the vessel has an overdue Condition of Class. Additionally, inspectors should record in the comments if the vessel has any Condition of Class, Flag Dispensation, Significant Recommendation and Memorandum.

	2.6	For the purpose of clarification about the ship handling course, RightShip clarified the definition of a vessel as 'large' or having 'considerable deadweight or length' and consider large vessel or vessel having considerable deadweight or length if it exceeds either: • 270 meters in length overall (LOA), or • 140,000 deadweight tons (DWT). It is now required that the Masters, with less than 36 months of sea service in rank, or those newly promoted, should complete a ship handling training course aligned with the objectives of STCW Code B-V/a. The acceptable type of training (onboard training or approved shore-based training facility) and the requirement of the training has been clarified.
	2.7	The application of this question has been clearly updated in the Guide to Inspection, stating: "If the ship is not carrying solid bulk cargo, the question should be answered N/A." Furthermore, it is now required that the relevant training course aligns with IMO Model Course 1.45, titled "Safe Handling and Transport of Solid Bulk Cargoes."
	2.8	The application of this question has been clearly updated in the Guide to Inspection, stating: "If the vessel is not handling packaged dangerous, hazardous, and harmful cargoes (dangerous goods) or was not carrying such cargo at the time of inspection or during the previous voyage, respond to this question with N/A" Furthermore, it is now required that the relevant training course aligns with IMO model course 1.10, "Dangerous, Hazardous, and Harmful Cargoes."
	2.9	The guidance has been updated and requires inspector to "Record in comments, the date of the last drug and alcohol test that was carried out on board either by an independent agency or under controlled conditions by ship's personnel with specimens being forwarded to an independent agency. Record in comments the date of the last unannounced alcohol test conducted on-board." The guideline has also urged the Ship manager with offices in multiple jurisdictions to seek legal advice, as there may be country-specific restrictions on the importation, possession, storage, or use of drugs and alcohol. Additionally, legal obligations concerning drug and alcohol testing may vary by location. Non-compliance with such regulations can result in legal consequences for both the employer and the employee. However, the above is an advice to ship managers only and guidelines clearly emphasis that the inspector is not required to verify whether the vessel's manager has sought legal advice in the preparation of the company's drug and alcohol policy and procedures during the physical inspection.
	2.10,2.11,2.12	The questions 2.10,2.11, and 2.12 have been deleted from the RISQ.
	2.13	The condition under which a finding should be recorded has been clarified in the guidance: "Inspectors should record a finding when the officer matrix has not been updated to reflect the officers who were on board at the time of the inspection. An allowance should be made for any officer who has changed within the previous seven days".
	2.14	The question has been revised to place greater emphasis on the familiarisation of the Master and Deck Officers with the ECDIS equipment. The guidance has also been updated to require ship managers to develop clear and specific procedures to ensure that the Master and all watchkeeping officers are effectively familiarised with the use of ECDIS equipment. Key points to be considered are clearly outlined in the Guide to Inspection under Question 2.14.
	2.15	Question 2.15 has been deleted from the RISQ.
Section Three	3.1	The inspection guide now requires navigation officers to be familiar with the actions needed in case of single or multiple ECDIS failures. This should be achieved through ECDIS familiarisation checklists or regular drills. Inspectors are expected to record a finding if the bridge team is not familiar with the company's contingency plans for such failures.
	3.2	To ensure alignment with RISQ expectations, the term bridge team has been added to the inspection guide. In this context, bridge team specifically refers to the Master and the Officer of the Watch. This clarification helps ensure consistency in understanding and application during inspections.

	3.3	The guideline has been updated to align with the IMO amendments to the 2021 Guidelines on the Shaft/Engine Power Limitation System, in accordance with the EEXI requirements and the use of a power reserve (Resolution MEPC.335(76), as amended by Resolution MEPC.375(80)). As a result, the following documents, outlined in the appendices to the Recommendation on the Provision and Display of Maneuvering Information on Board Ships (Annex, Resolution A.601(15)), must be updated to reflect the ship's maneuvering characteristics under two conditions: • When full shaft and engine power is available. • When shaft or engine power is limited. The documents to be updated are: • The Pilot Card • The Wheelhouse Poster • The Maneuvering Booklet
	3.10	The guideline has been updated to recognise the importance of carrying a spare magnetron onboard. If the radar magnetron has not been replaced according to the manufacturer's recommendation, but mitigation measures are in place—such as regular performance monitor tests per watch and maintaining a spare magnetron as essential spares, inspectors should not record a finding. Instead, they should record a comment noting the date of the last magnetron replacement and the mitigation measures in place.
	3.12	The inspection question has been revised to ensure navigation officers are familiar with the company's procedures for using the BNWAS. Inspectors should confirm that officers understand the procedure and that records are available showing BNWAS was operated while the ship was underway and at anchor and tested in accordance with company procedures. Vessel managers are now required to establish procedures for BNWAS operation, including when it should be active, who is responsible for activating it, and how it should be activated. The procedures must also include safeguards against unauthorised deactivation, actions to take if a stage 2 or 3 alarm is triggered, and regular testing to ensure proper system functionality.
	3.16	The guideline has been updated to accept software equivalents to the NP133C Admiralty ENC Maintenance Record Book. It now states that either the NP133C or equivalent software should be available onboard.
	3.18	The inspection question has been updated to confirm that the ECDIS is of an approved type, compliant with SOLAS requirements, and that both the Master and watchkeeping officers can demonstrate familiarity with its use. The guideline has been updated to reinforce SOLAS requirements for ECDIS used to meet chart carriage regulations. ECDIS must be type-approved, use up-to-date ENC's, comply with current IHO standards, and have independent backup arrangements. Inspectors should verify the installed IHO standard version and confirm it is current. If the manufacturer recommends specific spare parts, these should be available onboard.
	3.19	Question 3.19 has been deleted from the RISQ.
	3.24 and 3.25	Inspection questions 3.24 and 3.25 have been consolidated to address track monitoring during pilotage, with added emphasis on the use of parallel indexing techniques in coastal and pilotage waters, particularly during restricted visibility or nighttime conditions. This consolidation has also resulted in the deletion of question 3.25. The guideline of Q3.24 has been updated to support best practices for position verification and passage monitoring. Radar overlays should be used regularly, as defined by company SMS, across various navigation conditions (e.g., open waters, confined waters, fairways, pilotage). Traditional position plotting techniques, such as visual or radar fixes using lines of position, should be used as a cross-check. Compliance with the passage plan must be closely monitored by the Officer of the Watch (OOW), using all available means including visual/radar fixes, echo sounder, and verification of navigation equipment integrity.
Section Four	4.1	The inspection guideline for Question 4.1 has been expanded to include cargo operation requirements specific to each ship type under RISQ. Risks and operational requirements are now outlined for: • Solid bulk cargoes: liquefaction, fire/explosion hazards, and structural damage risks. • Grain cargoes: shifting, contamination, and toxic vapor exposure. • General cargoes: various industrial and packaged goods. • Container cargoes: stowage and securing practices, mis-declared weights, visibility limitations, and weather-related risks. • Self-unloading vessels: the SMS should include detailed procedures for safe operation, emergency handling, and equipment maintenance.

	4.2 and 4.12	The question has been updated to reflect the qualifications and responsibilities of the safety officer. The guideline has been updated to clarify expectations for Safety Officers and safety management onboard. On ships with five or more seafarers, companies must arrange for the election of safety representatives with at least two years of consecutive sea service after age 18. A Safety Committee, chaired by the Master, must be established to regularly discuss health and safety matters, especially following serious incidents. It should not be used for training purposes. Vessel managers should implement procedures for regular safety inspections across all accessible areas of the ship, conducted by the designated Safety Officer. A specific checklist should support the inspection of the main deck areas, forecandle, machinery space, and accommodation block. Question 4.12 has been deleted.
	4.3	The guidelines now include updated superintendent visit and inspection requirements, to ensure effective oversight. Key Requirements: Superintendent Inspections Conducted every six months (± 1 month tolerance), not exceeding seven months. Marine and Technical Superintendent visits must not exceed 14 months between successive inspections. Inspections must follow a systematic process and result in a standardised report covering vessel condition, defects, procedural weaknesses, and onboard management. Internal Audits • May count as superintendent visits, provided a separate inspection report is prepared in addition to the documented audit report. • Should be conducted by personnel with working knowledge of the vessel manager's Safety Management System. Remote Inspections • Permitted under a formal program with defined procedures, verification methods, and standardised reporting. Non-Qualifying Visits Visits by senior management or superintendents addressing specific issues without a full inspection/report do not qualify. The superintendent inspection should be conducted by personnel with working knowledge of the vessel's Safety Management System, and the use of a third-party company is not considered a qualifying inspection under these criteria.
	4.4	The guidelines now require that the periodic master review process be designed to ensure meaningful feedback. Master reviews should include both positive and negative observations, supported by evidence, and should not be treated as a tick-box exercise.
	4.5	The inspection guide has been thoroughly revised to comply with the requirements of IMO Resolution MSC.581(110), issued in June 2025.
	4.6	To ensure crew preparedness for rescuing a stevedore from the cargo hold, the guidelines now require that enclosed space entry/rescue drills include a scenario involving rescue of a person from the cargo hold.
	4.7	The guidelines now require that hot work conducted outside designated spaces should follow safety protocols. A written plan should be prepared, reviewed, and formally approved by the vessel management company prior to commencement. All personnel involved should be familiar with the permit-to-work system and comply with defined safety requirements throughout the operation.
	4.8	The revised question introduces the requirement for a Stop Work Authority policy and procedure, alongside effective implementation of a specific Permit to Work system and Lock-Out/Tag-Out (LOTO) controls. The guidelines now ask whether these systems have been formally introduced and are being used effectively to support safe operations. The guidelines now include underwater operations as an example of high-risk jobs, reinforcing the need for strict safety controls when such activities are undertaken. To ensure maximum safety when working with hazardous energy sources or in spaces containing such sources, the guidelines now require integration of the Lockout/Tagout (LOTO) system with the Permit to Work (PTW) process. Operating these systems independently may lead to critical safety gaps. Key Benefits of Integration: • Ensures all hazardous energy sources are isolated before work begins • Provides a unified workflow for managing high-risk tasks • Enhances accountability, traceability, and regulatory compliance • Reduces the risk of human error and miscommunication
	4.9 and 4.40	The guideline for Question 4.9 has been expanded to include specific guidance on the muster list, covering both scheduled drills and emergency shipboard situations. This enhancement ensures that muster list requirements are addressed within the broader context of emergency preparedness and crew response. Question 4.40 has been deleted from the RISQ.

	4.10	The revised question now requires procedures for incident reporting and investigation to include a summarised lessons learned bulletin or report for each incident, ensuring effective communication and continuous improvement. Inspectors are now required to record in the comments the number of near-miss reports submitted by the vessel over the past 12 months. The guidelines now require inspectors to review incident records from the past 12 months and verify whether an investigation report or lessons learned bulletin has been provided for each case marked as completed by the vessel's manager. Additionally, near-misses must be investigated under ISM Code provisions, and vessel managers should actively promote near-miss reporting to support safety awareness and continuous improvement.
	4.11	The guidelines now recognise various international safety helmet standards and emphasise the importance of following manufacturer instructions. Helmets should be replaced after significant impact or when they no longer meet required safety standards to ensure proper head protection. The guidelines now include specific PPE requirements for ships using ammonia as fuel. Suitable protective equipment, including eye protection meeting recognised national or international standards, must be provided for crew involved in ammonia fuel system operations. All required PPE must be stored in clearly marked, easily accessible lockers.
	4.13	The guideline now requires the ship-shore safety checklist to be shared with the stevedore representative when stevedores are on board. Both the Stevedore and the Master's representative are responsible for ensuring compliance, including Item 13 of the BLU checklist. Access hatches to enclosed spaces should remain locked, and effective access control should be maintained. The checklist should include Stevedore familiarisation with emergency procedures and verify cargo hold entry compliance at least once per watch.
	4.17	The guideline has been updated to allow for cases where manufacturer guidelines are unavailable. A finding should be recorded if the annual check of oxygen-acetylene welding equipment has not been carried out by a competent person per manufacturer guidelines, or if no evidence of such a check exists. In the absence of manufacturer guidelines, certified service providers should be consulted.
	4.18	The updated guideline now requires inspectors to assess crew familiarity with life raft davit operations under both normal and emergency power modes. Inspectors should also verify the functionality of the davit system using both power sources to ensure operational readiness and safety compliance.
	4.22	Question 4.22 has been deleted from the RISQ.
	4.23	The guideline now addresses the harmful effects of Perfluoro-Octane Sulfonic Acid (PFOS). Fire-extinguishing media containing PFOS must be disposed of at appropriate shore-based facilities no later than the first survey on or after 1 January 2026.
	4.27	To ensure a consistent approach, the guideline clarifies that inspectors are not required to witness a physical test of the water mist system head unless visible defects justify it. The guidelines now align with MSC.1/Circ.1432, requiring foam proportioners and mixing devices to be tested at least every five years to ensure they operate within the correct mixing ratio tolerance.
	4.30	The guideline clarifies that firefighter outfits must fully protect the skin, in line with FSS Code Chapter 3, Section 2.1.1. It also requires that two-way portable radios used by fire parties comply with IACS UI SC291 and be operable while wearing full firefighting gear, ensuring effective communication during emergencies.
	4.35	The guideline now addresses the use of suspended or floating accommodation ladders, acknowledging that such use may conflict with engineering design. However, it allows their use with strict mitigation measures, including risk assessment, signage, secure rigging, and compliance with relevant regulations. Inspectors should not record a finding if all specified safety measures are properly implemented.
	4.37	The guidelines now consider two approaches to mitigate the risk posed by counterfeit pilot ladders and certificates, which remain a serious safety concern in the industry. To address this risk, ship managers should adopt one or both of the following measures: Maintain accurate records Verify certificate authenticity.

	4.39	Guideline updated to require inspectors to obtain and attach the helicopter landing area strength confirmation letter to the RightShip inspection report under Question 4.39. Inspectors should also provide 360-degree photographs of the landing site and access, where practicable. Vessels designated as "Winch Only" are exempt from the strength letter requirement.
	4.43	The updated guideline recommends one-on-one deck officer familiarisation using a common working language. Companies should use BPG checklists C2.3 and C2.4 to create tailored checklists covering equipment and procedures relevant to each team member's role.
	New question	New inspection question (Q4.44) added to address stevedore safety during cargo operations. Vessel managers should have documented procedures to identify hazards and implement mitigation measures. The guideline outlines shared responsibilities between the vessel and stevedore company, including pre-work briefings, access control, supervision, risk assessments, and activation of stop-work authority when unsafe conditions arise.
Section Five	5.3 and 5.4	Guideline for Question 5.3 updated to require the use of scupper plugs during cargo operations involving residues harmful to the marine environment. Vessel managers should implement procedures to maintain deck containment integrity, including proper use of drain valves and controlled drainage of accumulated water. As a result of this update, Question 5.4 has been removed.
	5.9	Question 5.9 and its associated guideline have been revised to address the maintenance of emergency pumping arrangements for spaces located forward of the collision bulkhead, in alignment with SOLAS 74 (2020) requirements. The updated question asks whether these arrangements are in good order and prominently marked. Water ingress detection and alarm systems must be located on the navigation bridge, cover all relevant forward spaces (e.g., Fore Peak Tank, Bosun's Store, Forecastle Space), be of an approved type, and powered by two independent sources with alarm on primary power failure. Vessel managers must develop vessel-specific procedures, including use of pumps/eductors, valve markings, and emergency response instructions.
	5.10	Question 5.10 revised to include maintenance requirements for Ballast Water Treatment Systems (BWTS) in line with manufacturer recommendations. The guideline now addresses biological risks associated with systems using Active Substances (AS) and UV treatment. It highlights key system alarms (e.g., low/high TRO*, TRO communication) and their implications for treatment effectiveness and marine pollution. Vessel managers should ensure TRO sensors are calibrated and maintained, and that ballast water operations are correctly recorded in the Ballast Water Record Book (BWRB) in accordance with IMO guidelines. If using electronic record books, a ship-specific declaration must be carried onboard confirming compliance with BWM Convention requirements. TRO*: The Total Residual Oxidant.
	5.11	Question 5.11 has been deleted from the RISQ.
	5.15	Question 5.15 has been updated to assess engineering officers' familiarity with company procedures for operating oil filtering equipment, ensuring compliance with MARPOL Annex I and adherence to manufacturer instructions. The guideline now incorporates technical requirements from MEPC.107(49) for systems installed on or after 1 January 2005, including: • Representative effluent sampling to the 15ppm bilge alarm • Fail-safe arrangements to prevent discharge during alarm malfunction • Alarm activation and automatic stopping device response times • Operational testing procedures simulating overboard discharge • Inspection of valve sealing and sample line integrity RightShip aligns with AMSA's interpretation that failure of the 15ppm bilge alarm to activate the stopping device in the absence of a representative sample constitutes noncompliance. Findings should be recorded where systems fail to meet MEPC.107(49) standards. The guideline has been updated to include operational testing of the oily water separator. Inspectors are now required to verify system functionality by simulating discharge and observing activation of the 15ppm bilge alarm and automatic stopping device. Additional testing is required if sampling valves are not sealed by Class, including interruption of sample flow to confirm alarm and stopping device response times. Any failure to meet MEPC.107(49) requirements should be documented as a finding. Guideline updated to require specific warning signs at the oily water separator's overboard discharge valve. Effective sealing arrangements should be in place to prevent accidental opening, and the system should be designed to protect against unauthorized access or unintentional operation.
	5.16 and 5.17	Questions 5.16 and 5.17 have been deleted from the RISQ.

	5.19	Guideline updated to reinforce requirements for maintaining a Garbage Record Book in compliance with MARPOL Annex V. Applicable to ships ≥ 400 GT, vessels certified to carry ≥ 15 persons on international voyages, and all fixed or floating platforms. Key updates include: • Mandatory recording of all discharges, incinerations, and accidental losses • Entries must include date/time, position, garbage category, and estimated volume • Record Book must be retained onboard and preserved for at least two years • Entries must be in English, French, or Spanish, with the flag state's language prevailing in case of disputes • Garbage categories defined in Annex V Appendix (A–I), including plastics, food waste, e-waste, etc. • Volume estimates should be made in cubic meters, acknowledging variability due to processing. Requirement for cleaning garbage storage areas has been moved under Question 5.19. The updated guideline states that garbage collected throughout the ship should be delivered to designated processing or storage locations. Regular cleaning and disinfecting of garbage storage areas are essential for both preventative and remedial pest control.
	5.20	Q5.20 has been deleted from the RISQ.
	5.21	Guideline updated to reflect requirements for SEEMP Part III as a dynamic document that must be updated periodically with current fuel consumption data to track CII performance and energy efficiency measures. SEEMP Part III must undergo company audits at least every three years, as outlined in MEPC.347(78). Audits must be conducted within six months of issuing the Statement of Compliance. Classification Societies may perform audits on behalf of the flag Administration. Additional verification and audits are required for ships rated D for three consecutive years or rated E, to ensure corrective action plans are in place.
	5.22	Q5.22 has been deleted from the RISQ.
	5.23	Guideline updated to require reporting of EGCS (Exhaust Gas Cleaning System) malfunctions lasting over one hour to flag and port state administrations. Emphasis placed on proper maintenance and calibration of SO_2 monitoring equipment to avoid undetected failures. Zero or negative SO_2 readings should be scrutinized, especially under conditions that typically produce low emissions. Engineer officers are advised to perform plausibility checks and validation tests. Manufacturer-recommended corrective actions must be followed, including leak checks, calibration, and servicing.
	5.24	Q5.24 has been deleted from the RISQ.
	5.25	Question has been revised to include assessment of the ballast pumping system, its control panel, and associated instrumentation. The updated question asks whether the ballast pumping systems, including instruments, controls, valves, pipework, pressure and draft gauges, and remote-control systems, are in good working order, and whether there is recorded evidence of regular inspection.
Section Seven A (Oil Fuel)	7A.2	The guidelines under Question 7.2 have been updated to clarify the testing requirements for bunker transfer systems. Bunker pipelines are defined as any pipeline used for taking on, discharging, or internally transferring fuel for onboard consumption. The system scope includes the discharge pump and piping up to the vessel's manifold but excludes non-metallic hoses. Testing requirements now specify that an annual hydrostatic test should be conducted at 100% of the Maximum Allowable Working Pressure (MAWP), while a more rigorous hydrostatic test at 150% of MAWP should be performed twice every five years. The MAWP is determined either by the relief valve setting or, if no relief valve is fitted, by the maximum pump discharge pressure. All pipelines should be clearly marked with the test date and pressure. Pressure testing should be a hydrostatic test, pressure testing using compressed air is not acceptable.
	7A.4	The guideline has been clarified to specify that the height of any save-alls surrounding fuel, diesel, and lubricating oil tank vents should be lower than the vent heads themselves, including any fitted extension pieces. This change ensures that water ingress or water seal formation is prevented, which could otherwise obstruct vacuum pressure release during adverse weather conditions if the save-alls become filled with water.
	7A.5	The definition of fuel oil has been updated to improve clarity and ensure comprehensive coverage of fuel types used on ships. The revised definition states that fuel oil means any fuel delivered to and intended for combustion purposes for propulsion or operation on board a ship. This includes gas, distillate, and residual fuels.
	7A.6	Question 7A.6 has been deleted from RISQ.
	7A.7	A new requirement has been added to Question 7.7, stating that deck scuppers must be securely plugged during all bunkering operations to prevent the accidental discharge of fuel into the sea.

	7A.8	Question 7A.8 has been deleted from RISQ.
	7A.10	The updated guidelines require that in multi-engine installations using a shared fuel source, each engine must have independent fuel and spill line isolation that does not affect the operation of other engines and can be operated remotely in case of fire (SOLAS 74, 2020). AMSA Marine Notice 2024/05 recommends installing quick-closing valves between the fuel changeover valves and each engine, with remote actuators located elsewhere in the machinery space. For ships built after June 2009, valves or actuators should be at least 5 meters from the engine or otherwise protected. Refer to AMSA Notice 2024/05 and MSC.1/Circ.1321.
Section 7 C (Methanol)	7C.2	The guideline now requires a documented methanol bunkering procedure, which may be included in the ship's fuel handling manual. It should detail qualified personnel, their roles, system design parameters, de-bunkering steps, safety zones, SIMOPS assessment, nitrogen padding, hose pressure testing and purging, vapour recovery, operational limitations, PPE and firefighting equipment, gas detection tools, SSL use for ESD communication, QCDC use at the manifold, and emergency contacts. ESD: Emergency Stop Device SSL: Ship Shore Link QCDC: Quick Connect and Disconnect Coupling
	7C.3	The guidelines now require methanol and methyl/ethyl alcohol fuel-related drills and exercises to be included in the SMS, along with a schedule for regular drills.
	7C.4	The updated guideline requires the ship's fuel handling manual to include firefighting and emergency procedures, covering the operation, maintenance of fire systems, and use of extinguishing agents.
	7C.11	Ships using methanol or methyl/ethyl alcohol fuel are now recommended to be equipped with thermal cameras, lightning detectors, and CCTV at the bunkering station to support leak and fire prevention. Crew should be trained in the use of this equipment and familiarised through shipboard drills.
Section 7D (Ammonia)	Fuel Management (Alternative Fuel- Ammonia)	The entire section on Ammonia Fuel has been updated following consultation with subject matter experts. For full details of these changes, please refer to the RISQ 3.2 document.
Section 8A	8A.1	Question 8A.1 has been deleted from the RISQ.
	8A.5	Question 8A.5 has been deleted.
	8A.10	Under guideline Q8.10, ships must be provided with an approved stability and loading booklet written in a language understood by the ship's officers, ensuring safe and compliant vessel operation.
	8A.17	The inspection guidelines now include updated provisions for Direct Reduced Iron (DRI), following the 2025 amendment to the IMSBC Code which introduces DRI (D) as a new category. DRI (D) refers to by-product fines with moisture content $\geq 2\%$, differing from DRI (C), which must not exceed 0.3%. DRI (D) is classified as both Group A (prone to liquefaction) and Group B (Materials Hazardous only in Bulk), with the main hazard being flammable hydrogen gas generation. The IMSBC Code emphasizes hydrogen concentration control through surface ventilation.

	8A.18	The wording of Question 8.18 has been revised to assess whether measured parameters related to coal cargo are actively monitored to ensure ship and cargo safety. It now asks whether the ship is equipped with functioning instruments to monitor cargo temperature, hold atmosphere, and bilge pH levels, and whether these parameters are measured, recorded, and monitored, with appropriate actions taken in response to self-heating or excessive gas concentrations. Inspectors are now required to review the cargo logbook to confirm that all necessary records have been maintained. A finding should be recorded if the records are incomplete, or if no action was taken when methane (CH₄) levels reached or exceeded 20% of the Lower Explosive Limit (LEL), or carbon monoxide (CO) levels exceeded 50 parts per million (PPM). The guideline now advises that if methane (CH₄) levels are rising and exceed 20% LEL, ventilation should be prioritised due to the higher explosion risk, and expert advice should be sought. If carbon monoxide (CO) levels exceed 50 PPM in unventilated holds, this may indicate coal self-heating. In such cases, expert advice is recommended, and gas measurements should be taken at least every 12 hours until conditions stabilise.
Section 8B	8B.1	Question 8B.1 has been deleted from the RISQ.
	8B.3	Question 8B.3 has been deleted from the RISQ.
	8B.11	The guideline has been updated to require that ships carry an approved stability and loading booklet written in a language understood by the ship's officers.
	8B.25	Question 8B.25 and its associated guidelines have been revised to address situations where the fumigator-in-charge is unavailable at the discharge port and a gas-free certificate may not be issued. Given the acute toxicity of fumigants and the limitations of periodic atmospheric monitoring in detecting dangerous concentrations, it is critical to implement robust safety measures. In such cases, the Master should promptly notify the Port Authority and Ship Manager, clearly communicate the absence of certification and any safety concerns, and document all actions taken. A thorough risk assessment should be conducted prior to discharge operations, with appropriate PPE provided to guard against fumigant exposure and other hazards such as oxygen depletion. Hazardous or unattended areas should be secured to prevent unauthorized access, and atmospheric testing should be carried out frequently using calibrated equipment specific to the fumigant in use. Crew members should be empowered to intervene in any unsafe behavior. Vessel managers are responsible for ensuring that the SMS includes clear operational guidelines and for supporting the Master in implementing necessary safety protocols when fumigation company representatives are not present.
	8B.33	The guidelines have been updated to recognise certain types of cargo hold lighting such as LED and fluorescent lights, as safe for continuous use, provided specific conditions are met. These lights are considered low-risk due to their low heat emission and reduced ignition potential. If such lighting is installed, and there is objective evidence of regular maintenance in accordance with manufacturer recommendations, along with SMS procedures that address isolation requirements based on cargo type, and those procedures have been followed, the inspector should select "Yes" to the relevant question. Additionally, the inspector should record in the comments section the measures in place to ensure the safety of the cargo within the hold.
	8B.28	Question 8B.28 has been deleted from the RISQ.
Section 8C	8C.1	Question 8C.1 has been deleted from the RISQ.
	8C.3	The guidelines have been updated to incorporate key safety requirements from the AMSA Information Sheet regarding the carriage of Ammonium Nitrate. It reminds ship managers of the importance of stowing the cargo under deck in a clean space that can be accessed in an emergency, ensuring protection from heat sources, avoiding direct contact with metal engine room bulkheads, and permitting carriage in Flexible Intermediate Bulk Containers (FIBC). A link to the AMSA Information Sheet has been provided within the guidelines for reference.
	8C.6	The guidelines have been updated to include a new requirement stating that there should be a procedure in place for identifying and removing damaged lashing devices from service.
	8C.7	Question 8C.7 has been deleted from the RISQ.

	8C.20	The inspection question has been revised to include a new requirement regarding spare parts. It now asks whether, in addition to having procedures for monitoring the temperature of refrigerated containers and maintaining temperature records, the vessel also carries sufficient spare parts on board to support the continued safe operation of refrigerated units. The guidelines have been updated to reinforce the requirement for carrying additional spare parts for refrigerated containers. It now specifies that, in the event of reefer container breakdowns, ships should have adequate spare parts on board and the necessary skills to carry out emergency repairs. Furthermore, the ship must promptly notify relevant parties of any reefer malfunctions that cannot be repaired on board, ensuring timely intervention and minimising cargo risk.
	8C.21	Question 8C.21 has been deleted from the RISQ.
	8C.34	Question 8C.34 has been deleted from the RISQ.
	8C.37	The guidelines have been updated to recognise certain types of cargo hold lighting such as LED and fluorescent lights, as safe for continuous use, provided specific conditions are met. These lights are considered low-risk due to their low heat emission and reduced ignition potential. If such lighting is installed, and there is objective evidence of regular maintenance in accordance with manufacturer recommendations, along with SMS procedures that address isolation requirements based on cargo type, and those procedures have been followed, the inspector should select "Yes" to the relevant question. Additionally, the inspector should record in the comments section the measures in place to ensure the safety of the cargo within the hold.
Section 8D	8D.1	Question 8D.1 has been deleted from the RISQ.
	8D.2	The guidelines have been updated to incorporate key safety requirements from the AMSA Information Sheet regarding the carriage of Ammonium Nitrate. It reminds ship managers of the importance of stowing the cargo under deck in a clean space that can be accessed in an emergency, ensuring protection from heat sources, avoiding direct contact with metal engine room bulkheads, and permitting carriage in Flexible Intermediate Bulk Containers (FIBC). A link to the AMSA Information Sheet has been provided within the guidelines for reference.
Section 8E	8E.1	Question 8E.1 has been deleted from the RISQ.
Section 8F	Cargo Operation- Heavy Lift Vessel	A new section has been developed to cover the inspection of heavy lift vessels, specifically those designed to transport non-standardised heavy cargoes. These specialised ships are categorised into four main types: project cargo ships, open deck cargo ships, dock ships, and semi-submersible ships. The guideline clarifies that this section does not apply to general cargo ships that occasionally transport heavy lift items (i.e., heavy lift carriers).
Section 9A	9A.2	The guideline now requires vessel managers to establish a procedure for measuring hatch cover component clearances every six months. It also provides enhanced guidance for key components, rubber packing, bearing pads, locators, and stoppers, highlighting the importance of regular inspection and referencing manufacturer specifications to ensure hatch cover integrity and cargo safety. Each vessel should carry and maintain an up-to-date list of the minimum recommended spare parts for the safe operation and to support both planned and corrective maintenance of type of hatch cover, as specified by the manufacturer.
	9A.15	The criteria for recording a finding have been revised.
	9A.19	The guideline for crane inspections has been updated to require that rocking tests be conducted at least once every six months. Inspectors should now record findings if such tests have not been performed within this timeframe. Results should be recorded in the Registry of Lifting Appliances and compared against manufacturer specifications.

Section 9B	9B.16	A new guideline has been added under question 9B.16 recognising the use of radios as a valid means of communication during crane operations. It outlines key considerations, including the need to test radios before use to ensure clear transmission, the use of a separate working channel where possible, and equipping the crane operator's radio with a hands-free system. Radios should not be used for personal communication, and all personnel involved should be trained in the proper use of radio equipment to ensure safe and effective operation.
Section 10	Introduction	The introduction section has been updated to reference key regulatory and industry standards relevant to mooring equipment and safety. It now cites SOLAS Regulation II-1/3, MSC.1/619 and 1620, IACS Recommendation No. 10 Rev. 5, and specific sections of the Mooring Equipment Guidelines Edition 4 (MEG4), including Section 5 (Mooring Lines), Section 6 (Mooring Winch), and Appendix B (Guideline for the Purchasing and Testing of Mooring Lines and Tails).
	10.1	The guideline has been updated to align with 33 CFR 401.12, which requires that every vessel carry a minimum of two spare mooring lines that are available and ready for immediate use.
	10.2	Question 10.2 has been deleted from the RISQ.
	10.3	The criteria for recording findings and comments under Question 10.3 have been updated.
	10.4	The stowage requirements for mooring ropes have now been moved under question 10.4. The guideline specifies that mooring ropes should be stowed clear of the deck, preferably on a pallet, and the area must be free of obstructions that could hinder visibility of the mooring deck. Additionally, mooring stations should be painted with a non-slip treatment to enhance safety during operations.
	10.7	Question 10.7 has been deleted from the RISQ.
	10.10	The guideline has been updated to specify that the eyes of mooring pendants (tails) should be fitted with chafe protection. In cases where mooring pendants are not used, the mooring lines themselves should be protected with chafe gear to prevent wear and damage.
	10.11	Question 10.11 has been deleted from the RISQ.
	10.12	The question has been updated to acknowledge alternative acceptable types of heaving lines. It now asks whether the heaving lines are constructed with either a monkey's fist or a small, high-visibility soft pouch at one end, ensuring the total weight does not exceed 0.5 kg.
	10.13	The question now highlights mooring station safety, asking if the area is well maintained and clearly marked as hazardous. The guideline aligns by requiring stations to be well lit, clean, and free of oil leaks, with surfaces prepared to prevent slips.
	10.14	The inspection guideline for Question 10.14, concerning messenger lines, has been revised to align with OCIMF recommendations. It highlights the importance of using appropriate stoppers due to the varying properties of synthetic fibre lines. Lines made from polyamide, polypropylene, polyethylene blends, or high-modulus synthetic fibres (HMSF) are deemed unsuitable for use as stoppers. Ideally, stoppers should be polyester, used in double configuration, flexible yet stiff, about two meters long, thinner than the mooring line, and approximately half its strength.
	10.23	Question 10.23 has been deleted from the RISQ.
Section 11	11.1	Question 11.1 has been deleted from the RISQ.
Section 12	12.4	Question 12.4 has been deleted from the RISQ.
	12.5	Question 12.5 has been updated to include the requirements for security charts. It now asks whether updated security charts and relevant publications have been provided, and whether a voyage risk assessment has been completed, if the vessel has transited or is expected to transit a high-risk piracy area. The requirement for security charts has been moved under Question 12.5 to align with the updated question. ADMIRALTY Maritime Security Charts now form part of the expected documentation when transiting high-risk piracy areas. These charts provide verified, safety-critical information from UKHO, NATO, and other government bodies, covering threats such as piracy and terrorism, along with security procedures, reporting requirements, and weekly updates via Notices to Mariners and SRIM services. The guideline now directs readers to the Introduction to Best Management Practices for Maritime Security (BMP) publication for further details.

	12.7	The guideline now defines a "Cyber Security Expert Firm" as a professional entity with proven expertise in certifying management systems under the ISM Code and / or cyber security frameworks aligned with ISO/IEC 27001 standards. These firms are equipped with the necessary skills, technology, and training to protect sensitive enterprise data from internal and external threats.
Section 13	13.1	The guideline now recommends conducting OPL override drills every three months, or immediately after sailing if key crew members have changed. Drills should be realistic and cover the purpose, limitations, and reset procedures of the OPL. If a drill would breach regulations, a video demonstration may be used instead. Additionally, toolbox talks should be held daily and before major maintenance work to review tasks, identify risks, and encourage crew input on safety measures and PPE.
	13.4	The guideline clarifies that a finding should not be recorded if the vessel's machinery space is manned for operational reasons, such as transiting high-risk piracy areas, hold cleaning, post dry-dock (up to one month), or maneuvering, provided there are sufficient engineers and crew onboard to safely man the space.
	13.11 and 13.12	The requirement for critical spare parts has now been incorporated under Question 13.11, resulting in the deletion of Question 13.12.
	13.15 and 13.16	The condition of lighting in the engine room has been incorporated under Question 13.15, which now asks whether all areas of the machinery space are well illuminated, emergency escape routes clearly marked and unobstructed, and whether the crew is familiar with these routes. The guideline now requires that emergency escape routes be regularly checked. When an escape trunk is fitted, door seals and lighting must be operational, and doors should be easy to operate. While doors should not be locked in port, any security measures should still allow safe escape. Locking arrangements should support both escape and access, and doors requiring keys for escape are not permitted. Embarkation decks should remain accessible from open decks connected to escape routes. This update has led to the deletion of Question 13.16.
	13.7	The guideline now requires that emergency generator testing be conducted under load at least once every quarter.
	13.22	The question and inspection guideline have been updated to restrict the requirement for insulating matting to only the front and rear of main and emergency switchboards. Individual machinery starter boxes do not require matting. If composite insulating deck materials are used, documentation must verify their safe working voltage. Insulating matting should meet IEC 61111:2009 Class 0 (up to 1,000 volts) or higher for greater voltages, with acceptable evidence including manufacturer certification or markings.
	13.29 and 13.30	The requirement for engine room bilge cleanliness has been incorporated into Question 13.29, resulting in the deletion of Question 13.30. The updated question now covers housekeeping standards in the machinery space and steering gear room, including the condition of the engine room bilge.
	13.31	The requirement for testing high-level alarms in engine room bilges has been updated. These alarms should now be tested at least once daily when the engine room is manned, and as part of the pre-UMS checks before switching to Unmanned Machinery Space (UMS) mode.
	13.33	The requirement for battery maintenance and renewal has been incorporated into Question 13.33, along with updated guidance on boiler and cooling feedwater management. Inspectors should record a finding if feedwater management, including routine testing and documentation, is not being conducted. The guideline emphasises the use of distilled or properly treated water to prevent corrosion, scaling, and biofouling. It outlines risks associated with untreated water and details proper chemical treatment, monitoring, and boiler blowdown procedures. Both bottom and continuous blowdown methods are described, and log sheets for boiler and cooling systems should be maintained to ensure effective water management and system reliability. A detailed guideline on battery maintenance has been added, emphasising the risks of thermal runaway, especially in VRLA and AGM batteries, due to overcharging and poor ventilation. It outlines the characteristics, suitable applications, and replacement timelines for various battery types, including flooded, VRLA wet cell, gel, and AGM. Batteries should be stored in well-ventilated, explosion-proof compartments and maintained with documented service logs detailing type, location, service dates, inspections, and corrective actions. Safe stowage is critical to minimize fire or explosion risks. For reference, the ATSB accident report on air receiver pot explosion is available in the guidelines. The guideline now requires that ship-specific procedures for monitoring economizer water levels be documented within the Safety Management System (SMS). These procedures must include clearly defined corrective actions in the event of water loss. Inspectors should verify the presence and clarity of these procedures during inspections.
	13.37	The guideline now requires that the spare starter motor be tested periodically to ensure operational readiness. This routine check helps confirm the motor's reliability in emergency situations.
	13.43	The question has been revised to focus on the legibility and condition of the rudder angle indicator at the emergency steering position, removing the previous requirement for specific color coding. It now asks whether the indicator is in good order and clearly legible.

	13.44	Question 13.44 has been deleted from the RISQ.
Section 14	14.7	Question 14.7 has been deleted from the RISQ.
	14.9	The guideline has been updated to remove the instruction for inspectors to "record a finding when wire stays with sheathed plastic are used to secure mast heads". Instead, it now requires that the vessel's manager establish a procedure for the regular inspection and maintenance of such wire stays.
	14.10	The guideline now acknowledges the installation of an isolation transformer as an acceptable alternative method to enhance the safety of portable cargo lights. It has been updated to require inspector "Record a finding if the lamp holder, drip shield, and shade of portable cargo lights are not non-conductive, do not isolate the crew from electrical shock hazards by the installation of an isolation transformer, or have a voltage greater than 50V AC (1-1000Hz) or 120V DC."
	14.12	Question 14.12 now includes requirements for the safe storage of chemicals and gases. It asks whether chemicals and gases are stored correctly and whether areas like paint lockers, battery rooms, and flammable storage spaces are equipped with proper ventilation, explosion-proof lighting, and functioning safety fittings. The guideline also highlights the need to consult Safety Data Sheets (SDS) before using hazardous substances, ensure SDS are accessible in storage areas, and follow best practices for segregation, location, stowage, and stock control to minimize risks, especially for substances not covered by the IMDG Code.
	14.14	Question 14.14 has been deleted from the RISQ.
	14.15	The guideline for deep-fat fryers now includes SOLAS II-2 Regulation 10.6.4 requirements for ships built on or after 1 July 2002. These fryers must be equipped with a certified fire-extinguishing system, dual thermostats with alarms, automatic power shut-off, galley alarms, and clearly labelled manual controls for crew use.
Section 15	15.1 and 15.8	Question 15.1 has been updated to include the requirement for individual monthly statement payments. It now also covers the guidance previously provided in Question 15.8, which has therefore been deleted from the RISQ.
	15.2	Additional points have been added to the accommodation inspection guidelines to enhance safety and health measures. These include the fire risk associated with lint buildup inside and under dryers, which can be mitigated by regularly cleaning dryer vents, hoses, and filters. The guidelines also address the risk of Legionella bacteria proliferation, recommending weekly flushing of all freshwater outlets, particularly 'dead legs' (pipes leading to removed or rarely used outlets), infrequently used outlets such as showers in hospital rooms, and areas with long runs in the water distribution system.
	15.6 and 15.7	Question 15.6 has been updated to include requirements for providing medical facilities and free doctor visits at ports of call. It also emphasizes that health protection and medical care, including essential dental care, should be available free of charge to all seafarers. Medical logs and visit reports should be kept up to date using a standard, confidential medical report form. As a result, Question 15.7 has been deleted.
	15.9	Question 15.9 has been expanded to include guidelines addressing violence, harassment, bullying, and sexual assault on board. It now asks whether the vessel's manager has developed a policy and procedure covering these issues, whether the policy is communicated to crew in their native language, and whether crew are familiar with the complaint procedure. The guideline outlines a zero-tolerance approach, defines types of harassment, mandates training and reporting channels, and underscores the importance of confidentiality and support for affected crew.
	15.10	Question 15.10 has been revised to include a training requirement. It now asks whether the vessel has adequate policies addressing mental health and whether key onboard personnel—such as the Master, Chief Engineer, Chief Officer, and Second Engineer—have received training to help them recognise signs of mental health issues. The guidelines recommend that companies provide or arrange training for management-level personnel both onboard and ashore, covering how to identify mental health concerns, facilitate discussions, and support affected individuals.
	15.13	Question 15.13 has been deleted from the RISQ.
Section 16	16.3	The inspection guide has been updated to clarify the application of this question. Inspectors are instructed to mark it as N/A if the vessel is not operating in polar regions at the time of inspection. Comments should only be made if officers hold the relevant certificates.
Section 17	Introduction to Section 17	The introduction to Section 17 has been clarified to state that it applies only when the bulk carrier has conducted STS operations in the past 12 months, is currently involved, or intends to engage in such operations.
	New question	A new question (17.11) has been introduced with an inspection guide and a unit-specific questionnaire to help ship managers assess the condition and operational standards of units intended for STS operations. This enables risk evaluation and supports necessary arrangements, mitigation measures, and actions to enhance the safety of STS operations.

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The hyperlinks of the references in the bibliography list have been updated and verified for accessibility. If, following the document dissemination, any broken hyperlinks are discovered, please contact RISQ@rightship.com

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